

Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module to operate in a memory system with a memory controller, the memory system operating according to a system clock, comprising:

data buffers, and

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with a local clock signal received from the respective one of the data buffers.

2. The memory module of claim 1, further comprising:

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices.

3. The memory module of claim 2, wherein the respective buffer circuit includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.

4. The memory module of claim 2, wherein the plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time, and

wherein the respective buffer circuit is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.

5. The memory module of claim 1, wherein the respective buffer circuit includes one or more programmable registers controlling a phase of the local clock signal.

6. The memory module of claim 5, further comprising:

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the buffer circuit is configured to program the one or more registers according to the data buffer control signals.

7. The memory module of claim 1, wherein the respective buffer circuit is configured to regenerate the local clock signal from a received clock signal, and to phase-lock the local clock signal with the received clock signal.

8. The memory module of claim 1, wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks.

9. A method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising:

receiving a system clock signal;

generating a module clock signal;

transmitting the module clock signal to a plurality of data buffers coupled between the memory devices and the memory bus;

generating local clock signals associated with respective ones of the data buffers;

transmitting the local clock signals to the memory devices; and

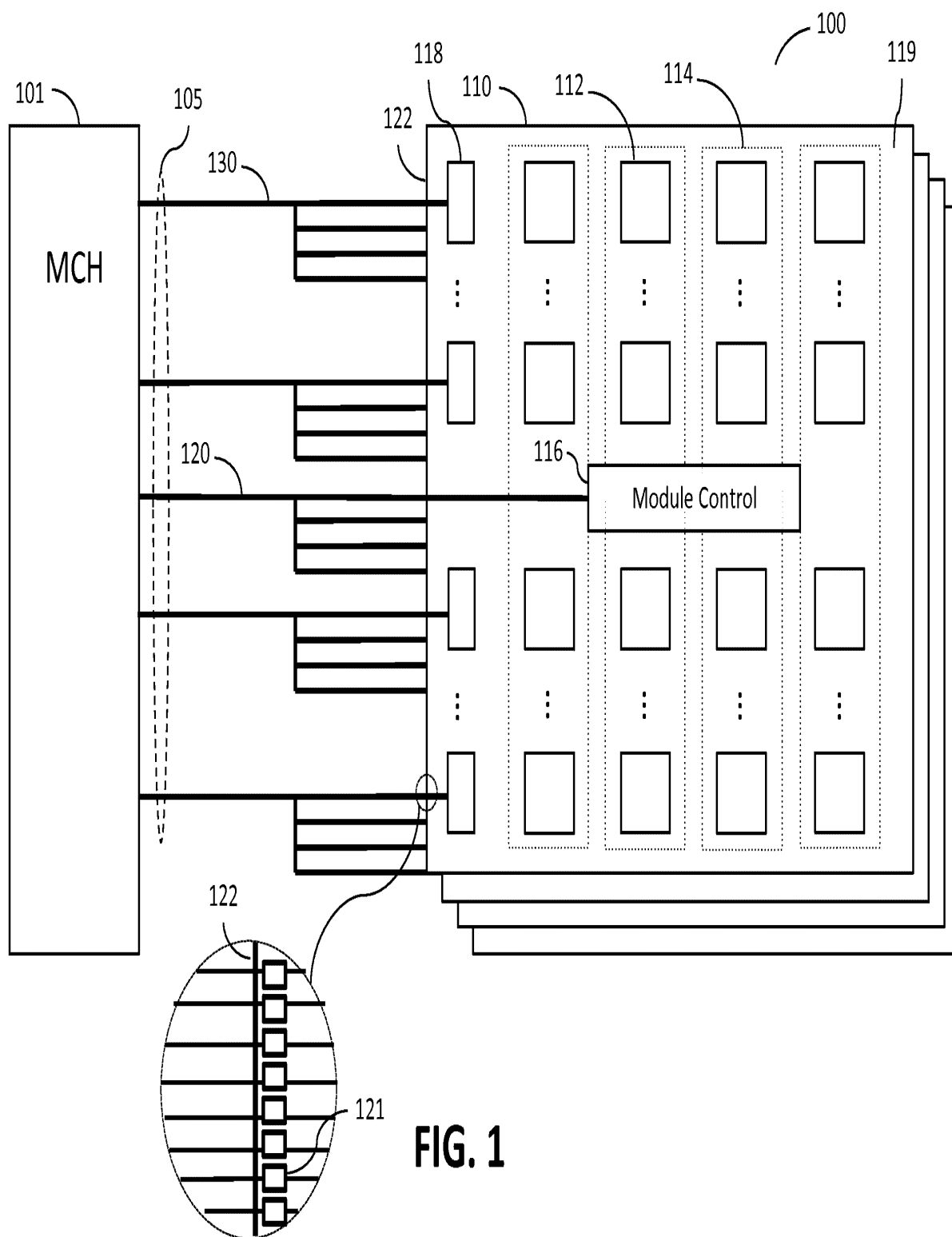
perform read/write operations a the memory devices in accordance with the local clock signals.

10. A method of claim 9, further comprising incorporating a programmable amount of delay to the local clock signals.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS

ABSTRACT OF THE DISCLOSURE

A memory module is operatable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals from the memory controller and to output module C/A signals and data buffer control signals. The module C/A signals are provided to memory devices organized in groups, each group including at least one memory device, while the data buffer control signals are provided to a plurality of buffer circuits to control data paths in the buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices. The plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. The plurality of buffer circuits include clock regeneration circuits to regenerate a clock signal received from the module control device and to provide regenerated clock signals to respective groups of memory devices.



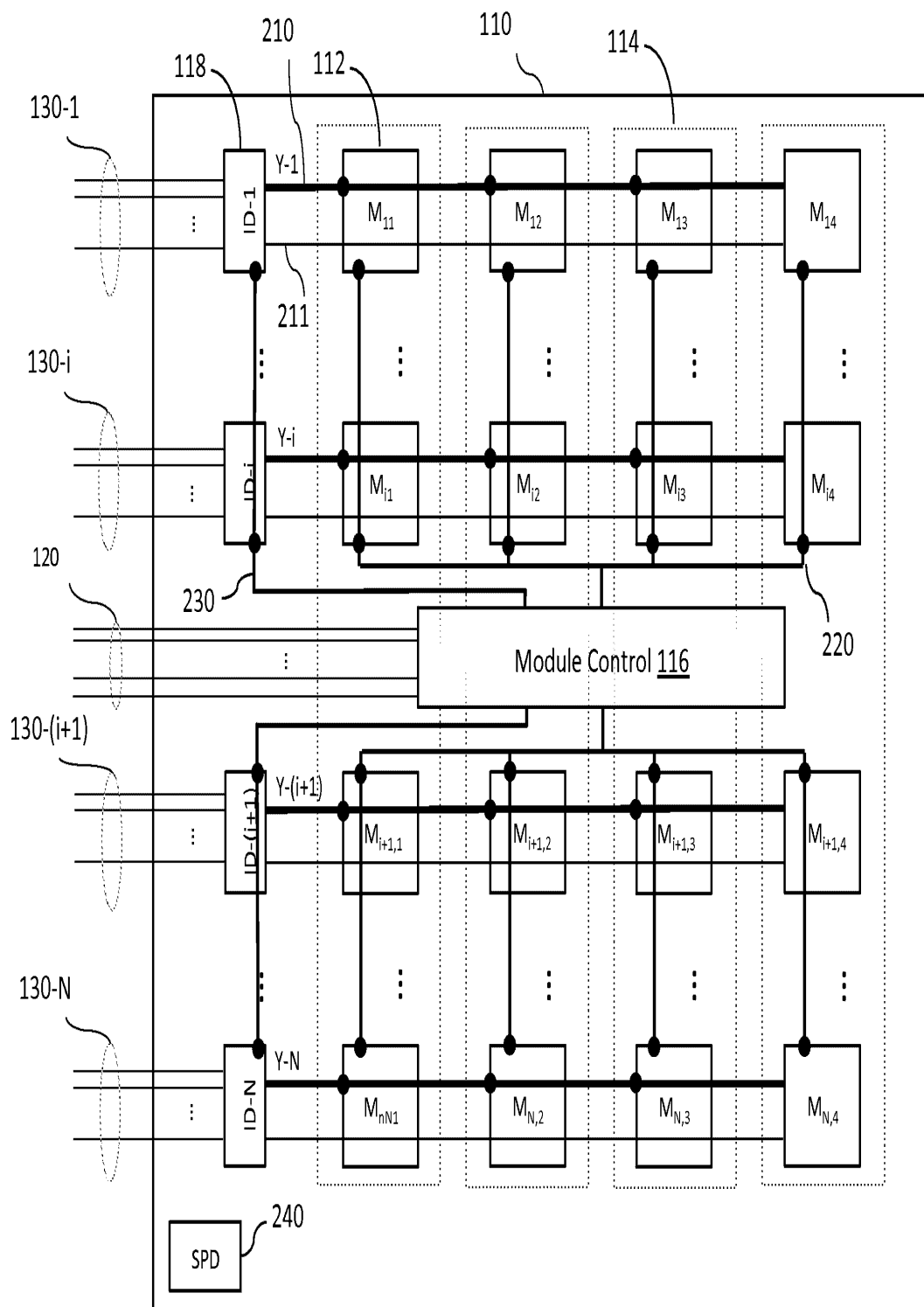


FIG. 2A

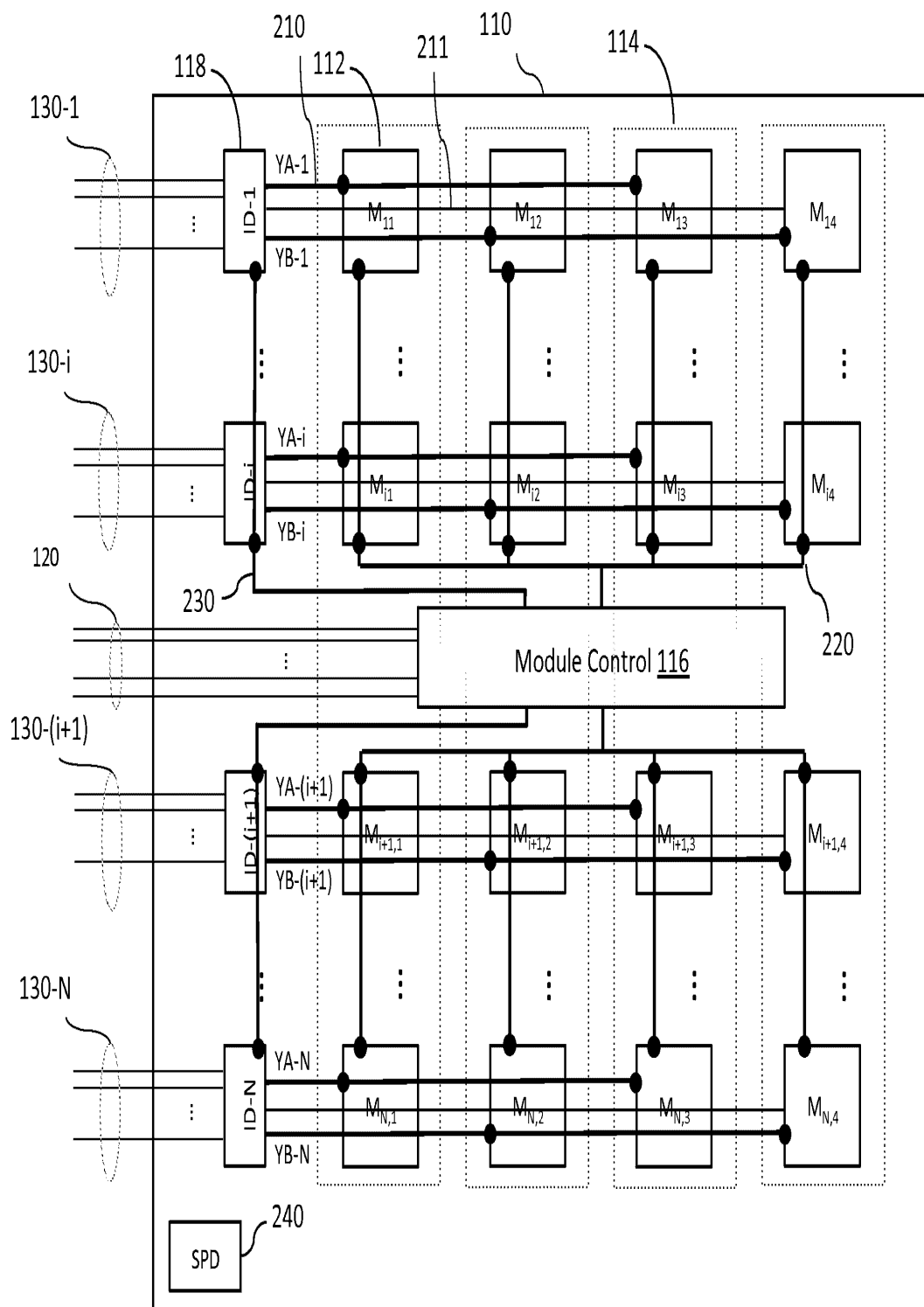


FIG. 2B

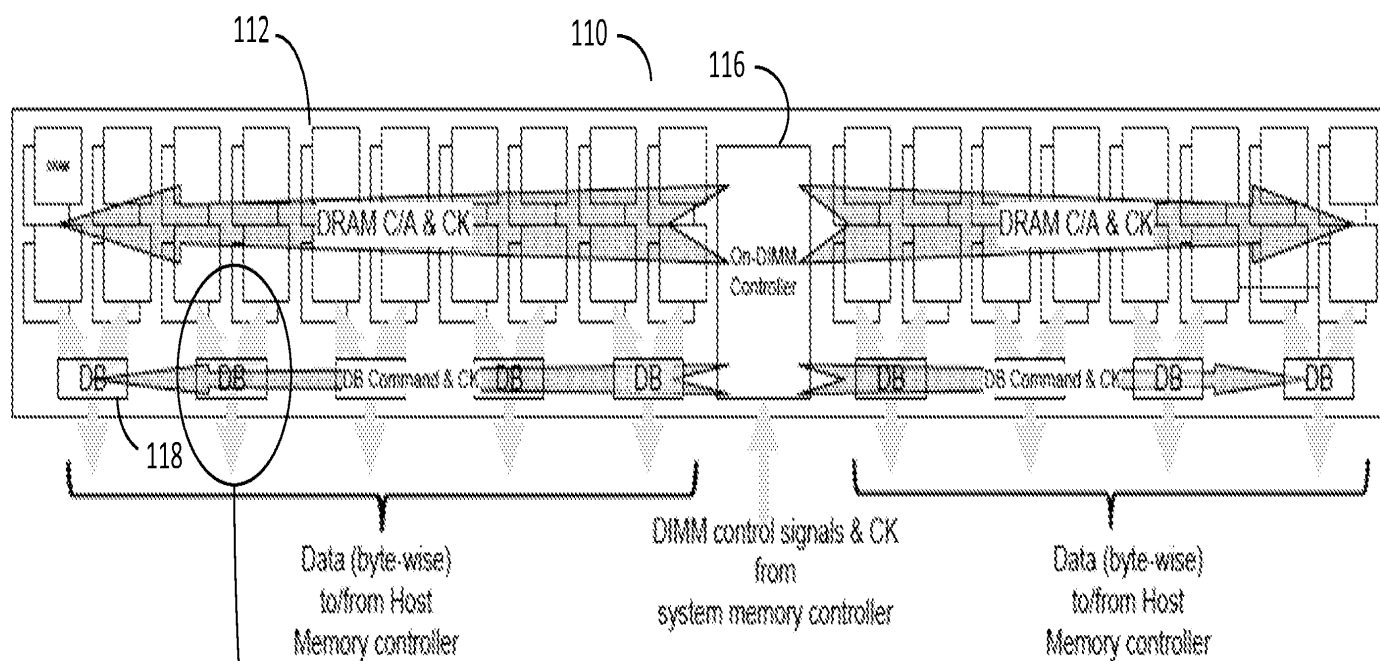


FIG. 2D

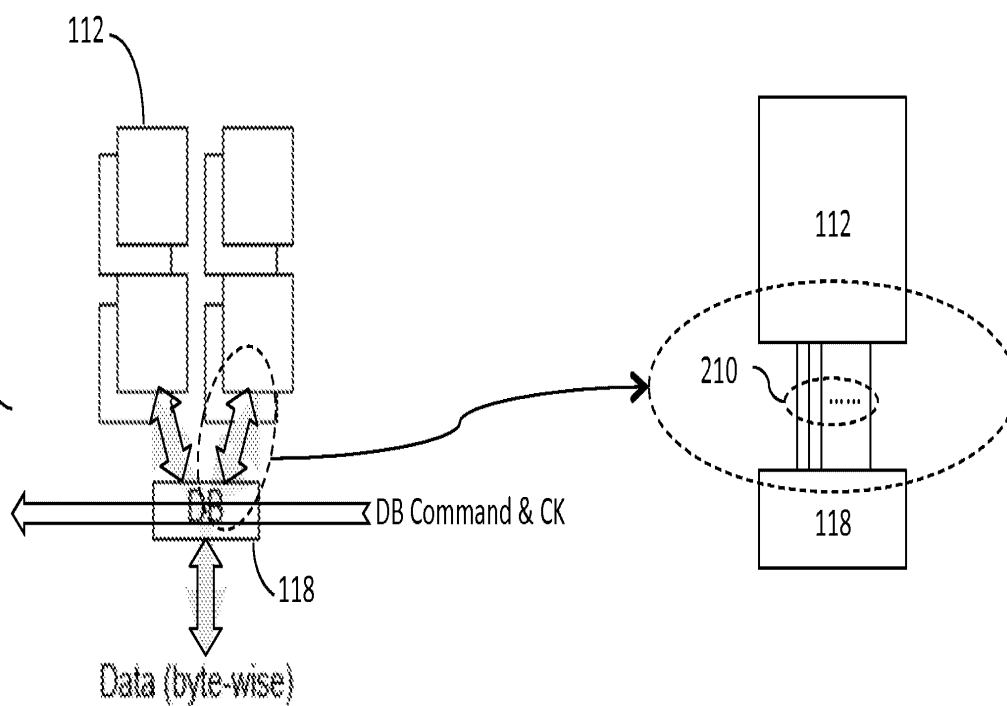


FIG. 2C

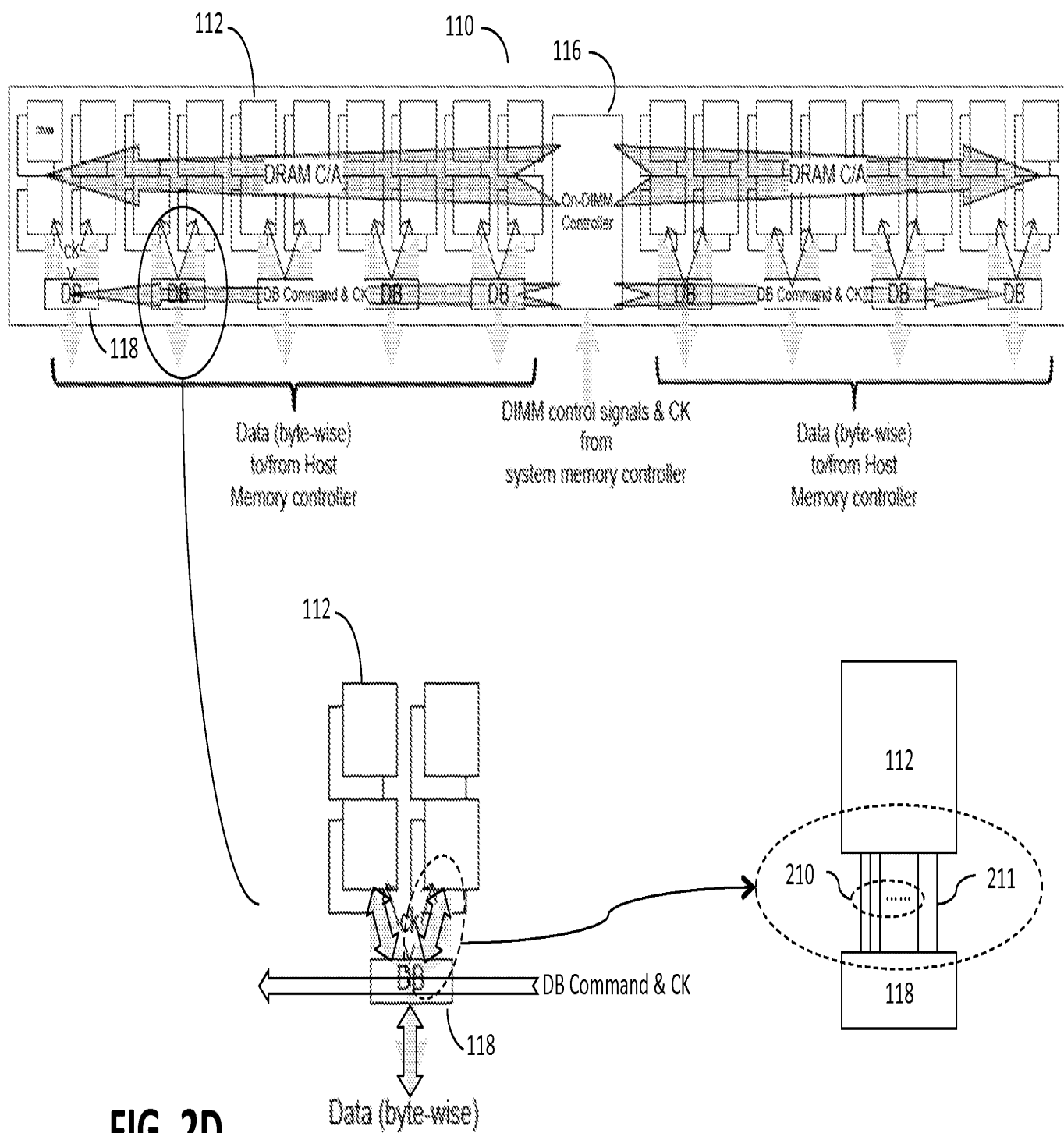


FIG. 2D

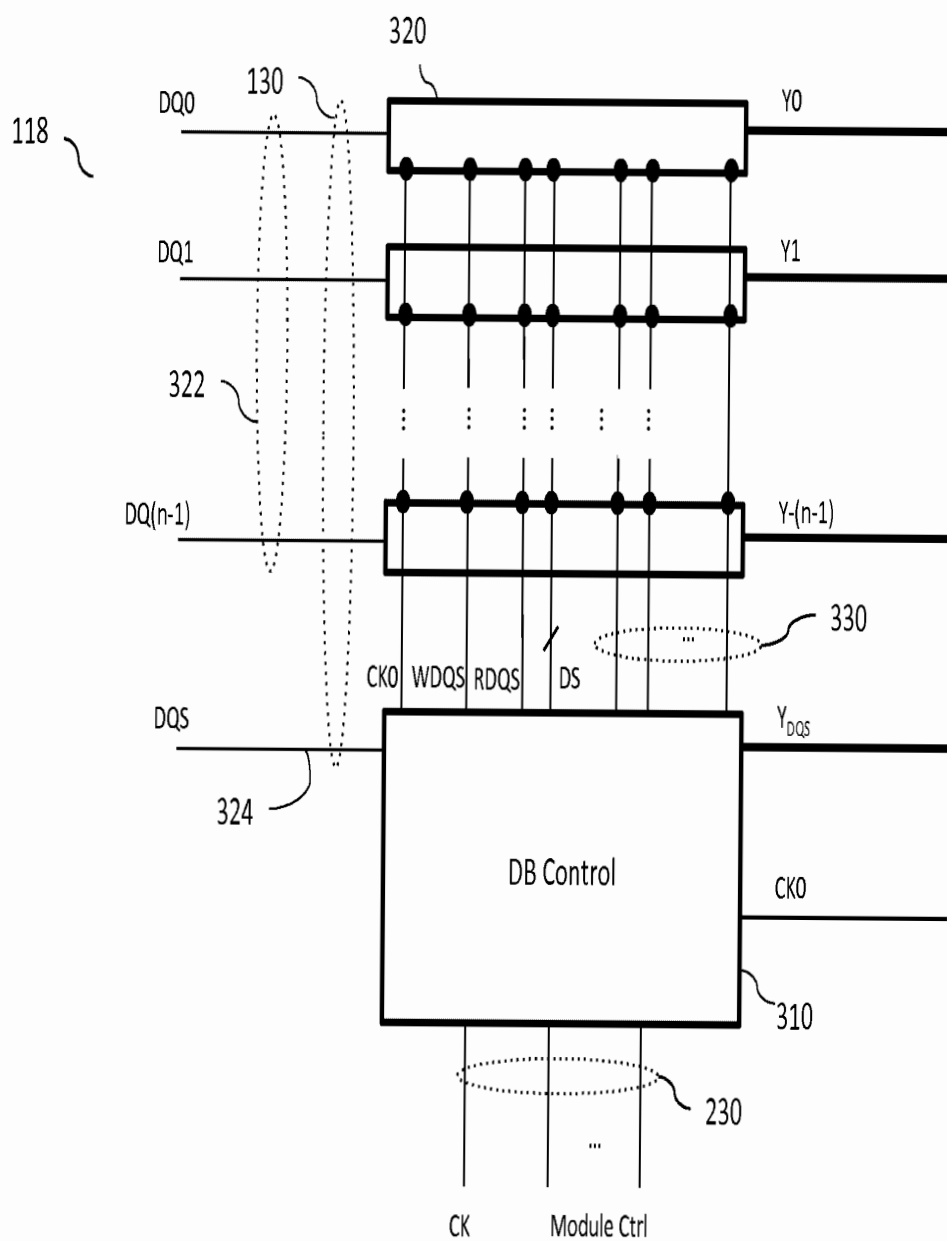


FIG. 3

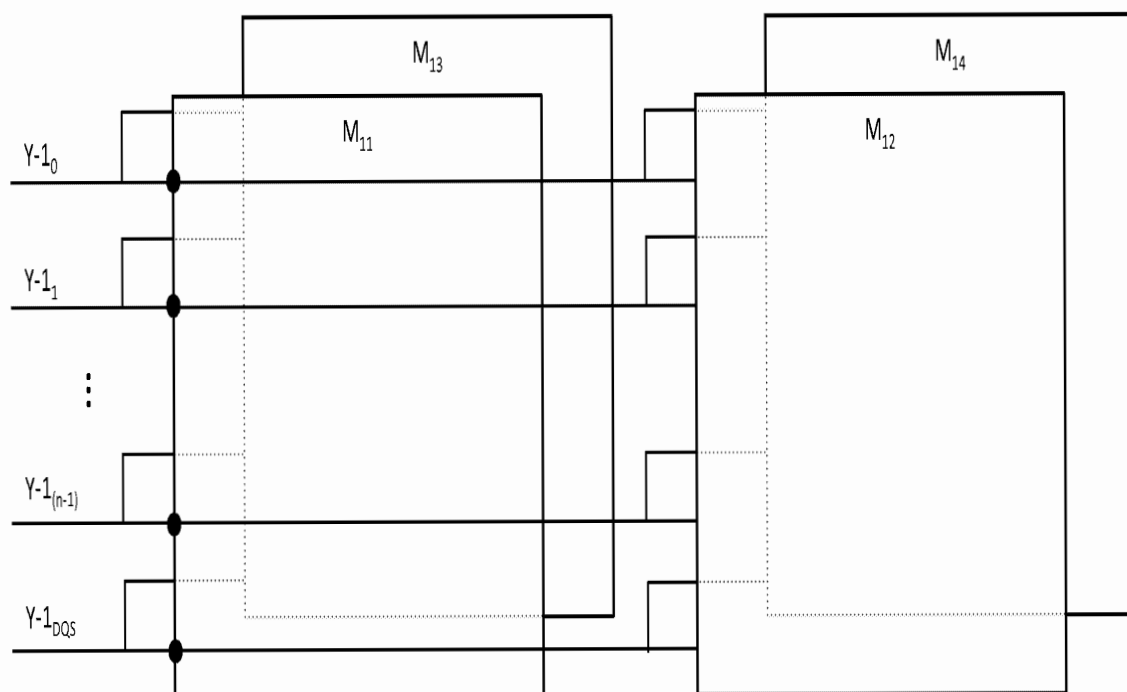


FIG. 4A

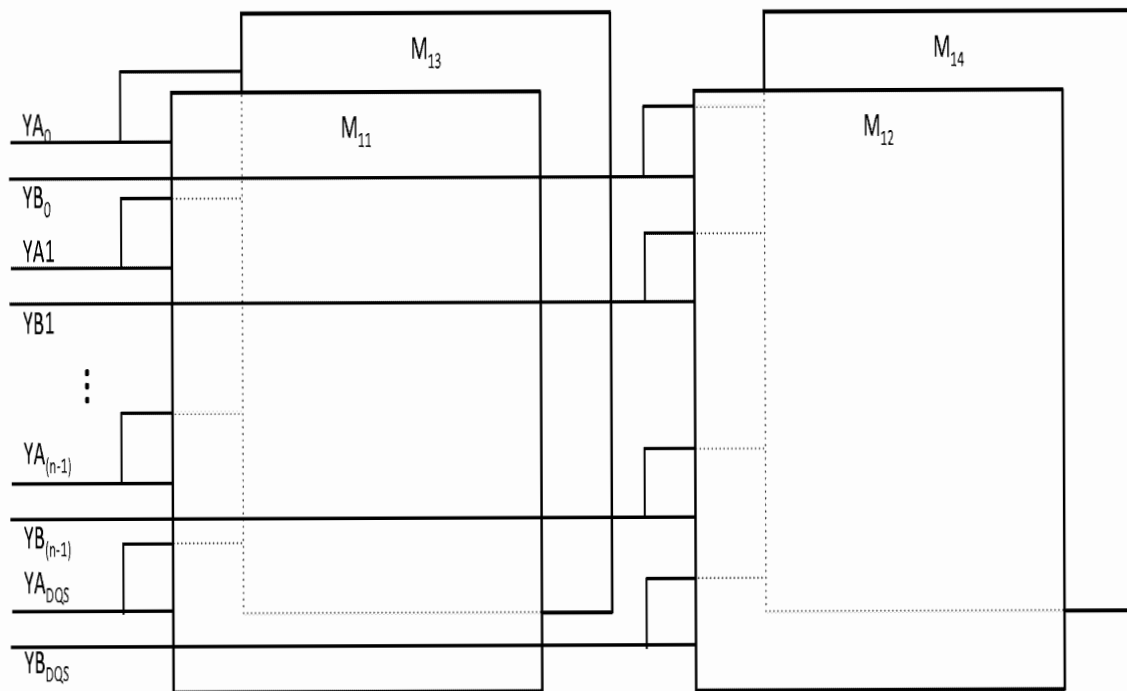


FIG. 4B

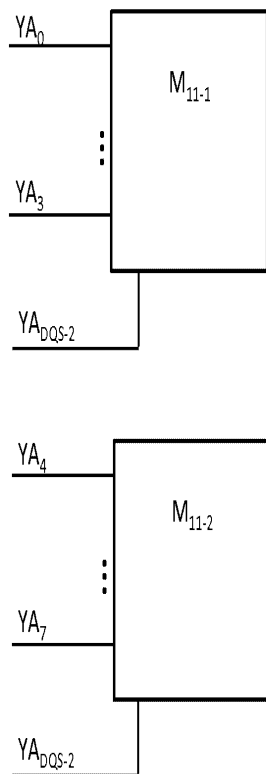


FIG. 5A

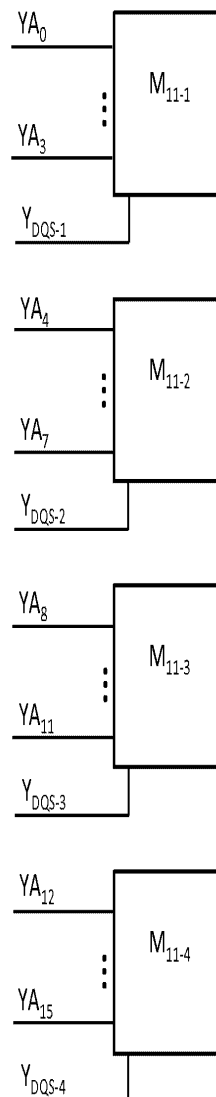


FIG. 5B

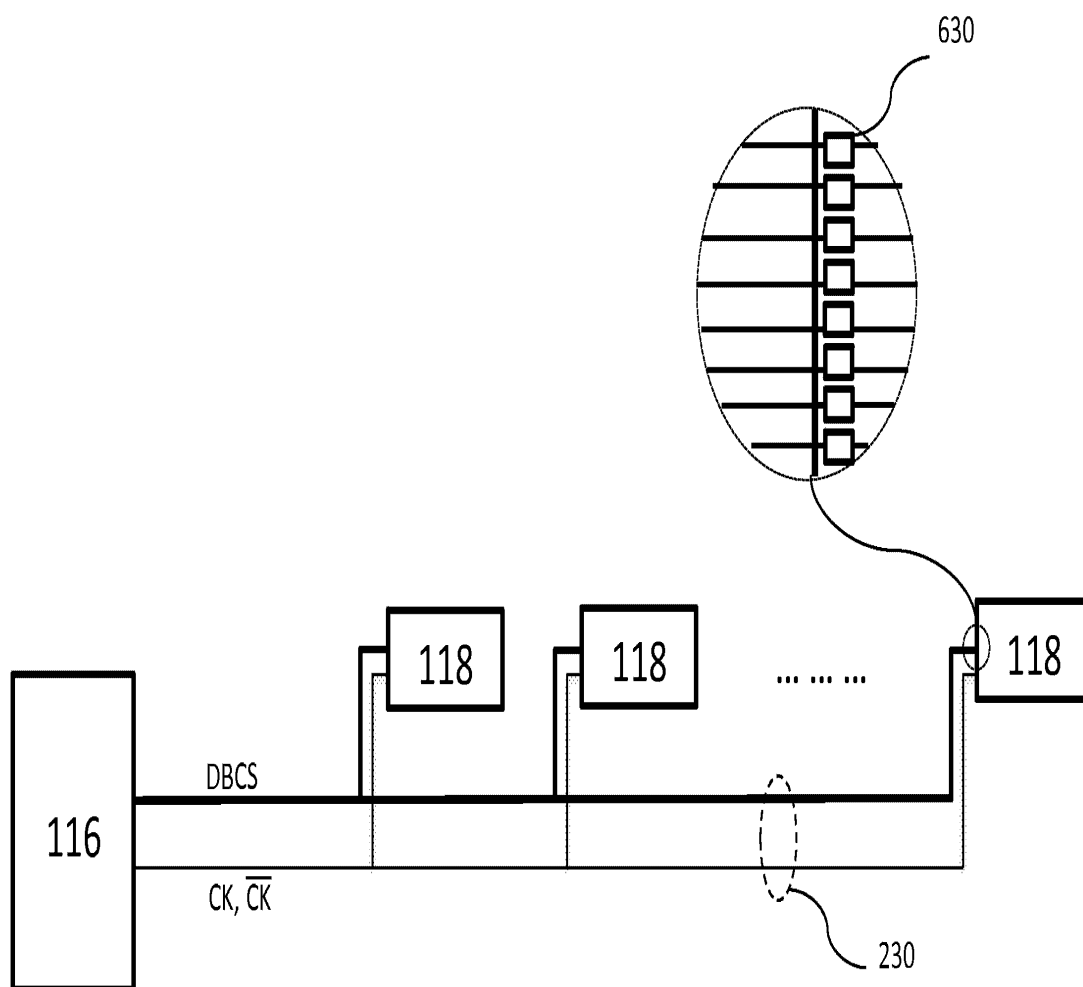


FIG. 6A

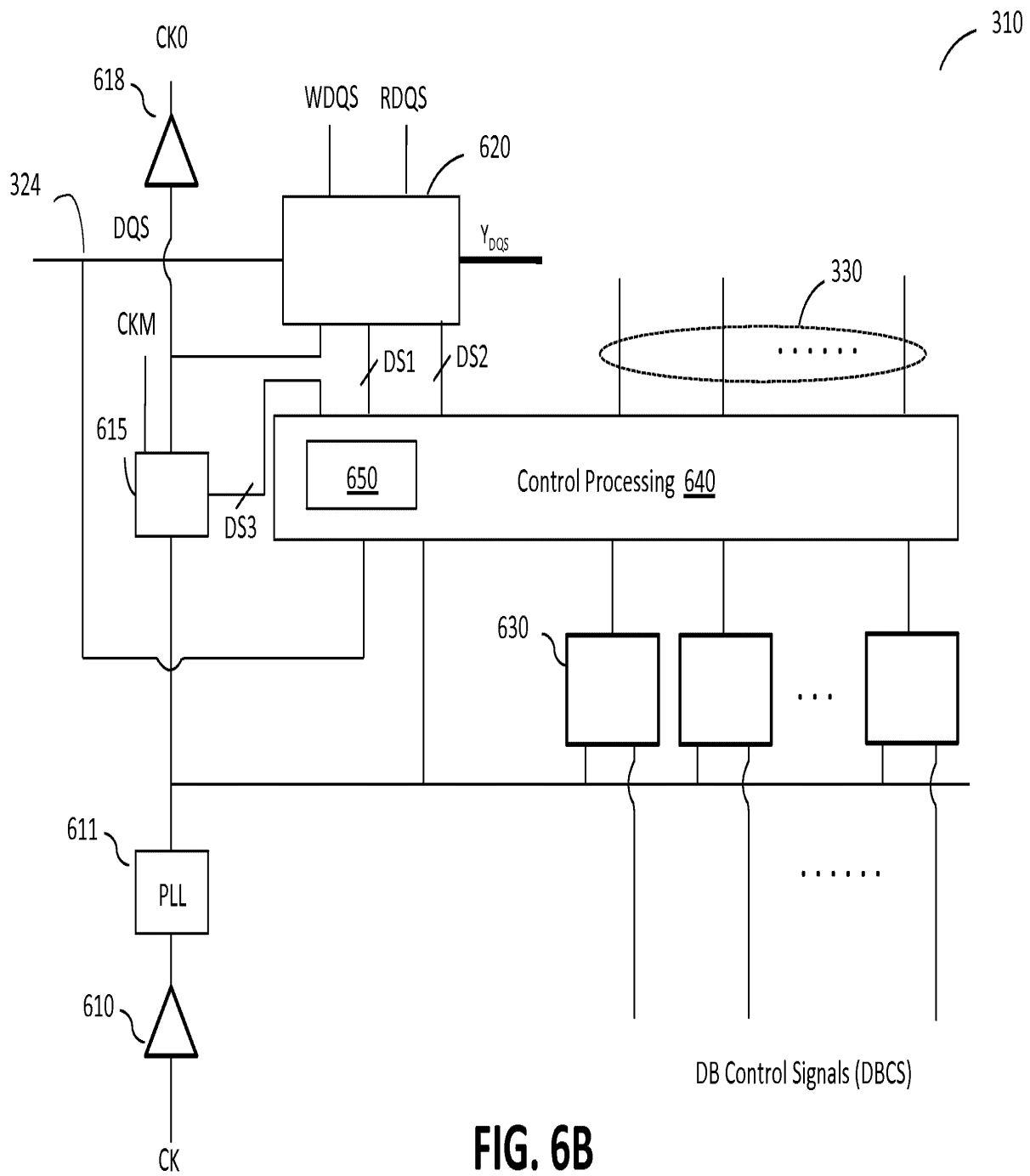
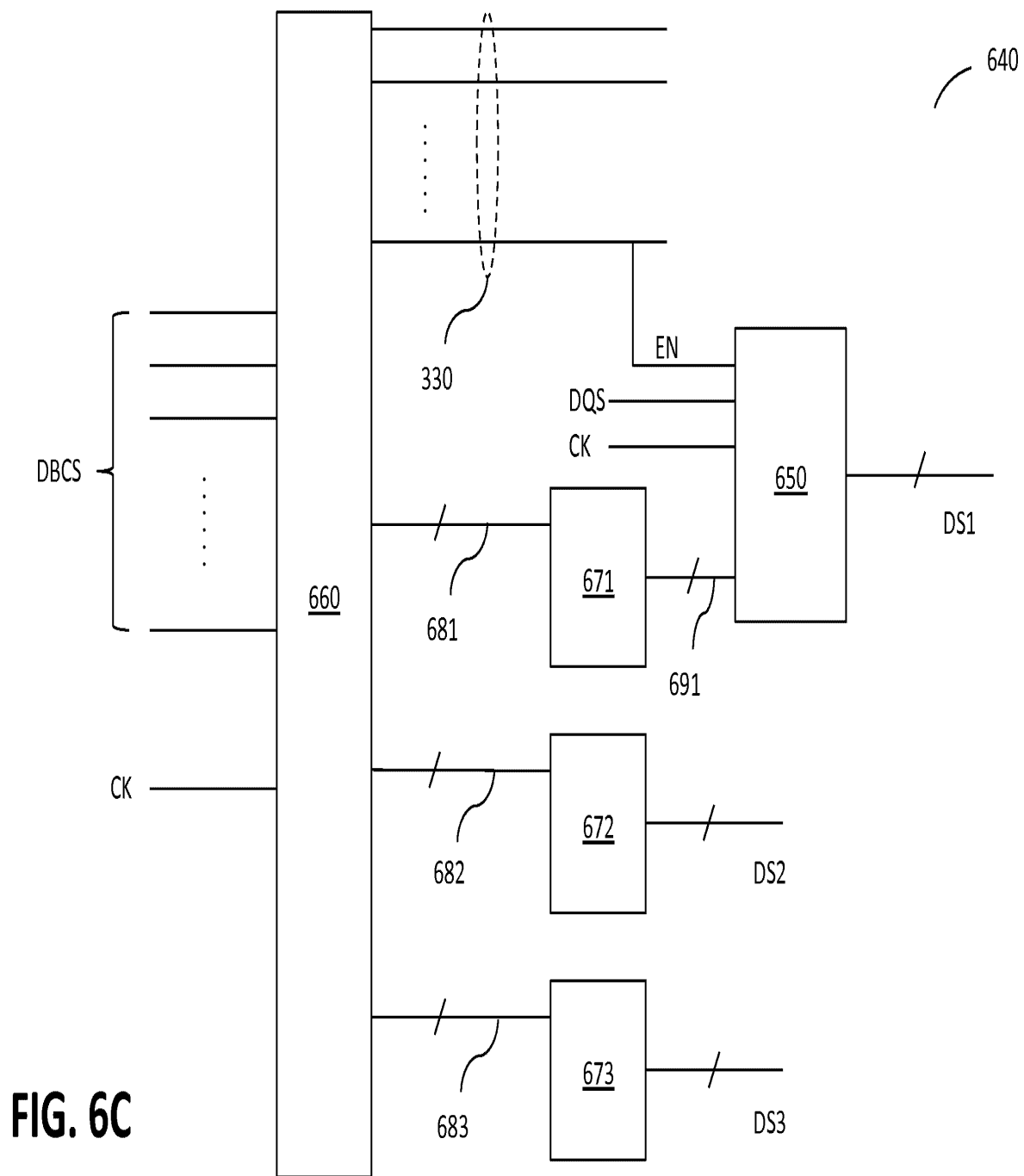


FIG. 6B



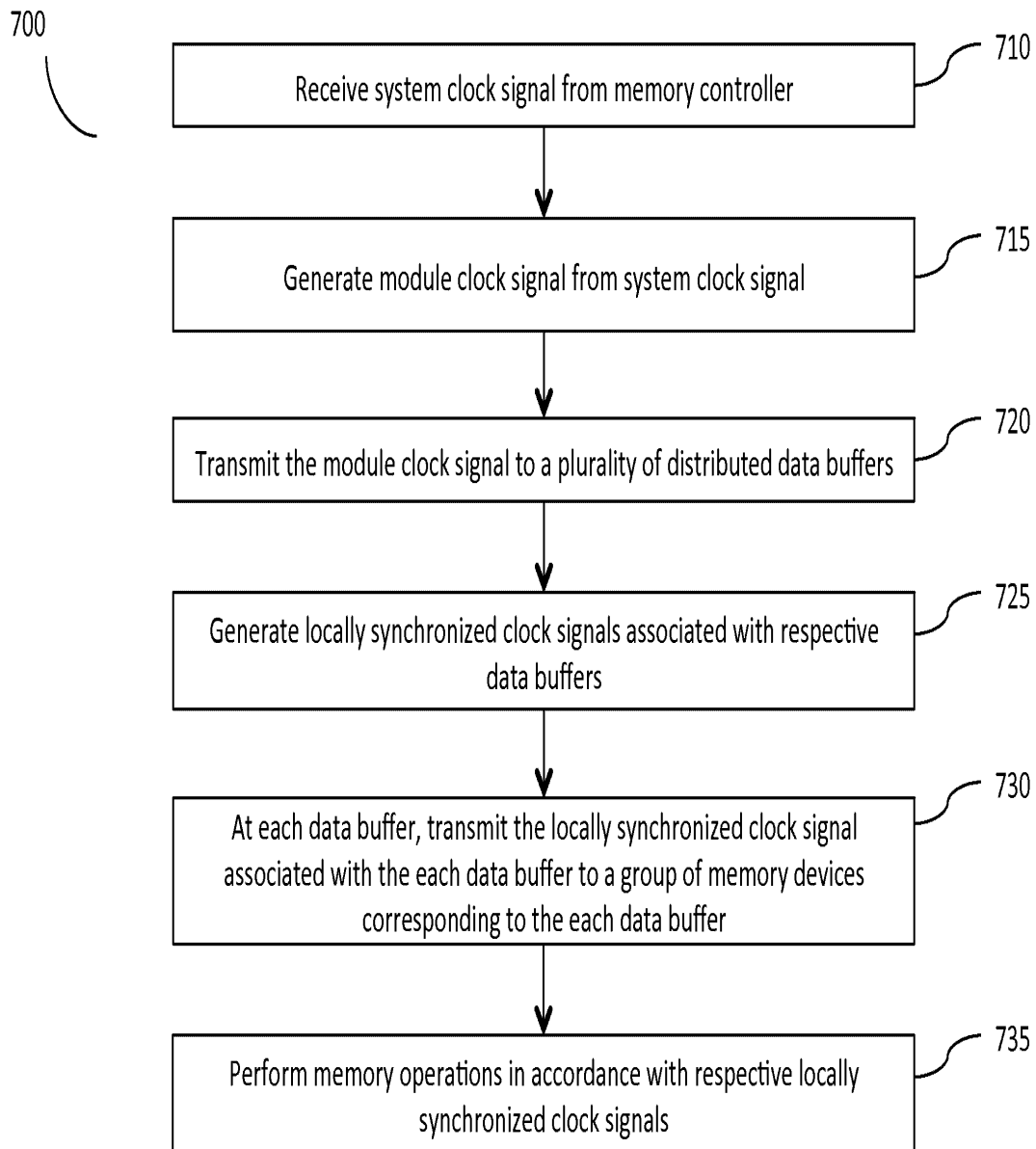


FIG. 7A

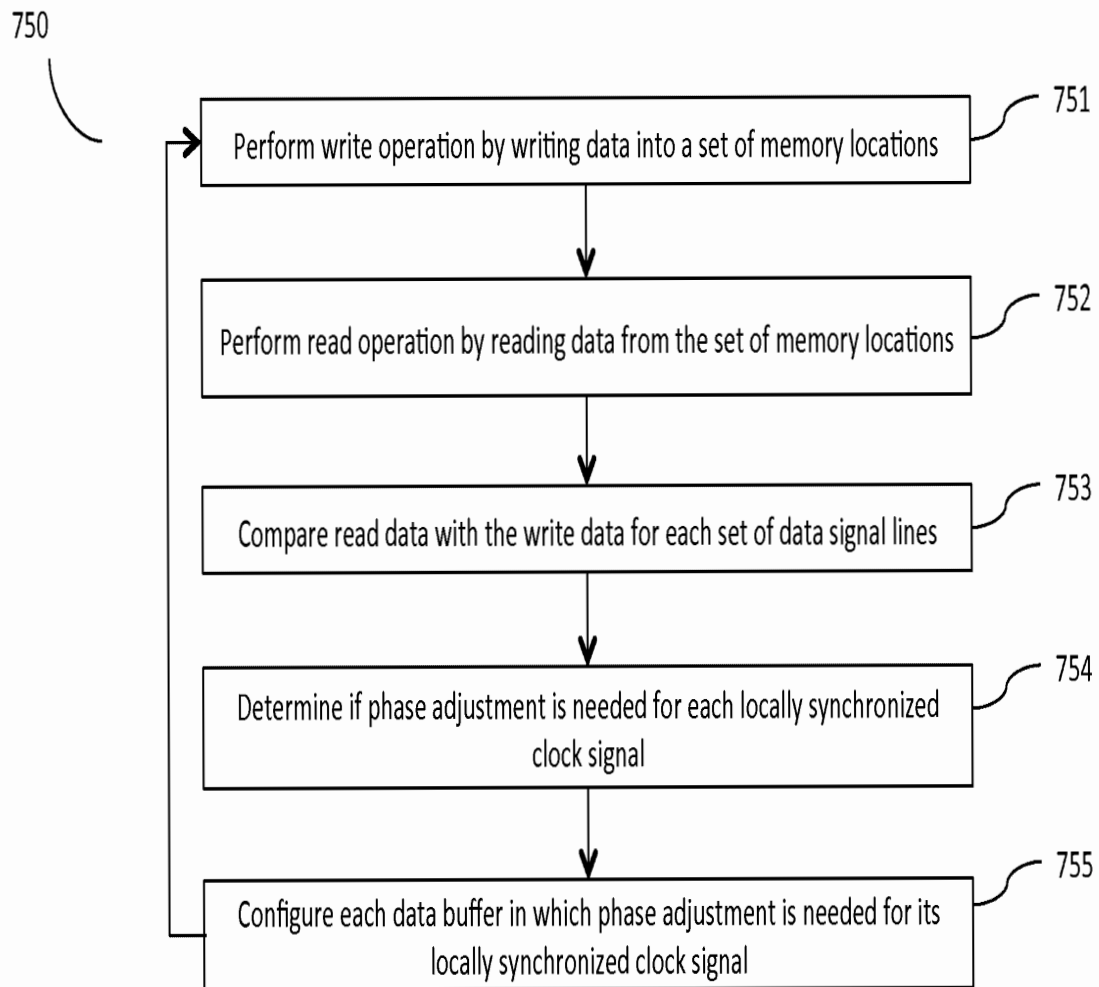


FIG. 7B

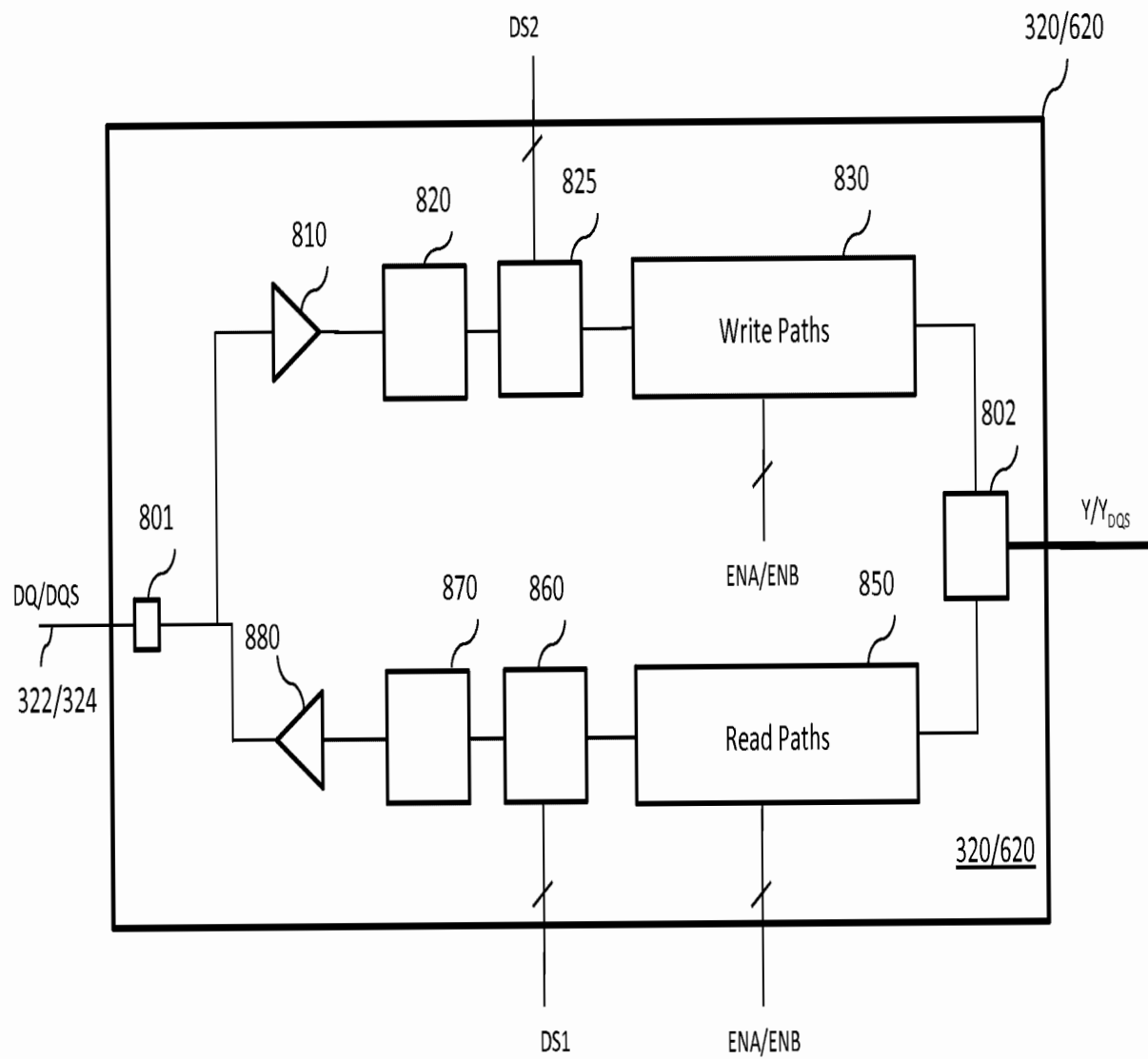


FIG. 8

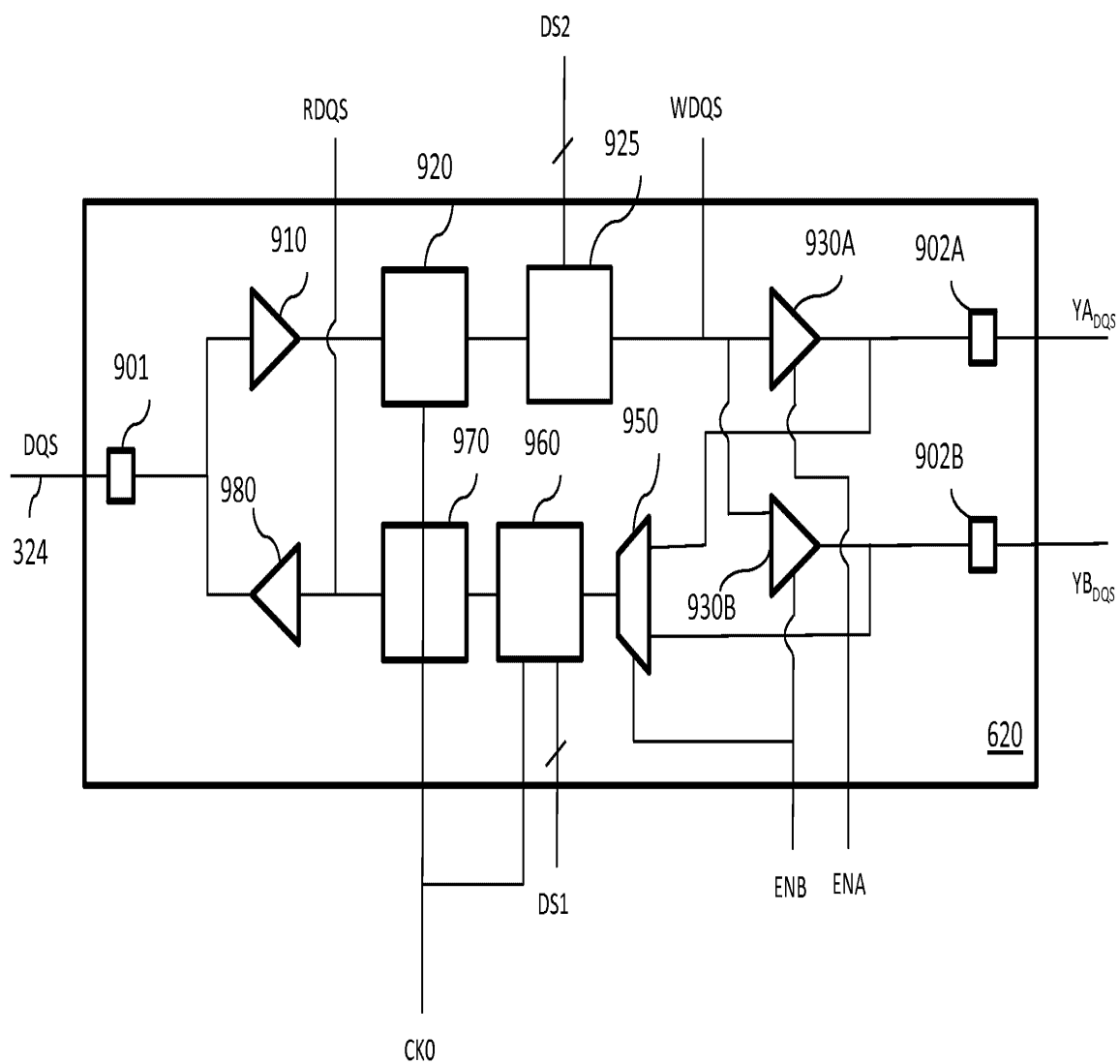


FIG. 9

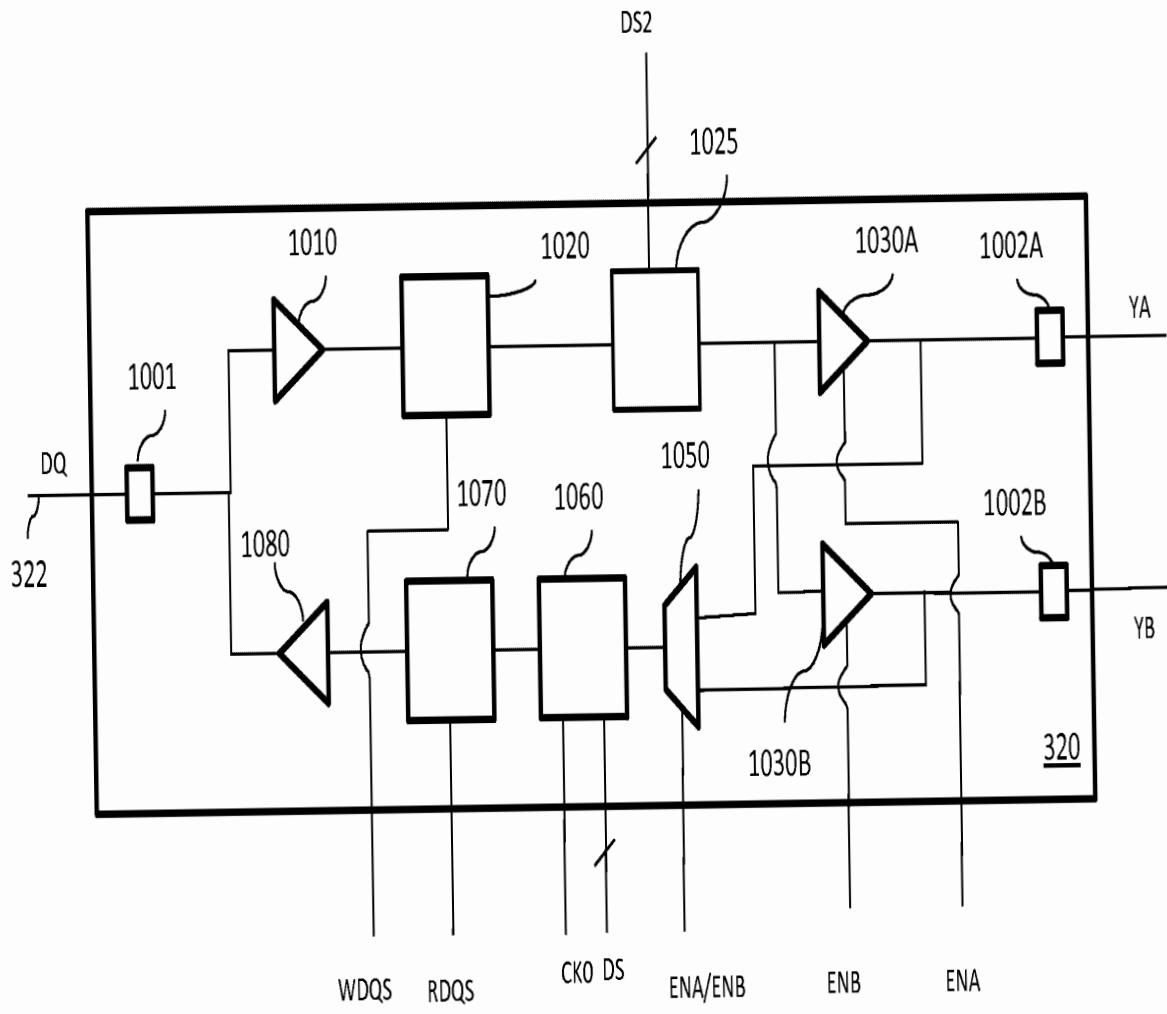


FIG. 10

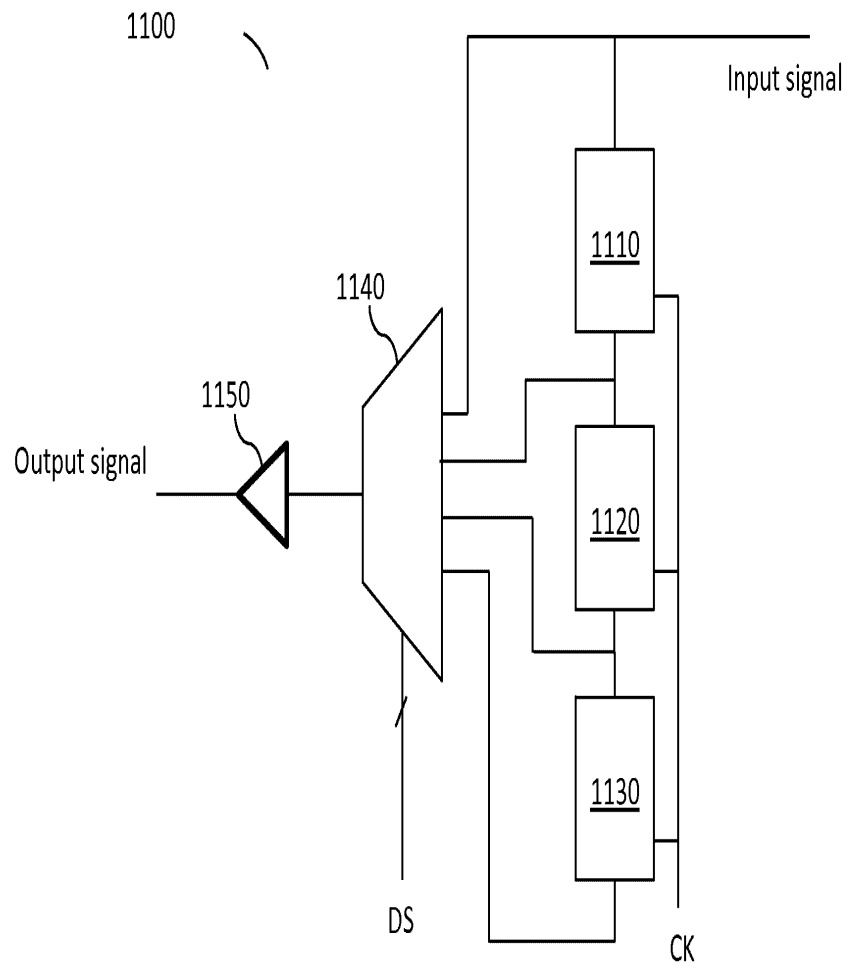


FIG. 11

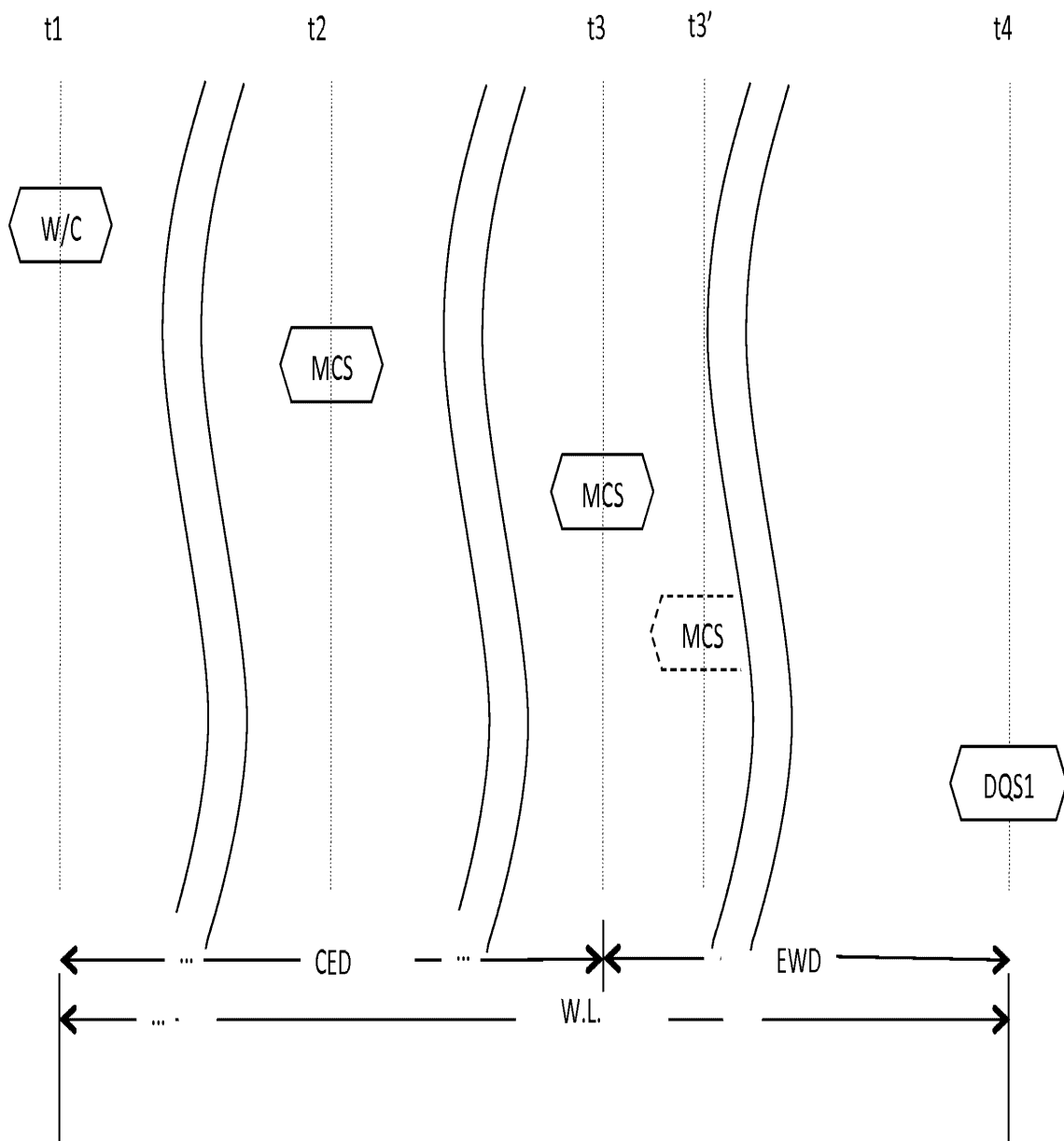


FIG. 12A

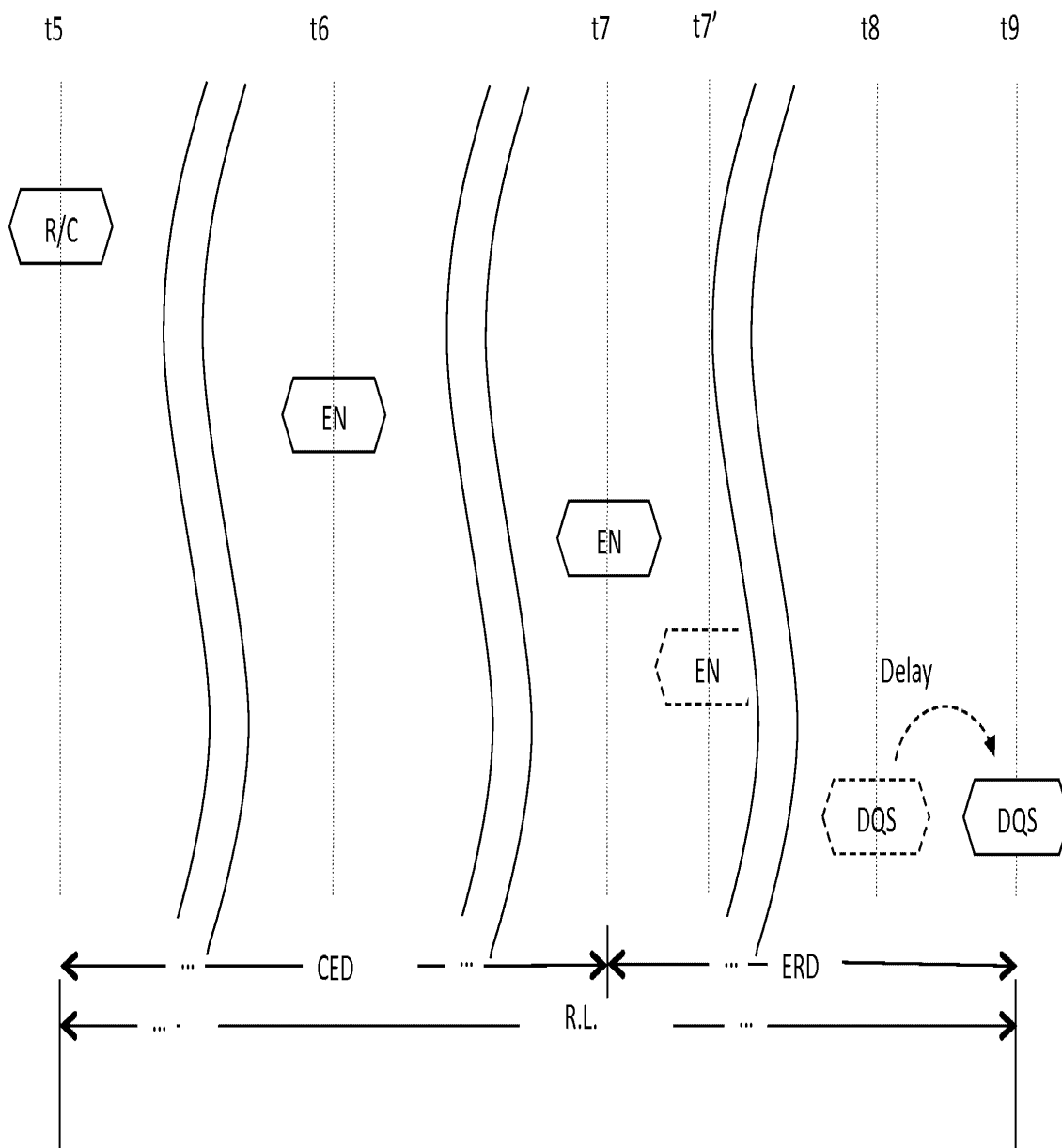


FIG. 12B

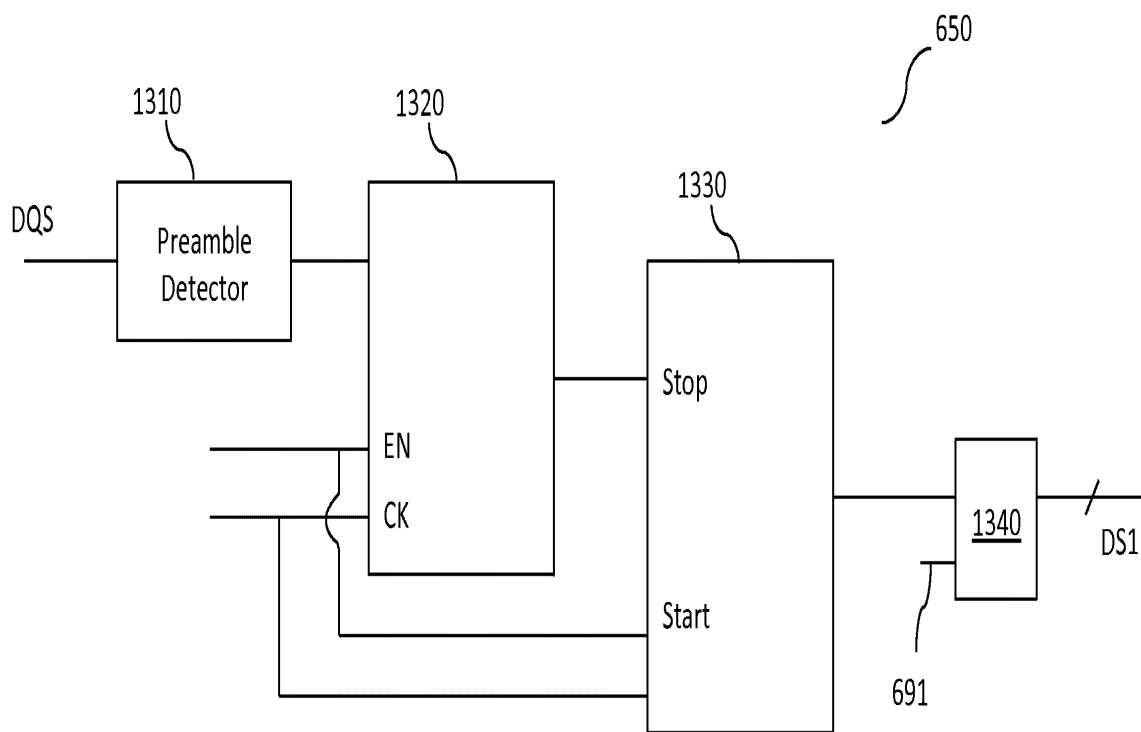


FIG. 13

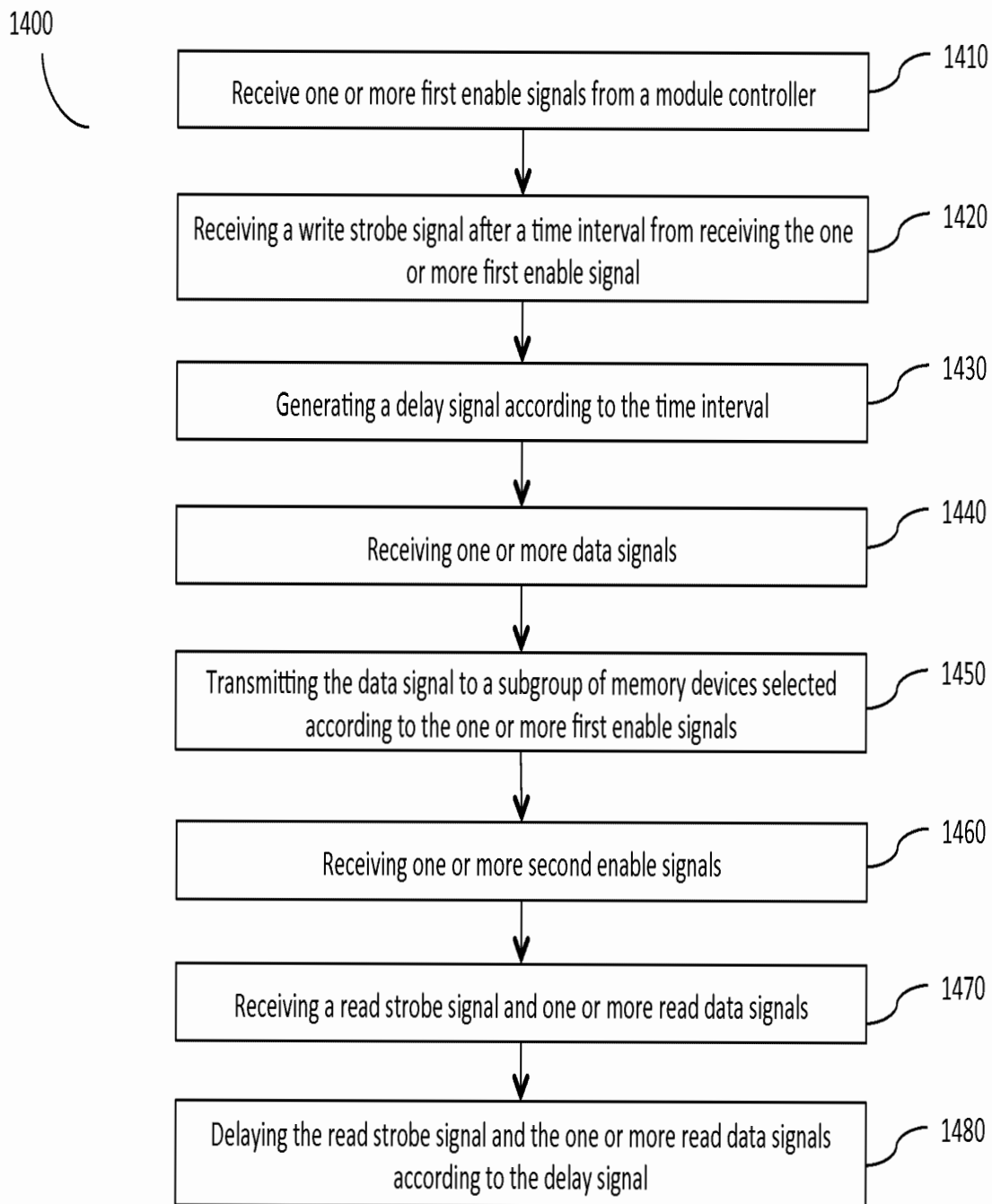


FIG. 14

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NT006US
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		
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	Jayesh	R.	Bhakta		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NT006US
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

☐ An Address is being provided for the correspondence information of this application.

Customer Number	79141		
Email Address	jz@jzpatent.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		
Attorney Docket Number	NT006US	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	2D

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61859215	2013-07-27

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NT006US
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

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			Remove
Application Number	Country i	Filing Date (YYYY-MM-DD)	Priority Claimed
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NT006US
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

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If the Applicant is an Organization check here. ☒					
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Phone Number		949-435-0025	Fax Number		
Email Address		nwhitley@netlist.com			
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NT006US
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

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First Name	Jamie	Last Name	Zheng	Registration Number
				51167

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Application Number:

Filing Date:

Title of Invention:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Jamie Jie Zheng

Attorney Docket Number:

NT006US

Filed as Small Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				730

Electronic Acknowledgement Receipt

EFS ID:	19707961
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	28-JUL-2014
Filing Date:	
Time Stamp:	23:03:49
Application Type:	Utility under 35 USC 111(a)

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Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2014-07-28_Specification_NT006US.pdf	268768 e6ee5603b73c09440ac1b67b0762018eb19a2cfa	yes	33

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Specification	1	29
	Claims	30	32
	Abstract	33	33

Warnings:

Information:

2	Drawings-other than black and white line drawings	2014-07-28_Drawings_NT006US.pdf	282244	no	22
			354d9e7a3f001f0297692b3f169a5c01388ba8cf		

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

3	Application Data Sheet	2014-07-28_ApplicationDataSheet_NT006US.pdf	1437188	no	6
			54128b6bef5de4d366292ec6d4d09e0d4b60b8d		

Warnings:

Information:

4	Fee Worksheet (SB06)	fee-info.pdf	32609	no	2
			90c4a1813e59bafa9b4facfd10fe1a1990f2745		

Warnings:

Information:

Total Files Size (in bytes):		2020809
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

SCORE Placeholder Sheet for IFW Content

Application Number: 14445035

Document Date: 07/28/2014

The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

- Drawing

At the time of document entry (noted above):

- USPTO employees may access SCORE content via eDAN using the Supplemental Content tab, or via the SCORE web page.
- External customers may access SCORE content via PAIR using the Supplemental Content tab.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/445,035	07/28/2014	Hyun Lee	NT006US

CONFIRMATION NO. 9700

FORMALITIES LETTER



OC000000070028711

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 08/08/2014

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **70** basic filing fee.
- \$ **70** surcharge.
- \$ **300** search fee.
- \$ **360** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **800** TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's

oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Hyun Lee
Jayesh R. Bhakta

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/ygizaw/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/445,035

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	10 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/445,035	07/28/2014	2189	0.00	NT006US	10	2

CONFIRMATION NO. 9700

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



OC000000070028710

Date Mailed: 08/08/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 08/06/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/445,035**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No**PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES**

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



**Courtesy Reminder for
Application Serial No: 14/445,035**

Attorney Docket No: NT006US
Customer Number: 79141
Date of Electronic Notification: 08/08/2014

This is a courtesy reminder that new correspondence is available for this application. If you have not done so already, please review the correspondence. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:
jz@jzpatent.com

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>. If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006US			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	2051	1	70	70
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				800

Electronic Acknowledgement Receipt

EFS ID:	20330244
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	04-OCT-2014
Filing Date:	28-JUL-2014
Time Stamp:	00:13:44
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 800
RAM confirmation Number	6116
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Fee Worksheet (SB06)	fee-info.pdf	36650 1d346ef0b66a420858e54612e9ff5fc039263866	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			36650		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/445,035

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	10 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/445,035	07/28/2014	2189	800	NT006US	10	2

CONFIRMATION NO. 9700

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

UPDATED FILING RECEIPT



Date Mailed: 12/09/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 08/06/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/445,035**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/445,035	07/28/2014	Hyun Lee	NT006US

CONFIRMATION NO. 9700

79141

The Law Office of Jamie Zheng, Ph.D Esq.

P.O. Box 60573

Palo Alto, CA 94306

NOTICE



Date Mailed: 12/09/2014

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Hyun Lee
Jayesh R. Bhakta



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	NT006US	9700

79141 7590 11/12/2015
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
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2132

NOTIFICATION DATE	DELIVERY MODE
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11/12/2015

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
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Office Action Summary	Application No. 14/445,035	Applicant(s) LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 07/28/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-10 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-10 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 07/28/2014 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

DETAILED ACTION

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This Office Action is in response to original application filed on 07/28/2014. Claims 1-10 are pending in this application.

Claim Objections

3. Claim 9 is objected to because of the following informalities: the last limitation of independent claim 9 recites "perform read/write operations a the memory devices in accordance with the local clock signals". There appears to be a typo in the language "a the". Appropriate correction is required.

Claim Rejections - 35 USC § 112

4. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claim 6 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.
6. Claim 6 recites the limitation "a module control device" in line 1. There is insufficient antecedent basis for this limitation in the claim.

Claim Rejections - 35 USC § 103

7. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

8. Claims 1-5, 7, and 9 are rejected under 35 U.S.C. 103 as being unpatentable over Sutardja et al. US 2007/0168781 (hereinafter Sutardja) in view of Kang et al. US 2012/0317352 (hereinafter Kang).

9. As per independent claim 1, Sutardja teaches “data buffers” (Figure 17 – collectively buffer modules 530, para 0133),

10. “memory devices including a plurality of groups of memory devices” (Figure 17 – groups of memory ICs 520-11 to 520-1Y through 520-Z1 to 520-ZY collectively memory ICs 520, para 0133),

11. “each respective group of memory devices communicating data with the memory controller via a respective one of the data buffers” (Figure 17 - the buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc., and address signals. Local control/address lines (not shown) are disposed on the memory modules 514 to locally distribute the buffered control and address signals to each memory IC 520 on the memory module 514, para 0136),

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12. “the each respective group of memory devices receiving and transmitting data signals in accordance with a local clock signal received from the respective one of the data buffers” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals. The buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533, para 0136).

13. Sutardja’s invention discloses all of the claimed limitations from above except for “A memory module to operate in a memory system with a memory controller, the memory system operating according to a system clock, comprising”.

14. However, Kang teaches “A memory module to operate in a memory system with a memory controller, the memory system operating according to a system clock, comprising” (Figure 13A depicts a memory system 4000 comprising controller 4100 and memory module 4200. The clock signal CLK is transmitted from the controller 4100 to the memory module 4200, paras 0138, 0140).

15. Given the teaching of Kang, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Sutardja with a memory module operating with a memory controller in a system and the system operating according to a system clock. The motivation would be that semiconductor memory devices are widely used in mobile devices such as laptop computers, mobile phones, etc., para 0004 of Kang.

16. As per dependent claim 2, Sutardja in combination with Kang discloses the memory module of claim 1. Sutardja teaches “the module control device transmitting the

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data buffer control signals and the module clock signal to the data buffers” (Figure 17 - the buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136),

17. “wherein the module control device does not transmit any clock signal to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

18. Sutardja’s invention discloses all of the claimed limitations from above except for “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal” and “the module control device transmitting the module C/A signals to at least some of the memory devices”.

19. However, Kang teaches “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal” (Figure 13A – register 4220 and PLL 4230 together are mapped to the module control device. Register 4220 receives and outputs command/address (CA) signals, PLL 4230 receives clock signal CLK and outputs one or more clock signals CLKs, para 0140),

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20. “the module control device transmitting the module C/A signals to at least some of the memory devices” (Figure 13A - the register 4220 buffers and provides the command/address CA signals to each DRAM chip 4210, para 0140).

21. The same motivation that was utilized for combining Sutardja and Kang as set forth in claim 1 is equally applicable to claim 2.

22. As per dependent claim 3, Sutardja in combination with Kang discloses the memory module of claim 2. Sutardja teaches “wherein the respective buffer circuit includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals” (Figure 17 - communication between the memory control module 510 and the memory modules 514 may be via serial and/or parallel signaling. A bus 531 may be used to support data flow between the memory control module 510 and the memory modules 514. A bus 533 may be used to support data flow between the memory modules 514 and the memory control module 510, para 0134).

23. As per dependent claim 4, Sutardja in combination with Kang discloses the memory module of claim 2. Sutardja teaches “wherein the plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time” (The memory ICs 520 and the buffer module 530 for each memory module 514 may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias, para 0135. It is well known in the art that in such an arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530),

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24. “wherein the respective buffer circuit is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

25. As per dependent claim 5, Sutardja in combination with Kang discloses the memory module of claim 1. Sutardja teaches “wherein the respective buffer circuit includes one or more programmable registers controlling a phase of the local clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. It is well known in the art that a PLL or a DLL (digital lock loop) contain programmable registers to control/adjust the phase and frequency of PLL or DLL).

26. As per dependent claim 7, Sutardja in combination with Kang discloses the memory module of claim 1. Sutardja teaches “wherein the respective buffer circuit is configured to regenerate the local clock signal from a received clock signal, and to phase-lock the local clock signal with the received clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

27. As per independent claim 9, Sutardja teaches “transmitting the module clock signal to a plurality of data buffers coupled between the memory devices and the memory bus” (Figure 17 - clock generator module 534 transmits clock signals to the buffers 530, para 0136),

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28. “generating local clock signals associated with respective ones of the data buffers” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136),
29. “transmitting the local clock signals to the memory devices” (Figure 17 - the buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136),
30. “perform read/write operations a the memory devices in accordance with the local clock signals” (Figure 17 - a bus 531 may be used to support data flow between the memory control module 510 and the memory modules 514. A bus 533 may be used to support data flow between the memory modules 514 and the memory control module 510. The buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, paras 0134, 0136).
31. Sutardja’s invention discloses all of the claimed limitations from above except for “A method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising” and “receiving a system clock signal” and “generating a module clock signal”.
32. However, Kang teaches “A method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising” (Figure 13A depicts a memory system 4000 comprising controller 4100 and memory module 4200. The memory bus comprises DQ, CA, and CLK, para 0138),

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33. “receiving a system clock signal” (Figure 13A - the clock signal CLK is transmitted from the controller 4100 to the memory module 4200, para 0140),

34. “generating a module clock signal” (Figure 13A – PLL 4230 receives clock signal CLK and outputs one or more clock signals CLKs, para 0140).

35. Given the teaching of Kang, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Sutardja with “A method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising” and “receiving a system clock signal” and “generating a module clock signal”. The motivation would be that semiconductor memory devices are widely used in mobile devices such as laptop computers, mobile phones, etc., para 0004 of Kang.

36. Claims 6 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over Sutardja in view of Kang and in further view of Carpenter et al. US 2009/0116312 (hereinafter Carpenter).

37. As per dependent claim 6, Sutardja in combination with Kang discloses the memory module of claim 5. Sutardja teaches “the module control device transmitting the data buffer control signals and the module clock signal to the data buffers” (Figure 17 - the buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS),

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column address strobe (CAS), etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136).

38. Sutardja's invention discloses all of the claimed limitations from above except for "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal" and "the module control device transmitting the module C/A signals to at least some of the memory devices" and "wherein the buffer circuit is configured to program the one or more registers according to the data buffer control signals".

39. However, Kang teaches "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal" (Figure 13A – register 4220 and PLL 4230 comprise the module control device. Register 4220 receives and outputs command/address (CA) signals, PLL 4230 receives clock signal CLK and outputs one or more clock signals CLKs, para 0140),

40. "the module control device transmitting the module C/A signals to at least some of the memory devices" (Figure 13A - the register 4220 buffers and provides the command/address CA signals to each DRAM chip 4210, para 0140).

41. The same motivation that was utilized for combining Sutardja and Kang as set forth in claim 5 is equally applicable to claim 6.

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42. Sutardja's invention modified by Kang discloses all of the claimed limitations from above except for "wherein the buffer circuit is configured to program the one or more registers according to the data buffer control signals".

43. However, Carpenter teaches "wherein the buffer circuit is configured to program the one or more registers according to the data buffer control signals" (The storage array can also be used as a general-purpose array with adjustable timing.

Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

44. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Sutardja as modified by Kang with buffer circuit configured to program the one or more registers according to the data buffer control signals. The motivation would be that timing margins for accessing storage cells must be evaluated because this would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

45. As per dependent claim 10, Sutardja in combination with Kang discloses the method of claim 9. Sutardja's invention modified by Kang discloses all of the claimed limitations from above except for "further comprising incorporating a programmable amount of delay to the local clock signals".

46. However, Carpenter teaches "further comprising incorporating a programmable amount of delay to the local clock signals" (The storage array can also be used as a

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general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

47. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Sutardja as modified by Kang with incorporating a programmable delay to the local clock signals. The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

48. Claim 8 is rejected under 35 U.S.C. 103 as being unpatentable over Sutardja in view of Kang and in further view of Sung et al. US 2011/0110168 (hereinafter Sung).

49. As per dependent claim 8, Sutardja in combination with Kang discloses the memory module of claim 1. Sutardja's invention modified by Kang discloses all of the claimed limitations from above except for " wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks".

50. However, Sung teaches "wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks" (Figures 1, 2 correspond specifically to a case in which the first memory unit 113 includes two memory ranks, and in this regard,

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two first chip selection signals are input to the first memory unit 113. The second memory unit 123 includes two memory ranks, and in this regard, two second chip selection signals are input to the second memory unit 123, paras 0046, 0049).

51. Given the teaching of Sung, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Sutardja as modified by Kang with each group of memory devices including at least one memory device from each of the one or more ranks. The motivation would be that the memory controller can determine which memory rank of which memory module is to be selected, para 0052 of Sung.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 8:30AM - 5:00PM Mon-Fri.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID X YI/
Supervisory Patent Examiner, Art Unit 2132

Notice of References Cited	Application/Control No. 14/445,035		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2007/0168781 A1	07-2007	Sutardja; Sehat	G06F11/1008	714/718
*	B	US-2012/0317352 A1	12-2012	Kang; Uk-Song	G11C11/40611	711/106
*	C	US-2009/0116312 A1	05-2009	Carpenter; Gary D.	G11C7/22	365/194
*	D	US-2011/0110168 A1	05-2011	Sung; Myung-hee	G11C5/04	365/189.09
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
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CONFIRMATION NO. 9700

SERIAL NUMBER 14/445,035	FILING or 371(c) DATE 07/28/2014 RULE	CLASS 711	GROUP ART UNIT 2132	ATTORNEY DOCKET NO. NT006US		
APPLICANTS NETLIST, INC., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; ** CONTINUING DATA ***** This appln claims benefit of 61/859,215 07/27/2013 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 08/06/2014						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/ZUBAIR AHMED/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 22	TOTAL CLAIMS 10	INDEPENDENT CLAIMS 2
ADDRESS The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES						
TITLE MEMORY MODULE WITH LOCAL SYNCHRONIZATION						
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

<i>Index of Claims</i> 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	11/02/2015								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
	7	✓								
	8	✓								
	9	✓								
	10	✓								

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16

S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2015/11/01 06:14

			IBM_TDB			
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12

11/ 2/ 2015 4:48:50 PM
C:\ Users\ zahmed\ Documents\ EAST\ Workspaces\ 14445035.wsp

Search Notes 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/08	10/31/2015	ZA

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
See attached search history	11/01/2015	ZA

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/ZUBAIR AHMED/ Examiner.Art Unit 2132	
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : NT006US

CERTIFICATE OF EFS WEB

TRANSMISSION

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

February 12, 2016

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

RESPONSE TO FINAL OFFICE ACTION DATED OCTOBER 22, 2015

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Non-Final Office Action dated November 12, 2015 for the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

1. (Currently Amended) A memory module ~~to operate operable~~ in a ~~memory~~ system with a memory controller, the ~~memory~~ system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal derived from the system clock, and

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with ~~the~~ a local clock signal ~~received from generated by~~ the respective one of the data buffers.

2. (Original) The memory module of claim 1, further comprising:

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices.

3. (Currently Amended) The memory module of claim 2, wherein the respective one of the N data buffers ~~data buffer~~ includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.

4. (Currently Amended) The memory module of claim 2, wherein the N data buffers ~~plurality of data buffers~~ are distributed across a surface of the memory module such that each data buffer control signal arrives at the N data buffers ~~plurality of data buffers~~ at different points in time, and

wherein the respective one of the N data buffers ~~data-buffer~~ is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.

5. (Currently Amended) The memory module of claim 2-4, wherein the respective one of the N data buffers ~~data-buffer~~ includes one or more programmable registers controlling a phase of the local clock signal.

6. (Currently Amended) The memory module of claim 5, ~~further comprising:~~
a ~~module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;~~

wherein the respective one of the N data buffers ~~data-buffer~~ is configured to program the one or more registers according to the data buffer control signals.

7. (Currently Amended) The memory module of claim 2-4, wherein the respective one of the N data buffers ~~data-buffer~~ is configured to regenerate the local clock signal from the module clock signal ~~a received clock signal~~, and to phase-lock the local clock signal with the module ~~received~~ clock signal.

8. (Original) The memory module of claim 1, wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks.

9. (Currently Amended) A method of operating a memory module coupled to a memory bus in a ~~memory~~ system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

receiving a system clock signal;

generating a module clock signal;
transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;
~~generating local clock signals associated with respective ones a~~ generating a respective local clock signal at each of the N data buffers;
transmitting the local clock signals to respective groups of the memory devices;
and
~~performing~~ perform read/write operations at a the memory devices in accordance with the local clock signals.

10. (Currently Amended) The ~~A~~ method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers ~~the local clock signals~~.

11. (New) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers;

12. (New) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (New) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of data buffers at different points in time, and
wherein the respective data buffer is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.

14. (New) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.
15. (New) The method of claim 11, further comprising:
programming one or more registers in each of the N data buffers according to the data buffer control signals.
16. (New) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.
17. (New) The method of claim 9, wherein the memory devices are organized in one or more ranks, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.
18. (New) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n integers larger than one, comprising:
a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;
 N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals;
and
memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local

clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (New) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board.

20. (New) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

REMARKS

This amendment responds to the Final Office Action dated November 12, 2015.

In the Office Action, the Examiner:

- objected to claim 9 for informalities;
- rejected claim 6 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite;
- rejected claims 1-5, 7, and 9 under 36 U.S.C. 103 as being unpatentable over Sutardja et al. US 2007/0168781 (hereafter Sutardja) in view of Kang et al. US 2012/0317352 (hereafter Kang);
- rejected claims 6 and 10 under 35 U.S.C. 103 as being unpatentable over Sutardja in view of Kang and in further view of Carpenter et al. US 2009/0116312 (hereafter Carpenter); and
- rejected claim 8 under 35 U.S.C. 103 as being unpatentable over Sutardja in view of Kang and in further view of Sung et al. US 2011/0110168 (hereafter Sung).

Claims 1, 3-7, 9-10 are amended. New claims 11-20 are presented. No new matter is added.

The amendment in claim 9 addresses the issue of informalities raised by the Examiner.

Claim 6 recites the limitation “a module control device,” not “the module control device.” Therefore, there should not be any antecedent basis issue. Nonetheless, claim 6 has been amended to remove the limitation. Therefore, the rejection thereto under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, is now moot.

Claim 1 as amended recites:

A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are positive integers, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal derived from the system clock, and

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers.

In Sutardja, each of the memory modules 514-1, 514-2, ..., 514-Z includes only one buffer and the data width of the one buffer is the same as the corresponding memory module. The Examiner tries to liken the multiple memory modules 514-1, 514-2, ..., 514-Z together as one memory module with multiple buffers, but even the multiple memory modules 514-1, 514-2, ..., 514-Z together do not make up the memory module in claim 1 as amended. As shown in FIG. 17 of Sutardja, all of the buffers 530-1, 530-2, ..., 530-Z are coupled to the memory control module 510 in series along the bus 531, and each of the buffers is coupled to the entire bus 531 instead of part of the bus 531. Thus, each of the buffers 530-1, 530-2, ..., 530-Z have the same data width as the corresponding memory module on which it sits, and the multiple memory modules 514-1, 514-2, ..., 514-Z together do not form a memory module with a wider data width than the data width of each of the memory buffers 530-1, 530-2, ..., 530-Z or each of the memory modules 514-1, 514-2, ..., 514-Z.

Kang does not make up for the deficiencies of Sutardja.

Thus, Sutardja in view of Kang, or any of the other cited references, does not teach or disclose, among other things, a $(N \times n)$ bit wide memory module including N data buffers each being n -bit wide but together make up the full data width of the memory module, and the data buffers each generating a respective local clock signal derived from the system clock signal for a respective group of the memory devices on the memory module.

Therefore, claim 1 as amended is patentable over Sutardja in view of Kang.

Claim 2, claim 3-7 as amended, and claim 8 depend from claim 1 and include further limitations in addition to the limitations in claim 1. Therefore, claim 2, claim 3-7 as amended, and claim 8 are patentable for at least the same reasons as claim 1 as amended is patentable.

The arguments regarding claim 1 as amended applies to claim 9 as amended. Therefore, claim 9 as amended is also patentable over Sutardja in view of Kang.

Claim 10 as amended and new claims 11-17 depend from claim 9 as amended and include further limitations in addition to the limitations in claim 9 as amended. Therefore, claim 10 as amended and new claims 11-17 are patentable for at least the same reasons claim 9 as amended is patentable.

The arguments regarding claim 1 as amended applies to new claim 18. Therefore, new claim 18 is also patentable over the cited references.

New claims 19-20 depend from new claim 18 and include further limitations in addition to the limitations in new claim 18. Therefore, new claims 19-20 are patentable for at least the same reasons new claim 18 is patentable

Applicant submits that claims 1-20 are in condition for allowance, and Applicant respectfully requests such action. The Examiner is invited to call the undersigned at (650) 521-4828 if such phone call would help resolve any remaining issues.

Respectfully submitted,

Date: February 12, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Acknowledgement Receipt

EFS ID:	24909516
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	12-FEB-2016
Filing Date:	28-JUL-2014
Time Stamp:	21:56:44
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2016-02-12_Amendment_NT006US.pdf	140824 2a1f0f3d83d300feffbbe1864f0d172f5af8d298	yes	9

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Claims	2	6
	Applicant Arguments/Remarks Made in an Amendment	7	9
Warnings:			
Information:			
Total Files Size (in bytes):		140824	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/445,035	Filing Date 07/28/2014	☐ To be Mailed
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 ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA		RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A		N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*		X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*		X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).				
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))					
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	02/12/2016	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0		x \$40 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0		x \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
TOTAL ADD'L FEE								0

(Column 1)

(Column 2)

(Column 3)

		CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=		X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=		X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
TOTAL ADD'L FEE								

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

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The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

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 /TAMMY d. MCBETH BROWN/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
				Attorney Docket Number	NT006US
Sheet	1	of	13		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
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				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
				Attorney Docket Number	NT006US
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				Examiner Name	AHMED, ZUBAIR
				Attorney Docket Number	NT006US
Sheet	10	of	13		

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EFS ID:	24909711
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	13-FEB-2016
Filing Date:	28-JUL-2014
Time Stamp:	02:05:42
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-02-12_IDS_Cover_NT006 US.pdf	62274 3ed04c53e14723fc01d14f71b47357b8ee93d33f	no	2

Warnings:

Information:

2	Information Disclosure Statement (IDS) Form (SB08)	2016-02-12_IDS_filed_NT006U S.pdf	404220 0d73f72d65d76433bf2fd3c4863755197463 c484	no	13
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
Total Files Size (in bytes):				466494	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/445,035
Filed	: July 28, 2014
For	: MEMORY MODULE LOCAL SYNCHRONIZATION
Examiner	: AHMED, ZUBAIR
Art Unit	: 2132
Conf. No.	: 9700
Docket No.	: NT006US

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 323 references. This same listing of the 323 references are also included in a Form PTO/SB/08a/b Equivalent filed on January 5, 2015 in related U.S. Patent Application No. 13/952,599, filed July 27, 2013. Copies of foreign documents and/or their translation and non-patent literature in the listing can be found in the file-wrapper of the above-referenced related application and are available on PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols,

Application No.: 14/445,035
Filing Date: July 28, 2014

results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

The Commissioner is authorized to charge any required fees to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: February 12, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51,167
(Reg. No.)

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
				Attorney Docket Number	NT006US
Sheet	1	of	2		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6754746	06-22-2004	Monolithic System Technology, Inc.	
	2.	6799252	09-28-2004	Unisys Corporation	
	3.	7464225	12-09-2008	Rambus Inc.	
	4.	7884619	02-08-2011	Altera Corporation	
	5.	8214616	07-03-2012	Rambus Inc.	
	6.	8565033	10-22-2013	Altera Corporation	
	7.	8689064	04-01-2014	Netlist, Inc.	
	8.	8949519	02-03-2015	Google Inc.	
	9.	9128632	09-08-2015	Netlist, Inc.	
	10.	20020038405	03-28-2002	Leddige, M., et al.	
	11.	20020039323	04-04-2002	Mitsubishi Denki Kabushiki Kaisha	
	12.	20040105292	06-03-2004	Elpida Memory, Inc.	
	13.	20070064462		Elpida Memory, Inc.	
	14.	20080037412	02-14-2008	Adc Telecommunications, Inc.	
	15.	20090198924	08-06-2009	Rambus Inc.	
	16.	20090248969	10-01-2009	Wu; Larry	
	17.	20090298503	12-03-2009	Chu, M., et al.	
	18.	20100125681	05-20-2010	Patel; Shwetal A.	
	19.	20100228891	09-09-2010	Talbot Gerald R	
	20.	20110085406	04-14-2011	Solomon, J. et al.	
	21.	20120256639	10-11-2012	Marco Pausini	

US PATENT APPLICATIONS					
Examiner Initials *	Cite No. ¹	Application Number	Filing Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
	22.	13970,606	08-20-2013	Netlist, Inc.	Entire Prosecution History
	23.	14229844	03-29-2014	Netlist, Inc.	Entire Prosecution History
	24.	14337168	07-21-2014	Netlist, Inc.	Entire Prosecution History
	25.	14715486	05-18-2015	Netlist, Inc.	Entire Prosecution History
	26.	14846993	09-07-2015	Netlist, Inc.	Entire Prosecution History

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
				Attorney Docket Number	NT006US
Sheet	2	of	2		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	27.	International Search Report and Written Opinion, PCT/US2014/048517, October 27, 2014.	
	28.	International Preliminary Report on Patentability, PCT/US2014/048517, February 2, 2016.	
	29.	Non-Final Office Action dated July 31, 2014, for U.S. Patent Application No. 13/952,599 filed July 27, 2013.	
	30.	Response to Non-Final Office Action July 31, 2014, for U.S. Patent Application No. 13/952,599, filed December 31, 2014.	
	31.	Notice of Allowance issued on April 22, 2015, for U.S. Patent Application No. 13/952,599 filed July 27, 2013.	
	32.	Non-Final Office Action dated November 23, 2015, for U.S. Patent Application No. 13/970,606 filed August 20, 2013.	
	33.	Inter Partes Review Case No. IPR2014-01369, Patent Owner's Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on December 16, 2014.	
	34.	Inter Partes Review Case No. IPR2014-01369, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued March 9, 2015.	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

To: Jamie J. Zheng
P.O. Box 60573
Palo Alto, CA 94306
United States of America

PCT

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

Date of mailing
(day/month/year)

27 OCT 2014

Applicant's or agent's file reference
NT006PCT

FOR FURTHER ACTION See paragraphs 1 and 4 below

International application No.
PCT/US14/48517

International filing date
(day/month/year) 28 July 2014 (28.07.2014)

Applicant
NETLIST, INC.

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.

How? Directly to the International Bureau of WIPO preferably through ePCT or on paper to, 34 chemin des Colombettes
1211 Geneva 20, Switzerland. Facsimile No.: +41 22 338 82 70

For more detailed instructions, see *PCT Applicant's Guide*, International Phase, paragraphs 9.004 – 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:
- ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.
- ☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. **Reminders**

The applicant may **submit comments on an informal basis on the written opinion of the International Searching Authority** to the International Bureau. These comments will be made available to the public after international publication. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established.

Shortly after the expiration of **18 months from the priority date, the international application will be published** by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90bis.1 and 90bis.3).

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for **entry into the national phase** before those designated Offices. In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months. For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.

Within **19 months from the priority date, the applicant may request that a supplementary international search be carried out** by a different International Searching Authority that offers this service (Rule 45bis.1). The procedure for requesting supplementary international search is described in the *PCT Applicant's Guide*, International Phase, paragraphs 8.006-8.032.

Name and mailing address of the ISA/
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Authorized officer

Shane Thomas

PCT Helpdesk: 571-272-4300
Telephone No. PCT OSP: 571-272-7774

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference NT006PCT	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US14/48517	International filing date (<i>day/month/year</i>) 28 July 2014 (28.07.2014)	(Earliest) Priority Date (<i>day/month/year</i>) 27 July 2013 (27.07.2013)
Applicant NETLIST, INC.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 2 sheets.

☐ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed.
☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (see Box No. II).

3. ☐ **Unity of invention is lacking** (see Box No. III).

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant.
☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☒ the text is approved as submitted by the applicant.
☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority.

6. With regard to the **drawings**,

- a. the figure of the **drawings** to be published with the abstract is Figure No. 2D
☒ as suggested by the applicant.
☐ as selected by this Authority, because the applicant failed to suggest a figure.
☐ as selected by this Authority, because this figure better characterizes the invention.
b. ☐ none of the figures is to be published with the abstract.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US14/48517

A. CLASSIFICATION OF SUBJECT MATTER

IPC(8) - G06F 12/02, 13/38, 13/42 (2014.01)

CPC - G11C 5/04, 5/025, 7/22

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC(8): G06F 12/00, 12/02, 13/38, 13/42; G11C 7/10 (2014.01);

CPC: G06F 12/00; G11C 5/00, 5/04, 5/025, 7/22; USPC: 711/115, 154, E12.001

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

MicroPatent (US-G, US-A, EP-A, EP-B, WO, JP-bib, DE-C,B, DE-A, DE-T, DE-U, GB-A, FR-A); Google; Google Scholar; ProQuest; IP.com; Espacenet; SEARCH TERMS: memory module, local clock, system clock, module clock, controller, buffer, cache, bus, time, arrive, rank.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X ----- Y	US 2002/0038405 A1 (LEDDIGE, M., et al.) March 28, 2002; figure 7; paragraphs [0015], [0037], [0039], and [0042].	1-3, 5-7, 9 ----- 4, 8, 10
Y	US 2009/0296503 A1 (CHU, M., et al.) December 03, 2009; paragraphs [0006] and [0007].	4, 10
Y	US 2011/0085406 A1 (SOLOMON, J., et al.) April 14, 2011; figures 1 and 4A; paragraphs [0045] and [0053].	8
A	US 2011/0125966 A1 (AMIDI, H., et al.) May 26, 2011; entire document.	1-20

☐ Further documents are listed in the continuation of Box C.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

14 October 2014 (14.10.2014)

Date of mailing of the international search report

27 OCT 2014

Name and mailing address of the ISA/US

Mail Stop PCT, Attn: ISA/US, Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450

Facsimile No. 571-273-3201

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300

PCT OSP: 571-272-7774

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To: Jamie J. Zheng
P.O. Box 60573
Palo Alto, CA 94306
United States of America

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43bis.1)

Date of mailing
(day/month/year)

27 OCT 2014

Applicant's or agent's file reference
NT006PCT

FOR FURTHER ACTION

See paragraph 2 below

International application No.

PCT/US14/48517

International filing date (day/month/year)

28 July 2014 (28.07.2014)

Priority date (day/month/year)

27 July 2013 (27.07.2013)

International Patent Classification (IPC) or both national classification and IPC

IPC(8) - G06F 12/02, 13/38, 13/42 (2014.01)

CPC - G11C 5/04, 5/025, 7/22

Applicant
NETLIST, INC.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☒ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1 bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/US
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Date of completion of this opinion

14 October 2014 (14.10.2014)

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:
 - a. (means)
 - ☐ on paper
 - ☐ in electronic form
 - b. (time)
 - ☐ in the international application as filed
 - ☐ together with the international application in electronic form
 - ☐ subsequently to this Authority for the purposes of search
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Claims	4, 8, 10	YES
	Claims	1-3, 5-7, 9	NO
Inventive step (IS)	Claims	NONE	YES
	Claims	1-10	NO
Industrial applicability (IA)	Claims	1-10	YES
	Claims	NONE	NO

2. Citations and explanations:

Claims 1-3, 5-7, and 9 lack novelty under PCT Article 33(2) as being anticipated by US 2002/0038405 A1 to LEDDIGE et al. (hereinafter "Leddige").

As per claim 1, Leddige discloses a memory module to operate in a memory system (a plurality of memory modules operating in a memory system; paragraph [0015]) with a memory controller (a memory module controller; paragraph [0039]), the memory system operating according to a system clock (memory devices (memory system) are enabled through clock signals (system clock); paragraph [0042]), comprising: data buffers (the system includes write buffer 712 and read buffer 738); figure 7; paragraph [0039]), and memory devices including a plurality of groups of memory devices (each memory module comprises a plurality of memory devices; paragraph [0015]), each respective group of memory devices communicating data with the memory controller via a respective one of the data buffers (the plurality of memory devices send and receive data to/from the memory controller through write buffer 712 and read buffer 738); figure 7; paragraph [0039]), the each respective group of memory devices receiving and transmitting data signals (the devices on the memory module are coupled to command and address buses which carry both address and control information for memory transactions; paragraph [0037]) in accordance with a local clock signal received from the respective one of the data buffers (clock signals are passed to the memory devices from clock buffer 716); figure 7; paragraph [0042]).

As per claim 2, Leddige discloses the memory module of claim 1. Leddige additionally discloses comprising: a module control device to receive command/address (C/A) signals (the devices on the memory module are coupled to command and address buses which may carry address and control transaction (receive) data (signals); paragraph [0037]) and a system clock from the memory controller (clock signals are passed from the memory controller; figure 7; paragraphs [0035] and [0042]) and to output module C/A signals (command and address buses carry address and control transaction data (output); paragraph [0037]), data buffer control signals (data is sent to and received from write buffer 712 and read buffer 738); figure 7; paragraph [0039]) and a module clock signal (delayed clock signals (module clock signal) are passed from the memory controller; figure 7; paragraph [0042]), the module clock signal being derived from the system clock signal (the delayed clock signal (module clock signal) is generated by a delay lock loop based on (derived from) clock signals from the system; paragraph [0042]), the module control device transmitting the module C/A signals to at least some of the memory devices (signals from the memory module controller passes command and address data (signals) to the memory devices of the respective module; paragraphs [0037]-[0039]), the module control device transmitting the data buffer control signals (the memory controller writes to and reads from write buffer 712 and read buffer 738, respectively); figure 7; paragraph [0040]) and the module clock signal to the data buffers (the delayed clock signal (module clock signal) is passed through the control logic to the read and write buffers; paragraphs [0040] and [0042]); wherein the module control device does not transmit any clock signal to the memory devices (module memory controller may omit (does not transmit) the clock signal for select memory devices; paragraphs [0037] and [0039]).

As per claim 3, Leddige discloses the memory module of claim 2. Leddige additionally discloses wherein a respective buffer circuit includes data paths for communicating data between the memory controller and the respective group of memory devices (more than one memory bus (data paths) may be used to carry (communicating) data signals from (between) the memory controller to the memory devices; paragraph [0035]), the data paths being controlled by at least one of the data buffer control signals (memory buses (data paths) pass (controlled by) clock signals from clock buffer 716 (data buffer control signals); figure 7; paragraph [0042]).

As per claim 5, Leddige discloses the memory module of claim 1. Leddige additionally discloses wherein a respective buffer circuit includes one or more programmable registers (the interface circuits may include control logic for register elements (programmable registers) to drive address lines; paragraph [0040]) controlling a phase of the local clock signal (delay lock loop 758 received a clock signal the system and generates a signal having a different frequency than the clock; figure 7; paragraph [0042]).

---Continued Within the Next Supplemental Box---

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

Claim 9 is objected to under PCT Rule 66.2(a)(iii) as containing the following defects in the form or contents thereof: Claim 9 recites in line 9, "operations a the memory devices". As best understood these are typographical errors and for the purposes of this opinion will be interpreted as being "operations at the memory devices".

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

Claims 3, 5, and 7 are objected to under PCT Rule 66.2(a)(v) as lacking clarity under PCT Article 6 because claims 3, 5 and 7 are indefinite for the following reason(s): In line 1 of claims 3, 5, and 7, respectively, the claims recite "the respective buffer circuit". There is a lack of antecedent basis for these limitations in the claims. For the purpose of this opinion, as best understood, the claims are interpreted to read "a respective buffer circuit".

Claim 4 is objected to under PCT Rule 66.2(a)(v) as lacking clarity under PCT Article 6 because claim 4 is indefinite for the following reason(s): line 1 of claim 4 recites "the plurality of buffer circuits". There is a lack of antecedent basis for this limitation in the claim. For the purpose of this opinion, as best understood, the claim is interpreted to read "a plurality of buffer circuits".

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

---Continued from Box V: Citations and Explanations---

As per claim 6, Leddige discloses the memory module of claim 5. Leddige additionally discloses comprising: a module control device to receive command/address (C/A) signals (the devices on the memory module are coupled to command and address buses which may carry address and control transaction data (receive); paragraph [0037]) and a system clock from the memory controller (clock signals are passed from the memory controller; figure 7; paragraphs [0035] and [0042]) and to output module C/A signals (command and address buses carry address and control transaction data (output); paragraph [0037]), data buffer control signals (data is sent to and receive from write buffer 712 and read buffer 738; figure 7; paragraph [0039]) and a module clock signal (delayed clock signals (module clock signal) are passed from the memory controller; figure 7; paragraph [0042]), the module clock signal being derived from the system clock signal (the delayed clock signal (module clock signal) is generated by a delay lock loop based on (derived) clock signals from the system; paragraph [0042]), the module control device transmitting the module C/A signals to at least some of the memory devices (signals from the memory module controller passes command and address data (signals) to the memory devices of the respective module; paragraphs [0037]-[0039]), the module control device transmitting the data buffer control signals (the memory controller writes to and reads from write buffer 712 and read buffer 738, respectively; figure 7; paragraph [0040]) and the module clock signal to the data buffers (the delayed clock signal (module clock signal) is passed through the control logic to the read and write buffers; paragraphs [0040] and [0042]); wherein the module control device does not transmit any clock signal to the memory devices (module memory controller may omit (does not transmit) the clock signal for select memory devices; paragraphs [0037] and [0039]).

As per claim 7, Leddige discloses the memory module of claim 1. Leddige additionally discloses wherein a respective buffer circuit is configured to regenerate the local clock signal from a received clock signal (the clock generator 710 receives a clock signal from the system, and in response, generates (regenerate) a delayed clock signal (local clock signal) therefrom; figure 7; paragraph [0042]), and to phase-lock the local clock signal with the received clock signal (a delay lock loop (phase-lock) is used to generate a delayed clock signal based on the clock signal received from the system; figure 7; paragraph [0042]).

As per claim 9, Leddige discloses a method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising: receiving a system clock signal (the module receives the system clock signal 724 through clock signal lines; figure 7; paragraph [0037]); generating a module clock signal (a delay lock loop generates a delayed clock signal; figure 7; paragraph [0042]); transmitting the module clock signal to a plurality of data buffers (the delayed clock signal is passed from clock generator 710 to write buffer 712 and read buffer 738; figure 7, paragraph [0039]) coupled between the memory devices and the memory bus (data is transmitted to the memory devices from the read and write buffers (coupled between) and the memory bus; figures 5 and 7; paragraphs [0037] and [0039]); generating local clock signals associated with respective ones of the data buffers (clock buffer 716 drives the clock signal to memory devices, based on the same clock signal (associated) transmitted to the read and write buffers; figure 7; paragraphs [0039] and [0042]); transmitting the local clock signals to the memory devices (a memory hub provides (transmitting) the clock signal 730 (local clock signals) to the memory devices; figure 7; paragraph [0037]); and perform read/write operations at the memory devices (command and address buses carry both address and control information for memory transactions (read/write operations) to the memory devices; paragraph [0037]) in accordance with the local clock signals (clock buffer 716 drives the clock signal (local clock signals) to memory devices; figure 7; paragraph [0042]).

Claims 4 and 10 lack an inventive step under PCT Article 33(3) as being obvious over Leddige in view of US 2009/0296503 A1 to CHU et al. (hereinafter "Chu").

As per claim 4, Leddige discloses the memory module of claim 2. Leddige additionally discloses the memory module wherein a respective buffer circuit is configured to regenerate the local clock signal from the module clock signal (the clock generator 710 receives a clock signal from the system, and in response, generates (regenerate) a delayed clock signal (local clock signal) therefrom; figure 7; paragraph [0042]), and provide the local clock signal to the respective group of memory devices (a memory hub provides (transmitting) the clock signal 730 (local clock signals) to the memory devices; figure 7; paragraph [0037]). Leddige does not disclose the memory module wherein a plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. Chu discloses the memory module wherein a plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time (the topology (distributed across a surface) of the memory devices results in the reception of the system clock signals (control signal arrives) by the memory devices (plurality of buffer circuits) at different times; paragraph [0007]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Chu, because by doing so, the memory module and method of Leddige would provide an efficient transfer of data from an input device to the core circuitry of the memory module, thereby providing faster and more efficient data transfers.

As per claim 10, Leddige discloses a method of claim 9. Leddige does not disclose the method further comprising incorporating a programmable amount of delay to the local clock signals. Chu discloses the method further comprising incorporating a programmable amount of delay to the local clock signals (the system calibrates a delay setting (a programmable amount of delay) in order to control the clock signals provided to the immediate registers (local clock signals); paragraph [0006]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Chu, because by doing so, the memory module and method of Leddige would provide an efficient transfer of data from an input device to the core circuitry of the memory module, thereby providing faster and more efficient data transfers.

---Continued Within the Next Supplemental Box---

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US14/48517

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

----Continued from Previous Supplemental Box----

Claim 8 lacks an inventive step under PCT Article 33(3) as being obvious over Leddige in view of US 2011/0085406 A1 to SOLOMON et al. (hereinafter "Solomon").

As per claim 8, Leddige discloses the memory module of claim 1. Leddige does not disclose the memory module wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks. Solomon discloses the memory module wherein memory devices are organized in one or more ranks (memory devices are arranged in four ranks; figure 1; paragraph [0045]), and wherein each group of memory devices including at least one memory device from each of the one or more ranks (memory device groups are arranged within ranks 32a, 32b, with each rank having a corresponding memory device coupled to a set of DQ data signal lines; figure 4A; paragraph [0053]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Solomon, because by doing so, the memory module and method of Leddige would provide a design wherein larger memory modules are possible, thereby providing a more robust memory module having greater memory capacities.

Claims 1-10 have industrial applicability as defined by PCT Article 33(4) because the subject matter can be made or used in industry.

PCT Recordation of Search History

Case/PCT Application Number: PCT/US14/48517

CLIN Number/Technical Field of PCT Application: 4

Date(s) during which the search was conducted: 13 October 2014 – 14 October 2014

Date of Completion of Recordation of Search History Form: 14 October 2014

Research Analyst Initials: TWM

Search Approval Official (SAO) Initials: HKP

Field of Search/Classification Information:

IPC(8) Classification(s): G06F 12/00, 12/02, 13/38, 13/42; G11C 7/10 (2014.01)

CPC Classification(s): G06F 12/00; G11C 5/00, 5/04, 5/025, 7/22

USPC Classification(s): 711/115, 154, E12.001

Database(s) Searched (Patent and Non-Patent Literature (NPL), Including Sub-Databases and Files Searched) and Search Terms Used:

MicroPatent (US-G, US-A, EP-A, EP-B, WO, JP-bib, DE-C,B, DE-A, DE-T, DE-U, GB-A, FR-A); Google; Google Scholar; ProQuest; IP.com; Espacenet;

SEARCH TERMS: memory module*, local clock, system clock, modul* clock, controller*, buffer*, cache*, bus*3, time*, arriv*, rank*

Database Search String Recordation, Including Dates of Searches):
Patent Database Search Strategy/Results:

#	comments	results	criteria	last run
20		<u>49 hits</u>	rank* AND (memory ADJ module*) Full patent spec. AND signal* NEAR5 (different* NEAR3 time*) Current IPC-R G06F001200 OR G06F001202 OR G06F001338 OR G06F001342 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-15
19		<u>149 hits</u>	(memory ADJ module*) AND signal* Full patent spec. NEAR5 (different* NEAR3 time*) Current IPC-R G06F001200 OR G06F001202 OR G06F001338 OR G06F001342 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-15
18		<u>845 hits</u>	(memory ADJ module*) AND signal* Full patent spec. NEAR5 (different* NEAR3 time*) Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-15
17		no hits	(memory ADJ module*) AND ((local Full patent spec. OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Current IPC-R G011C000710 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14

16		<u>254 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current IPC-R G06F001342 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
15		<u>81 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current IPC-R G06F001338 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
14		<u>120 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current IPC-R G06F001202 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
13		<u>611 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current IPC-R G06F001200 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
12		<u>99 hits</u>	Full patent spec. (memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*)	2014-10-14

			AND control*4 AND (buffer* OR cache* OR bus*3) Current US Class 711E12001 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	
11		<u>144 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current US Class 711154 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
10		<u>69 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Current US Class 711115 Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
9		<u>4960 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Full patent spec. Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
8		<u>145 hits</u>	Combined query 7 NOT 6	2014-10-14
7		<u>162 hits</u>	(memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR Claims, Title or Abstract	2014-10-14

			cache* OR bus*3) Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	
6		<u>17 hits</u>	Title or Abstract (memory ADJ module*) AND ((local OR system or modul*) NEAR2 clock*) AND control*4 AND (buffer* OR cache* OR bus*3) Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-14
5		<u>17 hits</u>	Assignee/Applicant Jayesh NEAR3 Bhakta Full patent spec. sync* and clock* and module* Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-13
4		<u>12 hits</u>	Assignee/Applicant Hyun NEAR2 Lee Full patent spec. local* and sync* and clock* and module* Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-13
3		<u>3068 hits</u>	Assignee/Applicant Hyun NEAR2 Lee Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-13
2		<u>26 hits</u>	Assignee/Applicant Netlist Full patent spec. local* AND sync* Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA	2014-10-13

			Years 2001-2014	
1		<u>84 hits</u>	Assignee/Applicant Netlist Databases USG USA EPA EPB WO JP DEG DEA DET DEU GBA FRA Years 2001-2014	2014-10-13

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4 items in query history list

☒ 1. **memory AND module AND sync* G11C5/025 OR G11C7/22**
 memory AND module AND sync* in the title or abstract AND G11C5/025 OR G11C7/22 as the Cooperative Patent Classification
 Approximately 45 results found in the Worldwide database on Tue, 14 Oct 2014 23:18

☒ 2. **memory AND module AND sync* G11C5/00 OR G11C5/04**
 memory AND module AND sync* in the title or abstract AND G11C5/00 OR G11C5/04 as the Cooperative Patent Classification
 28 results found in the Worldwide database on Tue, 14 Oct 2014 23:17

☒ 3. **memory AND module AND sync* G06F12/00 OR G06F12/02**
 memory AND module AND sync* in the title or abstract AND G06F12/00 OR G06F12/02 as the Cooperative Patent Classification.
 0 results found in the Worldwide database on Tue, 14 Oct 2014 23:17

☒ 4. **memory AND module AND synchroniz* G06F12/00 OR G06F12/02**
 memory AND module AND synchroniz* in the title or abstract AND G06F12/00 OR G06F12/02 as the Cooperative Patent Classification
 0 results found in the Worldwide database on Tue, 14 Oct 2014 23:16

Non-Patent Literature (NPL) Search Strategy/Results:

Google / Google Scholar: 14 October 2014

Memory module with local synchronization using distributed data buffers

Memory module with local synchronization

Memory buffer synchronization

memory module, one memory device from each of the one or more ranks

memory module, each control signal arrives at different points in time

ProQuest: 14 October 2014

Memory module with local synchronization using distributed data buffers

Memory module with local synchronization

Memory buffer synchronization

IP.com: 14 October 2014

Memory module with local synchronization using distributed data buffers

Memory module with local synchronization

Memory buffer synchronization

PATENT COOPERATION TREATY

PCT

INTERNATIONAL PRELIMINARY REPORT ON PATENTABILITY

(Chapter I of the Patent Cooperation Treaty)

(PCT Rule 44bis)

Applicant's or agent's file reference NT006PCT	FOR FURTHER ACTION	See item 4 below
International application No. PCT/US2014/048517	International filing date (<i>day/month/year</i>) 28 July 2014 (28.07.2014)	Priority date (<i>day/month/year</i>) 27 July 2013 (27.07.2013)
International Patent Classification (8th edition unless older edition indicated) See relevant information in Form PCT/ISA/237		
Applicant NETLIST, INC.		

1.	This international preliminary report on patentability (Chapter I) is issued by the International Bureau on behalf of the International Searching Authority under Rule 44 bis.1(a).																								
2.	This REPORT consists of a total of 8 sheets, including this cover sheet.																								
In the attached sheets, any reference to the written opinion of the International Searching Authority should be read as a reference to the international preliminary report on patentability (Chapter I) instead.																									
3.	This report contains indications relating to the following items: <table style="width: 100%;"> <tr> <td style="width: 10%; text-align: center;">☒</td> <td style="width: 20%;">Box No. I</td> <td>Basis of the report</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. II</td> <td>Priority</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. III</td> <td>Non-establishment of opinion with regard to novelty, inventive step and industrial applicability</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. IV</td> <td>Lack of unity of invention</td> </tr> <tr> <td style="text-align: center;">☒</td> <td>Box No. V</td> <td>Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. VI</td> <td>Certain documents cited</td> </tr> <tr> <td style="text-align: center;">☒</td> <td>Box No. VII</td> <td>Certain defects in the international application</td> </tr> <tr> <td style="text-align: center;">☒</td> <td>Box No. VIII</td> <td>Certain observations on the international application</td> </tr> </table>	☒	Box No. I	Basis of the report	☐	Box No. II	Priority	☐	Box No. III	Non-establishment of opinion with regard to novelty, inventive step and industrial applicability	☐	Box No. IV	Lack of unity of invention	☒	Box No. V	Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement	☐	Box No. VI	Certain documents cited	☒	Box No. VII	Certain defects in the international application	☒	Box No. VIII	Certain observations on the international application
☒	Box No. I	Basis of the report																							
☐	Box No. II	Priority																							
☐	Box No. III	Non-establishment of opinion with regard to novelty, inventive step and industrial applicability																							
☐	Box No. IV	Lack of unity of invention																							
☒	Box No. V	Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement																							
☐	Box No. VI	Certain documents cited																							
☒	Box No. VII	Certain defects in the international application																							
☒	Box No. VIII	Certain observations on the international application																							
4.	The International Bureau will communicate this report to designated Offices in accordance with Rules 44bis.3(c) and 93bis.1 but not, except where the applicant makes an express request under Article 23(2), before the expiration of 30 months from the priority date (Rule 44bis .2).																								

The International Bureau of WIPO 34, chemin des Colombettes 1211 Geneva 20, Switzerland Facsimile No. +41 22 338 82 70	Date of issuance of this report 02 February 2016 (02.02.2016) Authorized officer El Mostafa Moussaid e-mail: pt03.pct@wipo.int
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PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To: Jamie J. Zheng
P.O. Box 60573
Palo Alto, CA 94306
United States of America

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43bis.1)

Date of mailing
(day/month/year)

27 OCT 2014

Applicant's or agent's file reference
NT006PCT

FOR FURTHER ACTION

See paragraph 2 below

International application No.

PCT/US14/48517

International filing date (day/month/year)

28 July 2014 (28.07.2014)

Priority date (day/month/year)

27 July 2013 (27.07.2013)

International Patent Classification (IPC) or both national classification and IPC

IPC(8) - G06F 12/02, 13/38, 13/42 (2014.01)

CPC - G11C 5/04, 5/025, 7/22

Applicant NETLIST, INC.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☒ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1 bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/US
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Date of completion of this opinion

14 October 2014 (14.10.2014)

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US14/48517

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:

- ☒ the international application in the language in which it was filed.
- ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))

3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:

a. (means)

- ☐ on paper
- ☐ in electronic form

b. (time)

- ☐ in the international application as filed
- ☐ together with the international application in electronic form
- ☐ subsequently to this Authority for the purposes of search

4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.

5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Claims	4, 8, 10	YES
	Claims	1-3, 5-7, 9	NO
Inventive step (IS)	Claims	NONE	YES
	Claims	1-10	NO
Industrial applicability (IA)	Claims	1-10	YES
	Claims	NONE	NO

2. Citations and explanations:

Claims 1-3, 5-7, and 9 lack novelty under PCT Article 33(2) as being anticipated by US 2002/0038405 A1 to LEDDIGE et al. (hereinafter "Leddige").

As per claim 1, Leddige discloses a memory module to operate in a memory system (a plurality of memory modules operating in a memory system; paragraph [0015]) with a memory controller (a memory module controller; paragraph [0039]), the memory system operating according to a system clock (memory devices (memory system) are enabled through clock signals (system clock); paragraph [0042]), comprising: data buffers (the system includes write buffer 712 and read buffer 738); figure 7; paragraph [0039]), and memory devices including a plurality of groups of memory devices (each memory module comprises a plurality of memory devices; paragraph [0015]), each respective group of memory devices communicating data with the memory controller via a respective one of the data buffers (the plurality of memory devices send and receive data to/from the memory controller through write buffer 712 and read buffer 738); figure 7; paragraph [0039]), the each respective group of memory devices receiving and transmitting data signals (the devices on the memory module are coupled to command and address buses which carry both address and control information for memory transactions; paragraph [0037]) in accordance with a local clock signal received from the respective one of the data buffers (clock signals are passed to the memory devices from clock buffer 716); figure 7; paragraph [0042]).

As per claim 2, Leddige discloses the memory module of claim 1. Leddige additionally discloses comprising: a module control device to receive command/address (C/A) signals (the devices on the memory module are coupled to command and address buses which may carry address and control transaction (receive) data (signals); paragraph [0037]) and a system clock from the memory controller (clock signals are passed from the memory controller; figure 7; paragraphs [0035] and [0042]) and to output module C/A signals (command and address buses carry address and control transaction data (output); paragraph [0037]), data buffer control signals (data is sent to and received from write buffer 712 and read buffer 738); figure 7; paragraph [0039]) and a module clock signal (delayed clock signals (module clock signal) are passed from the memory controller; figure 7; paragraph [0042]), the module clock signal being derived from the system clock signal (the delayed clock signal (module clock signal) is generated by a delay lock loop based on (derived from) clock signals from the system; paragraph [0042]), the module control device transmitting the module C/A signals to at least some of the memory devices (signals from the memory module controller passes command and address data (signals) to the memory devices of the respective module; paragraphs [0037]-[0039]), the module control device transmitting the data buffer control signals (the memory controller writes to and reads from write buffer 712 and read buffer 738, respectively); figure 7; paragraph [0040]) and the module clock signal to the data buffers (the delayed clock signal (module clock signal) is passed through the control logic to the read and write buffers; paragraphs [0040] and [0042]); wherein the module control device does not transmit any clock signal to the memory devices (module memory controller may omit (does not transmit) the clock signal for select memory devices; paragraphs [0037] and [0039]).

As per claim 3, Leddige discloses the memory module of claim 2. Leddige additionally discloses wherein a respective buffer circuit includes data paths for communicating data between the memory controller and the respective group of memory devices (more than one memory bus (data paths) may be used to carry (communicating) data signals from (between) the memory controller to the memory devices; paragraph [0035]), the data paths being controlled by at least one of the data buffer control signals (memory buses (data paths) pass (controlled by) clock signals from clock buffer 716 (data buffer control signals); figure 7; paragraph [0042]).

As per claim 5, Leddige discloses the memory module of claim 1. Leddige additionally discloses wherein a respective buffer circuit includes one or more programmable registers (the interface circuits may include control logic for register elements (programmable registers) to drive address lines; paragraph [0040]) controlling a phase of the local clock signal (delay lock loop 758 received a clock signal the system and generates a signal having a different frequency than the clock; figure 7; paragraph [0042]).

---Continued Within the Next Supplemental Box---

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. VII Certain defects in the international application

The following defects in the form or contents of the international application have been noted:

Claim 9 is objected to under PCT Rule 66.2(a)(iii) as containing the following defects in the form or contents thereof: Claim 9 recites in line 9, "operations a the memory devices". As best understood these are typographical errors and for the purposes of this opinion will be interpreted as being "operations at the memory devices".

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

Claims 3, 5, and 7 are objected to under PCT Rule 66.2(a)(v) as lacking clarity under PCT Article 6 because claims 3, 5 and 7 are indefinite for the following reason(s): In line 1 of claims 3, 5, and 7, respectively, the claims recite "the respective buffer circuit". There is a lack of antecedent basis for these limitations in the claims. For the purpose of this opinion, as best understood, the claims are interpreted to read "a respective buffer circuit".

Claim 4 is objected to under PCT Rule 66.2(a)(v) as lacking clarity under PCT Article 6 because claim 4 is indefinite for the following reason(s): line 1 of claim 4 recites "the plurality of buffer circuits". There is a lack of antecedent basis for this limitation in the claim. For the purpose of this opinion, as best understood, the claim is interpreted to read "a plurality of buffer circuits".

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

-***-Continued from Box V: Citations and Explanations-***-

As per claim 6, Leddige discloses the memory module of claim 5. Leddige additionally discloses comprising: a module control device to receive command/address (C/A) signals (the devices on the memory module are coupled to command and address buses which may carry address and control transaction data (receive); paragraph [0037]) and a system clock from the memory controller (clock signals are passed from the memory controller; figure 7; paragraphs [0035] and [0042]) and to output module C/A signals (command and address buses carry address and control transaction data (output); paragraph [0037]), data buffer control signals (data is sent to and receive from write buffer 712 and read buffer 738; figure 7; paragraph [0039]) and a module clock signal (delayed clock signals (module clock signal) are passed from the memory controller; figure 7; paragraph [0042]), the module clock signal being derived from the system clock signal (the delayed clock signal (module clock signal) is generated by a delay lock loop based on (derived) clock signals from the system; paragraph [0042]), the module control device transmitting the module C/A signals to at least some of the memory devices (signals from the memory module controller passes command and address data (signals) to the memory devices of the respective module; paragraphs [0037]-[0039]), the module control device transmitting the data buffer control signals (the memory controller writes to and reads from write buffer 712 and read buffer 738, respectively; figure 7; paragraph [0040]) and the module clock signal to the data buffers (the delayed clock signal (module clock signal) is passed through the control logic to the read and write buffers; paragraphs [0040] and [0042]); wherein the module control device does not transmit any clock signal to the memory devices (module memory controller may omit (does not transmit) the clock signal for select memory devices; paragraphs [0037] and [0039]).

As per claim 7, Leddige discloses the memory module of claim 1. Leddige additionally discloses wherein a respective buffer circuit is configured to regenerate the local clock signal from a received clock signal (the clock generator 710 receives a clock signal from the system, and in response, generates (regenerate) a delayed clock signal (local clock signal) therefrom; figure 7; paragraph [0042]), and to phase-lock the local clock signal with the received clock signal (a delay lock loop (phase-lock) is used to generate a delayed clock signal based on the clock signal received from the system; figure 7; paragraph [0042]).

As per claim 9, Leddige discloses a method of operating a memory module coupled to a memory bus in a memory system, the memory module including memory devices, comprising: receiving a system clock signal (the module receives the system clock signal 724 through clock signal lines; figure 7; paragraph [0037]); generating a module clock signal (a delay lock loop generates a delayed clock signal; figure 7; paragraph [0042]); transmitting the module clock signal to a plurality of data buffers (the delayed clock signal is passed from clock generator 710 to write buffer 712 and read buffer 738; figure 7, paragraph [0039]) coupled between the memory devices and the memory bus (data is transmitted to the memory devices from the read and write buffers (coupled between) and the memory bus; figures 5 and 7; paragraphs [0037] and [0039]); generating local clock signals associated with respective ones of the data buffers (clock buffer 716 drives the clock signal to memory devices, based on the same clock signal (associated) transmitted to the read and write buffers; figure 7; paragraphs [0039] and [0042]); transmitting the local clock signals to the memory devices (a memory hub provides (transmitting) the clock signal 730 (local clock signals) to the memory devices; figure 7; paragraph [0037]); and perform read/write operations at the memory devices (command and address buses carry both address and control information for memory transactions (read/write operations) to the memory devices; paragraph [0037]) in accordance with the local clock signals (clock buffer 716 drives the clock signal (local clock signals) to memory devices; figure 7; paragraph [0042]).

Claims 4 and 10 lack an inventive step under PCT Article 33(3) as being obvious over Leddige in view of US 2009/0296503 A1 to CHU et al. (hereinafter "Chu").

As per claim 4, Leddige discloses the memory module of claim 2. Leddige additionally discloses the memory module wherein a respective buffer circuit is configured to regenerate the local clock signal from the module clock signal (the clock generator 710 receives a clock signal from the system, and in response, generates (regenerate) a delayed clock signal (local clock signal) therefrom; figure 7; paragraph [0042]), and provide the local clock signal to the respective group of memory devices (a memory hub provides (transmitting) the clock signal 730 (local clock signals) to the memory devices; figure 7; paragraph [0037]). Leddige does not disclose the memory module wherein a plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. Chu discloses the memory module wherein a plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time (the topology (distributed across a surface) of the memory devices results in the reception of the system clock signals (control signal arrives) by the memory devices (plurality of buffer circuits) at different times; paragraph [0007]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Chu, because by doing so, the memory module and method of Leddige would provide an efficient transfer of data from an input device to the core circuitry of the memory module, thereby providing faster and more efficient data transfers.

As per claim 10, Leddige discloses a method of claim 9. Leddige does not disclose the method further comprising incorporating a programmable amount of delay to the local clock signals. Chu discloses the method further comprising incorporating a programmable amount of delay to the local clock signals (the system calibrates a delay setting (a programmable amount of delay) in order to control the clock signals provided to the immediate registers (local clock signals); paragraph [0006]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Chu, because by doing so, the memory module and method of Leddige would provide an efficient transfer of data from an input device to the core circuitry of the memory module, thereby providing faster and more efficient data transfers.

-***-Continued Within the Next Supplemental Box-***-

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US14/48517

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

-Continued from Previous Supplemental Box--

Claim 8 lacks an inventive step under PCT Article 33(3) as being obvious over Leddige in view of US 2011/0085406 A1 to SOLOMON et al. (hereinafter "Solomon").

As per claim 8, Leddige discloses the memory module of claim 1. Leddige does not disclose the memory module wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks. Solomon discloses the memory module wherein memory devices are organized in one or more ranks (memory devices are arranged in four ranks; figure 1; paragraph [0045]), and wherein each group of memory devices including at least one memory device from each of the one or more ranks (memory device groups are arranged within ranks 32a, 32b, with each rank having a corresponding memory device coupled to a set of DQ data signal lines; figure 4A; paragraph [0053]). It would have been obvious to a person of ordinary skill in the art, at the time of the invention, to have modified the memory module and method of Leddige, with the teaching of Solomon, because by doing so, the memory module and method of Leddige would provide a design wherein larger memory modules are possible, thereby providing a more robust memory module having greater memory capacities.

Claims 1-10 have industrial applicability as defined by PCT Article 33(4) because the subject matter can be made or used in industry.

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	2806	1	90	90
Total in USD (\$)				90

Electronic Acknowledgement Receipt

EFS ID:	24910669
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	15-FEB-2016
Filing Date:	28-JUL-2014
Time Stamp:	00:28:55
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$90
RAM confirmation Number	7669
Deposit Account	
Authorized User	

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-02-14_IDS_Cover_NT006US.pdf	61619 a8c6fb2f30e040d878ba930e074575a8655f ebb1	no	2
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2016-02-14_IDS_filed_NT006US.pdf	119961 ea0658d29e52f353d5933c258db8c5d2eb2 47b92	no	2
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	2014-10-27_ISR_PCTUS1448517_NT006PCT.pdf	831933 2299288aa5e8948c9ec51f304321d0c4b0e 1a0e4	no	17
Warnings:					
Information:					
4	Non Patent Literature	2016-02-11_IPRP_NT006PCT.pdf	361856 6a130f6089946ffa5867fa97b88bc38cffc6 d31	no	8
Warnings:					
Information:					
5	Non Patent Literature	2014-07-31_NFOA_NT003US.pdf	713650 b28053250a5029a7aebc563b83f983944b5 227fb	no	18
Warnings:					
Information:					
6	Non Patent Literature	2014-12-31_Amendment_NT003US.pdf	132651 cfad8065bafb9d79ac14decc757e99d22550 e4f2	no	11
Warnings:					
Information:					
7	Non Patent Literature	2015-04-22_NOA_NT003US.pdf	1689696 147dec1439d73084e9a59ca6b60f3150defc 1f70	no	31
Warnings:					
Information:					

8	Non Patent Literature	2014-12-16_Preliminary_Response_01369.pdf	4298268 c1789e6892f6c7dde2ec41bcbebb40114671111a	no	62
Warnings:					
Information:					
9	Non Patent Literature	2015-03-09_Institution_Decision_01369.pdf	867692 0ac8262b507e89a629812f6a13849fa9c274493a	no	27
Warnings:					
Information:					
10	Fee Worksheet (SB06)	fee-info.pdf	30477 aad4f45336b3157041eb69b4c50cc288c88e7b37	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			9107803		
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INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/445,035
Filed	: July 28, 2014
For	: MEMORY MODULE LOCAL SYNCHRONIZATION
Examiner	: AHMED, ZUBAIR
Art Unit	: 2132
Conf. No.	: 9700
Docket No.	: NT006US

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 34 references. Copies of foreign documents and/or their translation and non-patent literature are submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 14/445,035
Filing Date: July 28, 2014

The Commissioner is authorized to charge any required fees to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: February 14, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51,167
(Reg. No.)

GENERAL POWER OF ATTORNEY BY APPLICANT FOR UNITED STATES PATENT

I hereby revoke all previous powers of attorney given in the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

I hereby appoint the practitioner associated with the following Customer Number
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79141

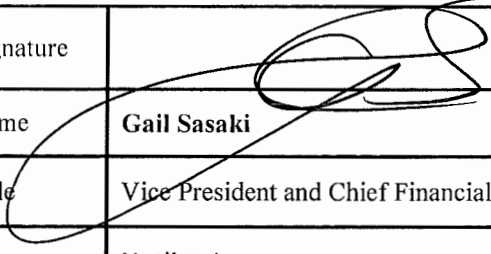
as my/our attorney or agent, and to transact all business in the United States Patent and Trademark Office in connection with the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

Please recognize or change the **correspondence address** for the application referenced in the attached transmittal letter to the address associated with the above-mentioned **Customer Number**.

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is currently being filed in this document)

SIGNATURE of Applicant for Patent

Signature		Date: January 9, 2013
Name	Gail Sasaki	
Title	Vice President and Chief Financial Officer	
Company	Netlist, Inc.	

Note: Signature - this form must be signed by the Applicant or a person authorized to act on behalf of Applicant.

Submit multiple forms for more than one signature
Total of ___ forms submitted.

Electronic Acknowledgement Receipt

EFS ID:	25041042
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	26-FEB-2016
Filing Date:	28-JUL-2014
Time Stamp:	22:09:38
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	2016-02-26_AIA82A_transmittal_POA_NT006US.pdf	195836 a05334792a1aa89a1eccc69b78da1b12cfed919a3	no	3

Warnings:

Information:

2	Power of Attorney	AIA_General_POA_Netlist.pdf	133545	no	1
			f8258889ee9edc63b1cc17705b3158faaed49e9e		
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
Total Files Size (in bytes):			329381		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

RESCISSION OF PREVIOUS NONPUBLICATION REQUEST (35 U.S.C. 122(b)(2)(B)(ii)) AND, IF APPLICABLE, NOTICE OF FOREIGN FILING (35 U.S.C. 122(b)(2)(B)(iii))	Application Number	14445035
	Filing Date	July 28, 2014
	First Named Inventor	Hyun Lee
	Title	Memory Module with Local Synchronization
	Atty Docket Number	NT006US
	Art Unit	2189
	Examiner	AHMED, ZUBAIR

Send completed form to:
Mail Stop PG Pub
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, VA 22313-1450
 FAX: (571) 273-8300

A request that the above-identified application not be published under 35 U.S.C. 122(b) (nonpublication request) was included with the above-identified application on filing pursuant to 35 U.S.C. 122(b)(2)(B)(i).
 I hereby **rescind** the previous nonpublication request.

If a notice of foreign or international filing is or will be required by 35 U.S.C. 122(b)(2)(B)(iii) and 37 CFR 1.213(c), I hereby provide such notice. This notice is being provided no later than forty-five (45) days after the date of such foreign or international filing.

If a notice of subsequent foreign or international filing required by 35 U.S.C. 122(b)(2)(B)(iii) and 37 CFR 1.213(c) was not filed within forty-five (45) days after the date of filing of the foreign or international application, the application is ABANDONED, and a petition to revive under 37 CFR 1.137(b) is required. See 37 CFR 1.137(f).

/Jamie J. Zheng/

February 26, 2016

Signature

Date

Jamie J. Zheng

51167

Typed or printed name

Registration Number, if applicable

650-521-4828

Telephone Number

This request must be signed in compliance with 37 CFR 1.33(b).

If information or assistance is needed in completing this form, please contact the Pre-Grant Publication Division at (703)605-4283 or by e-mail at PGPub@USPTO.gov.

CERTIFICATE OF MAILING OR TRANSMISSION

I hereby certify that this correspondence is being deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Mail Stop PG Pub, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450, or facsimile transmitted to the U.S. Patent and Trademark Office on the date shown below.

Signature

Name (Print/Type)

Date

This collection of information is required by 37 CFR 1.213(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 6 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Mail Stop PG Pub, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pet. Revive Abandon App, Delay Pymt-Resp	2453	1	850	850
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				850



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Decision Date : February 26, 2016

In re Application of :
Hyun Lee

DECISION ON PETITION

Application No: 14445035

Filed: 28-Jul-2014

Attorney Docket No: NT006US

This is an electronic decision on the petition, filed February 26, 2016, which is being treated as a petition under 37 CFR 1.137(a) to revive the instant nonprovisional application for failure to timely notify the U.S. Patent and Trademark Office (USPTO) of the filing of an application in a foreign country, or under a multinational treaty that requires publication of applications eighteen months after filing. See 37 CFR 1.137(f).

The petition is **GRANTED**.

Petitioner states that the present nonprovisional application is the subject of a foreign or international application filed on 07-28-2014

However, the USPTO was unintentionally not notified of this filing within 45 days subsequent to the filing of the subject application in a foreign country.

In view of the above, this application became abandoned pursuant to 35 U.S.C. § 122(b)(2)(B)(iii) and 37 CFR 1.213(c) for failure to timely notify the Office of the filing of an application in a foreign country, or under a multilateral international agreement that requires publication of applications 18 months after filing.

A petition under 37 CFR 1.137(f) must be accompanied by:

- (1) the reply which is met by the notification of such filing in a foreign country or under a multinational treaty;
- (2) the petition fee as set forth in 37 CFR 1.17(m); and
- (3) a statement that the entire delay in filing the required notice of a foreign or international filing from the due date for the required notice until the filing of a grantable petition under 37 CFR 1.137(a) was unintentional.

The instant petition has been found to be in compliance with 37 CFR 1.137(a). Accordingly, the failure to timely notify the USPTO of a foreign or international filing within 45 days after the date of filing of such foreign or international application as provided by 35 U.S.C. § 122(b)(2)(B)(iii) and 37 CFR 1.213(c) is accepted as having been unintentionally delayed.

The previous Request and Certification under 35 U.S.C. § 122(b)(2)(B)(i) has been rescinded. A Notice Regarding Rescission of Nonpublication Request which sets forth the projected publication date will be viewable in Private PAIR within one (1) business day.

It is not apparent whether the person signing the statement of unintentional delay was in a position to have firsthand or direct knowledge of the facts and circumstances of the delay at issue. Nevertheless, such statement is being treated as having been made as the result of a reasonable inquiry into the facts and circumstances of such delay. See 37 CFR 10.18(b) and Changes to Patent Practice and Procedure; Final Rule Notice, 62 Fed. Reg. 53131, 53178 (October 10, 1997), 1203 Off. Gaz. Pat. Office 63, 103 (October 21, 1997). In the event that such an inquiry has not been made, petitioner must make such an inquiry. If such inquiry results in the discovery that it is not correct that the entire delay in filing the required reply from the due date for the reply until the filing of a grantable petition pursuant to 37 CFR 1.137(a) was unintentional, petitioner must notify the Office.

This application file is being directed to the Office of Data Management.

Telephone inquiries concerning this decision should be directed to the Patent Electronic Business Center (EBC) at 866-217-9197.

Office of Petitions

Electronic Acknowledgement Receipt

EFS ID:	25041047
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	26-FEB-2016
Filing Date:	28-JUL-2014
Time Stamp:	22:18:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$850
RAM confirmation Number	7120
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Petition automatically granted by EFS	petition-request.pdf	32410	no	2
			f88e531fc01adf98a0d5a31cc16cd3a4a7ba6da2		

Warnings:

Information:

2	Rescind Nonpublication Request for Pre Grant Pub	2016-02-26_Rescind_NonPub_sb0036_NT006US.pdf	259335	no	2
			37677192bdc7c0d6f4df29eb4397a785b05899f		

Warnings:

Information:

3	Fee Worksheet (SB06)	fee-info.pdf	30493	no	2
			02d201e3680f16e169f8dc6faca648dadecc8363		

Warnings:

Information:

Total Files Size (in bytes):			322238
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Doc Code: PET.AUTO Document Description: Petition automatically granted by EFS-Web		PTO/SB/64a U.S. Patent and Trademark Office Department of Commerce	
Electronic Petition Request	PETITION FOR REVIVAL OF AN APPLICATION FOR PATENT ABANDONED FOR FAILURE TO NOTIFY THE OFFICE OF A FOREIGN OR INTERNATIONAL FILING UNDER 37 CFR 1.137(f)		
Application Number	14445035		
Filing Date	28-Jul-2014		
First Named Inventor	Hyun Lee		
Art Unit	2132		
Examiner Name	ZUBAIR AHMED		
Attorney Docket Number	NT006US		
Title	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		
The above-identified application became abandoned pursuant to 35 U.S.C. 122(b)(2)(B)(iii) for failure to timely notify the Office of the filing of an application in a foreign country or under a multinational treaty that requires publication of applications eighteen months after filing. The date of abandonment is the day after the expiration date of the forty-five (45) day period set in 35 U.S.C. 122(b)(2)(B)(iii). PURSUANT TO 37 CFR 1.137(f), APPLICANT HEREBY PETITIONS FOR REVIVAL OF THIS APPLICATION UNDER 37 CFR 1.137(a). A grantable petition requires the following items: (1) Petition fee; (2) Reply; (3) Statement that the entire delay was unintentional.			
Petition Fee			
☒ Small Entity			
☐ Micro Entity			
☐ Regular Undiscounted			
Notice of Foreign or International Filing (35 U.S.C. 122(b)(2)(B)(iii) and 37 CFR 1.213(c)) ☒ Subsequent to the filing of the above-identified application, an application was filed in another country, or under a multinational international treaty (e.g., filed under the Patent Cooperation Treaty), that requires publication of applications eighteen months after the filing. The filing date of the subsequently filed foreign or international application is 07-28-2014 ☒ The non-publication request has been filed on 28-Jul-2014 ☒ STATEMENT: The entire delay in filing the required notice of a foreign or international filing from the due date for the required notice until the filing of a grantable petition under 37 CFR 1.137(a) was unintentional.			

THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

- ☐ An attorney or agent registered to practice before the Patent and Trademark Office who has been given power of attorney in this application.
- ☒ An attorney or agent registered to practice before the Patent and Trademark Office, acting in a representative capacity.
- ☐ A sole inventor
- ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application
- ☐ A joint inventor; all of whom are signing this e-petition.

Signature	/Jamie J. Zheng/
Name	Jamie J. Zheng
Registration Number	51167

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B or equivalent) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5. If the Power of Attorney by Applicant form is not accompanied by this transmittal form or an equivalent, the Power of Attorney will not be recognized in the application.

Application Number	14/445,035
Filing Date	07/28/2014
First Named Inventor	HYUN LEE
Title	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
Art Unit	2189
Examiner Name	AHMED, ZUBAIR
Attorney Docket Number	NT006US

SIGNATURE of Applicant or Patent Practitioner

Signature	/Jamie J. Zheng/	Date	February 26, 2016
Name	Jamie J. Zheng	Telephone	650-273-6008
Registration Number	51,167		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in the attached transmittal letter.

- ☐ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

OR

- ☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

Name	Registration Number	Name	Registration Number

Please recognize or change the correspondence address for the application identified in the attached transmittal letter to:

- ☐ The address associated with the above-mentioned Customer Number.

OR

- ☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☐ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document)

SIGNATURE of Applicant for Patent

Signature

Date

Name

Telephone

Title and Company

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms for more than one signature, see below *.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: **Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/445,035	07/28/2014	Hyun Lee	NT006US

79141
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 9700
NONPUBLICATION RESCISSION
LETTER



Date Mailed: 02/29/2016

**Communication Regarding Rescission Of
Nonpublication Request and/or Notice of Foreign Filing**

Applicant's rescission of the previously-filed nonpublication request and/or notice of foreign filing is acknowledged. The paper has been reflected in the Patent and Trademark Office's (USPTO's) computer records so that the earliest possible projected publication date can be assigned.

The projected publication date is 06/09/2016.

If applicant rescinded the nonpublication request before or on the date of "foreign filing,"¹ then no notice of foreign filing is required.

If applicant foreign filed the application after filing the above application and before filing the rescission, and the rescission did not also include a notice of foreign filing, then a notice of foreign filing (not merely a rescission) is required to be filed within 45 days of the date of foreign filing. See 35 U.S.C. § 122(b)(2)(B)(iii), and Clarification of the United States Patent and Trademark Office's Interpretation of the Provisions of 35 U.S.C. § 122(b)(2)(B)(ii)-(iv), 1272 Off. Gaz. Pat. Office 22 (July 1, 2003).

If a notice of foreign filing is required and is not filed within 45 days of the date of foreign filing, then the application becomes abandoned pursuant to 35 U.S.C. § 122(b)(2)(B)(iii). In this situation, applicant should either file a petition to revive or notify the Office that the application is abandoned. See 37 CFR 1.137(f). Any such petition to revive will be forwarded to the Office of Petitions for a decision. Note that the filing of the petition will not operate to stay any period of reply that may be running against the application.

Questions regarding petitions to revive should be directed to the Office of Petitions at (571) 272-3282.

¹ Note, for purpose of this notice, that "foreign filing" means "filing an application directed to the same invention in another country, or under a multilateral international agreement, that requires publication of applications 18 months after filing".

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/eefswuser/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/445,035	07/28/2014	Hyun Lee	NT006US

79141
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 9700
POA ACCEPTANCE LETTER



Date Mailed: 03/10/2016

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 02/26/2016.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ltaba/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	NT006US	9700
79141	7590	03/21/2016		
Jamie J. Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306			EXAMINER AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			03/21/2016	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
eofficeaction@appcoll.com
infohub.j@gmail.com

Office Action Summary	Application No. 14/445,035	Applicant(s) LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 02/12/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-20 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 07/28/2014 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>2/15/16, 2/13/16</u> | 4) ☐ Other: _____ |

DETAILED ACTION

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This Office Action is responsive to amendment filed on 02/12/2016. In this amendment, new claims 11-20 have been added by the Applicant. Accordingly, claims 1-20 are pending in this application.

Response to Arguments

3. Applicant's arguments with respect to claims 1-10 have been considered but are moot because the arguments do not apply to any of the references being used in the current rejection.
4. In view of the new reference (Matsui US 2004/0105292), independent claims 1, 9, and 18 are not in a condition for allowance. Claims depending therefrom, either directly or indirectly, are also not in a condition for allowance.

Claim Rejections - 35 USC § 102

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

6. Claims 1, 9, 11-12, and 18-19 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Matsui US 2004/0105292 (hereinafter "Matsui").
7. As per independent claim 1, Matsui teaches "A memory module operable in a system with a memory controller, system operating according to a system clock, the

Art Unit: 2132

memory module having a data width of (N x n) bits, where N and n are integers larger than one, comprising” (Figure 5 – memory modules 103a-103d interfaced with a memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160),

8. “N data buffers each having a data width of n-bits and each generating a respective local clock signal derived from the system clock” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159),

9. “memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via

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a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159).

10. As per independent claim 9, Matsui teaches “A method of operating a memory module coupled to a memory bus in a system, the memory module having a data width of (N x n) bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising” (Figure 5 – memory modules 103a-103d interfaced with a memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-

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bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160),

11. “receiving a system clock signal” (Figure 4 – clock generator 102),
12. “generating a module clock signal” (Figure 5 – CLK/CLKB 113),
13. “transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n-bits” (Figure 5 – CLK/CLKB 113. Each data buffer 105a and 105b is coupled to a group of DRAM 110a and accordingly each data buffer may have a plurality of bits),
14. “generating a respective local clock signal at each of the N data buffers” (Figure 5 – CLK/CLKB 113’ generated by buffers 105a and 105b),
15. “transmitting the local clock signals to respective groups of the memory devices” (Figure 5 – CLK/CLKB 113’ transmitted to two groups of DRAMs 110),
16. “performing read/write operations at the memory devices in accordance with the local clock signals” (Figure 5 – DQ 111’ connected to the two groups of DRAMs 110 and synchronized by CLK/CLKB 113’).
17. As per dependent claim 11, Matsui discloses the method of claim 9. Matsui teaches “receiving command/address (C/A) signals from the memory controller” (Figure 5 – C/A 112),
18. “generating module C/A signals and data buffer control signals in response to the C/A signals” (Figure 5 – C/A 112 and CLK/CLKB 113. According to the instant

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specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035),

19. “transmitting the module C/A signals to at least some of the memory devices” (Figure 5 – C/A 112' transmitted to the two groups of DRAMs 110),

20. “transmitting the data buffer control signals to the N data buffers” (Figure 5 – CLK/CLKB 113. According to the instant specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035).

21. As per dependent claim 12, Matsui teaches the method of claim 11. Matsui teaches “further comprising controlling data paths in the N data buffers according to the data buffer control signals” (Figure 5 – DQ 111' is controlled by CLK/CLKB 113').

22. As per independent claim 18, Matsui teaches “A memory module having a width of N x n bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n integers larger than one, comprising” (Figure 5 – memory modules 103a-103d interfaced with a memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160),

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23. “a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals” (Figure 5 – C/A 112, CLK/CLKB 113 are received signals by the memory module 103a. C/A 112' and CLK/CLKB 113' are generated signals by the buffers 105a and 105b),

24. “N data buffers each n-bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals” (Figure 5 – memory modules 103a-103d interfaced with a memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. Figure 5 – buffers 105a and 105b generate CLK/CLKB 113' to control the two groups of DRAMs

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110 and data bus DQ 111' is coupled to the two groups of DRAMs 110 via the buffers 105a and 105b),

25. "memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal" (Figure 5 – buffers 105a and 105b are coupled to two groups of DRAMs 110 and C/A 112', data bus DQ 111' and CLK/CLKB 113' are connected to the two groups of DRAMs 110 via the buffers 105a and 105b).

26. As per dependent claim 19, Matsui teaches the memory module of claim 18. Matsui teaches "wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board" (Figure 3).

Claim Rejections - 35 USC § 103

27. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole

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would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

28. Claims 2-5, 7, 13-14, 16, and 20 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja et al. US 2007/0168781 (hereinafter "Sutardja").

29. As per dependent claim 2, Matsui discloses the memory module of claim 1. Matsui's invention discloses all of the claimed limitations from above except for "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers" and "wherein the module control device does not transmit any clock signal to the memory devices".

30. However, Sutardja teaches "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers" (Figure 17 - the buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on

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the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136),

31. “wherein the module control device does not transmit any clock signal to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

32. Given the teaching of Sutardja, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers” and “wherein the module control device does not transmit any clock signal to the memory devices”. The motivation would be that the buffer module may buffer signals between the memory control module and the memory modules, para 0136 of Sutardja.

33. As per dependent claim 3, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui teaches “wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one

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of the data buffer control signals” (Figure 5 - command/address signal and data are transmitted in the internal command/address wiring 112' and the internal data wiring 111', para 0157).

34. As per dependent claim 4, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for “wherein the N data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time” and “wherein the respective one of the N data buffers is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices”.

35. However, Sutardja teaches “wherein the N data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time” (The memory ICs 520 and the buffer module 530 for each memory module 514 may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias, para 0135. It is well known in the art that in such an arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530),

36. “wherein the respective one of the N data buffers is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

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37. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 4.

38. As per dependent claim 5, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal".

39. However, Sutardja teaches "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal" (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. It is well known in the art that a PLL or a DLL (digital lock loop) contain programmable registers to control/adjust the phase and frequency of PLL or DLL).

40. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 5.

41. As per dependent claim 7, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for "wherein the respective one of the N data buffers is configured to regenerate the local clock signal from the module clock signal clock signal, and to phase-lock the local clock signal with the module clock signal".

42. However, Sutardja teaches "wherein the respective one of the N data buffers is configured to regenerate the local clock signal from the module clock signal clock

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signal, and to phase-lock the local clock signal with the module clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

43. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 7.

44. As per dependent claims 13-14, these claims are rejected based on arguments provided for similar rejected dependent claims 4-5 respectively.

45. As per dependent claim 16, Matsui discloses the method of claim 9. Matsui’s invention discloses all of the claimed limitations from above except for “further comprising phase-locking the local clock signal with the module clock signal”.

46. However, Sutardja teaches “further comprising phase-locking the local clock signal with the module clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

47. Given the teaching of Sutardja, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with “further comprising phase-locking the local clock signal with the module clock signal”. The motivation would be that the buffer module may buffer signals between the memory control module and the memory modules, para 0136 of Sutardja.

48. As per dependent claim 20, this claim is rejected based on arguments provided above for similar rejected dependent claim 16.

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49. Claim 6 is rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereinafter “Carpenter”).

50. As per dependent claim 6, Matsui in combination with Sutardja discloses the memory module of claim 5. Matsui’s invention modified by Sutardja discloses all of the claimed limitations from above except for “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals”.

51. However, Carpenter teaches “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals” (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

52. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui as modified by Sutardja with “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals”. The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

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53. Claims 10 and 15 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Carpenter.

54. As per dependent claim 10, Matsui discloses the method of claim 9. Matsui's invention discloses all of the claimed limitations from above except for "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers".

55. However, Carpenter teaches "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers" (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

56. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

57. As per dependent claim 15, Matsui discloses the method of claim 11. Matsui's invention discloses all of the claimed limitations from above except for "programming

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one or more registers in each of the N data buffers according to the data buffer control signals”.

58. However, Carpenter teaches “programming one or more registers in each of the N data buffers according to the data buffer control signals” (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

59. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui with “programming one or more registers in each of the N data buffers according to the data buffer control signals”. The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

60. Claims 8 and 17 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sung et al. US 2011/0110168 (hereinafter “Sung”).

61. As per dependent claim 8, Matsui discloses the memory module of claim 1. Matsui’s invention discloses all of the claimed limitations from above except for “wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks”.

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62. However, Sung teaches “wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks” (Figures 1, 2 correspond specifically to a case in which the first memory unit 113 includes two memory ranks, and in this regard, two first chip selection signals are input to the first memory unit 113. The second memory unit 123 includes two memory ranks, and in this regard, two second chip selection signals are input to the second memory unit 123, paras 0046, 0049).

63. Given the teaching of Sung, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with “wherein memory devices are organized in one or more ranks, and wherein each group of memory devices including at least one memory device from each of the one or more ranks”. The motivation would be that the memory controller can determine which memory rank of which memory module is to be selected, para 0052 of Sung.

64. As per dependent claim 17, this claim is rejected based on arguments provided above for similar rejected dependent claim 8.

Conclusion

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 8:30AM - 5:00PM Mon-Fri.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID X YI/
Supervisory Patent Examiner, Art Unit 2132

Notice of References Cited	Application/Control No. 14/445,035		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2004/0105292 A1	06-2004	Matsui, Yoshinori	G06F13/4243	365/63
*	B	US-2007/0168781 A1	07-2007	Sutardja; Sehat	G06F11/1008	714/718
*	C	US-2009/0116312 A1	05-2009	Carpenter; Gary D.	G11C7/22	365/194
*	D	US-2011/0110168 A1	05-2011	Sung; Myung-hee	G11C5/04	365/189.09
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
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	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
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	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	11/02/2015	03/13/2016						
	1	✓	✓						
	2	✓	✓						
	3	✓	✓						
	4	✓	✓						
	5	✓	✓						
	6	✓	✓						
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	19		✓						
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16

S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2015/11/01 06:14

			IBM_TDB			
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S36	38	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 11:42
S37	8	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:13
S38	68	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:18
S39	4415	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S40	5	S39 and PLL	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27

Search Notes 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/08	03/13/2016	ZA

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
See attached search history	03/13/2016	ZA

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
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US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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	23.	14229844	03-29-2014	Netlist, Inc.	Entire Prosecution History
	24.	14337168	07-21-2014	Netlist, Inc.	Entire Prosecution History
	25.	14715486	05-18-2015	Netlist, Inc.	Entire Prosecution History
	26.	14846993	09-07-2015	Netlist, Inc.	Entire Prosecution History

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NON PATENT LITERATURE DOCUMENTS			
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	27.	International Search Report and Written Opinion, PCT/US2014/048517, October 27, 2014.	
	28.	International Preliminary Report on Patentability, PCT/US2014/048517, February 2, 2016.	
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	15.	5060188	10-22-1991	Zulian et al.	
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				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
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	269.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
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				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	11	of	13	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
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DATE CONSIDERED 03/13/2016

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	12	of	13	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS				
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EXAMINER /Zubair Ahmed/ (03/13/2016)

DATE CONSIDERED 03/13/2016

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
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	320.	Toal et al. "A 32-Bit SoPC Implementation of a P5." Proceedings of the Eighth IEEE International Symposium on Computers and Communications, 2003, pp. 1530-1346	
	321.	Tudruj, Marek. "Dynamically reconfigurable heterogenous multi-processor systems with transputer-controlled communication," Journal of Systems Architecture, vol. 43, 1997, pp. 27-32	
	322.	JEDEC Standard JESD79D, "Double Data Rate (DDR) SDRAM Specification," published Jan. 2004 (Feb. 2004)	
	323.	PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification JEDEC, Standard No. 21-C, Revision 1.3, Jan. 2002, pages 4.20.4-1	

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BIB DATA SHEET

CONFIRMATION NO. 9700

SERIAL NUMBER 14/445,035	FILING or 371(c) DATE 07/28/2014 RULE	CLASS 711	GROUP ART UNIT 2132	ATTORNEY DOCKET NO. NT006US		
APPLICANTS NETLIST, INC., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; ** CONTINUING DATA ***** This appln claims benefit of 61/859,215 07/27/2013 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 08/06/2014						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/ZUBAIR AHMED/</u> Examiner's Signature		☐ Met after Allowance Initials _____	STATE OR COUNTRY CA	SHEETS DRAWINGS 22	TOTAL CLAIMS 10	INDEPENDENT CLAIMS 2
ADDRESS Jamie J. Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES						
TITLE MEMORY MODULE WITH LOCAL SYNCHRONIZATION						
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
			☐ Credit			

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
Stylesheet Version v1.2

EPAS ID: PAT3833492

SUBMISSION TYPE:	NEW ASSIGNMENT
NATURE OF CONVEYANCE:	ASSIGNMENT
CONVEYING PARTY DATA	
Name	Execution Date
HYUN LEE	03/31/2016
JAYESH R. BHAKTA	03/31/2016
RECEIVING PARTY DATA	
Name:	NETLIST, INC.
Street Address:	175 TECHNOLOGY DR., SUITE#150
City:	IRVINE
State/Country:	CALIFORNIA
Postal Code:	92618
PROPERTY NUMBERS Total: 1	
Property Type	Number
Application Number:	14445035
CORRESPONDENCE DATA	
Fax Number:	
<i>Correspondence will be sent to the e-mail address first; if that is unsuccessful, it will be sent using a fax number, if provided; if that is unsuccessful, it will be sent via US Mail.</i>	
Phone:	6505214828
Email:	jz@jzpatent.com
Correspondent Name:	JAMIE J ZHENG
Address Line 1:	PO BOX 60573
Address Line 4:	PALO ALTO, CALIFORNIA 94306
ATTORNEY DOCKET NUMBER:	NT006US
NAME OF SUBMITTER:	JAMIE J. ZHENG
SIGNATURE:	/Jamie J. Zheng/
DATE SIGNED:	04/17/2016
This document serves as an Oath/Declaration (37 CFR 1.63).	
Total Attachments: 2	
source=2016-03-31_Declaration_Assignment_NT006US#page1.tif	
source=2016-03-31_Declaration_Assignment_NT006US#page2.tif	

**COMBINED DECLARATION AND ASSIGNMENT
FOR UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐ The attached application, or

☒ United States application or PCT international application
number 14/445,035 filed on July 28, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."

ASSIGNMENT

In consideration of the sum of One Dollar (\$1.00) and other good and valuable consideration paid, I hereby sell and assign to

Netlist, Inc.

(Name of Assignee)

of 175 Technology Dr., Suite #150, Irvine, CA 92618

(Address of Assignee)

(hereinafter designated as the Assignee) my entire right, title and interest for the United States as defined in 35 USC 100, for any invention set forth in the above-identified application.

I agree to execute all papers necessary in connection with this application for patent in the USPTO for the invention, and any continuation, divisional or reissue applications thereof and also to execute separate assignments in connection with such applications as the Assignee may deem necessary or expedient.

I agree to execute all papers necessary in connection with any post-grant proceeding which may occur in connection with this application or continuation, divisional or reissue thereof and to cooperate with the Assignee in every way possible in obtaining evidence and going forward with such post-grant proceeding.

I agree to execute all papers and documents and perform any act which may be necessary in connection with claims or provisions of the International Convention for Protection of Industrial Property or similar agreements.

I agree to perform all affirmative acts which may be necessary to obtain a grant of a valid United States patent to the

Assignee.

I hereby authorize and request the USPTO to issue any and all Letters Patents of the United States resulting from the application or any continuation, divisional or reissue applications thereof to the Assignee, as Assignee of my entire interest, and covenant that I have not executed, and will not execute, any agreement in conflict herewith.

First Inventor (Legal Name): Hyun Lee

Signature: 

Date: 3/31/2016

First Inventor (Legal Name): Jayesh R. Bhakta

Signature: 

Date: 3/31/2016

CERTIFICATION AND REQUEST FOR CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0		
Practitioner Docket No.: NT006US	Application No.: 14/445,035	Filing Date: 07/28/2014
First Named Inventor: Hyun Lee	Title: MEMORY MODULE WITH LOCAL SYNCHRONIZATION	
APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0 (AFCP 2.0) OF THE ACCOMPANYING RESPONSE UNDER 37 CFR 1.116. 1. The above-identified application is (i) an original utility, plant, or design nonprovisional application filed under 35 U.S.C. 111(a) [a continuing application (<i>e.g.</i>, a continuation or divisional application) is filed under 35 U.S.C. 111(a) and is eligible under (i)], or (ii) an international application that has entered the national stage in compliance with 35 U.S.C. 371(c). 2. The above-identified application contains an outstanding final rejection. 3. Submitted herewith is a response under 37 CFR 1.116 to the outstanding final rejection. The response includes an amendment to at least one independent claim, and the amendment does not broaden the scope of the independent claim in any aspect. 4. This certification and request for consideration under AFCP 2.0 is the only AFCP 2.0 certification and request filed in response to the outstanding final rejection. 5. Applicant is willing and available to participate in any interview requested by the examiner concerning the present response. 6. This certification and request is being filed electronically using the Office's electronic filing system (EFS-Web). 7. Any fees that would be necessary consistent with current practice concerning responses after final rejection under 37 CFR 1.116, <i>e.g.</i>, extension of time fees, are being concurrently filed herewith. [There is no additional fee required to request consideration under AFCP 2.0.] 8. By filing this certification and request, applicant acknowledges the following: • Reissue applications and reexamination proceedings are not eligible to participate in AFCP 2.0. • The examiner will verify that the AFCP 2.0 submission is compliant, <i>i.e.</i>, that the requirements of the program have been met (see items 1 to 7 above). For compliant submissions: ○ The examiner will review the response under 37 CFR 1.116 to determine if additional search and/or consideration (i) is necessitated by the amendment and (ii) could be completed within the time allotted under AFCP 2.0. If additional search and/or consideration is required but cannot be completed within the allotted time, the examiner will process the submission consistent with current practice concerning responses after final rejection under 37 CFR 1.116, <i>e.g.</i>, by mailing an advisory action. ○ If the examiner determines that the amendment does not necessitate additional search and/or consideration, or if the examiner determines that additional search and/or consideration is required and could be completed within the allotted time, then the examiner will consider whether the amendment places the application in condition for allowance (after completing the additional search and/or consideration, if required). If the examiner determines that the amendment does not place the application in condition for allowance, then the examiner will contact the applicant and request an interview. ▪ The interview will be conducted by the examiner, and if the examiner does not have negotiation authority, a primary examiner and/or supervisory patent examiner will also participate. ▪ If the applicant declines the interview, or if the interview cannot be scheduled within ten (10) calendar days from the date that the examiner first contacts the applicant, then the examiner will proceed consistent with current practice concerning responses after final rejection under 37 CFR 1.116.		
Signature /Jamie J. Zheng/	Date June 6, 2016	
Name (Print/Typed) Jamie J. Zheng	Practitioner Registration No. 51167	
Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.		
☐ * Total of _____ forms are submitted.		

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : NT006US

CERTIFICATE OF EFS WEB

TRANSMISSION

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

June 6, 16

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

RESPONSE TO FINAL OFFICE ACTION DATED MARCH 21, 2016

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Final Office Action dated March 21, 2016 for the above-identified patent application. Applicant requests consideration under the After Final Consideration Pilot Program 2.0 (AFCP 2.0) of this Amendment and the accompanying Remarks.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

1. (Cancelled)
2. (Currently Amended) ~~The memory module of claim 1, further comprising~~A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of ($N \times n$) bits, where N and n are integers larger than one, comprising:
 - N data buffers each having a data width of n -bits and each generating a respective local clock signal;
 - memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and
 - a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;
 - wherein the module control device does not transmit any clock signal to the memory devices; and
 - wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.
3. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.
4. (Currently Amended) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the N data buffers at different points in time, ~~and~~

~~wherein the respective one of the N data buffers generates is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.~~

5. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes one or more programmable registers controlling a phase of the local clock signal.

6. (Previously Presented) The memory module of claim 5,
wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals.

7. (Currently Amended) The memory module of claim 2, wherein the respective one of the N data buffers is configured to ~~regenerate the local clock signal from the module clock signal~~ and to phase-lock the local clock signal with the module clock signal.

8. (Currently Amended) The memory module of claim ~~2~~4, wherein memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices including ~~at least one~~ memory device from each of the one or more ranks.

9. (Currently Amended) A method ~~of operating performed by a~~ memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

receiving a system clock signal;

generating a module clock signal on the memory module from the system clock signal received by the memory module;

transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of

data buffers including N data buffers, each of the N data buffers having a data width of n -bits;

generating a respective local clock signal from the module clock signal at each of the N data buffers;

transmitting ~~the respective~~ local clock signals from the plurality of data buffers to respective groups of the memory devices; and

performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Previously Presented) The method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Previously Presented) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (Currently Amended) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of data buffers at different points in time, ~~and wherein the respective data buffer is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.~~

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.

15. (Previously Presented) The method of claim 11, further comprising:

programming one or more registers in each of the N data buffers according to the data buffer control signals.

16. (Previously Presented) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.

17. (Currently Amended) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include ~~at least~~ one memory device from each of the one or more ranks.

18. (Currently Amended) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the

corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Previously Presented) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board.

20. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

21. (New) The memory module of claim 2, wherein the module clock signal is transmitted by the module control device to the N data buffers in a fly-by arrangement.

22. (New) The memory module of claim 18, wherein the module clock signal is transmitted by the control circuit to the N data buffers in a fly-by arrangement.

23. (New) The memory module of claim 18, wherein the control circuit does not transmit any clock signal to the memory devices.

REMARKS

This amendment responds to the Final Office Action dated March 21, 2016. In the Office Action, the Examiner:

- rejected claims 1, 9, 11-12 and 18-19 under 36 U.S.C. 102(a)(1) as being anticipated by Matsui US 2004/0105292 (hereafter “Matsui”);
- rejected claims 2-5, 7, 13-14, 16, and 20 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”)); and
- rejected claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sung et al. US 2011/0110168 (hereafter Sung).

Claim 1 is cancelled without prejudice. Claims 2, 4, 8-9, 13, and 17-18 are amended. Claim 18 is amended to correct a typographical or grammatical error. New claims 21-23 are presented. Support for the amendments and for the new claims can be found in at least paragraphs [0049] and [0051], FIGS. 2D and 6A-6C, and FIGS. 2A-2B, 3, and 4A-4B. Thus, no new matter is added.

Claim 2 is amended to incorporate the limitations in claim 1. As amended, claim 2 recites:

A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal;

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices; and

wherein the each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.

Thus, the memory module in claim 2 as amended requires, among other things, a module control device on the memory module “to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal.” Claim 2 as amended further requires that “each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device,” and that “each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers.”

As acknowledged by the Examiner, Matsui does not disclose “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, ... wherein the module control device does not transmit any clock signal to the memory devices.” The Examiner then erred by stating that Sutardja teaches such a module control device by referring to Figure 17 and paragraph [0136] in Sutardja, where it is stated that the buffer module 530 may buffer signals between the memory control module 510, the memory module 514, and/or signals on the buses 531 and 533. The buffer module 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS_, etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136), ... the buffer module 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals.” The Examiner further stated that the motivation to combine Matsui with Sutardja “would be that the buffer module may buffer signals between the memory control module and the memory modules.”

It is unclear what in Sutardja the Examiner regards as the “module control device.” It cannot be the buffer module 530 because the buffer module 530 is used as a

buffer for the data signals in addition to the address, control, and clock signals between the memory module 531 and the memory control module 510. Therefore, there is no need for this buffer module 530 to control any other buffers on the memory module. Also, as recognized by the Examiner, the buffer module 530 generates local phase-adjusted clock signals for the memory devices. So the buffer module 530 in Sutardja cannot be the “module control device” in claim 2 as amended, which “does not transmit any clock signal to the memory devices.”

The clock generator module 534 in Sutardja also cannot be the “module control device” in claim 2 as amended. First of all, the clock generator module 534 is not even on the memory module 531 and it does not “receive command/address (C/A) signals and a system clock from the memory controller.” Furthermore, the clock generator module 534 does not “output module C/A signals, data buffer control signals and a module clock signal,” and it does not transmit “the module C/A signals to at least some of the memory devices.” As shown in Figure 17 in Sutardja, the only thing the clock generator module 534 does is to generate clock signals for the module control device 510 and the memory modules 531.

Therefore, Sutardja does not disclose or teach the module control device as recited in claim 2 as amended.

Moreover, there is no motivation for the suggested combination. As stated in Matsui, the embodiment of FIGS. 4 and 5, which the examiner used to reject claim 2, are provided to reduce the total input load of giving the clocks and the command/address signals to all the DRAMs 110 on each memory module 103 (paragraphs [0153] and [0154]). Thus, Matsui provides two buffers 105a and 105b connected to DRAMs 110 arranged on both left and right sides of the memory module 103.

As shown in FIGS. 4 and 5 of Matsui, the buffers 105a and 105b buffer the data, address, control and clock signals between the memory module 103 and memory controller 101. Thus, “the buffers 105a and 105b within each memory module 103 are connected to a memory controller 101 or the memory module of the next stage via data wirings 111, command/address wirings 112, and clock wirings 113.” (Paragraph [0156],

FIGS. 4-5) “In this manner, the data wirings 111, along with the command/address wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups.” (Paragraph [0158], FIGS. 4-5)

Thus, even if, for the sake of argument, Sutardja or any other reference teaches or discloses the module control device recited in claim 2 as amended, a person of ordinary skill in the art would not have any motivation to combine such teachings with Matsui. First of all, the combination would not add any benefit to Matsui because the buffers 105a and 105b are already buffering the control/address signals and the clock signal, so there is no need for another buffer to do the same. The combination would also require major changes in the functions of the buffers 105a and 105b and a major overhaul of the internal wirings of the memory module 103 and the wirings between the memory controller 101 and the memory module 103, because the combination would work against the teachings of Matsui, which stresses the importance of having both buffers 105a and 105b coupled to the memory controller 101 directly instead of through a module control device on the memory module.

Therefore, claim 2 as amended is patentable over Matsui in view of Sutardja.

Claim 3, claim 4 as amended, claims 5-6, claims 7-8 as amended and new claim 21 depend from claim 2 as amended. Therefore, claim 3, claim 4 as amended, claims 5-6, claims 7-8 as amended and new claim 21 are patentable for at least the same reasons claim 2 as amended is patentable.

Moreover, claims 3, claim 4 as amended, claims 5-6, claims 7-8 as amended and new claim 21 are patentable for the additional limitations recited therein. For example, Claim 3 recites the limitation that “the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.” Neither Matsui nor Sutardja discloses or teaches such limitations. The data buffers in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the memory module.

As a further example, claim 4 as amended recites that “the N data buffers are distributed across a surface of the memory module such that each data buffer control signal arrives at the N data buffers at different points in time.” As discussed above, Sutardja does not disclose multiple data buffers on a memory module. Matsui discloses two data buffers 105a and 105b on left and right sides of the memory module but these two data buffers 105a and 105b are connected to the memory controller 101 via “the data wirings 111, along with the command/address wirings 112 and the clock wirings 113 ... concentrically.” (Paragraph [0158], FIGS. 4-5). Thus, there are no data buffer control signal in either Matsui or Sutardja that arrives at the N data buffers on the same memory module at different points in time.

As a further example, claim 8 as amended recites that “the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices including one memory device from each of the one or more ranks.” This claim requires that each data buffer have the same data width as the memory devices. Clearly, this is not taught by either Matsui or Sutardja.

As a further example, new claim 21 recites that “the module clock signal is transmitted by the module control device to the N data buffers in a fly-by arrangement.” As described in paragraph [0049] and FIGS. 2D and 6A of the present application, in a fly-by signal routing configuration, signals are routed such that they connect to a row of devices in a daisy chain topology. This is clearly not disclosed in either Matsui or Sutardja. As discussed above, Sutardja does not disclose multiple data buffers on a memory module. Matsui discloses two data buffers 105a and 105b on left and right sides of the memory module that are connected to the memory controller 101 via “the data wirings 111, along with the command/address wirings 112 and the clock wirings 113 ... concentrically.” Thus, there is no fly-by signal routing to provide the module clock signal to the data buffers 105a and 105b in Matsui.

Claim 9 as amended recites:

A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory

module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

- receiving a system clock signal;
- generating a module clock signal on the memory module from the system clock signal received by the memory module;
- transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n-bits;
- generating a respective local clock signal from the module clock signal at each of the N data buffers;
- transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and
- performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

Claim 9 as amended cannot be rejected under 35 U.S.C. 102 (a)(1) by Matsui because Matsui does not disclose each and every limitation of the claim. For example, Matsui does not disclose “generating a module clock signal on the memory module from the system clock signal received by the memory module” and “transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus.” As discussed above, the buffers 105a and 105b on Matsui’s memory module 103 receive system the system clock signal CLK/CLKB from the memory controller 101 and provide buffered clock signals to the DRAMs 110a and NOT to other data buffers. Matsui also does not disclose “generating a respective local clock signal *from the module clock signal* at each of the N data buffers” and “performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals” (emphasis added). In Matsui, the buffers 105a and 105b relay the system clock signal CLK/CLKB received from the memory controller 101 to the memory devices instead of generating local clock signals based on a module clock signal received from a module control device on the memory module 103.

Therefore, claim 9 as amended is not anticipated by Matsui.

Claims 10-12, claim 13 as amended, claims 14-16, and claim 17 as amended depend from claim 9 as amended and includes further limitations in addition to the

limitations in claim 9 as amended. Therefore, claims 10-12, claim 13 as amended, claims 14-16, and claim 17 are patentable for at least the same reasons claim 9 as amended is patentable.

Moreover, claims 10-12, claim 13 as amended, claims 14-16, and claim 17 are patentable for the additional limitations recited therein. For example, claim 11 recites, among other things, “generating module C/A signals and data buffer control signals in response to the C/A signals” received from the memory controller and transmitting the data buffer control signals to the N data buffers.” As discussed above, the buffers in Matsui and Sutardja merely buffer the address/control signals and clock signals from the memory controller and provide the address/control signals and clock signals to the memory devices. Neither the buffers 105a and 105b in Matsui nor the buffer 531 in Sutardja transmits any data buffer control signals to some other data buffers.

As a further example, claim 12 recites “controlling data paths in the N data buffers according to the data buffer control signals.” As discussed above, the data buffers in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the memory module.

Claim 18 recites:

A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n-bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including

data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

Claim 18 cannot be rejected under 35 U.S.C. 102 (a)(1) by Matsui because Matsui does not disclose each and every limitation of claim 18. As recited, the control circuit and the data buffers in claim 18 have very distinct functions. On the other hand, Matsui's buffers 105a and 105b have identical functions and thus cannot be both the control circuit and the data buffers in claim 18. Furthermore, as discussed above, the data buffers in claim 18 receive the data buffer control signals and the module clock signal from the control circuit that is *on the memory module* and generate respective local clock signals in response to the module clock signal. In contrast, Matsui's buffers 105a and 105b do not communicate with each other or receive any signal from a control circuit *on the memory module*.

Therefore, claim 18 is not anticipated by Matsui.

Claims 19-20 and new claims 22-23 depend from claim 18 and include further limitations in addition to the limitations in claim 18. Therefore, claims 19-20 and new claims 22-23 are patentable for at least the same reasons claim 18 is patentable

Applicant submits that claims 2-23 are in condition for allowance and Applicant respectfully requests such action. This amendment is filed under the After Final Consideration Pilot Program 2.0 (AFCP 2.0). Therefore, Applicant appreciates the Examiner to call the undersigned at (650) 521-4828 to discuss any remaining issues in the claims.

Respectfully submitted,

Date: June 6, 2016

/Jamie J. Zheng/

51167

Jamie J. Zheng

(Reg. No.)

Customer No. 79141

Electronic Acknowledgement Receipt

EFS ID:	25985409
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
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Filing Date:	28-JUL-2014
Time Stamp:	23:17:53
Application Type:	Utility under 35 USC 111(a)

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	After Final Consideration Program Request	2016-05-27_sb0434_NT006US.pdf	226374 5205fa02a35ac323ade661cad1e2c8b530f7b7bf	no	2

Warnings:

Information:

2	Response After Final Action	2016-06-06_Amendment_NT006US.pdf	163249 a349a8db448e57b24dda364b0ffb7090f3e6026b	no	15
Warnings:					
Information:					
Total Files Size (in bytes):			389623		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/445,035	Filing Date 07/28/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	06/06/2016	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 22	Minus	** 20	= 2	x \$40 =	80
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	x \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
	TOTAL ADD'L FEE						80

(Column 1)

(Column 2)

(Column 3)

	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
	TOTAL ADD'L FEE					

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
/CAROL BARNES/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/445,035	07/28/2014	Hyun Lee	NT006US

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79141
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NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	NT006US	9700

79141 7590 06/16/2016
Jamie J. Zheng, Ph.D Esq.
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EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
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2132

NOTIFICATION DATE	DELIVERY MODE
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06/16/2016

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
eofficeaction@appcoll.com
infohub.j@gmail.com

Advisory Action Before the Filing of an Appeal Brief	Application No. 14/445,035	Applicant(s) LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

--The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

THE REPLY FILED 06 June 2016 FAILS TO PLACE THIS APPLICATION IN CONDITION FOR ALLOWANCE.

NO NOTICE OF APPEAL FILED

1. ☒ The reply was filed after a final rejection. No Notice of Appeal has been filed. To avoid abandonment of this application, applicant must timely file one of the following replies: (1) an amendment, affidavit, or other evidence, which places the application in condition for allowance;

(2) a Notice of Appeal (with appeal fee) in compliance with 37 CFR 41.31; or (3) a Request for Continued Examination (RCE) in compliance with 37 CFR 1.114 if this is a utility or plant application. Note that RCEs are not permitted in design applications. The reply must be filed within one of the following time periods:

a) ☒ The period for reply expires 3 months from the mailing date of the final rejection.

b) ☐ The period for reply expires on: (1) the mailing date of this Advisory Action; or (2) the date set forth in the final rejection, whichever is later. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of the final rejection.

c) ☐ A prior Advisory Action was mailed more than 3 months after the mailing date of the final rejection in response to a first after-final reply filed within 2 months of the mailing date of the final rejection. The current period for reply expires _____ months from the mailing date of the prior Advisory Action or SIX MONTHS from the mailing date of the final rejection, whichever is earlier.

Examiner Note: If box 1 is checked, check either box (a), (b) or (c). ONLY CHECK BOX (b) WHEN THIS ADVISORY ACTION IS THE FIRST RESPONSE TO APPLICANT'S FIRST AFTER-FINAL REPLY WHICH WAS FILED WITHIN TWO MONTHS OF THE FINAL REJECTION. ONLY CHECK BOX (c) IN THE LIMITED SITUATION SET FORTH UNDER BOX (c). See MPEP 706.07(f).

Extensions of time may be obtained under 37 CFR 1.136(a). The date on which the petition under 37 CFR 1.136(a) and the appropriate extension fee have been filed is the date for purposes of determining the period of extension and the corresponding amount of the fee. The appropriate extension fee under 37 CFR 1.17(a) is calculated from: (1) the expiration date of the shortened statutory period for reply originally set in the final Office action; or (2) as set forth in (b) or (c) above, if checked. Any reply received by the Office later than three months after the mailing date of the final rejection, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

NOTICE OF APPEAL

2. ☐ The Notice of Appeal was filed on _____. A brief in compliance with 37 CFR 41.37 must be filed within two months of the date of filing the Notice of Appeal (37 CFR 41.37(a)), or any extension thereof (37 CFR 41.37(e)), to avoid dismissal of the appeal. Since a Notice of Appeal has been filed, any reply must be filed within the time period set forth in 37 CFR 41.37(a).

AMENDMENTS

3. ☒ The proposed amendments filed after a final rejection, but prior to the date of filing a brief, will not be entered because

a) ☒ They raise new issues that would require further consideration and/or search (see NOTE below);

b) ☐ They raise the issue of new matter (see NOTE below);

c) ☐ They are not deemed to place the application in better form for appeal by materially reducing or simplifying the issues for appeal; and/or

d) ☐ They present additional claims without canceling a corresponding number of finally rejected claims.

NOTE: _____. (See 37 CFR 1.116 and 41.33(a)).

4. ☐ The amendments are not in compliance with 37 CFR 1.121. See attached Notice of Non-Compliant Amendment (PTOL-324).

5. ☐ Applicant's reply has overcome the following rejection(s): _____.

6. ☐ Newly proposed or amended claim(s) _____ would be allowable if submitted in a separate, timely filed amendment canceling the non-allowable claim(s).

7. ☒ For purposes of appeal, the proposed amendment(s): (a) ☒ will not be entered, or (b) ☐ will be entered, and an explanation of how the new or amended claims would be rejected is provided below or appended.

AFFIDAVIT OR OTHER EVIDENCE

8. ☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

9. ☐ The affidavit or other evidence filed after final action, but before or on the date of filing a Notice of Appeal will not be entered because applicant failed to provide a showing of good and sufficient reasons why the affidavit or other evidence is necessary and was not earlier presented. See 37 CFR 1.116(e).

10. ☐ The affidavit or other evidence filed after the date of filing the Notice of Appeal, but prior to the date of filing a brief, will not be entered because the affidavit or other evidence failed to overcome all rejections under appeal and/or appellant fails to provide a showing of good and sufficient reasons why it is necessary and was not earlier presented. See 37 CFR 41.33(d)(1).

11. ☐ The affidavit or other evidence is entered. An explanation of the status of the claims after entry is below or attached.

REQUEST FOR RECONSIDERATION/OTHER

12. ☒ The request for reconsideration has been considered but does NOT place the application in condition for allowance because:
See continuation sheet.

13. ☐ Note the attached Information *Disclosure Statement(s)*. (PTO/SB/08) Paper No(s). _____

14. ☒ Other: PTO-2323 attached.

STATUS OF CLAIMS

15. The status of the claim(s) is (or will be) as follows:

Claim(s) allowed: _____.

Claim(s) objected to: _____.

Claim(s) rejected: 2-23.

Claim(s) withdrawn from consideration: _____.

/ZUBAIR AHMED/
 Examiner, Art Unit 2132

Continuation of 12:

Further consideration and/or search is required. This AFCP 2.0 is treated under the pre-pilot program.

AFCP 2.0 Decision

Application No.

14/445,035

Applicant(s)

LEE ET AL.

Examiner

ZUBAIR AHMED

Art Unit

2132

This is in response to the After Final Consideration Pilot request filed 06 June 2016.

1. **Improper Request** – The AFCP 2.0 request is improper for the following reason(s) and the after final amendment submitted with the request will be treated under pre-pilot procedure.

- ☐ An AFCP 2.0 request form PTO/SB/434 (or equivalent document) was not submitted.
- ☐ A non-broadening amendment to at least one independent claim was not submitted.
- ☐ A proper AFCP 2.0 request was submitted in response to the most recent final rejection.
- ☐ Other:

2. **Proper Request**

A. After final amendment submitted with the request will not be treated under AFCP 2.0.

The after final amendment cannot be reviewed and a search conducted within the guidelines of the pilot program.

- ☒ The after final amendment will be treated under pre-pilot procedure.

B. Updated search and/or completed additional consideration.

The examiner performed an updated search and/or completed additional consideration of the after final amendment within the time authorized for the pilot program. The result(s) of the updated search and/or completed additional consideration are:

- ☐ 1. All of the rejections in the most recent final Office action are overcome and a Notice of Allowance is issued herewith.
- ☐ 2. The after final amendment would not overcome all of the rejections in the most recent final Office action. See attached interview summary for further details.
- ☐ 3. The after final amendment was reviewed, and it raises a new issue(s). See attached interview summary for further details.
- ☐ 4. The after final amendment raises new issues, but would overcome all of the rejections in the most recent final Office action. A decision on determining allowability could not be made within the guidelines of the pilot. See attached interview summary for further details, including any newly discovered prior art.
- ☐ 5. Other:

Examiner Note: Please attach an interview summary when necessary as described above.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : NT006US

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I hereby certify that this correspondence and
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June 6, 16

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

RESPONSE TO FINAL OFFICE ACTION DATED MARCH 21, 2016

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Final Office Action dated March 21, 2016 for the above-identified patent application. Applicant requests consideration under the After Final Consideration Pilot Program 2.0 (AFCP 2.0) of this Amendment and the accompanying Remarks.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
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For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

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CERTIFICATE OF EFS WEB

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June 21, 16

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Enclosed is an Amendment in response to the Final Office Action dated March 21, 2016 and a Request for Continued Examination for the above-identified patent application. This Amendment replaces the amendment filed on June 6, 2016, which was not entered.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

1. (Cancelled)
2. (Currently Amended) ~~The memory module of claim 1, further comprising~~A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of ($N \times n$) bits, where N and n are integers larger than one, comprising:
 - N data buffers each having a data width of n -bits and each generating a respective local clock signal;
 - memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and
 - a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;
 - wherein the module control device does not transmit any clock signal to the memory devices; and
 - wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.
3. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.
4. (Currently Amended) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that

each data buffer control signal arrives at some adjacent ~~the~~ N -data buffers at different points in time, and

~~wherein the respective one of the N data buffers generates is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.~~

5. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes one or more programmable registers controlling a phase of the local clock signal.

6. (Previously Presented) The memory module of claim 5, wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals.

7. (Currently Amended) The memory module of claim 2, wherein the respective one of the N data buffers is configured to ~~regenerate the local clock signal from the module clock signal~~ ~~clock signal~~, and to phase-lock the local clock signal with the module clock signal.

8. (Currently Amended) The memory module of claim ~~2~~ 4, wherein memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include ~~including at least one~~ memory device from each of the one or more ranks.

9. (Currently Amended) A method ~~of operating~~ performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

receiving a system clock signal;
generating a module clock signal on the memory module from the system clock signal received by the memory module;

transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;

generating a respective local clock signal from the module clock signal at each of the N data buffers;

transmitting ~~the respective~~ local clock signals from the plurality of data buffers to respective groups of the memory devices; and

performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Previously Presented) The method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Previously Presented) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (Currently Amended) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent ~~the plurality of~~ data buffers at different points in time, ~~and~~

~~wherein the respective data buffer is configured to regenerate the local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.~~

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.
15. (Previously Presented) The method of claim 11, further comprising:
programming one or more registers in each of the N data buffers according to the data buffer control signals.
16. (Previously Presented) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.
17. (Currently Amended) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices ~~include-at least~~ one memory device from each of the one or more ranks.
18. (Currently Amended) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:
- a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;
- N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and
- memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local

clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Previously Presented) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board.

20. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

21. (New) The memory module of claim 18, wherein the control circuit does not transmit any clock signals to the memory devices.

REMARKS

This amendment responds to the Final Office Action dated March 21, 2016. In the Office Action, the Examiner:

- rejected claims 1, 9, 11-12 and 18-19 under 36 U.S.C. 102(a)(1) as being anticipated by Matsui US 2004/0105292 (hereafter “Matsui”);
- rejected claims 2-5, 7, 13-14, 16, and 20 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”)); and
- rejected claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sung et al. US 2011/0110168 (hereafter Sung).

Claim 1 is cancelled without prejudice. Claims 2, 4, 8-9, 13, and 17-18 are amended. Claim 18 is amended to correct a typographical or grammatical error. New claim 21 is presented. Support for the amendments and for the new claims can be found in at least paragraphs [0049] and [0051], FIGS. 2D and 6A-6C, and FIGS. 2A-2B, 3, and 4A-4B. Thus, no new matter is added.

Claim 2 is amended to incorporate the limitations in claim 1. As amended, claim 2 recites:

A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal;

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices; and

wherein the each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.

Thus, the memory module in claim 2 as amended requires, among other things, a module control device on the memory module “to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal.” Claim 2 as amended further requires that “each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device,” and “each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers.”

As acknowledged by the Examiner, Matsui does not disclose “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, ... wherein the module control device does not transmit any clock signal to the memory devices.” The Examiner then erred by stating that Sutardja teaches such a module control device by referring to Figure 17 and paragraph [0136] in Sutardja, where it is stated:

“the buffer module 530 may buffer signals between the memory control module 510, the memory module 514, and/or signals on the buses 531 and 533. The buffer module 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS_, etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136), ... the buffer module 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals.”

The Examiner further stated that the motivation to combine Matsui with Sutardja “would be that the buffer module may buffer signals between the memory control module and the memory modules.”

It is unclear what in Sutardja the Examiner regards as the “module control device.” It cannot be the buffer module 530 because the buffer module 530 is used as a buffer for the data signals in addition to the address, control, and clock signals between

the memory module 531 and the memory control module 510. Therefore, this buffer module 530 does not control any other buffers on the memory module by generating data buffer control signals. Also, as recognized by the Examiner, the buffer module 530 generates local phase-adjusted clock signals for the memory devices. So the buffer module 530 in Sutardja cannot be the “module control device” in claim 2 as amended, which “does not transmit any clock signal to the memory devices.”

The clock generator module 534 in Sutardja also cannot be the “module control device” in claim 2 as amended. First of all, the clock generator module 534 is not even on the memory module 531 and it does not “receive command/address (C/A) signals and a system clock from the memory controller.” Furthermore, the clock generator module 534 does not “output module C/A signals, data buffer control signals and a module clock signal,” and it does not transmit “the module C/A signals to at least some of the memory devices.” As shown in Figure 17 in Sutardja, the only thing the clock generator module 534 does is to generate clock signals for the memory control module 510 and the memory modules 531.

Therefore, Sutardja does not disclose or teach the module control device as recited in claim 2 as amended.

Moreover, there is no motivation for the suggested combination. As stated in Matsui, the embodiment of FIGS. 4 and 5, which the examiner used to reject claim 2, are provided to reduce the total input load of giving the clocks and the command/address signals to all the DRAMs 110 on each memory module 103 (paragraphs [0153] and [0154]). Thus, Matsui provides two buffers 105a and 105b connected to DRAMs 110 arranged on both left and right sides of the memory module 103.

As shown in FIGS. 4 and 5 of Matsui, the buffers 105a and 105b buffer the data, address, control and clock signals between the memory module 103 and memory controller 101. Thus, “the buffers 105a and 105b within each memory module 103 are connected to a memory controller 101 or the memory module of the next stage via data wirings 111, command/address wirings 112, and clock wirings 113.” (Paragraph [0156], FIGS. 4-5) “In this manner, the data wirings 111, along with the command/address

wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups.” (Paragraph [0158], FIGS. 4-5)

Thus, even in hindsight, a person of ordinary skill in the art would not have any motivation to add the module control device recited in claim 2 as amended to Matsui. First of all, the buffers 105a and 105b in Matsui are already buffering the control/address signals and the clock signal, so there is no place for the module control device. The combination would also require major changes in the functions of the buffers 105a and 105b and a major overhaul of the internal wirings of the memory module 103 and the wirings between the memory controller 101 and the memory module 103. The combination would also work against the teachings of Matsui, which stresses the importance of having both buffers 105a and 105b coupled to the memory controller 101 directly instead of through a module control device on the memory module.

Therefore, claim 2 as amended is patentable over Matsui in view of Sutardja.

Claim 3, claim 4 as amended, claims 5-6, and claims 7-8 as amended depend from claim 2 as amended. Therefore, claim 3, claim 4 as amended, claims 5-6, and claims 7-8 are patentable for at least the same reasons claim 2 as amended is patentable.

Moreover, claims 3, claim 4 as amended, claims 5-6, claims 7-8 as amended and new claim 21 are patentable for the additional limitations recited therein. For example, Claim 3 recites the limitation that “the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.” Neither Matsui nor Sutardja discloses or teaches such limitations. The data buffers in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the memory module.

As a further example, claim 4 as amended recites that “the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in

time.” As discussed above, Sutardja does not disclose multiple data buffers on a memory module. Matsui discloses two data buffers 105a and 105b on left and right sides of the memory module but these two data buffers 105a and 105b are connected to the memory controller 101 via “the data wirings 111, along with the command/address wirings 112 and the clock wirings 113 ... concentrically.” (Paragraph [0158], FIGS. 4-5). Thus, there are no data buffers distributed across a surface of a memory module in a fly-by configuration, and there is no control signal in either Matsui or Sutardja that arrives at some adjacent data buffers on the same memory module at different points in time.

As a further example, claim 8 as amended recites that “the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices including one memory device from each of the one or more ranks.” This claim requires that each data buffer have the same data width as the memory devices. Clearly, this is not taught by either Matsui or Sutardja.

Claim 9 as amended recites:

A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

- receiving a system clock signal;
- generating a module clock signal on the memory module from the system clock signal received by the memory module;
- transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;
- generating a respective local clock signal from the module clock signal at each of the N data buffers;
- transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and
- performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

Claim 9 as amended cannot be rejected under 35 U.S.C. 102 (a)(1) by Matsui because Matsui does not disclose each and every limitation of the claim. For example, Matsui does not disclose “generating a module clock signal on the memory module

from the system clock signal received by the memory module” and “transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus.” As discussed above, the buffers 105a and 105b on Matsui’s memory module 103 receive system the system clock signal CLK/CLKB from the memory controller 101 and provide buffered clock signals to the DRAMs 110a and NOT to other data buffers. Matsui also does not disclose “generating a respective local clock signal *from the module clock signal* at each of the N data buffers” and “performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals” (emphasis added). In Matsui, the buffers 105a and 105b buffers the system clock signal CLK/CLKB received from the memory controller 101 instead of generating local clock signals based on a module clock signal received from a module control device on the memory module 103.

Therefore, claim 9 as amended is patentable over Matsui.

Claims 10-12, claim 13 as amended, claims 14-16, and claim 17 as amended depend from claim 9 as amended and includes further limitations in addition to the limitations in claim 9 as amended. Therefore, claims 10-12, claim 13 as amended, claims 14-16, and claim 17 are patentable for at least the same reasons claim 9 as amended is patentable.

Moreover, claims 10-12, claim 13 as amended, claims 14-16, and claim 17 are patentable for the additional limitations recited therein. For example, claim 11 recites, among other things, “generating module C/A signals and data buffer control signals in response to the C/A signals” received from the memory controller and transmitting the data buffer control signals to the N data buffers.” As discussed above, the buffers in Matsui and Sutardja merely buffer the address/control signals and clock signals from the memory controller and provide the address/control signals and clock signals to the memory devices. Neither the buffers 105a and 105b in Matsui nor the buffer 531 in Sutardja transmits any data buffer control signals to some other data buffers.

As a further example, claim 12 recites “controlling data paths in the N data buffers according to the data buffer control signals.” As discussed above, the data buffers

in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the same memory module.

Claim 18 recites:

A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

Claim 18 cannot be rejected under 35 U.S.C. 102 (a)(1) by Matsui because Matsui does not disclose each and every limitation of claim 18. As recited, the control circuit and the data buffers in claim 18 have very distinct functions. On the other hand, Matsui's buffers 105a and 105b have identical functions and thus cannot be both the control circuit and the data buffers in claim 18. Furthermore, as discussed above, the data buffers in claim 18 receive the data buffer control signals and the module clock signal from the control circuit that is *comprised by the memory module* and generate respective local clock signals in response to the module clock signal. In contrast, Matsui's buffers 105a

and 105b do not communicate with each other or receive any signal from a control circuit *comprised by the memory module*.

Therefore, claim 18 is not anticipated by Matsui.

Claims 19-20 and new claim 21 depend from claim 18 and include further limitations in addition to the limitations in claim 18. Therefore, claims 19-20 and new claim 21 are patentable for at least the same reasons claim 18 is patentable

Applicant submits that claims 2-21 are in condition for allowance and Applicant respectfully requests such action. The Examiner is invited to call the undersigned at (650) 521-4828 to discuss any remaining issues.

Respectfully submitted,

Date: June 21, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167

(Reg. No.)

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Request for Continued Examination (RCE) Transmittal

Address to:
Mail Stop RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	14/445,035
Filing Date	July 28, 2014
First Named Inventor	Hyun Lee
Art Unit	2132
Examiner Name	Ahmed, Zubair
Attorney Docket Number	NT006US

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. See Instruction Sheet for RCEs (not to be submitted to the USPTO on page 2.)

1. **Submission required under 37 CFR 1.114** Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

a. ☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

i. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

ii. ☐ Other _____

b. ☒ Enclosed

i. ☒ Amendment/Reply

iii. ☐ Information Disclosure Statement (IDS)

ii. ☐ Affidavit(s)/ Declaration(s)

iv. ☐ Other _____

2. Miscellaneous

a. ☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of _____ months. (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

b. ☐ Other _____

3. Fees

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

a. ☐ The Director is hereby authorized to charge the following fees, any underpayment of fees, or credit any overpayments, to Deposit Account No. _____.

i. ☐ RCE fee required under 37 CFR 1.17(e)

ii. ☐ Extension of time fee (37 CFR 1.136 and 1.17)

iii. ☐ Other _____

b. ☐ Check in the amount of \$ _____ enclosed

c. ☒ Payment by credit card (Form PTO-2038 enclosed)

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

Signature	/Jamie J. Zheng/	Date	June 21, 2016
Name (Print/Type)	Jamie J. Zheng	Registration No.	51167

CERTIFICATE OF MAILING OR TRANSMISSION

I hereby certify that this correspondence is being deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Mail Stop RCE, Commissioner for Patents, P. O. Box 1450, Alexandria, VA 22313-1450 or facsimile transmitted to the U.S. Patent and Trademark Office on the date shown below.

Signature			
Name (Print/Type)		Date	

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Mail Stop RCE, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Instruction Sheet for RCEs

(not to be submitted to the USPTO)

NOTES:

An RCE is not a new application, and filing an RCE will not result in an application being accorded a new filing date.

Filing Qualifications:

The application must be a utility or plant application filed on or after June 8, 1995. The application cannot be a provisional application, a utility or plant application filed before June 8, 1995, an international application that does not comply with the requirements of 35 U.S.C. 371, a design application, or a patent under reexamination. See 37 CFR 1.114(e). An international application does not comply with the requirements of 35 U.S.C. 371 until the requirements under 35 U.S.C. 371(c), including the requirement for the inventor's oath or declaration under 35 U.S.C. 371(c)(4), have been complied with.

Filing Requirements:

Prosecution in the application must be closed. Prosecution is closed if the application is under appeal, or the last Office action is a final action, a notice of allowance, or an action that otherwise closes prosecution in the application (e.g., an Office action under *Ex parte Quayle*). See 37 CFR 1.114(b).

A submission and a fee are required at the time the RCE is filed. If reply to an Office action under 35 U.S.C. 132 is outstanding (e.g., the application is under final rejection), the submission must meet the reply requirements of 37 CFR 1.111. If there is no outstanding Office action, the submission can be an information disclosure statement, an amendment, new arguments, or new evidence. See 37 CFR 1.114(c). The submission may be a previously filed amendment (e.g., an amendment after final rejection).

WARNINGS:

Request for Suspension of Action:

All RCE filing requirements must be met before suspension of action is granted. A request for a suspension of action under 37 C FR 1.103(c) does not satisfy the submission requirement and does not permit the filing of the required submission to be suspended.

Improper RCE will NOT toll Any Time Period:

Before Appeal - If the RCE is improper (e.g., prosecution in the application is not closed or the submission or fee has not been filed) and the application is not under appeal, the time period set forth in the last Office action will continue to run and the application will be abandoned after the statutory time period has expired if a reply to the Office action is not timely filed. No additional time will be given to correct the improper RCE.

Under Appeal - If the RCE is improper (e.g., the submission or the fee has not been filed) and the application is under appeal, the improper RCE is effective to withdraw the appeal. Withdrawal of the appeal results in the allowance or abandonment of the application depending on the status of the claims. If there are no allowed claims, the application is abandoned. If there is at least one allowed claim, the application will be passed to issue on the allowed claim(s). See MPEP 1215.01.

See MPEP 706.07(h) for further information on the RCE practice.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Request for Continued Examination	2801	1	600	600
Total in USD (\$)				600

Electronic Acknowledgement Receipt

EFS ID:	26134447
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	21-JUN-2016
Filing Date:	28-JUL-2014
Time Stamp:	21:36:32
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$600
RAM confirmation Number	8144
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2016-06-21_Amendment_RCE_NT006US.pdf	160266	yes	14
			23767b868f3106ef830b1669b162934b7a296423		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Request for Continued Examination (RCE)		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	14	
Warnings:					
Information:					
2	Transmittal Letter	2016-06-21_RCE_NT006US.pdf	212554	no	3
			b60fb894390d99e95a6ec53be0930ae9a75e2ed		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30358	no	2
			59203f995e210fa4042374a8ceecab25c71d9de7		
Warnings:					
Information:					
Total Files Size (in bytes):			403178		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/445,035	Filing Date 07/28/2014	☐ To be Mailed
---	---	----------------------------------	---------------------------------------

 ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	06/21/2016	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$40 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
 /SUSAN FORD/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NT006US
Sheet	1	of	3		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2002/0038405 A1	03-28-2002	Leddige, M. et al.	
	2.	2009/0296503 A1	12-03-2009	Chu, M. et al.	
	3.	2011/0085408 A1	04-14-2011	Solomon, J. et al.	
	4.	2008/0162790 A1	07-03-2008	IM	

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	5.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Related Matters, filed April 28, 2015	
	6.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner Response to Petition, filed February 22, 2016	
	7.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Objections to Evidence, filed February 29, 2016	
	8.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Reply, filed May 19, 2016	
	9.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner Request for Oral Argument, filed June 2, 2016	
	10.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	11.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	12.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, ORDER Trial Hearing, filed June 8, 2016	
	13.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	14.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	15.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Netlist's Submission, filed June 13, 2016	
	16.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	17.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	18.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Updated Exhibit List, filed June 24, 2016	
	19.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	20.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, May 10, 2016 Deposition of Carl Sechen, filed May 19, 2016	
	21.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Demonstrative Exhibits, filed June 24, 2016	
	22.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	23.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2002, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on February 12, 2016," filed February 22, 2016	
	24.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2003, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on April 20, 2015," filed February 22, 2016	
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2007, "Deposition of Dr. Srinivasan Jagannathan on May 25, 2016," filed June 2, 2016	
	26.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner Response to Petition, filed February 22, 2016	
	27.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Objections to Evidence, filed February 29, 2016	
	28.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Reply, filed May 19, 2016	
	29.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner Request for Oral Argument, filed June 2, 2016	
	30.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	31.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	32.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	33.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	34.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Patent Owner's Submission, filed June 13, 2016	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	35.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	36.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	
	37.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Updated Exhibit List, filed June 24, 2016	
	38.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	39.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	40.	International Search Report and Written Opinion, International Application No. PCT/US14/48517, filed July 28, 2014, dated October 27, 2014	
	41.	Inter Partes Review Case No. IPR2014-01029, Petitioner's Request for Rehearing pursuant to 37 C.F.R. § 42.71, filed on January 15, 2015.	
	42.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Request for Rehearing, Issued on March 3, 2015.	
	43.	Inter Partes Review Case No. IPR2014-01369, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued March 9, 2014.	
	44.	Examination Report, European Patent Application No. 10730021.2, dated April 14, 2014.	
	45.	Response to Examination Report, European Patent Application No. 10730021.2, dated June 4, 2014.	
	46.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	47.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	48.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	
	49.	Notice of Allowance, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, dated June 27, 2016	

EXAMINER

DATE CONSIDERED

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Electronic Acknowledgement Receipt

EFS ID:	26289722
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	07-JUL-2016
Filing Date:	28-JUL-2014
Time Stamp:	20:16:09
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-07_IDS_Cover_NT006 US.pdf	50359	no	2
			a48a79b6230238a6c32c5dbe4bdeb92ec6f39e9c		

Warnings:

Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2016-07-07_IDS_NT006US.pdf	132936 64a714e48a7522a0fdf5ea516c09441e24a77a17	no	3
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	11-04-2015_Rsps_ER_NT001EP.pdf	1201884 6ce107ca9f466a51d1aef5c2cff9cab7d18f5c7c	no	26
Warnings:					
Information:					
4	Non Patent Literature	2016-05-11_Eng_Trans_NGR_NT001KR.pdf	183171 1bd82675735359e2c4e2487f593949917116f0f0	no	5
Warnings:					
Information:					
5	Non Patent Literature	2016-06-27_NOA_NETL048CPC1.pdf	4054649 16c330249c95548259ee6c37db691c0381d933df	no	68
Warnings:					
Information:					
Total Files Size (in bytes):			5622999		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: NETLIST, INC.
App. No.	: 14/445,035
Filed	: July 28, 2014
For	: MEMORY MODULE WITH LOCAL SYNCHRONIZATION
Examiner	: UNKNOWN
Art Unit	: 2189
Conf No.	: 9700

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 49 references. Copies of some of the foreign documents and/or their translation and non-patent literature are submitted herewith. Copies of the rest of the foreign documents and/or their translation and non-patent literature have been submitted in commonly-owned patent applications. Specifically, Items 5-39 have been submitted in U.S. Pat. Appl. No. 14/715,486 (filed May 18, 2015) on June 29, 2016; Items 40-43 are submitted in U.S. Pat. Appl. No. 13/970,606 (filed August 20, 2013,) on March 18, 2015; Items 44-46 are submitted in U.S. Pat. Appl. No. 13/970,606 on October 27, 2015; all of which are accessible via PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional

Application No.: 14/445,035
Filing Date: July 28, 2014

information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: July 7, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167

(Reg. No.)

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6381140	04-30-2002	Liao, Sheng-Chi	
	2.	7254036	08-07-2007	Pauley et al.	
	3.	20090248969	10-01-2009	Wu; Larry	
	4.	20100125681	05-20-2010	Patel; Shwetal A.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials *	Cite No. ¹	Foreign Patent Document	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	5.	WO99/30240	06-17-1999	Intel Corporation		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	6.	Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated November 23, 2015	
	7.	Response to Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated March 23, 2016	
	8.	Inter Partes Review Case No. 2014-00882, Patent Owner Response, filed May 8, 2015	
	9.	Inter Partes Review Case No. 2014-00882, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	10.	Inter Partes Review Case No. 2014-00882, Petitioner's Request for Oral Argument, filed July 1, 2015	
	11.	Inter Partes Review Case No. 2014-00882, Patent Owner's Request for Oral Argument, filed July 3, 2015	
	12.	Inter Partes Review Case No. 2014-00882, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	13.	Inter Partes Review Case No. 2014-00882, Record of Oral Hearing, filed November 16, 2015	
	14.	Inter Partes Review Case No. 2014-00882, Final Written Decision 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	15.	Inter Partes Review Case No. 2014-00882, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	16.	Inter Partes Review Case No. 2014-00882, Exhibit 1022, 'Professor Carl Sechen Deposition Transcript,' filed June,19,2015	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	17.	Inter Partes Review Case No. 2014-00882, Exhibit 1023, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	18.	Inter Partes Review Case No. 2014-00882, Exhibit 2002, 'Declaration of Professor Carl Sechen,' filed May 8,2015	
	19.	Inter Partes Review Case No. 2014-00882, Exhibit 2003, 'Transcript of Videotaped Deposition of Srinivasan Jagannathan, Ph. D,' filed May 8,2015	
	20.	Microsoft Computer Dictionary, Fifth Edition, Microsoft Press, p. 334, 2002	
	21.	Jacob, Bruce, et al. "Memory Systems - Cache, DRAM, Disk," Elsevier Inc. 2008, Chapters 7 and 10	
	22.	U.S. District Court Northern District of California, Case No. 4.13-cn-05889-YGR, Netlist Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Joint Claim Construction and Prehearing Statement Pursuant to Patent L. T. 4-3, filed August 27, 2014, in 28 pages	
	23.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 349-411	
	24.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 133-265	
	25.	Selected Entries in Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/view/Entry/175039?redirectedFrom=selectively#eid	
	26.	Web search results from the Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/search?searchType=dictionary&q=select&_searchBin=Search	
	27.	Inter Partes Review Case No. 2014-00883, Patent Owner Response, filed May 8, 2015	
	28.	Inter Partes Review Case No. 2014-00883, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	29.	Inter Partes Review Case No. 2014-00883, Petitioner's Request for Oral Argument, filed July 1, 2015	
	30.	Inter Partes Review Case No. 2014-00883, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
	31.	Inter Partes Review Case No. 2014-00883, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	32.	Inter Partes Review Case No. 2014-00883, Record of Oral Hearing, filed November 16, 2015	
	33.	Inter Partes Review Case No. 2014-00883, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	34.	Inter Partes Review Case No. 2014-00883, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	35.	Inter Partes Review Case No. 2014-00883, Exhibit 1025, 'Professor Carl Sechen Deposition Transcript,' filed June 19,2015	

EXAMINER

DATE CONSIDERED

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	36.	Inter Partes Review Case No. 2014-00883, Exhibit 1026, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	37.	Inter Partes Review Case No. 2014-01011, Patent Owner Response, filed May 8, 2015	
	38.	Inter Partes Review Case No. 2014-01011, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	39.	Inter Partes Review Case No. 2014-01011, Petitioner's Request for Oral Argument, filed July 1, 2015	
	40.	Inter Partes Review Case No. 2014-01011, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
	41.	Inter Partes Review Case No. 2014-01011, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	42.	Inter Partes Review Case No. 2014-01011, Record of Oral Hearing, filed November 16, 2015	
	43.	Inter Partes Review Case No. 2014-01011, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	44.	Inter Partes Review Case No. 2014-01011, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	45.	Inter Partes Review Case No. 2014-01011, Exhibit 1027, 'Professor Carl Sechen Deposition Transcript,' filed June 19,2015	
	46.	Inter Partes Review Case No. 2014-01011, Exhibit 1028, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	

EXAMINER

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Electronic Acknowledgement Receipt

EFS ID:	26290378
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	08-JUL-2016
Filing Date:	28-JUL-2014
Time Stamp:	00:51:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-07_IDS_2_Cover_NT006US.pdf	49734 62a19011dc285f0ba7805143c78861edaf3b6d9a	no	2

Warnings:

Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2016-07-07_IDS_2_NT006US. pdf	94095 268de5acbdbe0e0764ac31c748f2b6708cf19c4a	no	3
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Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	2015-11-23_NFOA_NETL048CP C1.pdf	4155462 35b8c4f3b724d8b3e3f65fa2a1c1257016da bb25	no	79
Warnings:					
Information:					
4	Non Patent Literature	2016-03-23_Rsps_NFOA_NETL0 48CPC1.pdf	118678 0f3c0dba6a043802851f4422a0b58d52f88e 9d5a	no	9
Warnings:					
Information:					
Total Files Size (in bytes):			4417969		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: NETLIST, INC.
App. No.	: 14/445,035
Filed	: July 28, 2014
For	: MEMORY MODULE WITH LOCAL SYNCHRONIZATION
Examiner	: UNKNOWN
Art Unit	: 2189
Conf No.	: 9700

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 46 references. Copies of some of the foreign documents and/or their translation and non-patent literature are submitted herewith. Copies of the rest of the foreign documents and/or their translation and non-patent literature have been submitted in commonly-owned U.S. Pat. Appl. No. 14/715,486 (filed May 18, 2015) on February 21, 2016, which is accessible via PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 14/445,035
Filing Date: July 28, 2014

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: July 7, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	NT006US	9700

79141 7590 08/11/2016
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
----------	--------------

2132

NOTIFICATION DATE	DELIVERY MODE
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08/11/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
eofficeaction@appcoll.com
infohub.j@gmail.com

DETAILED ACTION

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This Office Action is responsive to RCE filed on 06/21/2016. In this RCE, Applicant has canceled claim 1 and added new claim 21. Accordingly, claims 2-21 are pending in this application.

Continued Examination Under 37 CFR 1.114

3. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 06/21/2016 has been entered.

Response to Arguments

4. Applicant's arguments filed 06/21/2016 have been fully considered but they are not persuasive.
5. In response to the argument of the Applicant that "it is unclear what in Sutardja the Examiner regards as the "module control device"" (pages 8-9 of the remarks), the Examiner respectfully submits that the "memory control module 510" and the "Clock generator module 534", both components of the memory module 500, together are mapped to the module control device of the claimed invention. Sutardja discloses "The buffer module 530 may buffer signals between the memory control module 510, the

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memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530", para 0136.

6. The Applicant further contends that "there is no motivation for the suggested combination" between Matsui and Sutardja (pages 9-10 of the remarks). The Examiner asserts that the placement of the "memory control module 510" and the "Clock generator module 534" of Sutardja in the memory module 103 of Matsui does not in any way change the operation of Matsui's device. Further, the Examiner is unable to find in Matsui's disclosure the Applicant's assertion that "the combination would also work against the teachings of Matsui, which stresses the importance of having both buffers 105a and 105b coupled to the memory controller 101 directly instead of through a module control device on the memory module" (page 10 of the remarks). The Examiner respectfully submits that Matsui makes no such assertion.

7. Regarding claim 3, Applicant asserts that neither Matsui nor Sutardja discloses or teaches "the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals." (Page 10 of the remarks). The Examiner respectfully disagrees. Matsui discloses in Figure 5 that "command/address signal and data are transmitted in the internal command/address wiring 112' and the internal data wiring 111'", para 0157.

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Figure 5 further shows a clock signal CLK/CLKB 113' controlling transmission of the signals CMD/ADD 112' and DQ 111' and these signals are generated by the memory controller 101. According to the instant specification, "data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal", para 0035.

8. Regarding claim 4 as amended, Applicant argues that "the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time" is not disclosed by Sutardja. The Examiner respectfully disagrees. Sutardja discloses "The memory ICs 520 and the buffer module 530 for each memory module 514 may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias", para 0135. It is well known in the art that in such an arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530. Further, referring to Figure 17, the buffer modules 530 are daisy-chained (fly-by configuration) resulting in a skew of the arrived signals.

9. Regarding claim 8, Applicant argues that "the memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices including one memory device from each of the one or more ranks." is not disclosed by Matsui or Sutardja (page 11 of the remarks). The Examiner respectfully submits that Sung was cited to teach this claim limitation. Figures 1, 2 of Sung correspond specifically to a case in which the first memory unit 113 includes two memory ranks, and in this regard, two first chip selection signals are input to the first memory unit 113. The

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second memory unit 123 includes two memory ranks, and in this regard, two second chip selection signals are input to the second memory unit 123, paras 0046, 0049. Each data buffer having the same data width as the memory devices is disclosed by Sutardja (see Figure 17).

10. With respect to claim 9 (pages 11-12 of the remarks), claim 9 is now unpatentable over Matsui in view of Sutardja (see current rejection below).

11. The Examiner respectfully disagrees with Applicant's remark that "generating module C/A signals and data buffer control signals in response to the C/A signals", as recited by claim 11, is not disclosed by Matsui or Sutardja (page 12 of the remarks).

Indeed Matsui discloses this feature in Figure 5. Figure 5 shows CLK/CLKB 113 transmitted from the memory controller 101. This clock signal is generated into clock signal CLK/CLKB 113' within the module buffers 105a and 105b. According to the instant specification, "data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal", para 0035.

12. The Examiner respectfully disagrees with Applicant's remark that "controlling data paths in the N data buffers according to the data buffer control signals", as recited by claim 12, is not disclosed by Matsui or Sutardja. Indeed in Figure 5 of Matsui DQ 111' is controlled by CLK/CLKB 113'. According to the instant specification, "data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal", para 0035.

13. With respect to claim 18 (pages 13-14 of the remarks), claim 18 is now unpatentable over Matsui in view of Sutardja (see current rejection below).

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14. In view of the foregoing remarks, independent claims 1, 9, and 18 are not in a condition for allowance. Claims depending therefrom, either directly or indirectly, are also not in a condition for allowance.

Claim Rejections - 35 USC § 103

15. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

16. Claims 2-5, 7, 9, 11-14, 16, and 18-21 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereinafter "Matsui") in view of Sutardja et al. US 2007/0168781 (hereinafter "Sutardja").

17. As per independent claim 2, Matsui teaches "A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising" (Figure 5 – memory modules 103a-103d interfaced with a memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight

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DRAMs 110a, para 0160. Referring to Figure 5, clock signals CLK/CLKB 113 operating at 1.33 GHz is received by the buffers 105a and 105b from the memory controller 101),

18. “N data buffers each having a data width of n-bits and each generating a respective local clock signal” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159),

19. “memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight

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DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159. Referring to Figure 5, the data bus DQ 111 is connected between the buffers 105a and 105b and the memory controller 101).

20. Matsui's invention discloses all of the claimed limitations from above except for "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers" and "wherein the module control device does not transmit any clock signal to the memory devices" and "wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device".

21. However, Sutardja teaches "a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers" (A functional block diagram of a memory module 500 is

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shown in Figure 17, para 0133 where a memory control module 510 is depicted. The buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530, para 0136),

22. “wherein the module control device does not transmit any clock signal to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136),

23. “wherein each of the *N* data buffers generates the respective local clock signal from the module clock signal received from the module control device” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. The clock generator module 534 is mapped as being part of the memory control module 510).

24. Given the teaching of Sutardja, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the

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module control device transmitting the data buffer control signals and the module clock signal to the data buffers” and “wherein the module control device does not transmit any clock signal to the memory devices” and “wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device”. The motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja.

25. As per dependent claim 3, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui teaches “wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals” (Figure 5 - command/address signal and data are transmitted in the internal command/address wiring 112' and the internal data wiring 111', para 0157).

26. As per dependent claim 4, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for “wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time”.

27. However, Sutardja teaches “wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time” (The memory ICs 520 and the buffer module 530 for each memory module 514

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may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias, para 0135. It is well known in the art that in such an arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530. Further, referring to Figure 17, the buffer modules 530 are daisy-chained (fly-by configuration) resulting in a skew of the arrived signals).

28. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 4.

29. As per dependent claim 5, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal".

30. However, Sutardja teaches "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal" (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. It is well known in the art that a PLL or a DLL (digital lock loop) contain programmable registers to control/adjust the phase and frequency of PLL or DLL).

31. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 5.

32. As per dependent claim 7, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations

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from above except for “wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal”.

33. However, Sutardja teaches “wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

34. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 7.

35. As per independent claim 9, most of the limitations of this claim is rejected based on arguments provided above for similar rejected independent claim 2.

36. As per other limitations of claim 9, Sutardja teaches “transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices” (Referring to Figure 17, it is disclosed that clock generator module 534 feeds clock signals to buffer modules 530 and memory modules 514. Sutardja does not explicitly state that the clock signal from the clock generator module 534 directly feed the memory devices 520. Instead, Sutardja explicitly teaches that “the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals”, para 0136),

37. “performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals” (Sutardja explicitly teaches that “the buffer modules 530 may include a phase locked loop (PLL) to

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generate local phase-adjusted clock signals”, para 0136. Accordingly, read/write operations are performed based on the local phase-adjusted clock signals).

38. As per dependent claim 11, Matsui in combination with Sutardja discloses the method of claim 9. Matsui teaches “receiving command/address (C/A) signals from the memory controller” (Figure 5 – C/A 112),

39. “generating module C/A signals and data buffer control signals in response to the C/A signals” (Figure 5 – C/A 112 and CLK/CLKB 113. According to the instant specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035),

40. “transmitting the module C/A signals to at least some of the memory devices” (Figure 5 – C/A 112’ transmitted to the two groups of DRAMs 110),

41. “transmitting the data buffer control signals to the N data buffers” (Figure 5 – CLK/CLKB 113. According to the instant specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035).

42. As per dependent claim 12, Matsui in combination with Sutardja teaches the method of claim 11. Matsui teaches “further comprising controlling data paths in the N data buffers according to the data buffer control signals” (Figure 5 – DQ 111’ is controlled by CLK/CLKB 113’).

43. As per dependent claims 13-14, these claims are rejected based on arguments provided for similar rejected dependent claims 4-5 respectively.

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44. As per dependent claim 16, Matsui in combination with Sutardja discloses the method of claim 9. Matsui's invention discloses all of the claimed limitations from above except for "further comprising phase-locking the local clock signal with the module clock signal".

45. However, Sutardja teaches "further comprising phase-locking the local clock signal with the module clock signal" (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

46. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 9 is equally applicable to claim 16.

47. As per independent claim 18, this claim is rejected based on arguments provided above for similar rejected independent claim 2.

48. As per dependent claim 19, Matsui in combination with Sutardja teaches the memory module of claim 18. Matsui teaches "wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board" (Figure 3).

49. As per dependent claim 20, this claim is rejected based on arguments provided above for similar rejected dependent claim 16.

50. As per dependent claim 21, Matsui in combination with Sutardja teaches the memory module of claim 18. Matsui's invention discloses all of the claimed limitations

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from above except for “wherein the control circuit does not transmit any clock signals to the memory devices”.

51. However, Sutardja teaches “wherein the control circuit does not transmit any clock signals to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

52. The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 18 is equally applicable to claim 21.

53. Claim 6, 10, and 15 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereinafter “Carpenter”).

54. As per dependent claim 6, Matsui in combination with Sutardja discloses the memory module of claim 5. Matsui’s invention modified by Sutardja discloses all of the claimed limitations from above except for “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals”.

55. However, Carpenter teaches “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals” (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

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56. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals”. The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

57. As per dependent claim 10, Matsui in combination with Sutardja discloses the method of claim 9. Matsui’s invention modified by Sutardja discloses all of the claimed limitations from above except for “further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers”.

58. However, Carpenter teaches “further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers” (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

59. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with “further comprising incorporating a programmable amount of delay to the respective local clock

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signal at each of the N data buffers". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

60. As per dependent claim 15, Matsui in combination with Sutardja discloses the method of claim 11. Matsui's invention modified by Sutardja discloses all of the claimed limitations from above except for "programming one or more registers in each of the N data buffers according to the data buffer control signals".

61. However, Carpenter teaches "programming one or more registers in each of the N data buffers according to the data buffer control signals" (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

62. Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with "programming one or more registers in each of the N data buffers according to the data buffer control signals". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

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63. Claims 8 and 17 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Sung et al. US 2011/0110168 (hereinafter "Sung").

64. As per dependent claim 8, Matsui in combination with Sutardja discloses the memory module of claim 1. Matsui's invention modified by Sutardja discloses all of the claimed limitations from above except for "wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks".

65. However, Sung teaches "wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks" (Figures 1, 2 correspond specifically to a case in which the first memory unit 113 includes two memory ranks, and in this regard, two first chip selection signals are input to the first memory unit 113. The second memory unit 123 includes two memory ranks, and in this regard, two second chip selection signals are input to the second memory unit 123, paras 0046, 0049).

66. Given the teaching of Sung, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui as modified by Sutardja with "wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks". The motivation would be that the memory controller can determine which memory rank of which memory module is to be selected, para 0052 of Sung.

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67. As per dependent claim 17, this claim is rejected based on arguments provided above for similar rejected dependent claim 8.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 8:30AM - 5:00PM Mon-Fri.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID X YI/
Supervisory Patent Examiner, Art Unit 2132



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APPLICANTS NETLIST, INC., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; ** CONTINUING DATA ***** This appln claims benefit of 61/859,215 07/27/2013 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 08/06/2014						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/ZUBAIR AHMED/</u> Examiner's Signature		☐ Met after Allowance Initials _____	STATE OR COUNTRY CA	SHEETS DRAWINGS 22	TOTAL CLAIMS 10	INDEPENDENT CLAIMS 2
ADDRESS Jamie J. Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES						
TITLE MEMORY MODULE WITH LOCAL SYNCHRONIZATION						
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				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6381140	04-30-2002	Liao, Sheng-Chi	
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	5.	WO99/30240	06-17-1999	Intel Corporation		

NON PATENT LITERATURE DOCUMENTS			
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	9.	Inter Partes Review Case No. 2014-00882, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
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	13.	Inter Partes Review Case No. 2014-00882, Record of Oral Hearing, filed November 16, 2015	
	14.	Inter Partes Review Case No. 2014-00882, Final Written Decision 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	15.	Inter Partes Review Case No. 2014-00882, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	16.	Inter Partes Review Case No. 2014-00882, Exhibit 1022, 'Professor Carl Sechen Deposition Transcript,' filed June,19,2015	

EXAMINER /ZUBAIR AHMED/

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	17.	Inter Partes Review Case No. 2014-00882, Exhibit 1023, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	18.	Inter Partes Review Case No. 2014-00882, Exhibit 2002, 'Declaration of Professor Carl Sechen,' filed May 8,2015	
	19.	Inter Partes Review Case No. 2014-00882, Exhibit 2003, 'Transcript of Videotaped Deposition of Srinivasan Jagannathan, Ph. D,' filed May 8,2015	
	20.	Microsoft Computer Dictionary, Fifth Edition, Microsoft Press, p. 334, 2002	
	21.	Jacob, Bruce, et al. "Memory Systems - Cache, DRAM, Disk," Elsevier Inc. 2008, Chapters 7 and 10	
	22.	U.S. District Court Northern District of California, Case No. 4.13-cn-05889-YGR, Netlist Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Joint Claim Construction and Prehearing Statement Pursuant to Patent L. T. 4-3, filed August 27, 2014, in 28 pages	
	23.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 349-411	
	24.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 133-265	
	25.	Selected Entries in Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/view/Entry/175039?redirectedFrom=selectively#eid	
	26.	Web search results from the Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/search?searchType=dictionary&q=select&_searchBin=Search	
	27.	Inter Partes Review Case No. 2014-00883, Patent Owner Response, filed May 8, 2015	
	28.	Inter Partes Review Case No. 2014-00883, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	29.	Inter Partes Review Case No. 2014-00883, Petitioner's Request for Oral Argument, filed July 1, 2015	
	30.	Inter Partes Review Case No. 2014-00883, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
	31.	Inter Partes Review Case No. 2014-00883, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	32.	Inter Partes Review Case No. 2014-00883, Record of Oral Hearing, filed November 16, 2015	
	33.	Inter Partes Review Case No. 2014-00883, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	34.	Inter Partes Review Case No. 2014-00883, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	35.	Inter Partes Review Case No. 2014-00883, Exhibit 1025, 'Professor Carl Sechen Deposition Transcript,' filed June 19,2015	

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	36.	Inter Partes Review Case No. 2014-00883, Exhibit 1026, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19, 2015	
	37.	Inter Partes Review Case No. 2014-01011, Patent Owner Response, filed May 8, 2015	
	38.	Inter Partes Review Case No. 2014-01011, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
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	44.	Inter Partes Review Case No. 2014-01011, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	45.	Inter Partes Review Case No. 2014-01011, Exhibit 1027, 'Professor Carl Sechen Deposition Transcript,' filed June 19, 2015	
	46.	Inter Partes Review Case No. 2014-01011, Exhibit 1028, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19, 2015	

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Search Notes 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/08	08/01/2016	ZA
G11C 7/1066	08/01/2016	ZA

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
See attached search history	08/01/2016	ZA

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/ZUBAIR AHMED/ Examiner.Art Unit 2132	
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<i>Index of Claims</i> 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected
=	Allowed

-	Cancelled
÷	Restricted

N	Non-Elected
I	Interference

A	Appeal
O	Objected

☐ Claims renumbered in the same order as presented by applicant					☐ CPA					☐ T.D.					☐ R.1.47				
CLAIM		DATE																	
Final	Original	11/02/2015	03/13/2016	08/01/2016															
	1	✓	✓	-															
	2	✓	✓	✓															
	3	✓	✓	✓															
	4	✓	✓	✓															
	5	✓	✓	✓															
	6	✓	✓	✓															
	7	✓	✓	✓															
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	17		✓	✓															
	18		✓	✓															
	19		✓	✓															
	20		✓	✓															
	21			✓															

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16

S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2015/11/01 06:14

			IBM_TDB			
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S36	38	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 11:42
S37	8	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:13
S38	68	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:18
S39	4415	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S40	5	S39 and PLL	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S41	0	G11C7/1066.cpc	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:00
S42	4534	G06F12/08.cpc.	US-PGPUB; USPAT;	OR	ON	2016/08/01

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			08:01
S43	0	S42 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:02

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US PATENT DOCUMENTS					
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		Number-Kind Code ² (if known)			
	1.	2002/0038405 A1	03-28-2002	Leddige, M. et al.	
	2.	2009/0296503 A1	12-03-2009	Chu, M. et al.	
	3.	2011/0085408 A1	04-14-2011	Solomon, J. et al.	
	4.	2008/0162790 A1	07-03-2008	IM	

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	5.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Related Matters, filed April 28, 2015	
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	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2007, "Deposition of Dr. Srinivasan Jagannathan on May 25, 2016," filed June 2, 2016	
	26.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner Response to Petition, filed February 22, 2016	
	27.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Objections to Evidence, filed February 29, 2016	
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	31.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	32.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	33.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	34.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Patent Owner's Submission, filed June 13, 2016	

EXAMINER /ZUBAIR AHMED/

DATE CONSIDERED 08/01/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /Z.A/

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NT006US

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	35.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	36.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	
	37.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Updated Exhibit List, filed June 24, 2016	
	38.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	39.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	40.	International Search Report and Written Opinion, International Application No. PCT/US14/48517, filed July 28, 2014, dated October 27, 2014	
	41.	Inter Partes Review Case No. IPR2014-01029, Petitioner's Request for Rehearing pursuant to 37 C.F.R. § 42.71, filed on January 15, 2015.	
	42.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Request for Rehearing, Issued on March 3, 2015.	
	43.	Inter Partes Review Case No. IPR2014-01369, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued March 9, 2014.	
	44.	Examination Report, European Patent Application No. 10730021.2, dated April 14, 2014.	
	45.	Response to Examination Report, European Patent Application No. 10730021.2, dated June 4, 2014.	
	46.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	47.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	48.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	
	49.	Notice of Allowance, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, dated June 27, 2016	

EXAMINER /ZUBAIR AHMED/

DATE CONSIDERED 08/01/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04.³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3).⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document.⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible.⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /Z.A/

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : NT006US

CERTIFICATE OF EFS WEB

TRANSMISSION

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

November 14, 2016

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION
DATED AUGUST 11, 2016

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Enclosed is an Amendment and Response to the Non-Final Office Action dated August 11, 2016.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

1. (Cancelled)
2. (Previously Presented) A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of ($N \times n$) bits, where N and n are integers larger than one, comprising:
 - N data buffers each having a data width of n -bits and each generating a respective local clock signal;
 - memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and
 - a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;
 - wherein the module control device does not transmit any clock signal to the memory devices; and
 - wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.
3. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.
4. (Currently Amended) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

5. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes one or more programmable registers controlling a phase of the local clock signal.
6. (Previously Presented) The memory module of claim 5,
wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals.
7. (Currently Amended) The memory module of claim 2, wherein the respective one of the N data buffers is configured to [[to]] phase-lock the local clock signal with the module clock signal.
8. (Previously Presented) The memory module of claim 2, wherein memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.
9. (Previously Presented) A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:
 - receiving a system clock signal;
 - generating a module clock signal on the memory module from the system clock signal received by the memory module;
 - transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;
 - generating a respective local clock signal from the module clock signal at each of the N data buffers;
 - transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and

performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Previously Presented) The method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;

transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Previously Presented) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (Previously Presented) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.

15. (Previously Presented) The method of claim 11, further comprising:
programming one or more registers in each of the N data buffers according to the data buffer control signals.

16. (Previously Presented) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.

17. (Previously Presented) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.

18. (Previously Presented) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Currently Amended) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors.

20. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.
21. (Previously Presented) The memory module of claim 18, wherein the control circuit does not transmit any clock signals to the memory devices.
22. (New) The memory module of claim 18, wherein the data buffer control signals are different from the module C/A signals.
23. (New) The memory module of claim 18, wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer.
24. (New) The memory module of claim 18, wherein each of the memory devices is n -bits wide.
25. (New) The memory module of claim 18, wherein each of the memory devices is $n/2$ -bits wide.

REMARKS

This amendment responds to the Non-Final Office Action dated August 11, 2016. In the Office Action, the Examiner:

- rejected claims 2-5, 7, 9, 11-14, 16, and 18-21 under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereafter “Matsui”) in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”);
- rejected claims 6, 10, and 15 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereafter “Carpenter”); and
- rejected claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and further in view of Sung et al. US 2011/0110168 (hereinafter “Sung”).

Claims 4 and 7 are amended to correct typographical errors. Claim 19 is amended for clarity. New claims 22-25 are presented. Support for the amendments and for the new claims can be found at least in paragraphs [0041] and [0061]-[0068], FIGS. 2A-2B, FIGS. 5A-5B, and 6A-6C. Thus, no new matter is added.

Claim 2 is rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja. In rejecting claim 2, the Examiner seems to ignore some of the limitations recited therein, for example, “a module control device *to receive* command/address (C/A) signals and a system clock *from the memory controller* and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals *to at least some of the memory devices*, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers.” (*Emphasis added*).

As acknowledged by the Examiner, Matsui does not disclose such a module control device. In fact, Matsui doesn’t even want such a module control device. The Examiner states that the motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals like Sutardja

(paragraph [0136] of Suturdja). But, Matsui teaches away from using a command/address buffer and a PLL chip to distribute commands/addresses and clocks, because doing so would require that “module command/address distributing wiring and module clock distributing wiring be drawn out in the longitudinal direction of the module board” (Matsui, paragraph [0006]).

In fact, Matsui deliberately avoids a command/address buffer and PLL chip by using two buffers 105a and 105b to produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data *from the memory controller* 101 into two or four parallel signals (Matsui, paragraph [0159], *emphasis added*). Since the buffers 105a and 105b are doing what a command/address buffer and a PLL chip would be doing, it does not make sense for Matsui to go back to its prior art, which it tries to improve upon, by adding a command/address buffer and a PLL chip. Therefore, the suggested combination would not defeat the purpose of Matsui’s module.

Furthermore, Sutardia does not disclose the module control device recited in claim 2. The Examiner erred by insisting that Sutardja’s memory control module 510 is such a module control device by referring to Figure 17 and paragraph [0133] in Sutardja, where it merely states:

“A memory control module 510 selectively sends data storing and data retrieval commands to one of a plurality of memory modules 514-1, 514-2, ..., and 514-Z (collectively memory modules 514).”

Nothing more is said in paragraph [0133] about the memory control module 510. And nowhere else in Sutardja shows Sutardja’s memory control module 510 receiving any command/address (C/A). As shown in FIG. 17 in Sutardja, Sutardja’s memory control module 510 does not receive any command/address (C/A) signals from a memory controller, as required by claim 2. In fact, Sutardja’s memory control module 510 *is itself* a memory controller and thus cannot possibly receive command/address (C/A) signals from a memory controller. As described in paragraph [0134] of Sutardja, Sutardja’s memory control module is nothing else but a memory controller that communicated data and control/address signals with the memory modules 514:

“Communication between the memory control module 510 and the memory modules 514 may be via serial and/or parallel signaling. A bus 531 may be used to support data flow between the memory control module 510 and the memory modules 514. A bus 533 may be used to support data flow between the memory modules 514 and the memory control module 510. Differential signaling may be used.”

Moreover, Sutardja’s memory control module 510 does not output any data buffer control signals in addition to module C/A signals and a module clock signal. As described in Sutardja (paragraph [0136]), Sutardja only discloses the buffer module 530 receiving incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc., which are all C/A signals:

“the buffer module 530 may buffer signals between the memory control module 510, the memory module 514, and/or signals on the buses 531 and 533. The buffer module 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS_, etc. Clock generator module 534 transmits clock signals to the buffers 530, para 0136), ... the buffer module 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals.”

Thus, there is no teaching in Sutardja about buffer control signals in addition to the C/A signals and the clock signal, as required by claim 2.

Sutardja’s memory control module 510 also does not transmit the module C/A signals to at least some of the memory devices. As cited above, control signals such as row access and precharge (RAS), column address strobe (CAS), etc. are transmitted from the memory control module 510 to the buffer module 530, instead of to any of the memory devices.

Therefore, Sutardja’s memory control module 510 cannot be the module control device recited in claim 2, and no other teachings in Sutardja come even close to the module control device recited in claim 2.

Therefore, claim 2 is patentable over Matsui in view of Suturdja.

Claims 3-8 depend from claim 2. Therefore, claims 3-8 are patentable for at least the same reasons claim 2 as amended is patentable.

Furthermore, claims 3-8 are patentable for the additional limitations recited therein. For example, Claim 3 recites the limitations that “the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.” Neither Matsui nor Sutardja discloses or teaches such limitations. The data buffers in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the memory module. The Examiner points out that in Matsui, command/address signal and data are transmitted in the internal command/address wiring 112’ and the internal data wiring 111’ (paragraph [0157]), but nowhere in Matsui or Sutardja teaches that these internal wirings are controlled by data buffer control signals. In fact, neither Matsui nor Sutardja discloses or teaches any data buffer control signals, which are in addition to the command/address signals and clock signals, as required by claim 2.

As a further example, claim 4 recites the limitations that “the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.” The Examiner rejects this claim by pointing out that Sutardja’s buffer modules 530 are daisy-chained. The Examiner neglects that claim 2, from which claim 4 depends, requires that each of the data buffers has a data width of n -bits, a fraction of the data width of the memory module, which is $(N \times n)$ bits. Claim 4 recites specifically that N such data buffers spanning the data width of a memory module are in a fly-by configuration. In contract, there is no teach in Sutardja that the Z buffer modules 530 span the data width of a memory module. In fact, Sutardja’s buffer modules 530 are each as wide as the memory modules 514, and they are disposed on different memory modules 514. Therefore, Sutardja does not even disclose the data buffers in claim 2, and cannot possibly disclose fly-by configuration of the data buffers in claim 4.

The arguments regarding claim 2 applies to claim 9. Therefore, claim 9 is also patentable over Matsui in view of Sutardja.

Claims 10-17 depend from claim 9 and include further limitations in addition to the limitations in claim 9. Therefore, claims 10-17 are patentable for at least the same reasons claim 9 is patentable.

The arguments regarding claim 2 applies to claim 18. Therefore, claim 18 is also patentable over Matsui in view of Sutardja.

Claims 19-21 and new claims 22-23 depend from claim 18 and include further limitations in addition to the limitations in claim 18. Therefore, claims 19-21 and new claim 22-23 are patentable for at least the same reasons claim 18 is patentable

Applicant submits that claims 2-25 are in condition for allowance and Applicant respectfully requests such action. The Examiner is invited to call the undersigned at (650) 521-4828 to discuss any remaining issues.

Respectfully submitted,

Date: November 14, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	2202	4	40	160
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	27509754
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006US
Receipt Date:	15-NOV-2016
Filing Date:	28-JUL-2014
Time Stamp:	02:18:10
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2016-11-14_Amendment_NT006US.pdf	150446 63b73e55cc1ae2b3a52f9c4590bdfa313e4705e3	yes	11

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	11	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30049	no	2
			e74b3802ec3f93a00d3985fb5227ac72569ac65d		
Warnings:					
Information:					
Total Files Size (in bytes):			180495		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/445,035	Filing Date 07/28/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	11/15/2016	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 24	Minus	** 20	= 4	X \$40 =	160
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						160	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
PEARLIE FENNELL

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14/445,035	07/28/2014	Hyun Lee	NT006US	9700

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EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
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2132

NOTIFICATION DATE	DELIVERY MODE
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01/09/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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Office Action Summary	Application No. 14/445,035	Applicant(s) LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 11/15/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 2-25 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 2-25 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 07/28/2014 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to amendment filed on 11/15/2016. Claim 1 was previously canceled. New claims 22-25 are added in this amendment. Accordingly, claims 2-25 are pending in this application.

Response to Arguments

Applicant's arguments filed 11/15/2016 have been fully considered but they are not persuasive.

Applicant's argument, pages 7-9 of the remarks, "Claim 2 is rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja. In rejecting claim 2, the Examiner seems to ignore some of the limitations recited therein, for example, "a module control device *to receive* command/address (C/A) signals and a system clock *from the memory controller* and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals *to at least some of the memory devices*, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers." (*Emphasis added*)" has been fully considered and is deemed not persuasive for the following reasons.

The Examiner respectfully submits that a person of ordinary skill in the art before the effective filing date of the claimed invention can modify the method and apparatus

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disclosed by Matsui with the teachings of Sutardja to arrive at the claimed invention as described herein. The claimed module control device receives two types of signals from the memory controller: command/address (C/A) and a system clock from the memory controller. As claimed, the module control device in turn transmits the C/A signals to at least some of the memory devices, and data buffer control signals and module clock signal to the data buffers.

Referring to Fig. 17 of Sutardja, a functional block diagram of a memory module 500 is shown where a memory control module 510 is depicted, para 0133. The memory control module 510 is mapped to the claimed module control device. Further shown in Fig. 17 of Sutardja are buffer modules 530 that may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. Still further, the buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. and clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530, para 0136. Now referring to Fig. 5 of Matsui, the memory control module 510 of Sutardja may be inserted between the memory controller 101 and the memory modules 103a-d interfacing with the buffers 105a and 105b. This modification, by a person of ordinary skill in the art, would neither change the mode of operation of Matsui's method and apparatus nor would it render Matsui's method and apparatus inoperable for the following reasons. Matsui already discloses the use of two buffers (buffers 105a and 105b, Fig. 5) in each memory module 103a-d. Each of the two buffers

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buffer DQ111, CLK113, and CMD/ADD112 (see Fig. 5 of Matsui). With the proposed modification of Matsui's method and apparatus, the memory control module 510 of Sutardja situated between memory controller 101 and memory modules 103a-d (Fig. 5 of Matsui) would only receive from the memory controller 101 the CMD/ADD112 and CLK113 signals. The DQ111 signals may be directly transmitted from the memory controller to the buffers 105a and 105b consistent with the claimed invention. The CMD/ADD112 signals would be directly transmitted to the memory devices as discussed further below. And the CLK113 signal would be transmitted from the memory control module 510 to the buffers 105a and 105b where it would be locally synchronized (possibly with phase-locked loops) and distributed to the memory devices. The motivation to modify Matsui's method and apparatus would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja. With respect to the data buffer control signals, the CLK113 is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal).

With respect Applicant's argument above that "the module control device transmitting the module C/A signals *to at least some of the memory devices*" this feature is well-known in the art. For example, see para 0010 of Saito et al. US 2010/0309706, paras 0039, 0048-0049 of Harashima, or paras 0007-0009 of Lee et al. US 2013/0242680.

Applicant's argument, page of the remarks, Furthermore, claims 3-8 are patentable for the additional limitations recited therein. For example, Claim 3 recites the limitations that "the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals." Neither Matsui nor Sutardja discloses or teaches such limitations. The data buffers in Matsui and Sutardja simply buffers data signals and do not include data paths that are controlled by data buffer control signals received from a module control device on the memory module. The Examiner points out that in Matsui, command/address signal and data are transmitted in the internal command/address wiring 112' and the internal data wiring 111' (paragraph [0157]), but nowhere in Matsui or Sutardja teaches that these internal wirings are controlled by data buffer control signals" has been fully considered and is deemed not persuasive.

With respect to the data buffer control signals, the clock signal is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal). See Fig. 17 of Sutardja describing how the clock signal by the clock generator module 534 controls each of the data buffers 530-1 through 530-Z.

Applicant's argument, page 10 of the remarks, "As a further example, claim 4 recites the limitations that "the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time." The Examiner rejects

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this claim by pointing out that Sutardja's buffer modules 530 are daisy-chained. The Examiner neglects that claim 2, from which claim 4 depends, requires that each of the data buffers has a data width of n -bits, a fraction of the data width of the memory module, which is $(N \times n)$ bits. Claim 4 recites specifically that N such data buffers spanning the data width of a memory module are in a fly-by configuration. In contrast, there is no teach in Sutardja that the Z buffer modules 530 span the data width of a memory module. In fact, Sutardja's buffer modules 530 are each as wide as the memory modules 514, and they are disposed on different memory modules 514. Therefore, Sutardja does not even disclose the data buffers in claim 2, and cannot possibly disclose fly-by configuration of the data buffers in claim 4" has been fully considered and is deemed not persuasive.

The Examiner respectfully submits that, as disclosed by Sutardja, the memory ICs 520 and the buffer module 530 for each memory module 514 may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias, para 0135. It is well known in the art that in such an arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530. Further, referring to Figure 17, the buffer modules 530 are daisy-chained (fly-by configuration) resulting in a skew of the arrived signals.

Accordingly, the Office has met its burden in finding the instant claims unpatentable over the prior art. Applicant is hereby reminded that the test for obviousness is not that the claimed invention must be expressly suggested in any one or all of the references, but rather that the test is what the combined teachings of the

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references would have suggested to those of ordinary skill in the art. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981). Hence, regarding the limitations of the claims addressed herein and on the record, the examiner respectfully submits that the record establishes a prima facie case that the combined teachings of the respective references cited would have suggested the claimed subject matter to those of ordinary skill in the art.

In view of the foregoing remarks, independent claims 1, 9, and 18 are not in a condition for allowance. Claims depending therefrom, either directly or indirectly, are also not in a condition for allowance.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 2-5, 7, 9, 11-14, 16, 18-22, and 24-25 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereinafter "Matsui") in view of Sutardja et al. US 2007/0168781 (hereinafter "Sutardja").

As per independent claim 2, Matsui teaches "A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising" (Figure 5 – memory modules 103a-103d interfaced with a

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memory controller 101. Each memory module comprises buffers 105a and 105b. In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. Referring to Figure 5, clock signals CLK/CLKB 113 operating at 1.33 GHz is received by the buffers 105a and 105b from the memory controller 101),

“ N data buffers each having a data width of n -bits and each generating a respective local clock signal” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159),

“memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock

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signal generated by the respective one of the data buffers” (Figure 5 – In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160. The buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals, para 0159. Referring to Figure 5, the data bus DQ 111 is connected between the buffers 105a and 105b and the memory controller 101).

Matsui's invention discloses all of the claimed limitations from above except for “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers” and “wherein the module control device does not transmit any clock signal to the memory devices” and “wherein each of the *N* data buffers generates the respective local clock signal from the module clock signal received from the module control device”.

However, Sutardja teaches “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers” (A functional block diagram of a memory module 500 is shown in Figure 17, para 0133 where a memory control module 510 is depicted. The buffer module 530 may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. The buffer modules 530 may buffer incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530, para 0136. With respect to the data buffer control signals, the CLK113 of Matsui is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal). With respect to the module control device transmitting the module C/A signals to at least some of the memory devices, this feature is well-known in the art. For example, see para 0010 of Saito et al. US 2010/0309706, paras 0039, 0048-0049 of Harashima, or paras 0007-0009 of Lee et al. US 2013/0242680),

“wherein the module control device does not transmit any clock signal to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136),

“wherein each of the *N* data buffers generates the respective local clock signal from the module clock signal received from the module control device” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. The clock generator module 534 is mapped as being part of the memory control module 510).

Given the teaching of Sutardja, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui with “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers” and “wherein the module control device does not transmit any clock signal to the memory devices” and “wherein each of the *N* data buffers generates the respective local clock signal from the module clock signal received from the module control device”. The motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja.

As per dependent claim 3, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui teaches “wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals” (Figure 5 - command/address signal and data are transmitted in the internal command/address wiring 112' and the internal data wiring 111', para 0157. With respect to the data buffer control signals, the clock signal is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal). See Fig. 17 of Sutardja describing how the clock signal by the clock generator module 534 controls each of the data buffers 530-1 through 530-Z).

As per dependent claim 4, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for “wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time”.

However, Sutardja teaches “wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at the N data buffers plurality of data buffers at different points in time” (The memory ICs 520 and the buffer module 530 for each memory module 514 may be mounted on one or both sides of a printed circuit board (PCB) having interconnecting traces and/or vias, para 0135. It is well known in the art that in such an

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arrangement, buffer modules 530 may have skew associated with signals reaching the buffer modules 530. Further, referring to Figure 17, the buffer modules 530 are daisy-chained (fly-by configuration) resulting in a skew of the arrived signals).

The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 4.

As per dependent claim 5, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal".

However, Sutardja teaches "wherein the respective one of the N data buffers data buffer includes one or more programmable registers controlling a phase of the local clock signal" (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136. It is well known in the art that a PLL or a DLL (digital lock loop) contain programmable registers to control/adjust the phase and frequency of PLL or DLL).

The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 5.

As per dependent claim 7, Matsui in combination with Sutardja discloses the memory module of claim 2. Matsui's invention discloses all of the claimed limitations from above except for "wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal".

However, Sutardja teaches “wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 2 is equally applicable to claim 7.

As per independent claim 9, most of the limitations of this claim is rejected based on arguments provided above for similar rejected independent claim 2.

As per other limitations of claim 9, Sutardja teaches “transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices” (Referring to Figure 17, it is disclosed that clock generator module 534 feeds clock signals to buffer modules 530 and memory modules 514. Sutardja does not explicitly state that the clock signal from the clock generator module 534 directly feed the memory devices 520. Instead, Sutardja explicitly teaches that “the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals”, para 0136),

“performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals” (Sutardja explicitly teaches that “the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals”, para 0136. Accordingly, read/write operations are performed based on the local phase-adjusted clock signals).

As per dependent claim 11, Matsui in combination with Sutardja discloses the method of claim 9. Matsui teaches “receiving command/address (C/A) signals from the memory controller” (Figure 5 – C/A 112),

“generating module C/A signals and data buffer control signals in response to the C/A signals” (Figure 5 – C/A 112 and CLK/CLKB 113. According to the instant specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035),

“transmitting the module C/A signals to at least some of the memory devices” (Figure 5 – C/A 112’ transmitted to the two groups of DRAMs 110),

“transmitting the data buffer control signals to the N data buffers” (Figure 5 – CLK/CLKB 113. According to the instant specification, “data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal”, para 0035).

As per dependent claim 12, Matsui in combination with Sutardja teaches the method of claim 11. Matsui teaches “further comprising controlling data paths in the N data buffers according to the data buffer control signals” (Figure 5 – DQ 111’ is controlled by CLK/CLKB 113’).

As per dependent claims 13-14, these claims are rejected based on arguments provided for similar rejected dependent claims 4-5 respectively.

As per dependent claim 16, Matsui in combination with Sutardja discloses the method of claim 9. Matsui’s invention discloses all of the claimed limitations from above

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except for “further comprising phase-locking the local clock signal with the module clock signal”.

However, Sutardja teaches “further comprising phase-locking the local clock signal with the module clock signal” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 9 is equally applicable to claim 16.

As per independent claim 18, this claim is rejected based on arguments provided above for similar rejected independent claim 2.

As per dependent claim 19, Matsui in combination with Sutardja teaches the memory module of claim 18. Matsui teaches “wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors” (Figure 3).

As per dependent claim 20, this claim is rejected based on arguments provided above for similar rejected dependent claim 16.

As per dependent claim 21, Matsui in combination with Sutardja teaches the memory module of claim 18. Matsui’s invention discloses all of the claimed limitations

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from above except for “wherein the control circuit does not transmit any clock signals to the memory devices”.

However, Sutardja teaches “wherein the control circuit does not transmit any clock signals to the memory devices” (Figure 17 - the buffer modules 530 may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).

The same motivation that was utilized for combining Matsui and Sutardja as set forth in claim 18 is equally applicable to claim 21.

As per dependent claim 22, Matsui in combination with Sutardja discloses the memory module of claim 18. Matsui teaches “wherein the data buffer control signals are different from the module C/A signals” (With respect to the data buffer control signals, the CLK113' (see Fig. 5) is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal)).

As per dependent claim 24, Matsui in combination with Sutardja discloses the memory module of claim 18. Matsui teaches “wherein each of the memory devices is n -bits wide” (In case of the memory system for transmitting/receiving data of a 32-bit width, the two x-8 configuration DRAMs 110a are placed on each side of each of the buffers 105a and 105b. When each memory module 103 is selected, the eight DRAMs 110a on each memory module 103 are activated by both buffers 105a and 105b so that data of a 64-bit width can be transmitted/received between the buffers 105a and 105b and the eight DRAMs 110a, para 0160).

As per dependent claim 25, this claim is rejected based on arguments provided above for similar rejected dependent claim 24. It would be obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to substitute “four x-4 configuration DRAMs 110a” on each side of the buffers 105a and 105b as described with respect to claim 24.

Claim 6, 10, and 15 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereinafter “Carpenter”).

As per dependent claim 6, Matsui in combination with Sutardja discloses the memory module of claim 5. Matsui’s invention modified by Sutardja discloses all of the claimed limitations from above except for “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals”.

However, Carpenter teaches “wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals” (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with “wherein the

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respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

As per dependent claim 10, Matsui in combination with Sutardja discloses the method of claim 9. Matsui's invention modified by Sutardja discloses all of the claimed limitations from above except for "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers".

However, Carpenter teaches "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers" (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with "further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

As per dependent claim 15, Matsui in combination with Sutardja discloses the method of claim 11. Matsui's invention modified by Sutardja discloses all of the claimed limitations from above except for "programming one or more registers in each of the N data buffers according to the data buffer control signals".

However, Carpenter teaches "programming one or more registers in each of the N data buffers according to the data buffer control signals" (The storage array can also be used as a general-purpose array with adjustable timing. Programmable delays are provided in the local clock buffer that times access to the cells by controlling the word-line assertion timing and pulse width and the global bit-line evaluate timing and pulse width, para 0022).

Given the teaching of Carpenter, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Masui as modified by Sutardja with "programming one or more registers in each of the N data buffers according to the data buffer control signals". The motivation would be that this would allow evaluation of timing margins for accessing storage cells which in turn would facilitate design improvement and determination of operating margins, para 0022 of Carpenter.

Claims 8 and 17 are rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Sung et al. US 2011/0110168 (hereinafter "Sung").

As per dependent claim 8, Matsui in combination with Sutardja discloses the memory module of claim 1. Matsui's invention modified by Sutardja discloses all of the

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claimed limitations from above except for “wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks”.

However, Sung teaches “wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks” (Figures 1, 2 correspond specifically to a case in which the first memory unit 113 includes two memory ranks, and in this regard, two first chip selection signals are input to the first memory unit 113. The second memory unit 123 includes two memory ranks, and in this regard, two second chip selection signals are input to the second memory unit 123, paras 0046, 0049).

Given the teaching of Sung, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify the scope of the invention of Matsui as modified by Sutardja with “wherein memory devices are organized in one or more ranks and are each n-bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks”. The motivation would be that the memory controller can determine which memory rank of which memory module is to be selected, para 0052 of Sung.

As per dependent claim 17, this claim is rejected based on arguments provided above for similar rejected dependent claim 8.

Claim 23 is rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Chu et al. US 2008/0291758 (hereinafter “Chu”).

As per dependent claim 23, Matsui in combination with Sutardja discloses the memory module of claim 18. Matsui's invention modified by Sutardja discloses all of the claimed limitations from above except for "wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer".

However, Chu teaches "wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer" (Referring to Fig. 7 and para 0049, Fig. 7 is a block diagram illustrating a capture register 720, leveling element 725, synchronization register 730, delay 710, delay-locked loop 740, and phase-locked loop 750. Data signal DQ is received by the capture register 720. A data strobe signal DQS is received by the delay 710 and phase shifted to generate a delayed DQS signal DDQS that clocks the capture registers 720. The output of the capture registers 720 is received by the leveling element 725, which in turn provides an output to the synchronization register 730. The synchronization register 730 is clocked by a synchronization clock provided by phase-locked loop 750. The synchronization register 730 provides an output DATAOUT to other circuitry (not shown). The delay-locked loop 740 synchronizes to a local clock generated by phase-locked loop 750 and provides a control signal COUNT to delay 710, such that the DDQS signal is properly phase shifted to clock the incoming data DQ).

Given the teaching of Chu, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to further modify

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the scope of the invention of Matsui as modified by Sutardja with "wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer". Matsui's buffer 105a and 105b may be modified to include Chu's interface circuit depicted in Fig. 7. The motivation would be that the synchronization register may become unstable and provide incorrect data on the DATAOUT line, para 0050 of Chu.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 8:30AM - 5:00PM Mon-Fri.

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If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID X YI/
Supervisory Patent Examiner, Art Unit 2132

Notice of References Cited	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2008/0291758 A1	11-2008	Chu; Michael H.M.	G11C7/1051	365/194
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16

S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2015/11/01 06:14

			IBM_TDB			
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S36	38	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 11:42
S37	8	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:13
S38	68	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:18
S39	4415	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S40	5	S39 and PLL	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S41	0	G11C7/1066.cpc	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:00
S42	4534	G06F12/08.cpc.	US-PGPUB; USPAT;	OR	ON	2016/08/01

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			08:01
S43	0	S42 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:02
S44	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:42
S45	553	S44 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:43
S46	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:48
S47	27	S46 and ((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:49
S48	0	((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module) same (program\$4 near3 register)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:50
S49	0	((local with (pll or (phase adj lock\$2 adj loop))) same (program\$4 near3 register)) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:52
S50	35	((local with (pll or (phase adj lock\$2 adj loop))) same register) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:53

1/ 3/ 2017 12:23:45 PM
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Search Notes 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/08	08/01/2016	ZA
G11C 7/1066	01/03/2017	ZA

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
See attached search history	01/03/2017	ZA

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/ZUBAIR AHMED/ Examiner.Art Unit 2132	
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<i>Index of Claims</i> 	Application/Control No. 14445035	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant						☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE									
Final	Original	11/02/2015	03/13/2016	08/01/2016	01/03/2017						
	1	✓	✓	-	-						
	2	✓	✓	✓	✓						
	3	✓	✓	✓	✓						
	4	✓	✓	✓	✓						
	5	✓	✓	✓	✓						
	6	✓	✓	✓	✓						
	7	✓	✓	✓	✓						
	8	✓	✓	✓	✓						
	9	✓	✓	✓	✓						
	10	✓	✓	✓	✓						
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	18		✓	✓	✓						
	19		✓	✓	✓						
	20		✓	✓	✓						
	21			✓	✓						
	22				✓						
	23				✓						
	24				✓						
	25				✓						

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : N3044.10018US01

CERTIFICATE OF EFS WEB

TRANSMISSION

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

April 10, 2017

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND RESPONSE TO FINAL OFFICE ACTION
DATED JANUARY 9, 2017

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Enclosed is an Amendment and Response to the Non-Final Office Action dated January 9, 2017.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

1. (Cancelled)
2. (Previously Presented) A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of ($N \times n$) bits, where N and n are integers larger than one, comprising:
 - N data buffers each having a data width of n -bits and each generating a respective local clock signal;
 - memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and
 - a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;
 - wherein the module control device does not transmit any clock signal to the memory devices; and
 - wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.
3. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.
4. (Previously Presented) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

5. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes one or more programmable registers controlling a phase of the local clock signal.
6. (Previously Presented) The memory module of claim 5,
wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals.
7. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal.
8. (Previously Presented) The memory module of claim 2, wherein memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.
9. (Previously Presented) A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:
 - receiving a system clock signal;
 - generating a module clock signal on the memory module from the system clock signal received by the memory module;
 - transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;
 - generating a respective local clock signal from the module clock signal at each of the N data buffers;
 - transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and

performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Previously Presented) The method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Previously Presented) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (Previously Presented) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.

15. (Previously Presented) The method of claim 11, further comprising:
programming one or more registers in each of the N data buffers according to the data buffer control signals.

16. (Previously Presented) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.

17. (Previously Presented) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.

18. (Previously Presented) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Previously Presented) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors.

20. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.
21. (Previously Presented) The memory module of claim 18, wherein the control circuit does not transmit any clock signals to the memory devices.
22. (Previously Presented) The memory module of claim 18, wherein the data buffer control signals are different from the module C/A signals.
23. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer.
24. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is n -bits wide.
25. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is $n/2$ -bits wide.

REMARKS

This amendment responds to the Non-Final Office Action dated January 9, 2017.

In the Office Action, the Examiner:

- rejected claims 2-5, 7, 9, 11-14, 16, 18-22 and 24-25 under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereafter “Matsui”) in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”);
- rejected claims 6, 10, and 15 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereafter “Carpenter”);
- rejected claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and further in view of Sung et al. US 2011/0110168 (hereinafter “Sung”); and
- rejected claim 23 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Chu et al. US 2008/0291758 (hereafter “Carpenter”).

No claims are amended, cancelled or added. Thus, no new matter is added.

Independent claim 2 is rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja.

As acknowledged by the Examiner, Matsui does not teach or suggest “a module control device.” The Office Action cites Sutardja as teaching this feature. In rejecting claim 2, the Examiner states: “The memory control module 510 [of Sutardja] is mapped to the claimed module control device” without showing **how** the memory control module 510 is performing all of the functions of the module control device, as claimed. (Office Action, p. 3). For example, the Office Action has not demonstrated how the memory control module 510 of Sutardja can “receive command/address (C/A) signals and a system clock from the memory controller,” “output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices,” and “transmitting the data buffer control signals and

the module clock signal to the data buffers,” as recited in claim 2. It is respectfully submitted that Sutardja does not teach or suggest at least these claimed features.

In spite of the above, the Office Action suggests that one having ordinary skill in the art at the time of the invention would be motivated to combine the deficient memory control module 510 of Sutardja with Matsui. (Office Action, p. 4). For example, on Page 4 of the Office Action, the Examiner states: “With the proposed modification of Matsui’s method and apparatus, the memory control module 510 of Sutardja situated between memory controller 101 and memory modules 103a-d (FIG. 5 of Matsui) would only.....” In other words, the Examiner is proposing to add another memory controller 510 of Sutardja between the memory controller 101 and memory modules 103a-d of Matsui, in order to meet the requirement of claim 2. There is no reason any person of ordinary skill in the art (POSITA) would want to do that because the added (and redundant) memory controller 510 would slow down the system without providing any benefit. Neither of the references provides any motivation for such a combination, which is made entirely by impermissible hindsight. Further, the resulting system of the Examiner’s proposed combination still would not meet the requirements of the claim because memory controller 510 would not perform the tasks of “transmitting the module C/A signals to at least some of the memory devices” and “transmitting the data buffer control signals and the module clock signal to the data buffers,” as required by claim 2

In fact, neither reference teaches or suggests the recited “buffer control signals.” Nonetheless, further in the same paragraph on page 4 of the Office Action, the Examiner tries to patch up the holes in the proposed combination by stating: “With respect to the data buffer control signals, the CLK113 is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal.” This does not make sense because the claim requires “the module control device transmitting the data buffer control signals **AND** the module clock signal to the data buffers.” (emphasis added). This requirement is due to the fact that the claimed memory module is very different from the references. Unlike the references, the data buffers in the claims do not receive the command/address signals. Instead, they rely on the module control signals from the module control device to control

its functions. Thus, the module control device in the claimed invention registers the command/address signals and controls the memory devices AND the data buffers accordingly.

Moreover, on Page 4 of the Office Action, the Examiner states: “the CLK113 of Matsui is considered as part of the data buffer control.” While the references need not describe the claimed features in exactly the same language, the teachings of the references in combination still need to meet the claim requirements of the “module control device ... transmitting the data buffer control signals **AND** the module clock signal to the data buffers.” (emphasis added). Neither of the cited references teach these features.

On page 11 of the Office Action, the Examiner states: “’wherein the module control device does not transmit any clock signal to the memory devices’ (Figure 17 – the buffer modules 530 [of Sutardja] may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136).” This assertion does not make sense. While not explicit, the Examiner now appears to map the buffer module 530 of Sutardja to the claimed module control device. The buffer module 530 may generate the local clock signal but it still sends the local clock signal to the memory devices. Thus, it does not meet the claim requirement that “the module control device does not transmit **any** clock signal to the memory devices.” (emphasis added).

Later on the same page of the Office Action, the Examiner states: “The motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja.” It is respectfully submitted that no person having ordinary skill in the art at the time of the invention in his/her right mind, for the sole purpose of providing a PLL, would slow down the memory module by adding another module control device in addition to the buffers of Matsui, which already registers and buffers all of the signals from the memory controller 101. The PLL can simply be provided in the buffers of Matsui.

Therefore, claim 2 is patentable over Matsui in view of Suturdja. Claims 3-8 depend from claim 2. Therefore, claims 3-8 are patentable for at least the same reasons claim 2 as amended is patentable.

Independent claim 9 recites similar features as claim 2 is and is patentable for similar reasons as those discussed above with respect to claim 2. Further, claim 9 is patentable for other reasons. For example, claim 9 recites “generating a module clock signal **on the memory module** from the system clock signal **received by the memory module**” and “transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus.” (emphasis added). It is submitted that neither Matsui nor Sutardja teach or suggest at least these features. Claims 10-17 depend from claim 9 and include further limitations in addition to the limitations in claim 9. Therefore, claims 10-17 are patentable for at least the same reasons claim 9 is patentable.

Independent claim 18 recites similar features as claim 2 is and is patentable for similar reasons as those discussed above with respect to claim 2. Further, claim 18 is patentable for other reasons. For example, claim 18 recites “*N* data buffers each *n*-bits wide and each configured to receive the module clock signal **AND** the data buffer control signals from the module control circuit” and “the respective data buffer including data paths and logic controlling the data paths **in response to the data buffer control signals**” (emphasis added). It is submitted that neither Matsui nor Sutardja teach or suggest at least these features. Claims 19-21 and new claims 22-23 depend from claim 18 and include further limitations in addition to the limitations in claim 18. Therefore, claims 19-21 and new claim 22-23 are patentable for at least the same reasons claim 18 is patentable.

Because the Office Action neither shows that the references teach or suggest each and every claim element of claims 2, 9 and 18 nor explains why the differences between the prior art and the claimed invention would have been obvious to a person of skill in the art, the Office Action fails to present a *prima facie* case that claims 2, 9 and 18 are obvious and the rejections are thus improper. Applicants therefore respectfully request

that Examiner withdraw the rejections under 35 U.S.C. § 103(a) of claims 2, 9 and 18 and of all claims which depend therefrom.

Applicant submits that claims 2-25 are in condition for allowance and Applicant respectfully requests such action. The Examiner is invited to call the undersigned at (650) 521-4828 to discuss any remaining issues.

Respectfully submitted,

Date: April 10, 2017

/Jamie J. Zheng/

51167

Jamie J. Zheng

(Reg. No.)

Customer No. 79141

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	N3044.10018US02			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
NOTICE OF APPEAL	2401	1	400	400
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				400

Electronic Acknowledgement Receipt

EFS ID:	28887140
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	N3044.10018US02
Receipt Date:	10-APR-2017
Filing Date:	28-JUL-2014
Time Stamp:	23:51:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 400
RAM confirmation Number	041117INTEFSW23572500
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Notice of Appeal Filed	2017-04-10_Notice-of-Appeal_N3044-10018US02.pdf	216210	no	2
			0b09cf25f42b1dfb973abc65342cc1ef06b59ba3		
Warnings:					
Information:					
2	Response After Final Action	2017-04-10_Amendment_N3044-10018US01.pdf	143251	no	11
			2a956429abec12454fd2fb55a22332037613eae2		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	29939	no	2
			2372bf298881215f08436fe1dedc078ee5caf1c3		
Warnings:					
Information:					
Total Files Size (in bytes):			389400		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

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NOTICE OF APPEAL FROM THE EXAMINER TO THE PATENT TRIAL AND APPEAL BOARD		Docket Number (Optional) N3044.10018US02			
I hereby certify that this correspondence is being facsimile transmitted to the USPTO EFS-Web transmitted to the USPTO, or or deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to "Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450" [37 CFR 1.8(a)] on _____ Signature _____ Typed or printed name _____		In re Application of Hyun Lee			
		<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px;">Application Number 14/445,035</td> <td style="width: 50%; padding: 5px;">Filed 07-28-2014</td> </tr> </table>		Application Number 14/445,035	Filed 07-28-2014
		Application Number 14/445,035	Filed 07-28-2014		
For MEMORY MODULE WITH LOCAL SYNCHRONIZATION					
		<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px;">Art Unit 2132</td> <td style="width: 50%; padding: 5px;">Examiner AHMED, ZUBAIR</td> </tr> </table>		Art Unit 2132	Examiner AHMED, ZUBAIR
Art Unit 2132	Examiner AHMED, ZUBAIR				

Applicant hereby **appeals** to the Patent Trial and Appeal Board from the last decision of the examiner.

The fee for this Notice of Appeal is (37 CFR 41.20(b)(1)) \$ 800.00

☒ Applicant claims small entity status. See 37 CFR 1.27. Therefore, the fee shown above is reduced by half, and the resulting fee is: \$ 400.00

☐ A check in the amount of the fee is enclosed.

☐ Payment by credit card. Form PTO-2038 is attached.

☐ The Director has already been authorized to charge fees in this application to a Deposit Account.

☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment to Deposit Account No. 50-5394.

☐ A petition for an extension of time under 37 CFR 1.136(a) (PTO/SB/22) is enclosed.

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

I am the

☐ applicant/inventor. ☐ assignee of record of the entire interest. See 37 CFR 3.71. Statement under 37 CFR 3.73(b) is enclosed. (Form PTO/SB/96) ☒ attorney or agent of record. Registration number <u>51167</u> ☐ attorney or agent acting under 37 CFR 1.34. Registration number if acting under 37 CFR 1.34. _____	<u>/Jamie J. Zheng/</u> Signature <u>Jamie J. Zheng</u> Typed or printed name <u>(435) 252-1360</u> Telephone number <u>April 10, 2017</u> Date
---	--

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

☒ *Total of <u>1</u> forms are submitted.

This collection of information is required by 37 CFR 41.31. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11, 1.14 and 41.6. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/445,035	Filing Date 07/28/2014	☐ To be Mailed
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 ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	04/10/2017	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 24	Minus	** 24	= 0	x \$40 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
	TOTAL ADD'L FEE						0

(Column 1)

(Column 2)

(Column 3)

		CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
	TOTAL ADD'L FEE						

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
 TAMMY D. MCBETH BROWN

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	N3044.10018US02	9700

79141 7590 04/21/2017
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
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2132

NOTIFICATION DATE	DELIVERY MODE
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04/21/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
infohub.j@gmail.com
d@mabr.com

Advisory Action Before the Filing of an Appeal Brief	Application No. 14/445,035	Applicant(s) LEE ET AL.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

--The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

THE REPLY FILED 10 April 2017 FAILS TO PLACE THIS APPLICATION IN CONDITION FOR ALLOWANCE.

NO NOTICE OF APPEAL FILED

1. ☒ The reply was filed after a final rejection. No Notice of Appeal has been filed. To avoid abandonment of this application, applicant must timely file one of the following replies: (1) an amendment, affidavit, or other evidence, which places the application in condition for allowance;
(2) a Notice of Appeal (with appeal fee) in compliance with 37 CFR 41.31; or (3) a Request for Continued Examination (RCE) in compliance with 37 CFR 1.114 if this is a utility or plant application. Note that RCEs are not permitted in design applications. The reply must be filed within one of the following time periods:

a) ☒ The period for reply expires 3 months from the mailing date of the final rejection.

b) ☐ The period for reply expires on: (1) the mailing date of this Advisory Action; or (2) the date set forth in the final rejection, whichever is later. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of the final rejection.

c) ☐ A prior Advisory Action was mailed more than 3 months after the mailing date of the final rejection in response to a first after-final reply filed within 2 months of the mailing date of the final rejection. The current period for reply expires _____ months from the mailing date of the prior Advisory Action or SIX MONTHS from the mailing date of the final rejection, whichever is earlier.

Examiner Note: If box 1 is checked, check either box (a), (b) or (c). ONLY CHECK BOX (b) WHEN THIS ADVISORY ACTION IS THE FIRST RESPONSE TO APPLICANT'S FIRST AFTER-FINAL REPLY WHICH WAS FILED WITHIN TWO MONTHS OF THE FINAL REJECTION. ONLY CHECK BOX (c) IN THE LIMITED SITUATION SET FORTH UNDER BOX (c). See MPEP 706.07(f).

Extensions of time may be obtained under 37 CFR 1.136(a). The date on which the petition under 37 CFR 1.136(a) and the appropriate extension fee have been filed is the date for purposes of determining the period of extension and the corresponding amount of the fee. The appropriate extension fee under 37 CFR 1.17(a) is calculated from: (1) the expiration date of the shortened statutory period for reply originally set in the final Office action; or (2) as set forth in (b) or (c) above, if checked. Any reply received by the Office later than three months after the mailing date of the final rejection, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

NOTICE OF APPEAL

2. ☒ The Notice of Appeal was filed on 10 April 2017. A brief in compliance with 37 CFR 41.37 must be filed within two months of the date of filing the Notice of Appeal (37 CFR 41.37(a)), or any extension thereof (37 CFR 41.37(e)), to avoid dismissal of the appeal. Since a Notice of Appeal has been filed, any reply must be filed within the time period set forth in 37 CFR 41.37(a).

AMENDMENTS

3. ☐ The proposed amendments filed after a final rejection, but prior to the date of filing a brief, will not be entered because

a) ☐ They raise new issues that would require further consideration and/or search (see NOTE below);

b) ☐ They raise the issue of new matter (see NOTE below);

c) ☐ They are not deemed to place the application in better form for appeal by materially reducing or simplifying the issues for appeal; and/or

d) ☐ They present additional claims without canceling a corresponding number of finally rejected claims.

NOTE: _____. (See 37 CFR 1.116 and 41.33(a)).

4. ☐ The amendments are not in compliance with 37 CFR 1.121. See attached Notice of Non-Compliant Amendment (PTOL-324).

5. ☐ Applicant's reply has overcome the following rejection(s): _____.

6. ☐ Newly proposed or amended claim(s) _____ would be allowable if submitted in a separate, timely filed amendment canceling the non-allowable claim(s).

7. ☒ For purposes of appeal, the proposed amendment(s): (a) ☐ will not be entered, or (b) ☒ will be entered, and an explanation of how the new or amended claims would be rejected is provided below or appended.

AFFIDAVIT OR OTHER EVIDENCE

8. ☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

9. ☐ The affidavit or other evidence filed after final action, but before or on the date of filing a Notice of Appeal will not be entered because applicant failed to provide a showing of good and sufficient reasons why the affidavit or other evidence is necessary and was not earlier presented. See 37 CFR 1.116(e).

10. ☐ The affidavit or other evidence filed after the date of filing the Notice of Appeal, but prior to the date of filing a brief, will not be entered because the affidavit or other evidence failed to overcome all rejections under appeal and/or appellant fails to provide a showing of good and sufficient reasons why it is necessary and was not earlier presented. See 37 CFR 41.33(d)(1).

11. ☐ The affidavit or other evidence is entered. An explanation of the status of the claims after entry is below or attached.

REQUEST FOR RECONSIDERATION/OTHER

12. ☒ The request for reconsideration has been considered but does NOT place the application in condition for allowance because:
See continuation sheet.

13. ☐ Note the attached Information *Disclosure Statement*(s). (PTO/SB/08) Paper No(s). _____

14. ☐ Other: _____.

STATUS OF CLAIMS

15. The status of the claim(s) is (or will be) as follows:

Claim(s) allowed: _____.

Claim(s) objected to: _____.

Claim(s) rejected: 2-25.

Claim(s) withdrawn from consideration: _____.

/DAVID X YI/ Supervisory Patent Examiner, Art Unit 2132	/ZUBAIR AHMED/ Examiner, Art Unit 2132
--	---

Continuation of 12: the reply filed on 04/10/2017 does NOT place the application in condition for allowance because:

First and foremost, the Examiner rejects Applicant's attempt to morph a specific terminology used by prior art of record Sutardja. For example, on page 8 of the remarks Applicant uses the terminology "memory controller 510" to denote the "memory control module 510" of Sutardja. This is misleading because a memory controller may be situated separately from a memory module (e.g., a DIMM) whereas a memory control module may be part of a memory module. Applicant should refrain from doing this.

Applicant alleges that the memory control module 510 of Sutardja does not teach "receive command/address (C/A) signals and system clock from the memory controller", "output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices," and "transmitting the data buffer control signals and the module clock signal to the data buffers," as recited in claim 2, pages 7-9 of the remarks.

The Examiner disagrees. It was described by the Examiner in the Office Action of 01/09/2017 that Matsui's method and apparatus may be modified by the teachings of Sutardja by a person having ordinary skill in the art to arrive at the claimed invention.

Referring to Fig. 17 of Sutardja, a functional block diagram of a memory module 500 is shown where a memory control module 510 is depicted, para 0133. The memory control module 510 is mapped to the claimed module control device. Further shown in Fig. 17 of Sutardja are buffer modules 530 that may buffer signals between the memory control module 510, the memory modules 514, and/or signals on the buses 531 and 533. Still further, a clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530, para 0136. Now referring to Fig. 5 of Matsui, the memory control module 510 of Sutardja may be inserted between the memory controller 101 and the memory modules 103a-d interfacing with the buffers 105a and 105b. This modification, by a person of ordinary skill in the art, would neither change the mode of operation of Matsui's method and apparatus nor would it render Matsui's method and apparatus inoperable for the following reasons. Matsui already discloses the use of two buffers (buffers 105a and 105b, Fig. 5) in each memory module 103a-d. Each of the two buffers buffer DQ111, CLK113, and CMD/ADD112 (see Fig. 5 of Matsui). With the proposed modification of Matsui's method and apparatus, the memory control module 510 of Sutardja situated between memory controller 101 and memory modules 103a-d (Fig. 5 of Matsui) would only receive from the memory controller 101 the CMD/ADD112 and CLK113 signals. The DQ111 signals may be directly transmitted from the memory controller to the buffers 105a and 105b consistent with the claimed invention. The CMD/ADD112 signals would be directly transmitted to the memory devices and this feature is well-known in the art. And the CLK113 signal would be transmitted from the memory control module 510 to the buffers 105a and 105b where it would be locally synchronized (possibly with phase-locked loops) and distributed to the memory devices. The motivation to modify Matsui's method and apparatus would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja. With respect to the data buffer control signals, the CLK113 is considered a group of clock/control signals (see FIG. 44 of Matsui, "GLOBAL CLOCKS" supplied to buffer 105) and these signals are mapped to the claimed module clock signal and the claimed data buffer control signals (see para 0035 of the instant specification).

With respect to Applicant's argument regarding claim 9 that neither Matsui nor Sutardja teach or suggest "generating a module clock signal on the memory module from the system clock signal received by the memory module" and "transmitting the module clock signal to a plurality of data buffers each couple between a respective group of the memory devices and the memory bus", page 10 of the remarks, the Applicant is referred to the above discussion where a memory control module 510 and a clock generator module 534 in combination generates module clock signal and distributes the module clock signals to the buffers 530 of Sutardja or the buffers 105 of Matsui.

With respect to Applicant's argument regarding claim 18 that neither Matsui nor Sutardja teach or suggest "the respective data buffer including data paths and logic controlling the data paths in response to data buffer control signals", page 10 of the remarks, the Examiner submits that Matsui in FIG. 44 explicitly teaches these features. For example (see FIG. 44), in response to "GLOBAL CLOCKS" received by buffer 105, these buffers generate WDPS (Write Data Phase Signal) to synchronize write operations with respect to the target DRAM.

In view of the foregoing remarks, independent claims 2, 9, and 18 are not in a condition for allowance. Claims depending therefrom, either directly or indirectly, are also not in a condition for allowance.

PATENT

Customer No. 79141

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION

Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : N3044.10018US01

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April 10, 2017

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND RESPONSE TO FINAL OFFICE ACTION
DATED JANUARY 9, 2017

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Enclosed is an Amendment and Response to the Non-Final Office Action dated January 9, 2017.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 7 of this paper.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14445035
	Filing Date	2014-07-28
	First Named Inventor	Hyun Lee
	Art Unit	2132
	Examiner Name	AHMED, ZUBAIR
	Attorney Docket Number	N3044.10018US02

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14445035
	Filing Date		2014-07-28
	First Named Inventor	Hyun Lee	
	Art Unit	2132	
	Examiner Name	AHMED, ZUBAIR	
	Attorney Docket Number	N3044.10018US02	

1	Extended European Search Report dated April 3, 2017 in European Patent Application No. 14831654.0.		
If you wish to add additional non-patent literature document citation information please click the Add button Add			
EXAMINER SIGNATURE			
Examiner Signature			Date Considered
*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14445035
Filing Date	2014-07-28
First Named Inventor	Hyun Lee
Art Unit	2132
Examiner Name	AHMED, ZUBAIR
Attorney Docket Number	N3044.10018US02

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/JAMIE J. ZHENG/	Date (YYYY-MM-DD)	2017-04-26
Name/Print	Jamie J. Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10018US02			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	2806	1	90	90
Total in USD (\$)				90

Electronic Acknowledgement Receipt

EFS ID:	29041036
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10018US02
Receipt Date:	26-APR-2017
Filing Date:	28-JUL-2014
Time Stamp:	19:30:17
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$90
RAM confirmation Number	042717INTEFSW19341500
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	2017-04-26_IDS_N3044-10018 US02.pdf	1035102	no	4
			134f0c6357cf7bb8a67af18501e6c6840b812832		
Warnings:					
Information:					
2	Non Patent Literature	EP143816540_EEUSR.pdf	1863592	no	7
			9c424a76841bea8808887fa11fa7848c4d6e7c84		
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3	Fee Worksheet (SB06)	fee-info.pdf	30591	no	2
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PATENT APPLICATION
Docket No. N3044.10018US02

**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE BEFORE
THE BOARD OF PATENT APPEALS AND INTERFERENCES**

Applicant : Netlist, Inc.
App. No. : 14/445,035
Filed : July 28, 2014
For : MEMORY MODULE WITH
LOCAL SYNCHRONIZATION
Examiner : AHMED, ZUBAIR
Art Unit : 2132
Conf. No. : 9700
Docket No. : N3044.10018US02

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June 12, 2017

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

The Honorable Commissioner of Patents and Trademarks
Washington, D.C. 20231

BRIEF OF APPELLANT

This is an appeal to the Board of Patent Appeals and Interferences (the “Board”) from the Final Office Action mailed January 9, 2017 (the “Office Action”) wherein the Examiner rejected claims 2-25. This Brief is being filed pursuant to the provisions of 37 C.F.R. § 41.37. The Commissioner is hereby authorized to charge any additional fees associated with this communication, or to credit any overpayment, to Deposit Account No. 50-5394.

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LIST OF REFERENCES

	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant
1	US 2004/0105292	06-2004	Matsui; Yoshinori
2	US 2007/0168781	07-2007	Sutardja; Sehat
3	US 2009/0116312	05-2009	Carpenter; Gary D.
4	US 2011/0110168	05-2011	Sung, Myung-hee
5	US 2008/0291758	11-2008	Chu; Michael H.M.

I. REAL PARTY IN INTEREST

The real party in interest comprises NETLIST, INC., by way of assignment from the named inventor(s), Hyun Lee and Jayesh R. Bhakata, who are captioned in the present application. The corresponding assignment document was recorded in the United States Patent and Trademark Office at Reel/Frame 038300 / 0101 on April 17, 2016.

II. RELATED APPEALS AND INTERFERENCES

There are no other related appeals and/or interferences with regard to the present application.

III. STATUS OF CLAIMS

Claims 2-25 are pending in this application. Claims 2-25 were rejected in the Final Office Action mailed January 9, 2017 (the “Office Action”). Claims 2-25 are the subject of this appeal.

IV. STATUS OF AMENDMENTS

The Appellant has not submitted any amendments subsequent to the Final Office Action.

V. SUMMARY OF CLAIMED SUBJECT MATTER

Claims 2, 9 and 18 are the independent claims in the present application. Claims 2, 9, and 18 are best understood with reference to FIGS. 2D and 6A-6C, and with the following citations that correlate claim elements to specific embodiments described in the specification of the present application. The following citations are provided to facilitate the Board’s understanding of the subject matter of this appeal and do not in any manner limit claim interpretation. (Note that the drawings of the present application contain an error in that FIG. 2C, the figure immediately follows FIG. 2B, is labeled with both “FIG. 2D” and “FIG. 2C.” Appellant will correct this error in a separate submission.)

Independent claim 2 is directed to a memory module operable in a system with a memory controller. (See, para [0028], lines 1-3). The system operates according to a system clock. (See,

e.g., page 5, lines 3-7). The memory module has a data width of $(N \times n)$ bits, where N and n are integers larger than one. (See, e.g., para [0037] and para [0055]). The memory module comprises N data buffers each having a data width of n -bits. (See, e.g., paras [0038], [0055] and [0061]). Each data buffer generates a respective local clock signal. (See, e.g., para [0058]). Memory devices on the memory module include a plurality of groups of memory devices, with each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers. (See, e.g., paras [0037] and [0044]). Each respective group of memory devices receive and transmit data signals in accordance with the local clock signal generated by the respective one of the data buffers. (See, e.g., para [0036], lines 1-4, para [0037], para [0049], lines 1-3, para [0044], and paras [0053]-[0059]).

The memory module of independent claim 2 further comprises a module control device. The module control device receives command/address (C/A) signals and a system clock from the memory controller and outputs module C/A signals, data buffer control signals and a module clock signal. (See, e.g., page 9, lines 2-7). The module clock signal is derived from the system clock signal. (See, e.g., para [0051], lines 5-7). The module control device transmits the module C/A signals to at least some of the memory devices. (See, e.g., page 9, lines 2-5 and 7-8, and para [0033], lines 4-5). The module control device transmits the data buffer control signals and the module clock signal to the data buffers. (See, e.g., page 9, lines 5-7). The module control device does not transmit any clock signal to the memory devices. (See, e.g., para [0049], lines 1-3). Each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device. (See, e.g., para [0058], lines 3-5).

Independent claim 9 is directed to a method performed by a memory module coupled to a memory bus in a system. (See, para [0028], lines 1-3). The memory module has a data width of $(N \times n)$ bits, where N and n are integers larger than one. (See, e.g., para [0037] and para [0055]). The memory module includes memory devices. The memory devices includes a plurality of groups of memory devices. (See, e.g., para [0037]). The method comprises receiving a system clock signal and generating a module clock signal on the memory module from the system clock signal received by the memory module. (See, e.g., para [0051], lines 5-7). The method further comprises transmitting the module clock signal to a plurality of data buffers. (See, e.g., page 9, lines 5-7). Each of the plurality of data buffers is coupled between a respective group of the

memory devices and the memory bus. (See, e.g., para [0037], lines 4-8). The plurality of data buffers include N data buffers, each of the N data buffers having a data width of n bits. (See, e.g., para [0038] and para [0061]).

The method further comprises generating a respective local clock signal from the module clock signal at each of the N data buffers, and transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices. (See, e.g., para [0058], lines 3-5, and para [0064], lines 1-7). The respective group of the memory devices perform read/write operations in accordance with the respective local clock signals. (See, e.g., page 22, lines 5-12).

Independent claim 18 is directed to a memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one. (See, para [0028], lines 1-4, para [0032], lines 8-11, and para [0055]). The memory module comprises a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines. (See, para [0032]). The control circuit generates a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals. (See, e.g., page 9, lines 2-7).

The memory module of independent claim 18 further comprises N data buffers each n-bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit. (See, e.g., para [0038] and paras [0061]-[0066]). A respective data buffer of the N data buffers is configured to operatively couple to a respective set of the N sets of data lines. (See, e.g., page 6, lines 2-5). The respective data buffer generates a respective local clock signal in response to the module clock signal. (See, e.g., para [0058], lines 3-5). The respective data buffer includes data paths and logic controlling the data paths in response to the data buffer control signals. (See, e.g., para [0056]).

The memory module of independent claim 18 further comprises memory devices including N groups of memory devices corresponding to respective ones of the N data buffers. (See, e.g., para [0037]). A respective group of the N groups of memory devices receive the

module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers. (See, e.g., page 22, lines 5-12). The respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal. (See, e.g., para [0036], lines 1-4 and paras [0037] and [0044]).

VI. GROUNDS OF REJECTION TO BE REVIEWED ON APPEAL

1. The rejection of claims 2-5, 7, 9, 11-14, 16, 18-22, and 24-25 under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereafter “Matsui”) in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”);
2. The rejection of claims 6, 10, and 15 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereafter “Carpenter”); and
3. The rejection of claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and further in view of Sung et al. US 2011/0110168 (hereinafter “Sung”).
4. The rejection of claim 23 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and further in view of Chu et al. US 2008/0291758 (hereinafter “Chu”).

VII. ARGUMENTS

The rejection of claim 2 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja

In *KSR*, the Supreme Court reaffirmed the doctrine that “[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *KSR*, 550 U.S. at 416, 82 USPQ2d at 1395. The rationale to support a conclusion that the claim would have been obvious is that all the claimed elements were known in the prior art, and one skilled in the art could have combined the elements as claimed by known methods with no change in their respective functions, if the combination yielded nothing more than predictable results to one of ordinary skill in the art. *Id.*; *Sakraida v. AG Pro, Inc.*, 425 U.S.

273, 282, 189 USPQ 449, 453 (1976); *Anderson's-Black Rock, Inc. v. Pavement Salvage Co.*, 396 U.S. 57, 62-63, 163 USPQ 673, 675 (1969); *Great Atl. & P. Tea Co. v. Supermarket Equip. Corp.*, 340 U.S. 147, 152, 87 USPQ 303, 306 (1950). “[I]t can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does.” *KSR*, 550 U.S. at 418, 82 USPQ2d at 1396. If any of these findings cannot be made, then this rationale cannot be used to support a conclusion that the claim would have been obvious to one of ordinary skill in the art.

Accordingly, independent claim 2 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja. First, Claim 2 recites elements not found in either Matsui or Sutardja, so claim 2 cannot be a combination of known elements. For example, claim 2 recites a “module control device,” data buffers and “memory devices.”

Specifically, claim 2 recites “a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals *and* a module clock signal.” (*Emphasis added*) Claim 2 further recites that “the module control device [transmit] the module C/A signals to at least some of the memory devices.” Claim 2 further recites that “the module control device [transmit] the data buffer control signals and the module clock signal to the data buffers.” Claim 2 also recites that “the module control device does not transmit any clock signal to the memory devices.”

Claim 2 further recites that each of the data buffers have a data width of n-bits and generate respective local clock signal. Claim 2 further recites that the memory devices on the memory module include a plurality of groups of memory devices, with each respective group of memory devices communicating data with the memory controller via a respective one of the data buffers. Claim 2 further recites that each respective group of memory devices receive and transmit data signals in accordance with the local clock signal generated by the respective one of the data buffers.

Matsui does not teach or suggest the module control device in claim 2. Matsui appears to teach two buffers 105a and 105b on left and right sides of the memory module. The buffers 105a and 105b buffer the data, address, control and clock signals between the memory module 103 and memory controller 101. (Matsui, paragraphs [0021] and [0159], FIGS. 4-5). “[T]he buffers

105a and 105b within each memory module 103 are connected to a memory controller 101 or the memory module of the next stage via data wirings 111, command/address wirings 112, and clock wirings 113.” (Matsui, paragraph [0156], FIGS. 4-5). “In this manner, the data wirings 111, along with the command/address wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups.” (Matsui, paragraph [0158], FIGS. 4-5).

Thus, neither buffer 105a nor buffer 105b in Matsui output data buffer control signals in addition to the module C/A signals and module clock signal as recited in claim 2. Further, neither buffer 105a nor buffer 105b satisfy the recitation of claim 2 that “the module control device does not transmit any clock signal to the memory devices,” because, in Matsui, the DRAMs 110 *a* within each memory module 103 are individually connected to the buffer 105 *a* or 105 *b* via the internal data wiring (DQ) 111', and further, commonly connected to the left or right side of the buffer 105 *a* or 105 *b* via the internal command/address wiring 112' and the internal clock wiring 113'. (Matsui, paragraph [0155], FIGS. 4-5).

Therefore, the Examiner is correct by acknowledging that Matsui does not teach the module control device in claim 2. (Office Action, page 9, first full paragraph).

The Examiner then errs by stating that Sutardja teaches the module control device in claim 2. (Office Action, page 10). The Examiner cites Figure 17 and paragraph [0133] in Sutardja, wherein a memory control module 510 is described. The Examiner then states that the memory control module 510 is mapped to the claimed module control device. (Office Action, page 3, lines 9). The Examiner notices the deficiencies of Sutardja's memory control module 510 because it does not output data buffer control signals, and it does not transmit the data buffer control signals to the data buffers. The Examiner then tries to make up the deficiencies of Sutardja by stating:

Clock generator module 534 part of the memory module 500 (mapped as being part of the memory control module 510 but shown separately in Figure 17) transmits clock signals to the buffers 530, para 0136. With respect to the data buffer control signals, the CLK 113 of Matsui is considered as part of the data buffer control (see para 0035 of the instant specification which describes that the module clock signal is considered part of the buffer control signal. (Office Action, page 10, lines 12-17).

These statements do not make sense. The Examiner seems to have forgotten that claim 2 recites that the module control device output the data buffer control signals *as well as* the module C/A signals and module clock signal, and that the module control device transmit “the data buffer control signals *and* the module clock signal to the data buffers.” (Claim 2, *Emphasis added*).

The Examiner also notices further deficiencies of Sutardja’s memory control module 510 because it does not transmit the module C/A signals to at least some of the memory devices. The Examiner then attempts to remedy the deficiencies of Sutardja’s memory control module 510 by stating:

With respect to the module control device transmitting the module C/A signals to at least some of the memory devices, this feature is well-known in the art. (Office Action, page 10, 5th lines from bottom of page).

Here, the Examiner seems to have forgotten that, right before this statement in the same paragraph, the Examiner relies on that fact that Sutardja has a buffer module 530 that buffers “incoming control signals such as row access and precharge (RAS), column address strobe (CAS), etc. Thus, the module C/A signals in Sutardja are transmitted to the memory devices from the buffer module 530. Therefore, the memory control module 510 cannot possibly transmit the module C/A signals to at least some of the memory devices.

Furthermore, by mapping Sutardja’s memory control module 510 to the claimed module control device, the Examiner ignores the features in claim 2 that the module control device needs to receive command/address (C/A) signals and a system clock from the memory controller. In contrast, Sutardja’s memory control module 510, together with the clock generator module 534, *generates* the command/address (C/A) signals and a system clock, instead of *receiving* them, as recited in claim 2.

Therefore, the Examiner errs by stating that Sutardja teaches the module control device in claim 2.

Moreover, contrary to the assertion made by the Examiner on page 11 of the Office Action, a person having ordinary skill in the art (PHOSITA) would not be motivated to modify

the scope of the invention of Matsui with the memory control module 510 in Sutardja. The Examiner states that:

The motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja.

This statement is very confusing. If a PHOSITA is motivated to modify the scope of the invention of Matsui with the memory control module 510 in Sutardja, the modification should provide improvement to something in Matsui, and not to the buffer module that is in Sutardja. The Examiner seems to infer here that the buffer module 530 in Sutardja, instead the memory control module 510, is now mapped to the module control device in claim 2, and the Examiner is now trying to use the buffer module 530 in Sutardja to improve Matsui, in order to produce the memory module of claim 2.

Sutardja's buffer module 530, however, is similar to the buffer 105a or 105b in Matsui, which, as discussed above, cannot be mapped to the module control device in claim 2. Like Matsui's buffers 105a and 105b, Sutardja's buffer module 530 buffers all of the signals, data, C/A, and clock, between the memory devices and the memory control module 510, which is really the system memory controller in Sutardja. (Sutardja, paragraph [0136], Figure 17). Since the buffer 105a or 105b in Matsui performs all the functions that may be performed by Sutardja's buffer module 530, adding Sutardja's buffer module 530 to Matsui would not improve upon Matsui.

As to the Examiner's alleged motivation of using a phase-locked loop (PLL) to generate local phase-adjusted clock signals, Matsui has no need of that, because the buffers 105a and 105b are already doing what a command/address buffer and a PLL chip would be doing. Thus, adding another command/address buffer and a PLL chip to Matsui's memory module would only slow down the command/address signal paths by at least one clock cycle without any other benefits.

In fact, Matsui teaches away from using a command/address buffer and a PLL chip to distribute commands/addresses and clocks, because doing so would require that "module command/address distributing wiring and module clock distributing wiring be drawn out in the

longitudinal direction of the module board” (Matsui, paragraph [0006]). In Matsui, “the data wirings 111, along with the command/address wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups.” (Matsui, paragraph [0158], FIGS. 4-5). This is needed because a command/address signal in Matsui is transferred at a speed that is m times a command/address signal frequency of the DRAM (i.e., memory devices). (Matsui, paragraph [0020]).

Therefore, claim 2 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja.

The rejection of claim 9 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja

Independent claim 9 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja because claim 9 recites limitations not found in either Matsui or Sutardja. For example, neither Matsui nor Sutardja teaches or suggests “generating a module clock signal *on the memory module* from the system clock signal *received by the memory module*” and then “transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n bits.” (*Emphasis added*)

As discussed above, the buffers 105a and 105b in Matsui buffer the data, address, control and clock signals directly between the memory module 103 and memory controller 101. (Matsui, paragraphs [0021] and [0159], FIGS. 4-5). Thus, Matsui does not and has no need to generate a module clock signal *on the memory module* from the system clock signal *received by the memory module* and then transmit the module clock signal to the data buffers 105a and 105b. In fact, each of the buffers 105a and 105b in Matsui operates independently of each other under direct control of the memory controller 101 because “the data wirings 111, along with the command/address wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups.” (Matsui, paragraph [0158], FIGS. 4-5). This is needed because a command/address signal in Matsui is transferred at a speed that is m times a command/address signal frequency of the DRAM (i.e., memory devices). (Matsui, paragraph [0020]).

Sutardja does not teach or suggest “a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n bits,” let alone “generating a module clock signal *on the memory module* from the system clock signal *received by the memory module*” and then “transmitting the module clock signal to a plurality of data buffers.” Sutardja uses only one buffer module 530 for the full data width of its memory module and that one buffer buffers command/address, data and clock signals between the memory control module 510, which is the memory controller in Sutardja, and the memory devices. (Sutardja, paragraph [0133] and Figure 17).

Therefore, claim 9 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja.

The rejection of claim 18 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja

Independent claim 18 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja because claim 18 recites limitations not found in either Matsui or Sutardja. For example, as discussed above in the arguments with regard to claim 2, neither Matsui nor Sutardja teaches or suggests “a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals *and* data buffer control signals in response to the system clock and the input C/A signals.” (*Emphasis added*)

Furthermore, neither Matsui nor Sutardja discloses N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and *logic controlling the data paths in response to the data buffer control signals*.” As discussed above in the arguments with regard to claim 2, neither Matsui nor Sutardja teaches any data buffer control signals *in addition to* the module clock signal, module

C/A signals, let alone any logic that controls the data paths in response to the data buffer control signals.

Therefore, claim 18 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja.

The rejection of claims 2-8, 9-17, and 19-25

Claims 3-8 are improperly rejected for at least the same reasons claim 2 is improperly rejected.

Claims 10-17 are improperly rejected for at least the same reasons claim 9 is improperly rejected.

Claims 19-25 are improperly rejected for at least the same reasons claim 18 is improperly rejected.

CONCLUSION

Based on the foregoing, Appellant respectfully submits that a *prima facie* case of obviousness has not been set forward in the Office Action. Accordingly, Appellant respectfully requests that the Board reverse the Examiner's rejections of claims 2-25 pending in this application and thereby place this application in condition for allowance.

Dated this 12th day of June, 2017.

Respectfully submitted,

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APPENDIX

A. CLAIMS ON APPEAL

1. (Cancelled)

2. (Previously Presented) A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal;

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices; and

wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.

3. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals.

4. (Previously Presented) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.
5. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers includes one or more programmable registers controlling a phase of the local clock signal.
6. (Previously Presented) The memory module of claim 5,
wherein the respective one of the N data buffers is configured to program the one or more registers according to the data buffer control signals.
7. (Previously Presented) The memory module of claim 2, wherein the respective one of the N data buffers is configured to phase-lock the local clock signal with the module clock signal.
8. (Previously Presented) The memory module of claim 2, wherein memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.
9. (Previously Presented) A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:
 - receiving a system clock signal;
 - generating a module clock signal on the memory module from the system clock signal received by the memory module;
 - transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;

generating a respective local clock signal from the module clock signal at each of the N data buffers;

transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and

performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Previously Presented) The method of claim 9, further comprising incorporating a programmable amount of delay to the respective local clock signal at each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Previously Presented) The method of claim 11, further comprising controlling data paths in the N data buffers according to the data buffer control signals.

13. (Previously Presented) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.

15. (Previously Presented) The method of claim 11, further comprising:
programming one or more registers in each of the N data buffers according to the data buffer control signals.

16. (Previously Presented) The method of claim 9, further comprising phase-locking the local clock signal with the module clock signal.

17. (Previously Presented) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit wide, and wherein each group of memory devices include one memory device from each of the one or more ranks.

18. (Previously Presented) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit to generate a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data with the memory controller via the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Previously Presented) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective

ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors.

20. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

21. (Previously Presented) The memory module of claim 18, wherein the control circuit does not transmit any clock signals to the memory devices.

22. (Previously Presented) The memory module of claim 18, wherein the data buffer control signals are different from the module C/A signals.

23. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data being communicated via the respective data buffer.

24. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is n -bits wide.

25. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is $n/2$ -bits wide.

B. EVIDENCE APPENDIX

None.

C. RELATED PROCEEDING APPENDIX

None.

Electronic Acknowledgement Receipt

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Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
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Customer Number:	79141
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APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR / PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
14/445,035	28 July, 2014	LEE ET AL.	N3044.10018US02

Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098		EXAMINER	
		ZUBAIR AHMED	
		ART UNIT	PAPER
		2132	20170802

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IDS filed on 04/26/2017 has been considered by the Examiner and has been entered.

/DAVID X YI/
Supervisory Patent Examiner, Art Unit 2132

/ZUBAIR AHMED/
Examiner, Art Unit 2132

Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

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	Filing Date	2014-07-28
	First Named Inventor	Hyun Lee
	Art Unit	2132
	Examiner Name	AHMED, ZUBAIR
	Attorney Docket Number	N3044.10018US02

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14445035
Filing Date	2014-07-28
First Named Inventor	Hyun Lee
Art Unit	2132
Examiner Name	AHMED, ZUBAIR
Attorney Docket Number	N3044.10018US02

1	Extended European Search Report dated April 3, 2017 in European Patent Application No. 14831654.0.
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EXAMINER SIGNATURE

Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/02/2017
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
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Application Number	14445035		
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Art Unit	2132		
Examiner Name	AHMED, ZUBAIR		
Attorney Docket Number	N3044.10018US02		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

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A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/JAMIE J. ZHENG/	Date (YYYY-MM-DD)	2017-04-26
Name/Print	Jamie J. Zheng	Registration Number	51167

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	N3044.10018US02	9700
79141	7590	08/11/2017	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
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BEFORE THE PATENT TRIAL AND APPEAL BOARD

Application Number: 14/445,035
Filing Date: July 28, 2014
Appellant(s): LEE ET AL.

Jamie J. Zheng
For Appellant

EXAMINER'S ANSWER

This is in response to the appeal brief filed 06/13/2017 appealing from the Office action mailed 01/09/2017.

Examiner's Answer

This is in response to the appellant's brief on appeal filed 06/13/2017.

(1) Real Party in Interest

A statement identifying by name the real party in interest is contained in the brief.

(2) Related Appeals and Interferences

The examiner is not aware of any related appeals, interferences, or judicial proceedings which will directly affect or be directly affected by or have a bearing on the Board's decision in the pending appeal.

(3) Status of Claims

The status of the claims outlined in the brief is correct.

(4) Status of Amendments

The Appellant has not submitted any amendments subsequent to the Final Office Action.

(5) Summary of Claimed Subject Matter

The summary of claimed subject matter contained in the brief is correct.

(6) Grounds of Rejection to be Reviewed on Appeal

The appellant's statement of the grounds of rejection to be reviewed on appeal is correct.

(7) Response to Argument

Appellant's remarks **VII.**, regarding patentability of claim **2** pages 7-12 of the appeal brief have been considered and are deemed not persuasive for the following reasons.

(A) Appellant argues "Accordingly, independent claim 2 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja. First, Claim 2 recites elements not found in either Matsui or Sutardja, so claim 2 cannot be a combination of known elements" **(Recited from page 8 of the Appeal Brief)**.

In response to **(A)**, the Examiner respectfully submits that elements recited in claim 2 are commonly known in the relevant art and these elements may be

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found alone or in combination in Matsui and Sutardja. The Examiner further submits that a person having ordinary skill would be motivated to combine the teachings of Matsui and Sutardja to arrive at the claimed invention as proposed by the Examiner in the Final Office Action (01/09/2017, pages 2-4, 10-11). The Examiner will refrain from rebutting each and every argument of the Appellant on pages 8-10 of the brief as these arguments are directed to the teachings of Matsui and Sutardja alone and not in a motivated combination. Instead, Appellant's arguments directed to the combination of Matsui and Sutardja are rebutted below.

(B) Appellant argues, "Moreover, contrary to the assertion made by the Examiner on page 11 of the Office Action, a person having ordinary skill in the art (PHOSITA) would not be motivated to modify the scope of the invention of Matsui with the memory control module 510 in Sutardja. The Examiner states that:

The motivation would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals, para 0136 of Sutardja.

This statement is very confusing. If a PHOSITA is motivated to modify the scope of the invention of Matsui with the memory control module 510 in Sutardja, the modification should provide improvement to something in Matsui, and not to the buffer module that is in Sutardja. The Examiner seems to infer here that the buffer module 530 in Sutardja, instead the memory control module 510, is now mapped to the module control device in claim 2, and the Examiner is now trying

to use the buffer module 530 in Sutardja to improve Matsui, in order to produce the memory module of claim 2.” **(Recited from pages 10-11 of the Appeal Brief).**

In response to **(B)**, the Examiner first briefly outlines the proposed combination of Matsui and Sutardja as was outlined in the Final Office Action of 01/09/2017 (pages 2-4 and 10-11). The instant invention claims a “module control circuit” which allegedly is not found in Matsui and Sutardja. The module control circuit is shown in FIG. 1 of the drawings (reproduced below).

FIG. 1 of the claimed invention

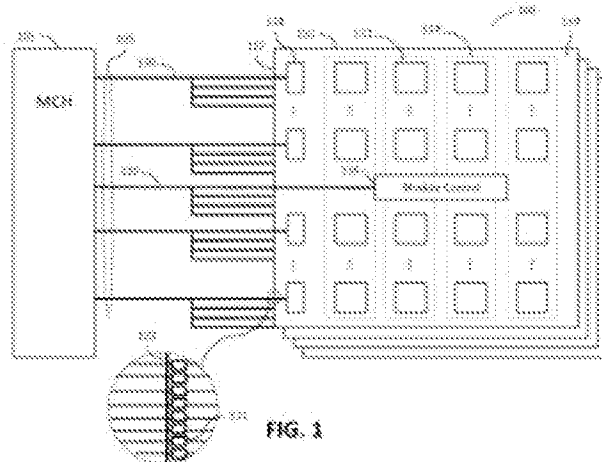


FIG. 1 of the drawing depicts a system 100 with host memory controller (MCH) 101, one or more memory modules 110 coupled to the MCH 101, and each memory module 110 includes a “module control circuit” 116 and a plurality of buffer circuits or data buffers 118, para 0028 of the instant specification.

Matsui discloses a memory system with a plurality of memory modules 103a to 103d (FIG. 5 of Matsui, reproduced below). Each memory module is provided with a plurality of buffers (two buffers 105a and 105b in FIG. 5 of Matsui). Further, the buffers 105a and 105b within each memory module 103 are connected to a memory controller 101, paras 0154-0156 of Matsui.

FIG. 5 of Matsui

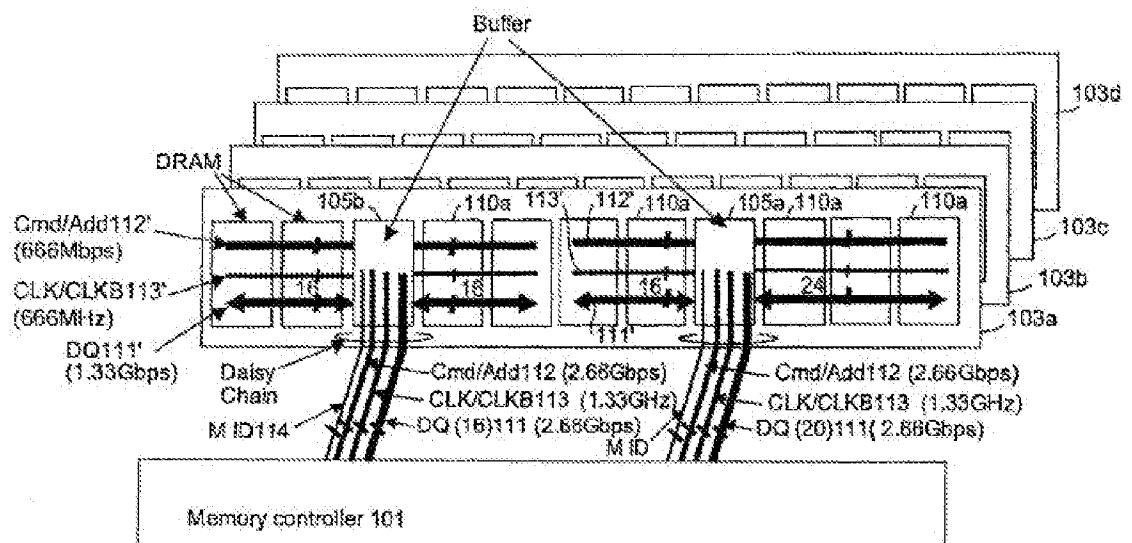


FIG. 5

It was admitted that Matsui alone does not teach the claimed invention because Matsui has no teaching of a “module control circuit”. However, Sutardja, an analogous art in the same field of endeavor, teaches a module control circuit

(see memory control module 510 in FIG. 17 of Sutardja, reproduced below). It was proposed in the Final Office Action (01/09/2017, pages 2-4 and 10-11) that the memory control module 510 together with the clock generation module 534 of Sutardja may be inserted between the memory controller 101 and the memory modules 103a-d interfacing with the buffers 105a and 105b [of Matsui]. This modification, by a person of ordinary skill in the art, would neither change the mode of operation of Matsui's method and apparatus nor would it render Matsui's method and apparatus inoperable for the following reasons.

FIG. 17 of Sutardja

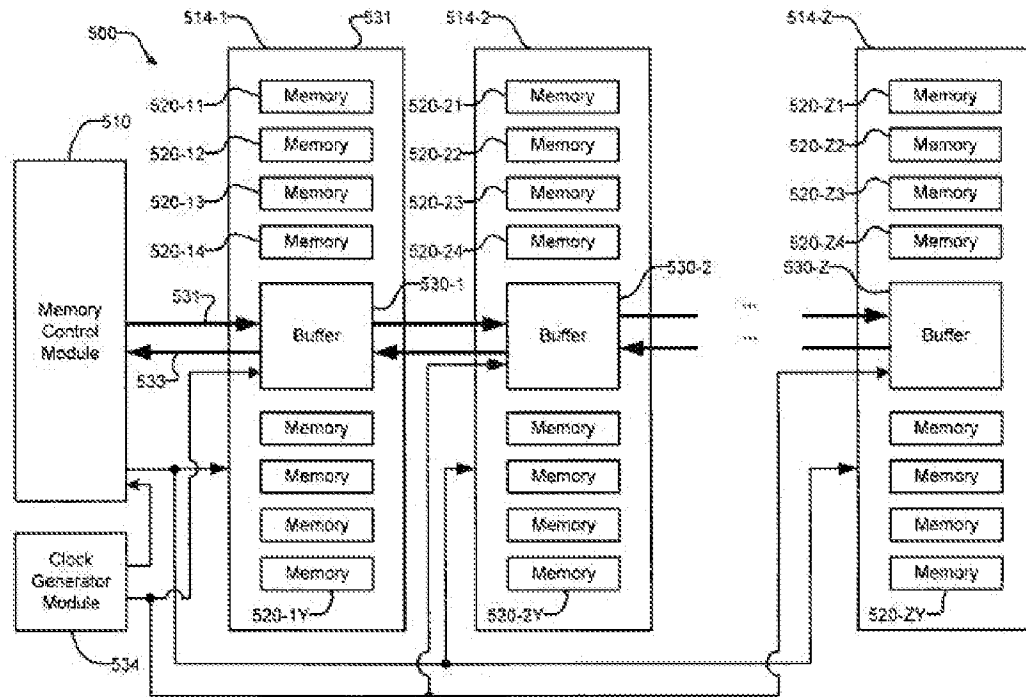


FIG. 17

It is respectfully submitted that Matsui already discloses the use of two buffers (buffers 105a and 105b, Fig. 5) in each memory module 103a-d. Each of the two buffers buffer DQ111, CLK113, and CMD/ADD112 (see Fig. 5 of Matsui). With the proposed modification of Matsui's method and apparatus, the memory control module 510 and the clock generation module 534 of Sutardja situated between memory controller 101 and memory modules 103a-d (Fig. 5 of Matsui) would only receive from the memory controller 101 the CMD/ADD112 and CLK113 signals. The DQ111 signals may be directly transmitted from the memory controller 101 to the buffers 105a and 105b consistent with the claimed invention and as disclosed by Matsui's FIG. 5. The CMD/ADD112 signals would be transmitted from the memory control module 510 (which is already combined with the clock generation module 534) to the memory devices and this feature is well-known in the art (a one-stage buffering of command/address signals are emphasized, see para 0010 of Saito et al. US 2010/0309706, paras 0039, 0048-0049 of Harashima US 2013/0063998, or paras 0007-0009 of Lee et al. US 2013/0242680). A one-stage buffering of the command/address signals is common in the art. It would be uncommon and detrimental to the performance of a memory system if one were to use two or more stages of buffering of the command/address signals. Thus in the proposed combination of Matsui and Sutardja, it would be standard if the command/address signals are transmitted from the memory controller 101 to the memory control module 510 and from the memory control module 510 to the DRAM chips. It would be unheard of if these

signals were transmitted from the memory control module 510 to the buffers 105a and 105b and then transmitted from these buffers to the DRAM chips because such an arrangement would incur an extra clock cycle.

The CLK113 signal (clock signal from memory controller 101 and mapped to the claimed system clock) would be received by and transmitted from the memory control module 510 (which is already combined with the clock generation module 534) to the buffers 105a and 105b where it would be locally synchronized (with phase-locked loops disclosed by Sutardja) and distributed to the memory devices. It is noted that when CLK113 is output from the memory control module 510 and clock generation module 534, it is mapped to the claimed module clock signal. With respect to the “data buffer control signals” of the claimed invention, the CLK113 output from the memory controller 101 of Matsui and input to the memory control module 510 and clock generation module 534 generates “global clocks” for transmission to buffers 105a and 105b. Global clocks is considered a group of clock/control signals (see FIG. 44 of Matsui, “GLOBAL CLOCKS” supplied to buffer 105) and the control signals in this group of “global clocks” are mapped to the claimed data buffer control signals (see para 0035 of the instant specification).

Continuing with the response to **(B)**, the above few paragraphs describe that one or more elements of Sutardja may be used to modify the method and apparatus of Matsui to arrive at the claimed invention. And this modification would be a

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motivated modification by a person having ordinary skill to improve upon the method and apparatus of Matsui as will be described shortly. But for the record it needs to be stated for clarity that the Appellant appears to be confused between the memory control module 510 and buffer module 530 of Sutardja and using these two components interchangeably without recognizing the potential implications of interchanging these modules. The Examiner believes that the modification to Matsui's invention was stated clearly in unambiguous terms.

(C) Appellant argues, "Sutardja's buffer module 530, however, is similar to the buffer 105a or 105b in Matsui, which, as discussed above, cannot be mapped to the module control device in claim 2. Like Matsui's buffers 105a and 105b, Sutardja's buffer module 530 buffers all of the signals, data, C/A, and clock, between the memory devices and the memory control module 510, which is really the system memory controller in Sutardja. (Sutardja, paragraph [0136], Figure 17). Since the buffer 105a or 105b in Matsui performs all the functions that may be performed by Sutardja's buffer module 530, adding Sutardja's buffer module 530 to Matsui would not improve upon Matsui." **(Recited from page 11 of the Appeal Brief).**

In response to **(C)**, the Examiner submits that although Sutardja's buffer module 530 is similar to the buffers 105a and 105b, there are notable differences between the two. In particular, Sutardja advocates the use of a PLL (phase-locked loop) in the buffer module 530 for local clock synchronization and clock

Art Unit: 2132

frequency multiplication, see para 0136 of Sutardja, whereas Matsui uses a DLL (delay-locked loop) in buffers 105 for the same purpose, see FIG. 24 of Matsui DLL 305 used for clock reproduction and frequency multiplication. When used for clock frequency multiplication, a PLL offers advantages over a DLL as widely known in the art. For example, “DLLs cannot easily generate different output frequencies, unlike a PLL where we can just change the divide ratio”. The problem of frequency multiplication with DLL is further exacerbated because “DLLs have the potential to delay lock to undesired multiples of the reference cycles, necessitating additional lock detect circuitry with a wide delay range delay line.” (These notable differences between a PLL and a DLL are widely available in the literature. For example, see page 6 of “Network Theory: Broadband Circuit Design, Lecture 15: Delay-Locked Loops (DLLs)”, Sam Palermo, Analog & Mixed-Signal Center, Texas A&M University).

Accordingly, the Examiner strongly disagrees with the assertion of the Appellant that “adding Sutardja’s buffer module 530 to Matsui would not improve upon Matsui” as stated in (C). On the contrary, the Examiner firmly believes that a person having ordinary skill would recognize that the PLL in Sutardja’s buffer module 530 incorporated in Matsui’s buffer 105 in place of the DLL would significantly improve upon Matsui because it would provide stability, ease of design and test, and predictability of Matsui’s buffer modules (105a and 105b).

(D) Appellant argues, "In fact, Matsui teaches away from using a command/address buffer and a PLL chip to distribute commands/addresses and clocks, because doing so would require that "module command/address distributing wiring and module clock distributing wiring be drawn out in the longitudinal direction of the module board" (Matsui, paragraph [0006]). In Matsui, "the data wirings 111, along with the command/address wirings 112 and the clock wirings 113, are connected to the two buffers 105a and 105b concentrically, i.e., in groups." (Matsui, paragraph [0158], FIGS. 4-5). This is needed because a command/address signal in Matsui is transferred at a speed that is m times a command/address signal frequency of the DRAM (i.e., memory devices). (Matsui, paragraph [0020])." **(Recited from pages 11-12 of the Appeal Brief).**

In response to (D), the Examiner respectfully submits that the Appellant is not distinguishing between the memory control module 510 from the buffer module 530 of Sutardja and where they are situated in the modified apparatus of Matsui depicted in FIG. 5. Matsui already uses a DLL (DLL 305 in buffer 105, FIG. 24) for clock reproduction and clock frequency multiplication. The proposed modification replaces this DLL 305 with a PLL of buffer module 530 of Sutardja. Accordingly, the proposed modification does not change the clock wire distribution, DQ wire distribution, C/A wire distribution for each of the memory modules as described in FIG. 5 of Matsui.

Appellant's remarks **VII.**, regarding patentability of claim **9** pages 12-13 of the appeal brief have been considered and are deemed not persuasive for the following reasons.

(A) Appellant argues, "Independent claim 9 is improperly rejected under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja because claim 9 recites limitations not found in either Matsui or Sutardja. For example, neither Matsui nor Sutardja teaches or suggests "generating a module clock signal *on the memory module* from the system clock signal *received by the memory module*" and then "transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including *N* data buffers, each of the *N* data buffers having a data width of *n* bits." *{Emphasis added}*" **(Recited from page 12 of the Appeal Brief).**

In response to **(A)**, the Examiner submits that Matsui's method and apparatus modified by elements of Sutardja discloses the subject matter claimed in independent claim 9. With the proposed modification of Matsui's method and apparatus, the memory control module 510 and the clock generation module 534 of Sutardja situated between memory controller 101 and memory modules 103a-d (Fig. 5 of Matsui) would only receive from the memory controller 101 the CMD/ADD112 and CLK113 signals. The DQ111 signals may be directly transmitted from the memory controller 101 to the buffers 105a and 105b

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consistent with the claimed invention and as disclosed by Matsui's FIG. 5. The CMD/ADD112 signals would be transmitted from the memory control module 510 (which is already combined with the clock generation module 534) to the memory devices and this feature is well-known in the art and this feature of one-stage buffering of command/address has already been discussed in reference to independent claim 2 above. And the CLK113 signal (clock signal from memory controller 101 and mapped to the claimed system clock) would be received by and transmitted from the memory control module 510 (which is already combined with the clock generation module 534) to the buffers 105a and 105b where it would be locally synchronized (with phase-locked loops disclosed by Sutardja) and distributed to the memory devices. It is noted that when CLK113 is output from the memory control module 510 and clock generation module 534 it is mapped to the claimed module clock signal.

Appellant's remarks **VII.**, regarding patentability of claim **18** pages 13-14 of the appeal brief have been considered and are deemed not persuasive for the following reasons.

(A) Appellant argues, "Furthermore, neither Matsui nor Sutardja discloses N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective

local clock signal in response to the module clock signal, the respective data buffer including data paths and *logic controlling the data paths in response to the data buffer control signals.*" As discussed above in the arguments with regard to claim 2, neither Matsui nor Sutardja teaches any data buffer control signals *in addition to* the module clock signal, module C/A signals, let alone any logic that controls the data paths in response to the data buffer control signals." **(Recited from pages 12-13 of the Appeal Brief).**

In response to **(A)**, the Examiner submits that Matsui's method and apparatus modified by elements of Sutardja discloses the subject matter claimed in independent claim 18. With the proposed modification of Matsui's method and apparatus, the memory control module 510 and the clock generation module 534 of Sutardja situated between memory controller 101 and memory modules 103a-d (Fig. 5 of Matsui) would only receive from the memory controller 101 the CMD/ADD112 and CLK113 signals. The DQ111 signals may be directly transmitted from the memory controller 101 to the buffers 105a and 105b consistent with the claimed invention and as disclosed by Matsui's FIG. 5. The CMD/ADD112 signals would be transmitted from the memory control module 510 (which is already combined with the clock generation module 534) to the memory devices and this feature is well-known in the art as discussed above in reference to independent claim 2. And the CLK113 signal (clock signal from memory controller 101 and mapped to the claimed system clock) would be received by and transmitted from the memory control module 510 (which is already combined

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with the clock generation module 534) to the buffers 105a and 105b where it would be locally synchronized (with phase-locked loops disclosed by Sutardja) and distributed to the memory devices. It is noted that when CLK113 is output from the memory control module 510 and clock generation module 534 it is mapped to the claimed module clock signal. With respect to the “data buffer control signals” of the claimed invention, the CLK113 output from the memory controller 101 of Matsui and input to the memory control module 510 and clock generation module 534 generates “global clocks” for transmission to buffers 105a and 105b. Global clocks is considered a group of clock/control signals (see FIG. 44 of Matsui, “GLOBAL CLOCKS” supplied to buffer 105) and the control signals from this group of “global clocks” are mapped to the claimed data buffer control signals (see para 0035 of the instant specification). The Examiner strongly disagrees with Appellant’s contention that neither Matsui nor Sutardja discloses *“logic controlling the data paths in response to the data buffer control signals”*. In fact FIG. 44 of Matsui shows generation of a WDPS (write data phase signal) signal controlling the write operation to DRAMs based on the GLOBAL CLOCKS (control signals of GLOBAL CLOCKS are mapped to the claimed data buffer control signals).

Appellant’s remarks **VII.**, regarding patentability of claims **3-8, 10-17, and 19-25** page 14 of the appeal brief have been considered and are deemed not persuasive for the following reasons.

Art Unit: 2132

The Examiner respectfully submits that claims 3-8, 10-17, and 19-25 are properly rejected and their respective independent claims 2, 9, and 18 are also properly rejected as described above.

For the above reasons, it is believed that the rejections should be sustained.

Respectfully submitted,

/ZUBAIR AHMED/

Examiner, Art Unit 2132

Conferees:

/KEVIN ELLIS/

Primary Examiner

/DAVID X YI/

Supervisory Patent Examiner, Art Unit 2132

Requirement to pay appeal forwarding fee. In order to avoid dismissal of the instant appeal in any application or ex parte reexamination proceeding, 37 CFR 41.45 requires payment of an appeal forwarding fee within the time permitted by 37 CFR 41.45(a), unless appellant had timely paid the fee for filing a brief required by 37 CFR 41.20(b) in effect on March 18, 2013.

ELECTRONICALLY FILED

PATENT APPLICATION
Docket No. N3044.10018US02

**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE BEFORE
THE BOARD OF PATENT APPEALS AND INTERFERENCES**

In re application of	Hyun Lee	
Application No.:	14/445,035	Art Unit 2132
Filed:	July 28, 2014	
Conf. No.:	9700	
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION	
Examiner:	AHMED, ZUBAIR	
Customer No.:	79141	
Attorney Docket No.:	N3044.10018US02	

The Honorable Commissioner of Patents and Trademarks
Washington, D.C. 20231

REPLY BRIEF UNDER 37 C.F.R. § 41.50 (a)(2)(ii)

This Reply Brief is being filed in response to the Examiner's Answer dated August 11, 2017. This Reply Brief is being filed in accordance with the requirements of 37 C.F.R. § 41.50 (a)(2)(ii) to maintain the appeal of the rejection of the above-identified patent application. Commissioner is hereby authorized to charge any required fees associated with this communication, or to credit any overpayment, to Deposit Account No. 50-5394.

STATUS OF CLAIMS

Claims 2-25 are pending in this application. Claims 2-25 were rejected in the Final Office Action mailed January 9, 2017 (the “Final Office Action”). Claims 2-25 are the subject of this appeal. A complete copy of claims 2-25 is provided in Appendix A of the Appeal Brief filed on June 12, 2017.

GROUND OF REJECTION TO BE REVIEWED ON APPEAL

The first ground of rejection to be reviewed on appeal is the rejection of claims 2-5, 7, 9, 11-14, 16, and 18-21 under 35 U.S.C. 103 as being unpatentable over Matsui US 2004/0105292 (hereafter “Matsui”) in view of Sutardja et al. US 2007/0168781 (hereafter “Sutardja”);

The second ground of rejection to be reviewed on appeal is the rejection of claims 6, 10, and 15 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereafter “Carpenter”); and

The third ground of rejection to be reviewed on appeal is the rejection of claims 8 and 17 under 35 U.S.C. 103 as being unpatentable over Matsui in view of Sutardja and further in view of Sung et al. US 2011/0110168 (hereinafter “Sung”).

ARGUMENTS

The Examiner’s Answer admits on page 7, third line from bottom: “It was admitted that Matsui does not teach the claimed invention because Matsui has no teaching of a ‘module control circuit.’” The Examiner then states on page 8, line 1: “It was proposed in the Final Office Action (01/09/2017, pages 2-4 and 10-11) that the memory control module 510 together with the clock generation module 534 of Sutardja may be inserted between the memory controller 101 and the memory modules 103a-d interfacing with the buffers 105a and 105b [of Matsui].”

The combination suggested by the Examiner can be illustrated using Matsui’s FIG. 5, as shown below, where the changes/modifications are illustrated in red and include eliminating one set of Cmd/Add112 (2.66Gps) and CLK/CLKB113 (1.33GHz) signals output from the memory controller 101 and rerouting the other set of Cmd/Add 112 (2.66Gps) and CLK/CLKB113 (1.33GHz) signals to Sutardja’s memory controller/clock generator combo 510+534, in order to meet the claim limitations of “a module control device to receive command/address (C/A) signals and a system clock from the memory controller.”

The changes/modifications further include adding command/address signal lines to provide the command/address signals Cmd/Add112’ (666Mbps) from Sutardja’s memory

controller/clock generator combo 510+534 to the DRAMS, and adding clock signal lines to transmit the clock signals CLK/CLKB113 (1.33GHz) from Sutardja's memory controller/clock generator combo 510+534 to Matsui's buffers 105a, 105b, in order to meet the claim limitations of "the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting ... the module clock signal to the data buffers." The Cmd/Add112 (2.66Gps) inputs to Matsui's buffers 105a, 105b and the Cmd/Add112' (666Mbps) outputs from Matsui's buffers 105a, 105b would then be eliminated.

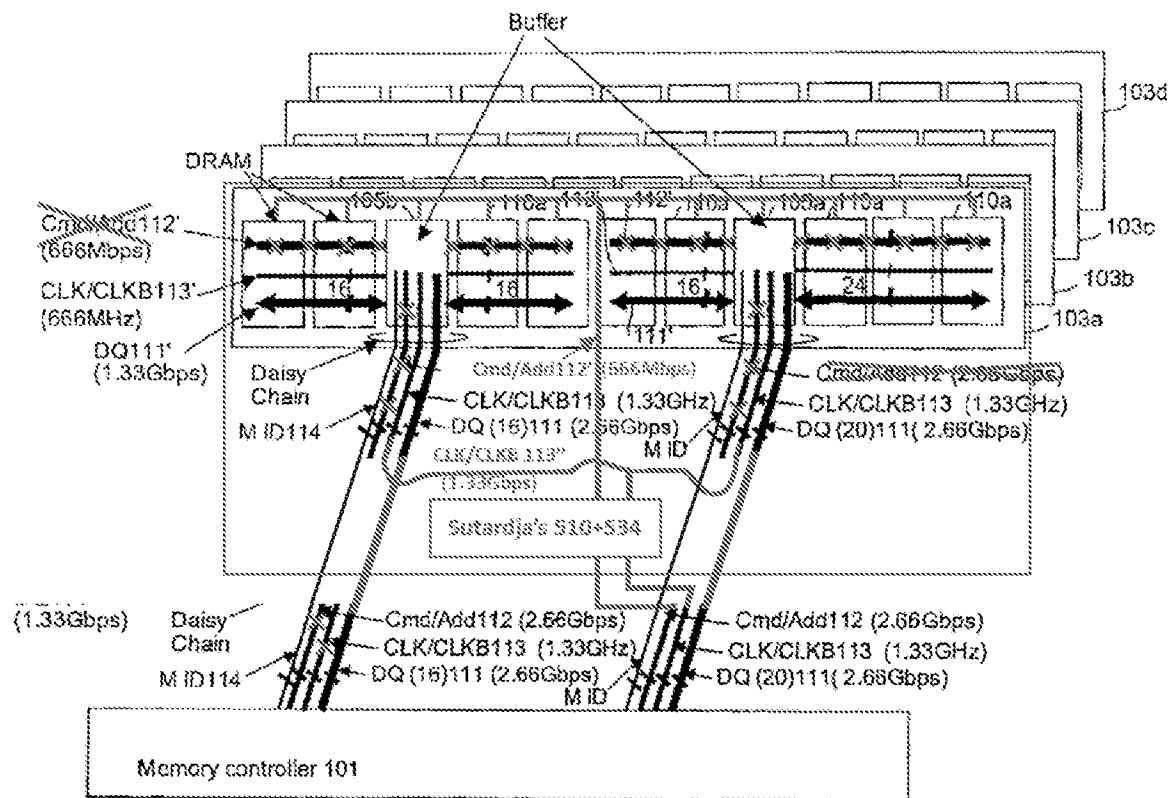


FIG. 5 (Modified)

These suggested changes/modifications to Matsui cannot be obvious to a person having ordinary skill in the art (PHOSITA) because, first of all, they would not make sense to the PHOSITA. Matsui requires that "the command/address wiring 112 is connected point-to-point between the memory controller 101 and the memory module 103 and between the adjacent memory modules 103" (Matsui, paragraphs [0020], [0147], [0156]). Matsui also requires that *"the buffer 105 also has a function of transferring or transmitting a command/address signal*

to the cascade-connected memory module 103 of the next stage” (Matsui, paragraphs [0021], [0149], *emphasis added*). This required function for the buffer 105 in Matsui would be eliminated if the Cmd/Add112 (2.66Gps) inputs to Matsui’s buffers 105a, 105b are rerouted to Sutardja’s memory controller/clock generator combo 510+534, which are “inserted between the memory controller 101 and the memory modules 103a-d interfacing with the buffers 105a and 105b,” as suggested by the Examiner. Therefore, the combination suggested by the Examiner is inconsistent with the teachings of Matsui.

Furthermore, the changes/modifications suggested by the Examiner require Sutardja’s memory control module 510 to have functions not taught or disclosed in Sutardja. First of all, Sutardja’s memory control module 510 is a memory controller. It generates command/address signals in response to commands from an operating system of a computer, instead of receiving them from another memory controller. Likewise, Sutardja’s clock generator 534 generates clock signal, instead of receiving them from a memory controller.

Moreover, the command/address signals Cmd/Add112 (2.66Gps) output from Matsui’s memory controller have the speed of 2.66 Gbps, much higher than what can be received by its DRAMs. (See Matsui, paragraph [0151]. “Therefore, it is seen that each of the buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data *by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals.*” (See Matsui, paragraph [0159], *emphasis added.*) Sutardja’s 510+534 combo does not provide such functions. For example, Sutardja’s memory control module 510 does not down-convert the Cmd/Add112 (2.66Gps) signals from the memory controller 101 to the Cmd/Add112’ (666Mbps) signals needed by the Matsui’s DRAMs, as required by Matsui’s memory module. Therefore, Sutardja’s memory control module 510 cannot be used “to receive command/address (C/A) signals” and to transmit “the module C/A signals to at least some of the memory devices” in Matsui’s memory module.

Moreover, since Matsui’s buffers 105a, 105b down-convert the clock signal CLK/CLKB113 (1.33GHz) they receive to the clock signal CLK/CLKB113’ (666MHz) needed by Matsui’s DRAMs, the suggested Sutardja’s memory controller/clock generator combo

510+534 would need to keep the clock signal at the same speed but down-convert the command/address signal, a function not disclosed in neither Sutardja nor Matsui.

Thus, in rejecting the claims under 35 U.S.C. 103 over Matsui in view of Sutardja, the Examiner is creating functions for Sutardja's memory control module 510 and clock generator 534 that are not taught or disclosed in Sutardja. The rejection is therefore improperly made and based mostly on hindsight and.

Not only are the suggested modifications based on hindsight, they also do not improve upon Matsui. Matsui already uses the buffers 105a and 105b to perform the functions the Examiner is trying to put in Sutardja's memory controller/clock generator combo 510+534. Also, Matsui in FIG. 4 shows that it already has a clock generator 102 coupled to its memory controller 101, and that the memory controller 101 provides the clock signal CLK/CLKB113 (1.33GHz) to the memory module. So, there is no need to insert another clock generator 534 between the memory controller and the buffers in Matsui. Matsui at paragraph [0159] teaches that "each of the buffers 105a and 105b can produce internal clocks, internal command/address signals, and internal data by converting the clocks, the command/address signal, and the data from the memory controller 101 into two or four parallel signals." So, there is no need for another PLL to phase-lock the clock on the module.

Instead of improving Matsui, the suggested combination would actually create the problems Matsui is trying to solve, because it requires Sutardja's memory controller 510 to transmit the command/address signals to the memory devices 110 on Matsui's memory module. Paragraph [0153] in Matsui teaches that "inasmuch as the clocks and the command/address signals are given to all the DRAMs 110 on each memory module 103, the total input load is large, and therefore, it can be considered that a problem may be raised upon high-frequency operation." The embodiment in FIG. 5 of Matsui, which the Examiner uses to reject the claims, is presented to overcome this problem by providing two buffers 105a and 105b, such that "each of the buffers 105a and 105b of each of the memory modules 103a and 103b is connected to a plurality of DRAMs 110a *arranged on both left and right sides thereof via internal data wiring (DQ) 111', internal command/address wiring 112', and internal clock wiring 113'.*" (See Matsui, paragraph [0154], *emphasis added.*) The two buffers are *coupled point-to-point to the*

memory controller via the command/address wiring 111, so that the two buffers are used to reduce input load at the internal command/address wiring 112' and internal clock wiring 113'. (See Matsui paragraph [0163]). By eliminating the internal command/address wiring 112' so as to move the function of buffering the command/address signals from Matsui's two buffers to Sutardja's memory controller 510, the suggested modifications would undo the improvements Matsui has made by using two buffers coupled point-to-point to the memory controller via the command/address wiring 112 and coupled to a plurality of DRAMs 110a *arranged on both left and right sides thereof via internal command/address wiring 112'*.

Thus, in rejecting the claims under 35 U.S.C. 103 over Matsui in view of Sutardja, the Examiner tore up the system in Matsui, inserted an unwanted memory controller and clock generator, and produced a strange combination that would require a lot of engineering work to patch it up and still would not work and/or would defeat the purpose of Matsui. In the end, the suggested combination does not even teach every feature in the claim because it never discloses the module control signals. Contrary to the Examiner's assertion, the module control signals are not clock signals. They are in addition to the clock signal, as recited in the claims of the present application.

Based on the foregoing, Appellant respectfully submits that a *prima facie* case of obviousness has not been set forward in the Office Action. Accordingly, Appellant respectfully requests that the Board reverse the Examiner's rejections of claims 2-25 pending in this application and thereby place this application in condition for allowance.

Dated this 10th day of October, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

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Attorney for Applicant
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Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10018US02			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
APPEAL FORWARDING FEE	2413	1	1000	1000
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	30617531
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10018US02
Receipt Date:	10-OCT-2017
Filing Date:	28-JUL-2014
Time Stamp:	18:43:41
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	101117INTEFSW18451800
Deposit Account	505394
Authorized User	Brian Gonzalez

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37 CFR 1.21 (Miscellaneous fees and charges)

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Reply Brief Filed	2017-10-10_Reply-Brief_N3044-10018US02.pdf	766266	no	7
			ba06aec86c16982182f2e97383030ea65757f292		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30326	no	2
			c94db089a7d247949de82c09f68d62370c1c5cfd		

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	N3044.10018US02	9700
79141	7590	10/16/2017	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			10/16/2017	ELECTRONIC

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Appeal No: 2018-000326
Application: 14/445,035
Appellant: Hyun Lee et al.

Patent Trial and Appeal Board Docketing Notice

Application 14/445,035 was received from the Technology Center at the Board on October 12, 2017 and has been assigned Appeal No: 2018-000326.

In all future communications regarding this appeal, please include both the application number and the appeal number.

The mailing address for the Board is:

PATENT TRIAL and APPEAL BOARD
UNITED STATES PATENT AND TRADEMARK OFFICE
P.O. BOX 1450
ALEXANDRIA, VIRGINIA 22313-1450

Telephone inquiries can be made by calling 571-272-9797 and referencing the appeal number listed above.

By order of the Patent Trial and Appeal Board.

EMS



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

79141 7590 09/27/2018
Morgan Lewis Bockius LLP / Jamie Zheng Ph.D.
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER

AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
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2132

NOTIFICATION DATE	DELIVERY MODE
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09/27/2018

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

Ex parte HYUN LEE and JAYESH R. BHAKTA

Appeal 2018-000326
Application 14/445,035
Technology Center 2100

Before JOSEPH L. DIXON, JOYCE CRAIG, and SCOTT E. BAIN,
Administrative Patent Judges.

BAIN, *Administrative Patent Judge.*

DECISION ON APPEAL

Appellants¹ appeal under 35 U.S.C. § 134(a) from the Examiner's Final Rejection of claims 2–25, which constitute all claims pending in the application. Claim 1 has been canceled. We have jurisdiction under 35 U.S.C. § 6(b).

We reverse.

¹ Appellants identify Netlist, Inc. as the real party in interest. App. Br. 4.

STATEMENT OF THE CASE

The Claimed Invention

Appellants' invention relates to a memory module including a "module control device" and distributed data buffers. Abstract; Spec. ¶¶ 2–3. The data buffers "regenerate a clock signal received from the module control device and [] provide regenerated clock signals to respective groups of memory devices." Abstract. This helps ensure "proper timing" of control and/or data signals in a memory circuit, particularly as operating speed and memory density increase. Spec. ¶ 8.

Claims 2, 9, and 18 are independent. Claim 2 is illustrative of the invention and the subject matter of the appeal, and reads as follows:

2. A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits and each generating a respective local clock signal;

memory devices including a plurality of groups of memory devices, each respective group of memory devices communicating data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal generated by the respective one of the data buffers; and

a module control device to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices; and

wherein each of the N data buffers generates the respective local clock signal from the module clock signal received from the module control device.

App. Br. 15 (emphases added).

The Rejections on Appeal

Claims 2–5, 7, 9, 11–14, 16, 18–22, 24, and 25 stand rejected under 35 U.S.C. § 103(a) as unpatentable over Matsui (US 2004/0105292 A1; June 3, 2004) and Sutardja et al. (US 2007/0168781 A1; July 19, 2007) (“Sutardja”). Final Act. 7–18.

Claims 6, 10, and 15 stand rejected under 35 U.S.C. § 103(a) as unpatentable over Matsui, Sutardja, and Carpenter et al. (US 2009/0116312 A1; May 7, 2009) (“Carpenter”). Final Act. 18–20.

Claims 8 and 17 stand rejected under 35 U.S.C. § 103(a) as unpatentable over Matsui, Sutardja, and Sung et al. (US 2011/0110168 A1; May 12, 2011) (“Sung”). Final Act. 20–21.

Claim 23 stands rejected under 35 U.S.C. § 103(a) as unpatentable over Matsui, Sutardja, and Chu et al. (US 2008/0291758 A1; Nov. 27, 2008) (“Chu”). Final Act. 21–22.

ANALYSIS

We have reviewed the Examiner’s rejections in light of the arguments raised in the briefs, on the record before us. For the reasons set forth below, we do not sustain the Examiner’s rejections.

Appellants argue the Examiner erred in finding Matsui and Sutardja teach or suggest a memory control module that “output[s] the data buffer

control signals as well as the module C/A signals and module clock signal,” as recited in independent claim 2.² App. Br. 9–10; Reply Br. 4–5.

Appellants contend the Examiner acknowledges Matsui “does not teach the module control device,” and that combining Matsui with Sutardja does not cure the deficiencies of Matsui because the Examiner still does not identify how the combination would “possibly transmit the [required] C/A signals to at least some of the memory devices” recited in claim 2. App. Br. 10.

Appellants also argue that a person of ordinary skill in the art would have no rationale or motivation for combining Matsui with Sutardja, that doing so would render Matsui inoperable for its intended purpose, and that Matsui teaches away from the proposed combination. App. Br. 10–12; Reply Br. 4–6. Appellants assert that the Examiner must have relied on hindsight in combining the references. App. Br. 11–12. Because we are persuaded, on this record, that the Examiner has not articulated a sufficient rationale for combining the references, we are persuaded of error.

In rejecting claim 2, the Examiner concedes that the “memory system” described in Matsui does not include a “module control device” which “receiv[es] command/address (C/A) signals and a system clock from the memory controller” and “output[s] module C/A signals, data buffer control signals and a module clock signal” in the manner recited in claim 2. Ans. 9–10. The Examiner, however, finds these elements in the combination of Sutardja’s “memory control module” with Matsui’s memory system. *Id.* at 10–12. The Examiner finds it “would have been obvious to a person of ordinary skill in the art” to “modify the scope of the invention of Matsui”

² Independent claims 9 and 18 have limitations commensurate in scope with claim 2. *See infra* at 6.

with Sutardja’s memory control module, including a clock generator module.” *Id.* at 11. The “motivation,” the Examiner finds, “would be that the buffer modules may include a phase locked loop (PLL) to generate local phase-adjusted clock signals [citing Sutardja paragraph 136].” *Id.* The Examiner further explains in the Answer that “Sutardja advocates [for] use of a PLL [] in the buffer module [] for local clock synchronization and clock frequency multiplication[,] whereas Matsui uses a DLL (delay-locked loop) in buffers [] for the same purpose.” Ans. 10–11. The Examiner explains that “PLL offers advantages over a DLL as widely known in the art.” *Id.* at 11. Thus, the Examiner finds, incorporating Sutardja’s memory control module in Matsui would “improve upon” Matsui. *Id.*

As Appellants argue, however, the Examiner’s explanation for combining the references is “confusing” at best. App. Br. 11. The memory control module described in Sutardja is a “controller” that “generates” command and address signals, not receives them from another controller (i.e., the one in Matsui) as in the Examiner’s proposed combination. Reply Br. 4–5. Similarly, the clock *generator* in Sutardja, which the Examiner cites as teaching part of the “module control device” recited in claim 2, *generates* clock signals, whereas combining this element with Matsui in the manner explained by the Examiner would result in the element *receiving* clock signals from Matsui’s memory controller. *Id.* at 5.

Moreover, the Examiner does not explain how finding “notable differences between” the buffer module in Sutardja and the memory module buffers in Matsui supports, rather than *teaches away from*, the proposed combination. Ans. 10–11; *In re Gurley*, 27 F.3d 551, 553 (Fed. Cir. 1994) (reference teaches away from combination “when a person of ordinary skill,

upon reading the reference . . . would be discouraged from following the path set out in the reference”). If the Examiner relied on some other explanation for combining the references, it is not apparent in the record before us.

Accordingly, on this record, we find the Examiner has not sufficiently “articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007). Specifically, the Examiner has not rationally explained why one of ordinary skill would be led to apply Sutardja’s teachings regarding a memory control module and clock generator, to Matsui’s memory system which already has a memory controller. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007).

For the foregoing reasons, we do not sustain the Examiner’s rejection of independent claim 2 as unpatentable over Matsui and Sutardja. For the same reasons, we also do not sustain the same rejection of independent claims 9 and 18, which include limitations commensurate in scope with claim 2. We also do not sustain the rejections of the remaining dependent claims, each of which relies (in whole or in part) on the Matsui-Sutardja combination.

DECISION

The Examiner’s decision rejecting claims 2–25 is reversed.

REVERSED



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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 10/03/2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER

AHMED, ZUBAIR

ART UNIT

PAPER NUMBER

2132

DATE MAILED: 10/03/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	01/03/2019

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 10/03/2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	01/03/2019

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 _____

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

4b. Payment of Fee(s): (**Please first reapply any previously paid issue fee shown above**)

☐ Issue Fee

☐ A check is enclosed.

☐ Publication Fee (No small entity discount permitted)

☐ Payment by credit card. Form PTO-2038 is attached.

☐ Advance Order - # of Copies _____

☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700
79141	7590	10/03/2018	EXAMINER	
Morgan Lewis Bockius LLP (PA)-J. Zheng			AHMED, ZUBAIR	
1400 Page Mill Road			ART UNIT	
Palo Alto, CA 94304			PAPER NUMBER	
			2132	
DATE MAILED: 10/03/2018				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/445,035	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the decision rendered by the patent board on 09/27/2018.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-25. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date ____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material ____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to the Patent Board Decision of 09/27/2018.

Claim 1 was previously canceled.

Response to Arguments

The 35 U.S.C. 103 rejection of claims 2-5, 7, 9, 11-14, 16, 18-22, and 24-25 as being unpatentable over Matsui US 2004/0105292 (hereinafter "Matsui") in view of Sutardja et al. US 2007/0168781 (hereinafter "Sutardja"), the 35 U.S.C. 103 rejection of claims 6, 10, and 15 as being unpatentable over Matsui in view of Sutardja and in further view of Carpenter et al. US 2009/0116312 (hereinafter "Carpenter"), the 35 U.S.C. 103 rejection of claims 8 and 17 being unpatentable over Matsui in view of Sutardja and in further view of Sung et al. US 2011/0110168 (hereinafter "Sung"), and the 35 U.S.C. 103 rejection of claim 23 as being unpatentable over Matsui in view of Sutardja and in further view of Chu et al. US 2008/0291758 (hereinafter "Chu") are withdrawn because of the decision rendered by the patent board. Therefore, a NOTICE OF ALLOWANCE is being administered.

Allowable Subject Matter

Claims 2-25 are allowed.

Reasons for Allowance

This communication warrants no examiner's reason for allowance, as the board's decision makes evident the reason for allowance, satisfying the record "as a whole" as required by rule 37 CFR 1.104 (e). Accordingly, the reason for allowance is in all

probability evident from the record and no statement for examiner's reason for allowance is necessary (see MPEP 1302.14).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 September 2018 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	28 September 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 08
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	119					

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 September 2018 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	28 September 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

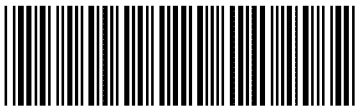
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	2	10	11	19	20										
2	3	11	12	20	21										
3	4	12	13	21	22										
4	5	13	14	22	23										
5	6	14	15	23	24										
6	7	15	16	24	25										
7	8	16	17												
8	9	17	18												
9	10	18	19												

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 September 2018 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	28 September 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Index of Claims</i> 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	11/02/2015	03/13/2016	08/01/2016	01/03/2017	09/28/2018				
	1	✓	✓	-	-	-				
	2	✓	✓	✓	✓	=				
	3	✓	✓	✓	✓	=				
	4	✓	✓	✓	✓	=				
	5	✓	✓	✓	✓	=				
	6	✓	✓	✓	✓	=				
	7	✓	✓	✓	✓	=				
	8	✓	✓	✓	✓	=				
	9	✓	✓	✓	✓	=				
	10	✓	✓	✓	✓	=				
	11		✓	✓	✓	=				
	12		✓	✓	✓	=				
	13		✓	✓	✓	=				
	14		✓	✓	✓	=				
	15		✓	✓	✓	=				
	16		✓	✓	✓	=				
	17		✓	✓	✓	=				
	18		✓	✓	✓	=				
	19		✓	✓	✓	=				
	20		✓	✓	✓	=				
	21			✓	✓	=				
	22				✓	=				
	23				✓	=				
	24				✓	=				
	25				✓	=				

<i>Search Notes</i> 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/08	08/01/2016	ZA
G11C 7/1066	01/03/2017	ZA
G11C 7/1066, 7/1093, 29/023, 29/028, 5/04	09/28/2018	ZA
G06F 12/0802	09/28/2018	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached search history	09/28/2018	ZA

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G11C	120/7/1066	09/28/2018	ZA

	/ZUBAIR AHMED/ Examiner.Art Unit 2132
--	--

EAST Search History

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S60	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:56
S61	1028	S60 and DRAM	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S62	259	S61 and clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S63	8	S62 and regenerat\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S64	1	"7,532,537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S65	1	"8,001,434".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S66	278156	memory adj controller	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S67	136	S66 same (local adj clock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S68	113142	memory adj module	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S69	53080	S68 same control\$4	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S70	7645	S69 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S71	42	S69 same (local adj clock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S72	6023	S70 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S73	1823	S70 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S74	177	S73 same (phase adj lock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S75	416	S73 same buffer	US-PGPUB;	OR	ON	2018/09/28 06:00

			USPAT			
S76	416	S73 same "buffers"	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S77	1	"7,532,537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S78	1	S77 and rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S79	1	S77 and group	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S80	1	"8,001,434".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S81	1	S80 and group	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S82	21655	group with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S83	1480	S82 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S84	0	S83 same each	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S85	378	S83 same (each or respective or correspond\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S86	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S87	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S88	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S89	2143	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S90	431	S89 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S91	8293	programmable adj delay	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S92	3386	S91 same clock	US-PGPUB; USPAT	OR	ON	2018/09/28 06:04
S93	780	S92 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:04
S94	61	((((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same	US-PGPUB;	OR	ON	2018/09/28 06:04

		group\$3	USPAT			
S95	61	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S96	12	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S97	80	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S98	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S99	14	S98 and PLL	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S100	0	G11C7/1066.cpc	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S101	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S102	0	S101 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S103	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S104	605	S103 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S105	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S106	27	S105 and ((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S107	0	(local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module) same (program\$4 near3 register)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S108	0	((local with (pll or (phase adj lock\$2 adj loop))) same (program\$4 near3 register)) and (memory near2 module)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S109	38	((local with (pll or (phase adj lock\$2 adj loop))) same register) and (memory near2 module)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S110	1	"20130242680".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S111	2650	Harashima	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S112	547	Harashima and memory and (command\$1address or c\$1a)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:09
S113	33	Harashima and (memory adj module) and (command\$1address or c\$1a)	US-PGPUB;	OR	ON	2018/09/28 06:09

			USPAT			
S114	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:09
S115	44	(module adj controller) with (command\$1address or C\$1A)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S116	1	"7532537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S117	1651	(711/120 711/119).ccls.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S118	5	S117 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10

9/28/2018 6:13:08 AM

C:\Users\zahmed\Documents\EAST\Workspaces\14445035.wsp

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14

S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16
S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04

S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:14
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12

S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S36	38	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 11:42
S37	8	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:13
S38	68	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:18
S39	4415	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S40	5	S39 and PLL	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S41	0	G11C7/1066.cpc	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:00
S42	4534	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:01
S43	0	S42 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:02
S44	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:42
S45	553	S44 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:43
S46	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:48

S47	27	S46 and ((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:49
S48	0	(local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module) same (program\$4 near3 register)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:50
S49	0	((local with (pll or (phase adj lock\$2 adj loop))) same (program\$4 near3 register)) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:52
S50	35	((local with (pll or (phase adj lock\$2 adj loop))) same register) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:53
S51	2	"20130242680".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:18
S52	7122	Harashima	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:20
S53	482	Harashima and memory and (command\$1address or c\$1a)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:22
S54	27	Harashima and (memory adj module) and (command\$1address or c\$1a)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:23
S55	27229	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/16 07:44
S56	38	(module adj controller) with (command\$1address or C\$1A)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/16 07:45
S57	3	"7532537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/07/31 07:45
S58	2310	(711/120 711/119).cls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/09/28 05:54

S59	5	S58 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/09/28 05:54
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9/ 28/ 2018 6:12:56 AM
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28 2014	Hyun Lee	129980-5073US	9700

79141 7590 10 25 2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER
AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
2132	

NOTIFICATION DATE	DELIVERY MODE
10/25/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Application No. : 14445035
Applicant : Lee
Filing Date : 07/28/2014
Date Mailed : 10/25/2018

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The informalities requiring correction are indicated in the attachment(s). If the informality pertains to the abstract, specification (including claims) or drawings, the informality must be corrected with an amendment in compliance with 37 CFR 1.121 (or, if the application is a reissue application, 37 CFR 1.173). Such an amendment may be filed after payment of the issue fee if limited to correction of informalities noted herein. See Waiver of 37 CFR 1.312 for Documents Required by the Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004). In addition, if the informality is not corrected until after payment of the issue fee, for purposes of 35 U.S.C. 154(b)(1)(iv), "all outstanding requirements" will be considered to have been satisfied when the informality has been corrected. A failure to respond within the above-identified time period will result in the application being ABANDONED.

See attachment(s).

*A copy of this notice **MUST** be returned with the reply. Please address response to
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P.O. Box 1450, Alexandria, VA 22313-1450".*

/Chris Harmon/
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IDENTIFICATION OF DRAWING DEFICIENCIES

- ☐ There is a hole or the image thereof within the illustration. FIG(s)
- ☐ The illustration is penetrated or traversed by a solid or broken line that is not intended to be part of the drawing, such as a dark line caused by a flaw in the copying process. FIG(s)
- ☐ An ink stamp or the image thereof obscures part of the illustration. FIG(s)
- ☐ The drawing is marred by black smudges, obliterations, or fax/copier marks (for example, speckles or dots in a substantial portion of the drawing). FIG(s)
- ☒ Figure numbers are duplicated or missing. FIG(s) 2D is duplicated
- ☐ Drawing sheet or figure is missing. FIG(s)
- ☐ Numbers, letters, or reference characters in the drawing have been crossed out or are illegibly handwritten. FIG(s)
- ☐ The character of the lines, numbers, and letters is poor. FIG(s)
- ☐ The drawing's background shows that the original drawing was made on graph paper or other paper with a pattern or decoration. FIG(s)
- ☐ The FIG. number label is placed in a location that causes the drawing to be read upside down. FIG(s)
- ☐ Data, a reference number, or part of the drawing is truncated or missing, or a lead line has no reference number. FIG(s)
- ☐ The drawing and/or the FIG. label contain(s) foreign language. FIG(s)
- ☐ This utility application contains a photograph of a view that is capable of being illustrated as a line drawing. FIG(s)
- ☐ A petition under 37 CFR 1.84(a)(2) to accept color drawings has been granted, but the brief description of the drawings in the specification does not contain (or has not been amended to contain) the paragraph required by 37 CFR 1.84(a)(2)(iii).
- ☐ This reissue application contains added and/or amended drawings that are not labeled as "New" or "Amended" or "Canceled" as required by 37 CFR 1.173(b)(3). FIG(s)
- ☐ This Design reissue application contains a drawing that is labeled as "Canceled" but is not surrounded by brackets, or a drawing that is surrounded by brackets but is not labeled as "Canceled." See 37 CFR 1.173(b)(3). FIG(s)
- ☐ OTHER:
- ☐ COMMENTS:

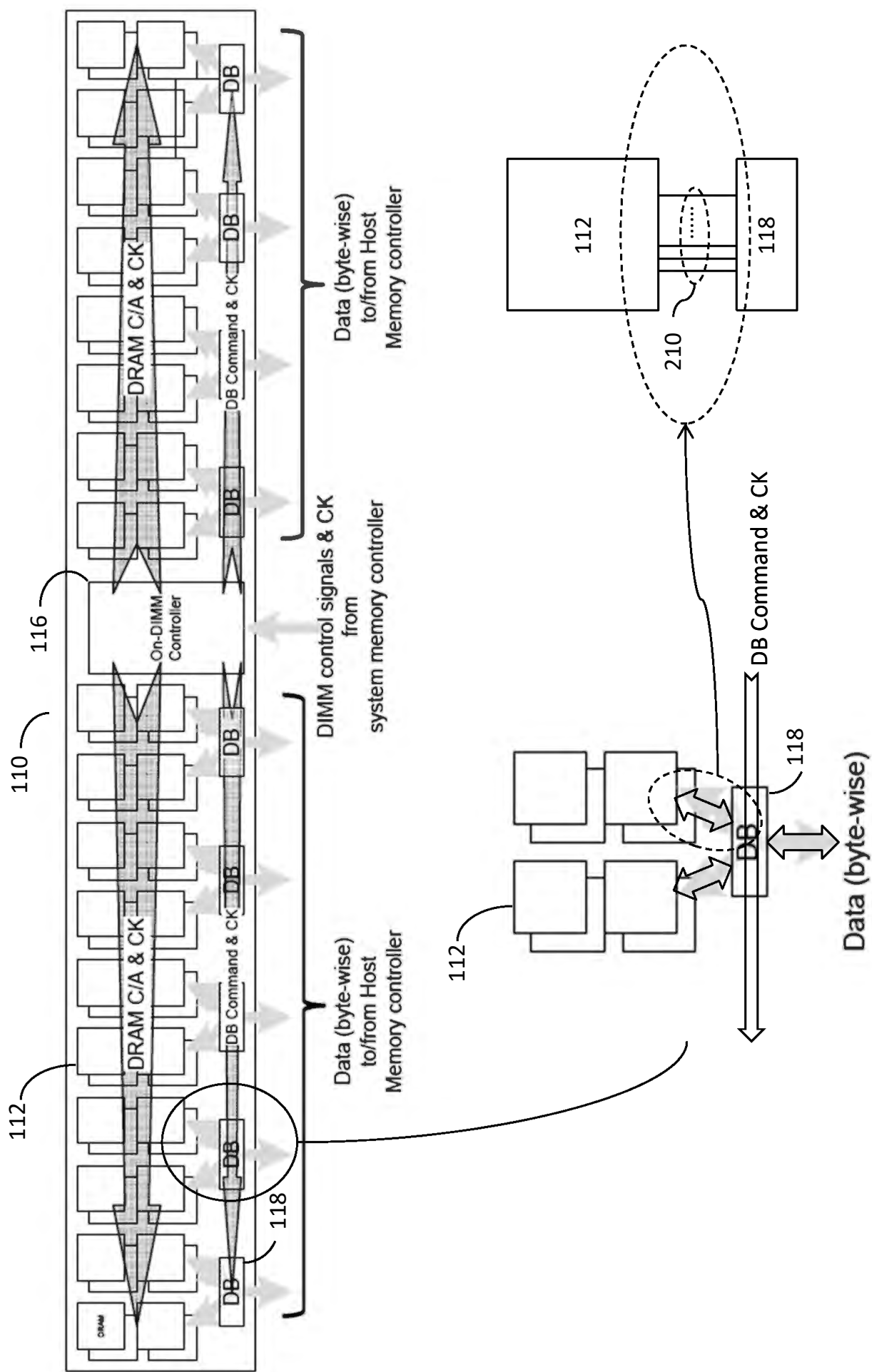


FIG. 2C

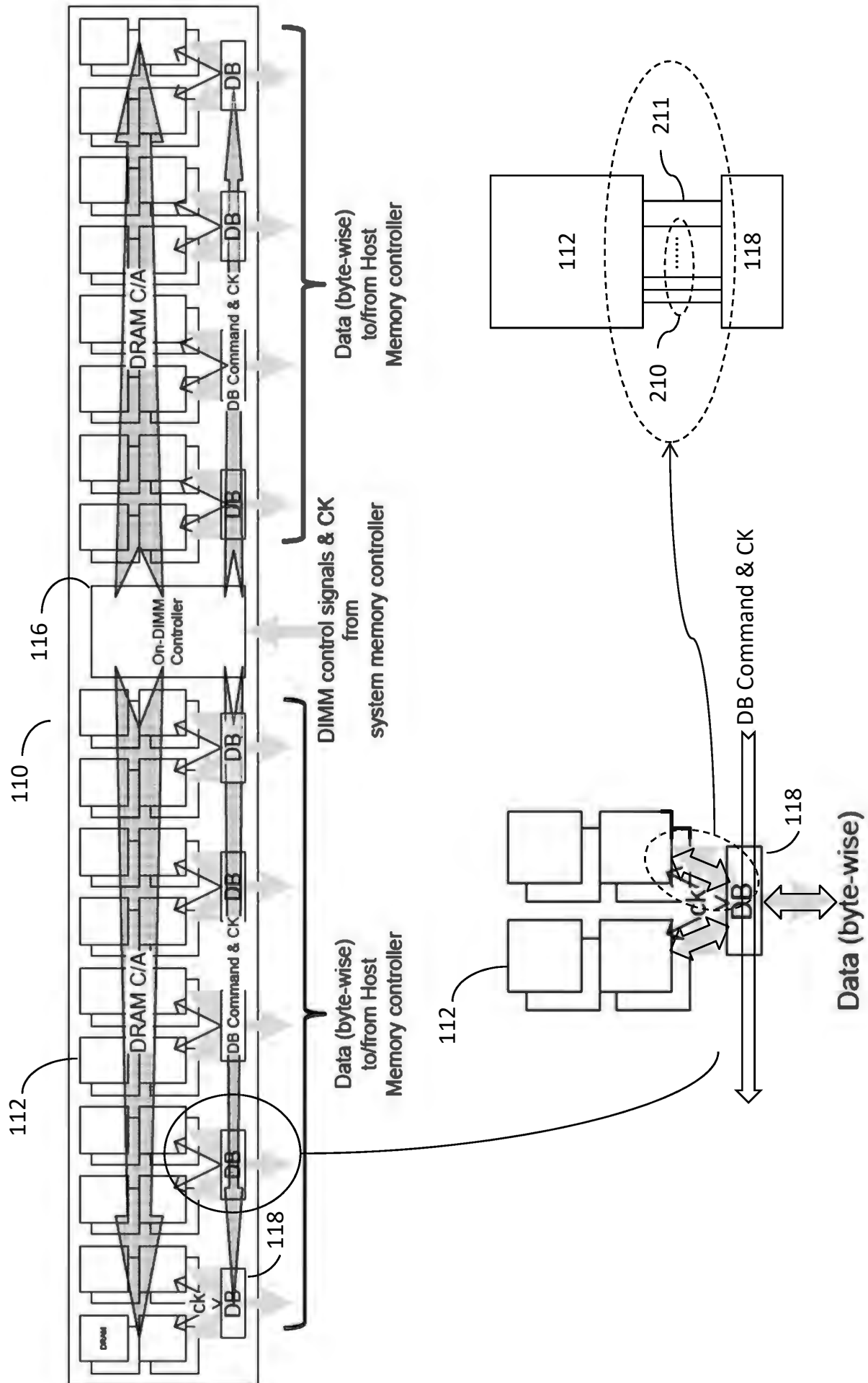


FIG. 2D

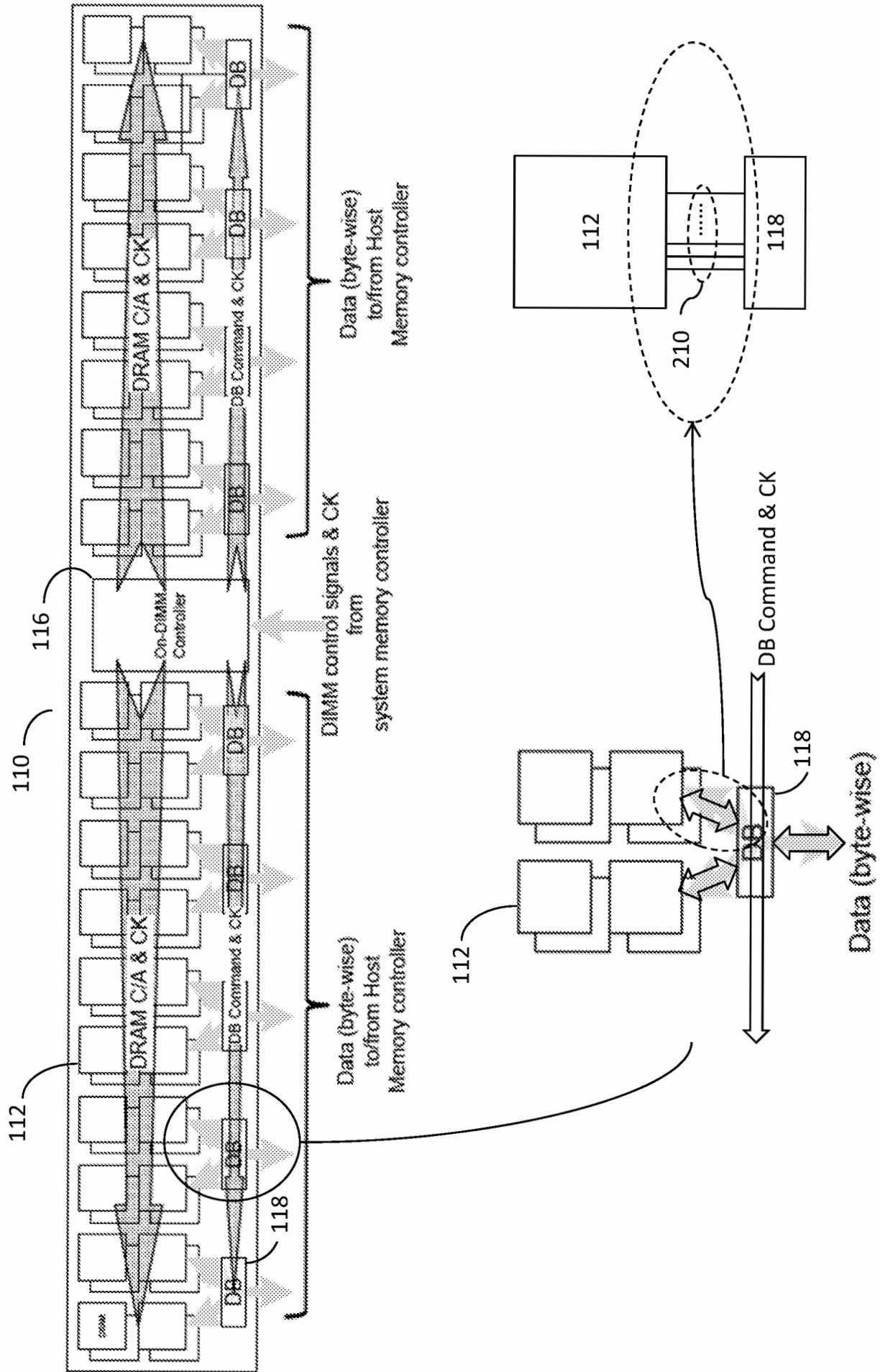


FIG. 2C

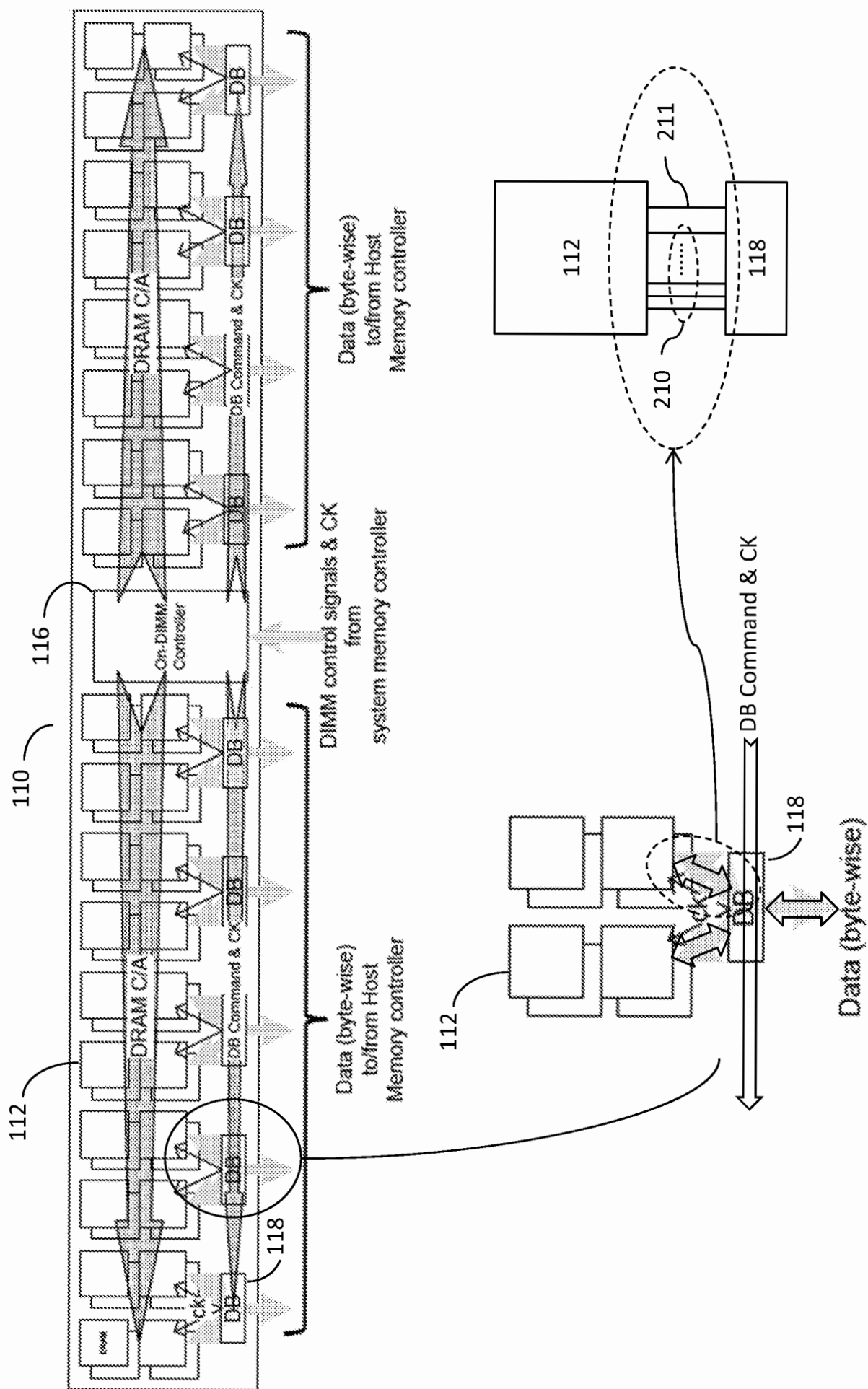


FIG. 2D

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5073US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	2251	1	100	100
Miscellaneous:				
Total in USD (\$)				100

Electronic Acknowledgement Receipt

EFS ID:	34756674
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073US
Receipt Date:	03-JAN-2019
Filing Date:	28-JUL-2014
Time Stamp:	20:17:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 100
RAM confirmation Number	010419INTEFSW00005585500310
Deposit Account	
Authorized User	

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980_5073_US_RES_DRWGS.pdf	780504 38bffc8fb71dbbd4ea77a7ff7314bfb15053261	yes	7
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Miscellaneous Incoming Letter		1	5	
	Drawings-only black and white line drawings		6	7	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30631 37ad01f428d5b63df276bcc51e21d306d4af748	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			811135		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

**REQUEST FOR
CONTINUED EXAMINATION (RCE)
TRANSMITTAL**

Address to:
Mail Stop RCE
Commissioner of Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Electronically filed	January 3, 2019
Application Number	14/445,035
Filing Date	July 28, 2014
First Named Inventor	Hyun Lee
Art Unit	2132
Examiner Name	AHMED, ZUBAIR
Attorney Docket No.	129980-5073-US

This is a Request for Continued Examination (RCE) under 37 C.F.R. § 1.114 of the above-identified application. Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application.

1. Submission required under 37 C.F.R. § 1.114

NOTE: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

- a. ☐ Previously submitted. If a final Office Action is outstanding, any amendments filed after the final Office Action may be considered as a submission.
- i. ☐ Request for non-entry of amendment: Please **do not** entered the amendment submitted on _____
- ii. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed _____
- iii. ☐ Other _____
- b. ☒ Enclosed
- i. ☒ Amendment/Reply
- ii. ☐ Affidavit(s)/Declaration(s)
- iii. ☒ Information Disclosure Statement (IDS)
- iv. ☐ Other _____

2. Miscellaneous

- a. ☐ Suspension of action on the above-identified application is requested under 37 C.F.R. § 1.103(c) for a period of _____ months.
(Period of suspension shall not exceed 3 months; Fee under 37 C.F.R. § 1.17(i) required)
- b. ☐ Other _____

3. Fees

The RCE fee under 37 C.F.R. § 1.17(e) is required by 37 C.F.R. § 1.114 when the RCE is filed.

- a. ☒ The Director is hereby authorized to charge the following fees, any underpayment of fees, or credit any overpayments, to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US).
- i. ☐ (1st request) RCE fee required under 37 C.F.R. § 1.17(e)(1), estimated to be \$1300 (large entity) or \$650 (small entity)
- ii. ☒ (2nd or subsequent request) RCE fee required under 37 C.F.R. § 1.17(e)(2), estimated to be \$1900 (large entity) or \$950 (small entity)
- iii. ☐ Extension of time fee required under 37 C.F.R. §§ 1.136 and 1.17, estimated to be \$ _____ for a _____ month(s) extension, the request for which is being made herewith
- iv. ☐ Other _____

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

Name (Print/Type)	Jamie J. Zheng	Registration No. (Attorney/Agent)	51,167
Signature	/Jamie J. Zheng/	Date	January 3, 2018

CERTIFICATE OF MAILING OR TRANSMISSION

I hereby certify that this correspondence is being deposited with the United States Postal Service as first class mail in an envelope addressed to: Mail Stop RCE, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450, or by facsimile transmitted to fax no. 1-703- _____ to the U.S. Patent and Trademark Office on the date indicated below.

Name (Print/Type)		Registration No. (Attorney/Agent)	
Signature		Date	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	9700
Serial No.:	14/445,035	Art Unit:	2132
Filed:	July 28, 2014	Examiner:	AHMED, ZUBAIR
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	129980-5073-US
		Date:	January 3, 2019

AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION

Mail Stop: AF

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450
Sir:

The enclosed Amendment is in response to the Notice of Allowance dated October 3, 2018 for the above identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Currently Amended) A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, comprising:

N data buffers each having a data width of n -bits, ~~and each~~ of the N data buffers ~~generating~~ outputting a respective local clock signal;

memory devices including a plurality of groups of memory devices, each respective group of memory devices ~~communicating~~ configured to communicate data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal ~~generated~~ output by the respective one of the data buffers; and

a module control device configured to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers;

wherein the module control device does not transmit any clock signal to the memory devices; and

wherein the each of the N data buffers ~~generates~~ is configured to derive the respective local clock signal from the module clock signal received from the module control device.

3. (Currently Amended) The memory module of claim 2, wherein the respective one of the N data buffers includes data paths for communicating data and data strobes between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the data buffer control signals, at least one of the data paths including a programmable delay element.

4. (Currently Amended) The memory module of claim 2, wherein the N data buffers are distributed across a surface of the memory module, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal ~~in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.~~
5. (Currently Amended) The memory module of claim ~~4~~2, wherein the respective one of the N data buffers includes one or more programmable registers controlling the programmable delay ~~a phase of the local clock signal~~, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective one of the N data buffers is configured to program the one or more programmable registers during the configuration mode.
6. (Currently Amended) The memory module of claim 5, wherein the respective one of the N data buffers is configured to program the one or more registers according to ~~the data buffer control signals~~ output from the module control device during the configuration mode.
7. (Currently Amended) The memory module of claim 2, wherein the ~~each~~ respective one of the N data buffers is configured to receive the module clock signal to generate a received clock signal, to phase-lock the local-received clock signal with the module clock signal, and to delay the received clock signal with a respective programmable delay to generate the respective local clock signal.
8. (Currently Amended) The memory module of claim 2, wherein the memory devices are organized in one or more ranks and are each n -bit or $n/2$ -bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.
9. (Currently Amended) A method performed by a memory module coupled to a memory bus in a system, the memory module having a data width of $(N \times n)$ bits, where N and n are integers larger than one, the memory module including memory devices, the memory devices including a plurality of groups of memory devices, comprising:

receiving a system clock signal;
generating a module clock signal on the memory module from the system clock signal received by the memory module;
transmitting the module clock signal to a plurality of data buffers each coupled between a respective group of the memory devices and the memory bus, the plurality of data buffers including N data buffers, each of the N data buffers having a data width of n -bits;
generating a respective local clock signal from the module clock signal at each of the N data buffers, the respective local clock signal being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal;
transmitting respective local clock signals from the plurality of data buffers to respective groups of the memory devices; and
performing read/write operations at the respective groups of the memory devices in accordance with the respective local clock signals.

10. (Currently Amended) The method of claim 9, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, the method further comprising: using the N data buffers to buffer data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and programming a set configuration registers in the each of the N data buffers during the configuration mode, the set of configuration registers controlling the ~~incorporating a programmable amount of delay to the respective local clock signal at~~ the each of the N data buffers.

11. (Previously Presented) The method of claim 9, further comprising:
receiving command/address (C/A) signals from the memory controller;
generating module C/A signals and data buffer control signals in response to the C/A signals;
transmitting the module C/A signals to at least some of the memory devices; and
transmitting the data buffer control signals to the N data buffers.

12. (Currently Amended) The method of claim 11, further comprising providing data paths in the N data buffers to buffer data and data strobe signals between respective groups of memory

devices and the memory controller, controlling the data paths in the N data buffers according to the data buffer control signals, and delaying a data strobe in a data path using a programmable delay element.

13. (Previously Presented) The method of claim 11, wherein the plurality of data buffers are distributed across a surface of the memory module in a fly-by configuration such that each data buffer control signal arrives at some adjacent data buffers at different points in time.

14. (Previously Presented) The method of claim 9, further comprising controlling a phase of the respective local clock signal at each of the N data buffers.

15. (Currently Amended) The method of claim 11, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, the method further comprising:

programming one or more registers in the each of the N data buffers according to the data buffer control signals in the configuration mode, the one or more registers at the each of the N data buffers controlling the respective programmable delay from the module clock signal.

16. (Currently Amended) The method of claim 9, further comprising: receiving the module clock signal at each of the N data buffers to generate a received clock signal, phase-locking the local-received clock signal with the module clock signal, and delaying the received clock signal to generate the respective local clock signal according to the respective programmable delay.

17. (Currently Amended) The method of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit or $n/2$ -bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.

18. (Currently Amended) A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory

controller via the control signal lines, the control circuit ~~to generate~~ being further configured output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each n -bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective local clock signal being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal; the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals, at least one of the data paths including a programmable delay element controlled by the logic; and

memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data and data strobes with the memory controller via the data paths in the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

19. (Previously Presented) The memory module of claim 18, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors.

20. (Currently Amended) The memory module of claim 18, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to delay the local clock signal by

the programmable delay, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

21. (Previously Presented) The memory module of claim 18, wherein the control circuit does not transmit any clock signals to the memory devices.

22. (Previously Presented) The memory module of claim 18, wherein the data buffer control signals are different from the module C/A signals.

23. (Previously Presented) The memory module of claim 18, wherein the respective data buffer includes registers configurable by the logic to time the respective local clock signal as well as the data and data strobes being communicated via the respective data buffer.

24. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is n -bits wide.

25. (Previously Presented) The memory module of claim 18, wherein each of the memory devices is $n/2$ -bits wide.

REMARKS

This amendment responds to the Notice of Allowance dated October 3, 2018, in which the Examiner allowed claims 2-25.

REMARKS CONCERNING CLAIMS

Claims 2-5-10, 12, 15-18, and 20 have been amended. Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0058] and [0065]-[0077]. Therefore, no new matter has been added.

REMARKS CONCERNING INFORMATION DISCLOSURE STATEMENTS

Filed herewith are Information Disclosure Statements listing references cited in a related patent application, U.S. Patent Application No. 15/470,856, filed March 27, 2017, entitled “Memory Module with Controlled Byte-Wise Buffers”.

CONCLUDING REMARKS

By responding in the foregoing remarks only to particular positions asserted by the examiner, the Applicant does not necessarily acquiesce in other positions that have not been explicitly addressed. In addition, the Applicant’s arguments for the patentability of a claim should not be understood as implying that no other reasons for the patentability of that claim exist.

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 843-4000, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: January 2, 2019

/Jamie J. Zheng/

51,167

Jamie J. Zheng

(Reg. No.)

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

Phone: (650) 843-4000

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee et al.	Confirmation No.:	9700
Serial No.:	14/445,035	Art Unit:	2132
Filed:	July 28, 2014	Examiner:	Ahmed, Zubair
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION	Attorney Docket No.:	129980-5073US

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 28 references. Copies of foreign documents and/or their translation and non-patent literature were submitted on commonly-owned U.S. Pat. App. No. 15/470,856, filed March 27, 2017, which is accessible via PAIR.

Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.

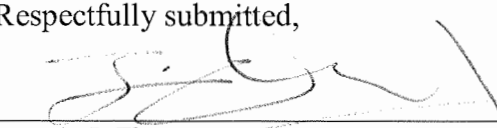
The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed with the filing of a Request for Continued Examination, and presumably no fee is required. The Commissioner is authorized to charge any required fees to Deposit Account No. 50-0310 (order no. 129980-5073-US).

Respectfully submitted,

Date: January 3, 2019


Jamie J. Zheng

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1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

51,167
(Reg. No.)

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed January 3, 2019</i>	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	Ahmed, Zubair
Sheet	1	of	2	Attorney Docket Number	129980-5073-US

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
		2004/0105292	06-2004	Matsui	
		2006/0077731 A1	04-2006	Ware	
		2007/0217559 A1	09-2007	Stott	
		2010/0271092 A1	10-2010	Zerbe	
		2014/0029370 A1	01-30-2014	Koshizuka	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ – Number ⁴ – Kind Code ⁵ (if known)				
		CN102576565B	09-30-2015	Netlist, Inc.		

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Netlist, Inc., Examination Decision of Request for Invalidation, CN201080039043.0, 30MAY2018, 43 pgs.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Record of Oral Hearing, filed June 19, 2018.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Final Written Decision, filed July 5, 2018.
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00667, Final Written Decision, filed July 18, 2018
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00668, Final Written Decision, filed July 18, 2018.
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, ORDER, Conduct of the Proceedings, filed August 2, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, Decision Granting Institution of <i>Inter Partes</i> Review, filed August 6, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Mehran Arjomand,' filed July 30, 2018

Examiner's Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed January 3, 2019</i>	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	Ahmed, Zubair
Sheet	2	of	2	Attorney Docket Number	129980-5073-US

		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Michael D. Hatcher,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, and Case No. IPR2018-00365, Decision Granting Institution of <i>Inter Partes</i> Review, filed June 29, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, Patent Owner's Objections to Petition Evidence, filed August 20, 2018
		Netlist, Inc., Examination Decision of Request for Invalidation, CN201080039043.0, 30MAY2018, 43 pgs.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Record of Oral Hearing, filed June 19, 2018.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Final Written Decision, filed July 5, 2018.
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00667, Final Written Decision, filed July 18, 2018
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00668, Final Written Decision, filed July 18, 2018.
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, ORDER, Conduct of the Proceedings, filed August 2, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, Decision Granting Institution of Inter Partes Review, filed August 6, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Mehran Arjomand,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Michael D. Hatcher,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, and Case No. IPR2018-00365, Decision Granting Institution of Inter Partes Review, filed June 29, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, Patent Owner's Objections to Petition Evidence, filed August 20, 2018

Examiner's Signature		Date Considered	
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Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5073US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 2ND AND SUBSEQUENT REQUEST	2820	1	950	950
Total in USD (\$)				950

Electronic Acknowledgement Receipt

EFS ID:	34756660
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073US
Receipt Date:	03-JAN-2019
Filing Date:	28-JUL-2014
Time Stamp:	20:24:56
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$950
RAM confirmation Number	010419INTEFSW00005617500310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980_5073_RCE_AMEND.pdf	284586	yes	10
			06e0637bf2071c0680a1ae0a24015072a708747d		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Request for Continued Examination (RCE)		1	1	
	Amendment/Req. Reconsideration-After Non-Final Reject		2	2	
	Claims		3	8	
	Applicant Arguments/Remarks Made in an Amendment		9	10	
Warnings:					
Information:					
2		129980-5073US_IDS.pdf	292170	yes	4
			87f5b2d21d4420d18ef5b2ae5c1a20986b9b0f57		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	2	
	Information Disclosure Statement (IDS) Form (SB08)		3	4	
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30439	no	2
			bfd468839a244028b07bcaa638beb4431b3fdf2e		
Warnings:					
Information:					
Total Files Size (in bytes):			607195		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	9700
Serial No.:	14/445,035	Art Unit:	2132
Filed:	July 28, 2014	Examiner:	AHMED, ZUBAIR
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	129980-5073-US

RESPONSE TO NOTICE TO FILE CORRECTED APPLICATION PAPERS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The enclosed is being filed to address drawings informalities identified on a Notice to File Corrected Application Papers dated October 25, 2018 (the "Notice"). A copy of the Notice is attached.

Petition for Extension of Time under 37 CFR 1.136. It is respectfully requested that the time for response to the Office Action dated October 25, 2018 be extended for a period of one month(s) from December 25, 2018 to January 25, 2019.

Applicant believes that no additional fees are necessitated by the present response, however, if any such fees are due, the Commissioner is hereby authorized to charge them to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US).

REMARKS

OVERVIEW OF CHANGES TO THE DRAWINGS

The drawings have been revised to address the “identification of drawing deficiencies” noted in the Notice. In particular, Figures 2C-2D have been revised to address the identified deficiencies. No new matter has been added.

CONCLUDING REMARKS

Entry of this amendment is respectfully requested.

The Examiner may contact the undersigned attorney by email (jamie.zheng@morganlewis.com) or by telephone (650.843.7249) to arrange a suitable time for discussing any remaining items.

Respectfully submitted,

Date: January 3, 2019

/Jamie J. Zheng/

Jamie J. Zheng

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000

51,167

(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28 2014	Hyun Lee	129980-5073US	9700

79141 7590 10 25 2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER
AHMED, ZUBAIR

ART UNIT	PAPER NUMBER
2132	

NOTIFICATION DATE	DELIVERY MODE
10/25/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Application No. : 14445035
Applicant : Lee
Filing Date : 07/28/2014
Date Mailed : 10/25/2018

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The informalities requiring correction are indicated in the attachment(s). If the informality pertains to the abstract, specification (including claims) or drawings, the informality must be corrected with an amendment in compliance with 37 CFR 1.121 (or, if the application is a reissue application, 37 CFR 1.173). Such an amendment may be filed after payment of the issue fee if limited to correction of informalities noted herein. See Waiver of 37 CFR 1.312 for Documents Required by the Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004). In addition, if the informality is not corrected until after payment of the issue fee, for purposes of 35 U.S.C. 154(b)(1)(iv), "all outstanding requirements" will be considered to have been satisfied when the informality has been corrected. A failure to respond within the above-identified time period will result in the application being ABANDONED.

See attachment(s).

*A copy of this notice **MUST** be returned with the reply. Please address response to
"Mail Stop Issue Fee, Commissioner for Patents,
P.O. Box 1450, Alexandria, VA 22313-1450".*

/Chris Harmon/
Publication Branch
Office of Data Management
(571) 272-4200

IDENTIFICATION OF DRAWING DEFICIENCIES

- ☐ There is a hole or the image thereof within the illustration. FIG(s)
- ☐ The illustration is penetrated or traversed by a solid or broken line that is not intended to be part of the drawing, such as a dark line caused by a flaw in the copying process. FIG(s)
- ☐ An ink stamp or the image thereof obscures part of the illustration. FIG(s)
- ☐ The drawing is marred by black smudges, obliterations, or fax/copier marks (for example, speckles or dots in a substantial portion of the drawing). FIG(s)
- ☒ Figure numbers are duplicated or missing. FIG(s) 2D is duplicated
- ☐ Drawing sheet or figure is missing. FIG(s)
- ☐ Numbers, letters, or reference characters in the drawing have been crossed out or are illegibly handwritten. FIG(s)
- ☐ The character of the lines, numbers, and letters is poor. FIG(s)
- ☐ The drawing's background shows that the original drawing was made on graph paper or other paper with a pattern or decoration. FIG(s)
- ☐ The FIG. number label is placed in a location that causes the drawing to be read upside down. FIG(s)
- ☐ Data, a reference number, or part of the drawing is truncated or missing, or a lead line has no reference number. FIG(s)
- ☐ The drawing and/or the FIG. label contain(s) foreign language. FIG(s)
- ☐ This utility application contains a photograph of a view that is capable of being illustrated as a line drawing. FIG(s)
- ☐ A petition under 37 CFR 1.84(a)(2) to accept color drawings has been granted, but the brief description of the drawings in the specification does not contain (or has not been amended to contain) the paragraph required by 37 CFR 1.84(a)(2)(iii).
- ☐ This reissue application contains added and/or amended drawings that are not labeled as "New" or "Amended" or "Canceled" as required by 37 CFR 1.173(b)(3). FIG(s)
- ☐ This Design reissue application contains a drawing that is labeled as "Canceled" but is not surrounded by brackets, or a drawing that is surrounded by brackets but is not labeled as "Canceled." See 37 CFR 1.173(b)(3). FIG(s)
- ☐ OTHER:
- ☐ COMMENTS:

SCORE Placeholder Sheet for IFW Content

Application Number: 14445035

Document Date: 01/03/2019

The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

- Drawing

At the time of document entry (noted above):

- USPTO employees may access SCORE content via eDAN using the Supplemental Content tab, or via the SCORE web page.
- External customers may access SCORE content via PAIR using the Supplemental Content tab.



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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 02/14/2019
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER

2132

DATE MAILED: 02/14/2019

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	05/14/2019

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 02/14/2019
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
 1400 Page Mill Road
 Palo Alto, CA 94304

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	05/14/2019

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700
79141	7590	02/14/2019	EXAMINER	
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)			AHMED, ZUBAIR	
1400 Page Mill Road			ART UNIT	
Palo Alto, CA 94304			PAPER NUMBER	
			2132	
DATE MAILED: 02/14/2019				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/445,035	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to RCE filed on 01/03/2019.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-25. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>01/03/2019</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
---	--

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to RCE filed on 01/03/2019. Claim 1 was previously canceled.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 01/03/2019 was filed after the mailing date of the Notice of Allowance on 10/03/2018. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Allowable Subject Matter

Claims 2-25 are allowed.

Reasons for Allowance

This communication warrants no examiner's reason for allowance, as the record makes evident the reason for allowance, satisfying the record "as a whole" as required by rule 37 CFR 1.104 (e). Accordingly, the reason for allowance is in all probability evident from the record and no statement for examiner's reason for allowance is necessary (see MPEP 1302.14).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	07 February 2019 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	07 February 2019 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 08
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	119					

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	07 February 2019 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	07 February 2019 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

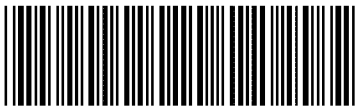
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	2	10	11	19	20										
2	3	11	12	20	21										
3	4	12	13	21	22										
4	5	13	14	22	23										
5	6	14	15	23	24										
6	7	15	16	24	25										
7	8	16	17												
8	9	17	18												
9	10	18	19												

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	07 February 2019 (Date)	Total Claims Allowed: 24	
/DAVID YI/ SPE, Art Unit 2132 (Primary Examiner)	07 February 2019 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Index of Claims</i> 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	11/02/2015	03/13/2016	08/01/2016	01/03/2017	09/28/2018	02/07/2019			
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	3	✓	✓	✓	✓	=	=			
	4	✓	✓	✓	✓	=	=			
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	25				✓	=	=			

<i>Search Notes</i> 	Application/Control No. 14/445,035	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/08	08/01/2016	ZA
G11C 7/1066	01/03/2017	ZA
G11C 7/1066, 7/1093, 29/023, 29/028, 5/04	09/28/2018	ZA
G06F 12/0802	09/28/2018	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached search history	02/07/2019	ZA
NPL Search	02/07/2019	ZA

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G11C	120/7/1066	09/28/2018	ZA

	/ZUBAIR AHMED/ Examiner.Art Unit 2132
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EAST Search History

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S60	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:56
S61	1028	S60 and DRAM	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S62	259	S61 and clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S63	8	S62 and regenerat\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S64	1	"7,532,537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S65	1	"8,001,434".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S66	278156	memory adj controller	US-PGPUB; USPAT	OR	ON	2018/09/28 05:57
S67	136	S66 same (local adj clock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S68	113142	memory adj module	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S69	53080	S68 same control\$4	US-PGPUB; USPAT	OR	ON	2018/09/28 05:58
S70	7645	S69 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S71	42	S69 same (local adj clock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S72	6023	S70 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S73	1823	S70 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 05:59
S74	177	S73 same (phase adj lock\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S75	416	S73 same buffer	US-PGPUB;	OR	ON	2018/09/28 06:00

			USPAT			
S76	416	S73 same "buffers"	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S77	1	"7,532,537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:00
S78	1	S77 and rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S79	1	S77 and group	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S80	1	"8,001,434".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S81	1	S80 and group	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S82	21655	group with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:01
S83	1480	S82 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S84	0	S83 same each	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
S85	378	S83 same (each or respective or correspond\$3)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:02
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S87	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S88	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S89	2143	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S90	431	S89 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S91	8293	programmable adj delay	US-PGPUB; USPAT	OR	ON	2018/09/28 06:03
S92	3386	S91 same clock	US-PGPUB; USPAT	OR	ON	2018/09/28 06:04
S93	780	S92 same memory	US-PGPUB; USPAT	OR	ON	2018/09/28 06:04
S94	61	((((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same	US-PGPUB;	OR	ON	2018/09/28 06:04

		group\$3	USPAT			
S95	61	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S96	12	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S97	80	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S98	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:05
S99	14	S98 and PLL	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S100	0	G11C7/1066.cpc	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S101	2669	G06F12/08.cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S102	0	S101 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S103	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:06
S104	605	S103 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S105	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S106	27	S105 and ((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:07
S107	0	(local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module) same (program\$4 near3 register)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S108	0	((local with (pll or (phase adj lock\$2 adj loop))) same (program\$4 near3 register)) and (memory near2 module)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S109	38	((local with (pll or (phase adj lock\$2 adj loop))) same register) and (memory near2 module)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S110	1	"20130242680".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S111	2650	Harashima	US-PGPUB; USPAT	OR	ON	2018/09/28 06:08
S112	547	Harashima and memory and (command\$1address or c\$1a)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:09
S113	33	Harashima and (memory adj module) and (command\$1address or c\$1a)	US-PGPUB;	OR	ON	2018/09/28 06:09

			USPAT			
S114	14982	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:09
S115	44	(module adj controller) with (command\$1address or C\$1A)	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S116	1	"7532537".pn.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S117	1651	(711/120 711/119).ccls.	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S118	5	S117 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT	OR	ON	2018/09/28 06:10
S127	1	"20060077731".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S128	1	"20070217559".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S129	1	"20100271092".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S130	1	"20140029370".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S131	1	"8516185".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S132	1	"7532537".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S133	1	"9606907".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06
S134	1	"20040105292".pn.	US-PGPUB; USPAT	OR	ON	2019/02/07 09:06

2/ 7/ 2019 9:07:51 AM
C:\ Users\ zahmed\ Documents\ EAST\ Workspaces\ 14445035.wsp

Memory module having buffer for isolating stacked memory devices

JB Halbert, RM Bonella - US Patent 6,487,102, 2002 - Google Patents

... Multi-rank partial width **memory modules**. WO2012030955A3 (en) *, 2010-08-31, 2012-04-19, Micron Technology, Inc ... US8250295B2 (en), 2004-01-05, 2012-08-21, Smart **Modular** Technologies, Inc ... System and method utilizing distributed byte-wise **buffers** on a **memory module** ...

☆ 77 Cited by 204 Related articles All 2 versions »

System featuring a controller device and a memory module that includes an integrated circuit buffer device and a plurality of integrated circuit memory devices

RE Perego, S Sidiropoulos, E Tsem - US Patent 7,010,642, 2006 - Google Patents

... system of claim 9, wherein the first substrate is a motherboard substrate and the second substrate is a **memory module** substrate ... 14. The **memory** system of claim 1, wherein the first **plurality** of integrated circuit **memory devices** are dynamic random access **memories** ...

☆ 77 Cited by 140 Related articles All 4 versions »

Buffering and interleaving data transfer between a chipset and memory modules

JB Halbert, JM Dodd, C Lam, RM Bonella - US Patent 7,249,232, 2007 - Google Patents

... in particular, the **memory modules** may be dynamic random access **memories** (DRAM), double ... US8250295B2 (en) *, 2004-01-05, 2012-08-21, Smart **Modular** Technologies, Inc ... 17, Intel Corporation, **Memory** system including a **memory module** having a **memory module controller** ...

☆ 77 Cited by 80 Related articles All 4 versions »

... and method featuring a controller device and a memory module that includes an integrated circuit buffer device and a plurality of integrated circuit memory devices

RE Perego, S Sidiropoulos, E Tsem - US Patent 7,000,062, 2006 - Google Patents

... G06F13/00—Interconnection of, or transfer of information or other signals between, **memories**, input/output ... the point-to-point link and situated on **buffer device** 350, on a **memory module** substrate and ... 2B) and connect to an I/O interface (not shown) of each **memory device** via a ...

☆ 77 Cited by 130 Related articles All 4 versions »

Memory module having buffer for isolating stacked memory devices

JB Halbert, RM Bonella - US Patent 6,747,887, 2004 - Google Patents

... US8990489B2 (en), 2004-01-05, 2015-03-24, Smart **Modular** Technologies, Inc ... System featuring a master **device**, a **buffer device** and a **plurality** of integrated circuit **memory devices** ... Configurable width **buffered module** having flyby elements ...

☆ 77 Cited by 106 Related articles All 4 versions »

Memory module and method for operating a memory module in a data memory system

H Ruckerbauer, G Braun - US Patent 7,275,189, 2007 - Google Patents

... Eight DRAMs 2 as user **data memories** and one further DRAM 3 as error **data** ... **Memory module** including a **plurality** of integrated circuit **memory devices** and a **plurality** of **buffer** ... US20080046802A1 (en) *, 2006-08-18, 2008-02-21, Fujitsu Limited, **Memory controller** and ...

☆ 77 Cited by 82 Related articles All 4 versions »

Memory expansion module including multiple memory banks and a bank control circuit

T Wong, J Carrillo, J Robinson, C Fang - US Patent 6,414,868, 2002 - Google Patents

... US20060117152A1 (en) *, 2004-01-05, 2006-06-01, Smart **Modular** Technologies Inc., A California Corporation, Transparent four rank **memory module** for ... Multi-rank partial width **memory modules** ... System and method utilizing distributed byte-wise **buffers** on a **memory module** ...

☆ 77 Cited by 135 Related articles All 2 versions »

Semiconductor **memory module**

A Jakobs, H Ruckerbauer, M Kuzmerka - US Patent 7,061,784, 2006 - Google Patents

... Refresh management of **memory modules** ... Method and apparatus for stackable **modular** integrated circuits. DE102008010544A1 (en) *, 2008-02-22, 2009-09-17, Qimonda Ag, **Memory module** for use in eg personal computer, to store digital **data** in dynamic RAM, has termination ...

☆ ⓘ Cited by 85 Related articles All 4 versions ⓘ

Memory module arranged for **data** and parity bits

JH Neal, KA Poteet - US Patent 5,228,132, 1993 - Google Patents

... Despite rapid increases in the capacities of DRAMs, Static Random Access **Memories** (SRAMs) and nonvolatile ... 1998-02-13, 2005-11-22, Intel Corporation, **Memory module** having a **memory module controller controlling memory** transactions for a **plurality of memory devices** ...

☆ ⓘ Cited by 70 Related articles All 2 versions ⓘ

Buffer to multiply **memory** interface

JM Dodd, MW Williams, JB Halberd, RM Bonella ... - US Patent ..., 2003 - Google Patents

... However, in particular, the **memory modules** may be dynamic random access **memories** (DRAM), double **data** ... US20060117152A1 (en) *, 2004-01-05, 2006-06-01, Smart **Modular** Technologies Inc., A California Corporation, Transparent four rank **memory module** for standard ...

☆ ⓘ Cited by 212 Related articles All 2 versions ⓘ

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed January 3, 2019</i>	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	Ahmed, Zubair
Sheet	1	of	2	Attorney Docket Number	129980-5073-US

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
		2004/0105292	06-2004	Matsui	
		2006/0077731 A1	04-2006	Ware	
		2007/0217559 A1	09-2007	Stott	
		2010/0271092 A1	10-2010	Zerbe	
		2014/0029370 A1	01-30-2014	Koshizuka	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)				
		CN102576565B	09-30-2015	Netlist, Inc.		

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Netlist, Inc., Examination Decision of Request for Invalidation, CN201080039043.0, 30MAY2018, 43 pgs.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Record of Oral Hearing, filed June 19, 2018.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Final Written Decision, filed July 5, 2018.
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00667, Final Written Decision, filed July 18, 2018
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00668, Final Written Decision, filed July 18, 2018.
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, ORDER, Conduct of the Proceedings, filed August 2, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, Decision Granting Institution of <i>Inter Partes</i> Review, filed August 6, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Mehran Arjomand,' filed July 30, 2018

Examiner's Signature	/ZUBAIR AHMED/	Date Considered	02/07/2019
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed January 3, 2019</i>	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	Ahmed, Zubair
Sheet	2	of	2	Attorney Docket Number	129980-5073-US

		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Michael D. Hatcher,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, and Case No. IPR2018-00365, Decision Granting Institution of <i>Inter Partes</i> Review, filed June 29, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, Patent Owner's Objections to Petition Evidence, filed August 20, 2018
		Netlist, Inc., Examination Decision of Request for Invalidation, CN201080039043.0, 30MAY2018, 43 pgs.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Record of Oral Hearing, filed June 19, 2018.
		Inter Partes Review of US Patent No. 8,516,185 B2, Case No. IPR2017-00577, Final Written Decision, filed July 5, 2018.
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00667, Final Written Decision, filed July 18, 2018
		Inter Partes Review of US Patent No. 7,532,537 B2, Case No. IPR2017-00668, Final Written Decision, filed July 18, 2018.
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, ORDER, Conduct of the Proceedings, filed August 2, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, Decision Granting Institution of Inter Partes Review, filed August 6, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Mehran Arjomand,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00362, and Case No. IPR2018-00363, EXHIBIT 3001, 'email from Michael D. Hatcher,' filed July 30, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, and Case No. IPR2018-00365, Decision Granting Institution of Inter Partes Review, filed June 29, 2018
		Inter Partes Review of US Patent No. 9,606,907 B2, Case No. IPR2018-00364, Patent Owner's Objections to Petition Evidence, filed August 20, 2018

Examiner's Signature	/ZUBAIR AHMED/	Date Considered	02/07/2019
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	4201	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S2	683	S1 and DRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:13
S3	161	S2 and clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S4	4	S3 and regenerat\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:14
S5	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:16
S6	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:20
S7	231360	memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:24
S8	131	S7 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 09:25
S9	112860	memory adj module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14
S10	52438	S9 same control\$4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:14

S11	7381	S10 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:15
S12	37	S10 same (local adj clock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:16
S13	5209	S11 and (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:18
S14	1811	S11 same (generat\$3 or re\$1generat\$3 or synthesi\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:19
S15	177	S14 same (phase adj lock\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:20
S16	397	S14 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:42
S17	397	S14 same "buffers"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/10/31 18:43
S18	3	"7,532,537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S19	2	S18 and rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:00
S20	1	S18 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:03
S21	14	"8,001,434".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04
S22	1	S21 and group	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:04

S23	17609	group with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S24	1174	S23 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:05
S25	0	S24 same each	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:06
S26	294	S24 same (each or respective or correspond\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:08
S27	0	(group\$3 or unit) with from with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:14
S28	0	(group\$3 or unit) with each with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:15
S29	0	(group\$3 or unit) with from with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S30	1940	(group\$3 or unit) with (each or respective or correspond\$3) with rank	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:16
S31	331	S30 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 06:17
S32	8948	programmable adj delay	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S33	3551	S32 same clock	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:11
S34	804	S33 same memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12

S35	153	S34 same buffer	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2015/11/01 07:12
S36	38	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 11:42
S37	8	((memory or DRAM or SRAM) adj module) with (data adj buffer\$3)) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:13
S38	68	((memory or DRAM or SRAM) adj module) with buffer\$3) same group\$3 same clock\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 12:18
S39	4415	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S40	5	S39 and PLL	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/03/13 19:27
S41	0	G11C7/1066.cpc	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:00
S42	4534	G06F12/08.cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:01
S43	0	S42 and ((module near2 control) same (local near2 clock))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/08/01 08:02
S44	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:42
S45	553	S44 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 05:43
S46	25012	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:48

S47	27	S46 and ((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:49
S48	0	((local with (pll or (phase adj lock\$2 adj loop))) same (memory near2 module) same (program\$4 near3 register)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:50
S49	0	((local with (pll or (phase adj lock\$2 adj loop))) same (program\$4 near3 register) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:52
S50	35	((local with (pll or (phase adj lock\$2 adj loop))) same register) and (memory near2 module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/01/03 10:53
S51	2	"20130242680".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:18
S52	7122	Harashima	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:20
S53	482	Harashima and memory and (command\$1address or c\$1a)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:22
S54	27	Harashima and (memory adj module) and (command\$1address or c\$1a)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/15 09:23
S55	27229	(G11C7/1066 G11C7/1093 G06F12/0802 G11C29/023 G11C29/028 G11C5/04).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/16 07:44
S56	38	(module adj controller) with (command\$1address or C\$1A)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/16 07:45
S57	3	"7532537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/07/31 07:45
S58	2310	(711/120 711/119).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/09/28 05:54

S59	5	S58 and ((memory near2 module) with (C\$1A or (command adj address)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/09/28 05:54
S119	3	"20060077731".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:08
S120	2	"20070217559".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:15
S121	2	"20100271092".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:17
S122	3	"20140029370".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:21
S123	4	"8516185".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:27
S124	3	"7532537".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:28
S125	8	"9606907".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:29
S126	3	"20040105292".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2019/02/05 13:37

2/ 7/ 2019 9:07:42 AM
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 7, 2019</i>		
				Application Number	14/445,035	
				Filing Date	July 28, 2014	
				First Named Inventor	Hyun Lee	
				Art Unit	2132	
Examiner Name	Ahmed, Zubair					
Sheet	1	of	1	Attorney Docket Number	129980-5073-US	

PATENT DOCUMENTS						
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	
FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
NON PATENT LITERATURE DOCUMENTS						
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published				
		Netlist Inc., Examination Report, CN201480043599.3, 07DEC2018, 6 pgs. -NO TRANSLATION AVAILABLE-				

Examiner's Signature		Date Considered	
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Electronic Acknowledgement Receipt

EFS ID:	35359359
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Edith Fuentes
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073US
Receipt Date:	07-MAR-2019
Filing Date:	28-JUL-2014
Time Stamp:	18:50:19
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment		no			
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980-5073US_IDS.pdf	202881	yes	4
			86bae9bd81a28c8052a93ff2fbfa083d193bc75		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Transmittal Letter	1	3
	Information Disclosure Statement (IDS) Form (SB08)	4	4

Warnings:

Information:

2	Non Patent Literature	Netlist_Examination-Report-CN201480043599-3_07DEC2018.PDF	384142	no	6
			45f1af351a3f667497bffd034b8b54b3ead3637e		

Warnings:

Information:

Total Files Size (in bytes):		587023
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee et al.	Confirmation No.:	9700
Serial No.:	14/445,035	Art Unit:	2132
Filed:	July 28, 2014	Examiner:	Ahmed, Zubair
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION	Attorney Docket No.:	129980-5073-US

**INFORMATION DISCLOSURE STATEMENT
SUBMITTED UNDER 37 C.F.R. § 1.97(i)**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information that the applicant considers cumulative with respect to references already of record in the subject U.S. patent application or otherwise not material to the examination of the application. It is requested that the references cited herein be made of record in the application in accordance with rule 37 C.F.R. §1.97(i).

1. Enclosures accompanying this Information Disclosure Statement are:
 - 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
 - 1b. A legible copy of :
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
 - 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
 - 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.
2. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
 - ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);

- ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.
3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b or 3c)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. _____).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. _____).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

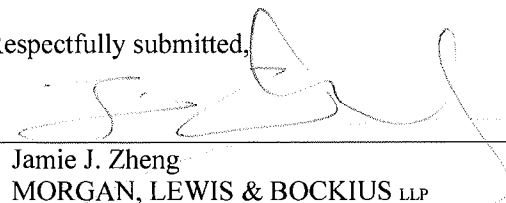
(Check either Item 4a and 4b or 4c)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. _____).
- 4c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. _____).
5. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(i) as it does not comply with § 1.97(a)-(d), Applicants request that this Information Disclosure Statement be placed in the file by the office.
6. ☐ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.

7. ☐ This application is a continuing application under 37 C.F.R. §1.53(b) or (d).
(Check appropriate Items 7a, 7b and/or 7c)
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☐ Copies of publications listed on Form PTO-1449 from prior application Serial No. _____, filed on _____, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☐ Copies of the publications listed on Form PTO-1449 were not previously cited in prior application Serial No. _____, filed on _____, and are provided herewith.
8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on _____. A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on _____.
9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:
(Check Item 9a, 9b, or 9c)
- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: March 7, 2019



Jamie J. Zheng
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

51,167
(Reg. No.)



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073US	9700

7590 03/12/2019
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER
AHMED, ZUBAIR

ART UNIT PAPER NUMBER
2132

NOTIFICATION DATE DELIVERY MODE
03/12/2019 ELECTRONIC

NOTICE OF NON-COMPLIANT INFORMATION DISCLOSURE STATEMENT

An Information Disclosure Statement (IDS) filed 3.7.19 in the above-identified application fails to meet the requirements of 37 CFR 1.97(d) for the reason(s) specified below. Accordingly, the IDS will be placed in the file, but the information referred to therein has not been considered.

The IDS is not compliant with 37 CFR 1.97(d) because:

- ☒ The IDS lacks a statement as specified in 37 CFR 1.97(e).
- ☐ The IDS lacks the fee set forth in 37 CFR 1.17(p).
- ☐ The IDS was filed after the issue fee was paid. Applicant may wish to consider filing a petition to withdraw the application from issue under 37 CFR 1.313(c) to have the IDS considered. See MPEP 1308.

571-272-4200 or 1-888-786-0101
Application Assistance Unit
Office of Data Management

PART B - FEE(S) TRANSMITTAL

Electronically filed May 10, 2019

Complete and send this form with applicable fee(s), to: MailMail Stop ISSUE FEE
Commissioner of Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
(571) 273-2885or Fax

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advanced orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS**MORGAN, LEWIS & BOCKIUS LLP**
1400 Page Mill Road
Palo Alto, CA 94304

Note: A certificate of mailing below can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing.

Certificate of Mailing

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)_____
(Signature)_____
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTY'S DOCKET NO.	CONFIRMATION NO.
14/445,035	07/28/2014	Hyun Lee	129980-5073-US	9700

TITLE OF INVENTION MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	1000	0	1000	05/14/2019

EXAMINER	ART UNIT	CLASS-SUB CLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list
(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed

1. **Morgan, Lewis & Bockius LLP**

2. _____

3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE:
NETLIST, INC.(B) RESIDENCE: (CITY and STATE OR COUNTRY)
Irvine, CAPlease check the appropriate assignee category or categories (will not be printed on the patent) ☐ Individual ☒ Corporation or other private group entity ☐ Government**4a. The following fee(s) are submitted:**☒ Issue Fee☐ Advanced Order - # of Copies _____**4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above.)**☐ A check in the amount of the fee(s) enclosed☐ Payment by credit card. Form PTO-2038 is attached.☒ The Director is hereby authorized to charge the required fee(s), or credit any overpayment, to Deposit Account Number 50-0310 (order no. 129980-5073-US).**5. Change in Small Entity Status (from status indicated previously)**☐ a. Applicant certifying micro entity status. See 37 CFR 1.29.

NOTE: Absent a valid certification of Micro Entity (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

☐ Applicant asserting small entity status. See 37 CFR 1.27.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

☒ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Jamie J. Zheng/

Date: May 10, 2019

Printed Name: Jamie J. Zheng

Registration No. 51,167

Electronic Patent Application Fee Transmittal

Application Number:	14445035			
Filing Date:	28-Jul-2014			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5073US			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1000	1000

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	35982860
Application Number:	14445035
International Application Number:	
Confirmation Number:	9700
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073US
Receipt Date:	10-MAY-2019
Filing Date:	28-JUL-2014
Time Stamp:	17:11:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	051319INTEFSW00004107500310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	129980_5073US_Issue_Fee.pdf	112837	no	1
			2f2b35298cb4d4ee478097cb200f5c021be4c114		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30179	no	2
			7e7b93b4c03eb16559dddb3773a338fc3fbcb30		

Warnings:

Information:

Total Files Size (in bytes):			143016
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	5	of	13	Attorney Docket Number	NT006US

US PATENT DOCUMENTS

Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	138.	7007175	02-28-2006	Chang et al.	
	139.	7046538	05-16-2006	Kinsley et al.	
Change(s) applied to document, /C.H.J/ 10/20/2018	140.	7047361	05-16-2006	Phison Electronics Corp	Chong et al.
	141.	7054179	05-30-2006	Cogdill et al.	
	142.	7065626	06-20-2006	Schumacher et al.	
	143.	7072231	07-04-2006	Pax	
	144.	7073041	07-04-2006	Dwyer et al.	
	145.	7078793	07-18-2006	Ruckerbauer et al.	
	146.	7093066	08-15-2006	Klein	
	147.	7120727	10-10-2006	Lee et al.	
	148.	7124260	10-17-2006	LaBerge et al.	
	149.	7127584	10-24-2006	Thompson et al.	
	150.	7130952	10-31-2006	Nanki et al.	
	151.	7133960	11-07-2006	Thompson et al.	
	152.	7133972	11-07-2006	Jeddeloh	
	153.	7142461	11-28-2006	Janzen	
	154.	7149841	12-12-2006	LaBerge	
	155.	7167967	01-23-2007	Bungo et al.	
	156.	7181591	02-20-2007	Tsai	
	157.	7200021	04-03-2007	Raghuram	
	158.	7227910	06-05-2007	Lipka	
	159.	7266639	09-04-2007	Raghuram	
	160.	7272709	09-18-2007	Zitlaw et al.	
	161.	7281079	10-09-2007	Bains et al.	
/Zubair Ahmed/ (03/13/2016)	162.	7286436	10-23-2007	Bhakta et al.	
	163.	7289386	10-30-2007	Bhakta et al.	
	164.	7346750	03-18-2008	Ishikawa	
	165.	7356639	04-08-2008	Perego et al.	
	166.	7370238	05-06-2008	Billick et al.	
	167.	7437591	10-14-2008	Wong	
	168.	7461182	12-02-2008	Fukushima et al.	
	169.	7464225	12-09-2008	Tsern	
	170.	7471538	12-30-2008	Hofstra	
	171.	7532537	05-12-2009	Solomon et al.	
	172.	7619912	11-17-2009	Bhakta et al.	

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DATE CONSIDERED 03/13/2016

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	4	of	13	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	102.	6658509	12-02-2003	Bonella et al.	
	103.	6674684	01-06-2004	Shen	
	104.	6681301	01-20-2004	Mehta et al.	
	105.	6683372	01-27-2004	Wong et al.	
	106.	6697888	02-24-2004	Halbert et al.	
	107.	6705877	03-16-2004	Li et al.	
	108.	6717885	04-06-2004	Lai	
	109.	6721843	04-13-2004	Lexar Media Inc. Estakhri	
	110.	6721860	04-13-2004	Klein	
	111.	6738880	05-18-2004	Lai et al.	
	112.	6742098	05-25-2004	Halbert et al.	
	113.	6754797	06-22-2004	Wu et al.	
	114.	6785189	08-31-2004	Jacobs et al.	
	115.	6788592	09-07-2004	Nakata et al.	
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	117.	6807650	10-19-2004	Lamb et al.	
	118.	6813196	11-02-2004	Park et al.	
	119.	6832284	12-14-2004	Perego et al.	
	120.	6832303	12-14-2004	Tanaka	
	121.	6834014	12-21-2004	Yoo et al.	
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	123.	6880094	04-12-2005	LaBerge	
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	125.	6912615	06-28-2005	Nicolai	
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	127.	6925028	08-02-2005	Hosokawa et al.	
	128.	6944694	09-13-2005	Pax	
	129.	6950366	09-27-2005	Lapidus et al.	
	130.	6954281	10-11-2005	Fukuda et al.	
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	133.	6982892	01-03-2006	Lee et al.	
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	136.	6996686	02-07-2006	Doblar et al.	
	137.	7007130	02-28-2006	Holman	

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	3	of	13	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	67.	6154418	11-28-2000	Li	
	68.	6154419	11-28-2000	Shakkarwar	
	69.	6185654	02-06-2001	Van Doren	
	70.	6188641	02-13-2001	Uchida	
	71.	6205516	03-20-2001	Usami	
	72.	6209074	03-01-2001	Dell et al.	
	73.	6226709	05-01-2001	Goodwin et al.	
	74.	6226736	05-01-2001	Niot	
	75.	6233650	05-15-2001	Johnson et al.	
	76.	6247088	06-12-2001	Seo et al.	
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	78.	6349051	02-19-2002	Micron Technology Inc	Klein
	79.	6400637	06-18-2002	Akamatsu et al.	
	80.	6408356	06-18-2002	Dell	
	81.	6414868	07-02-2002	Wong et al.	
	82.	6415374	07-02-2002	Faue et al.	
	83.	6438062	08-20-2002	Karabatsos (Curtis)	
	84.	6446158	09-03-2002	Karabatsos	
	85.	6446184	09-03-2002	Dell et al.	
	86.	6453381	09-17-2002	Yuan et al.	
	87.	6470417	10-22-2002	Kolor et al.	
	88.	6480439	11-12-2002	Tokutome et al.	
	89.	6487102	11-26-2002	Halbert et al.	
	90.	6502161	12-31-2002	Perego et al.	
	91.	6518794	02-11-2003	Coteus et al.	
	92.	6526473	02-25-2003	Kim	
	93.	6530007	03-04-2003	Olarig et al.	
	94.	6530033	03-04-2003	Raynham et al.	
	95.	6553450	04-22-2003	Dodd et al.	
	96.	6618320	09-09-2003	Hasegawa et al.	
	97.	6621496	09-16-2003	Ryan	
	98.	6625081	09-23-2003	Roohparvar et al.	
	99.	6625687	09-23-2003	Halbert et al.	
	100.	6636935	10-21-2003	Ware et al.	
	101.	6646949	11-11-2003	Ellis et al.	

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/Zubair Ahmed/ (03/13/2016)

DATE CONSIDERED

03/13/2016

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	2	of	13	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	34.	5602999	02-11-1997	Hyatt; Gilbert P.	
Change(s) applied to document, /C.H.J./ 10/20/2018	35.	5638534	06-10-1997	Samsung Electronics Co Ltd	Mote, Jr.
	36.	5655153	08-05-1997	EMC CORP	Sandorfi
	37.	5699542	12-16-1997	Mehta et al.	
	38.	5702984	12-30-1997	Bertin et al.	
	39.	5703826	12-30-1997	Hush et al.	
	40.	5745914	04-28-1998	Connolly et al.	
	41.	5764590	06-09-1998	Mitsubishi Electric Corp	Iwamoto et al.
	42.	5784705	07-21-1998	Leung, Wingyu	
	43.	5784705	07-21-1998	Jr Albert Pesak	
	44.	5802395	09-01-1998	Connolly et al.	
	45.	5805520	09-08-1998	Anglada et al.	
	46.	5822251	10-13-1998	Bruce et al.	
	47.	5909388	06-01-1999	Siemens Ag	Mueller
	48.	5926827	07-20-1999	Dell et al.	
	49.	5926839	07-20-1999	IBM	Katayama
	50.	5953215	09-14-1999	Karabatsos, Chris	
	51.	5953280	09-14-1999	Matsui	
	52.	5958025	09-01-1999	Nippon Electric Co	Sonobe
	53.	5959930	09-28-1999	Sakurai	
	54.	5963464	10-05-1999	Dell et al.	
	55.	5966736	10-12-1999	Gittinger et al.	
	56.	5974493	10-26-1999	Mitsubishi Electric Corp	Okumura et al.
	57.	6011710	01-04-2000	Wiggers	
	58.	6018787	01-25-2000	Ip	
	59.	6044032	03-28-2000	Li	
	60.	6070217	05-30-2000	Connolly et al.	
	61.	6070227	05-30-2000	Rokicki	
	62.	6097652	08-01-2000	Roh	
	63.	6108745	08-22-2000	Gupta et al.	
	64.	6115278	09-05-2000	Deneroff et al.	
	65.	6134638	10-17-2000	Olarig et al.	
	66.	6151271	11-21-2000	Lee	

EXAMINER

/Zubair Ahmed/ (03/13/2016)

DATE CONSIDERED 03/13/2016

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /Z.A./

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	1	of	13	Attorney Docket Number	NT006US

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
Change(s) applied to document, /C.H.J./ 10/20/2018	1.	4218740	08-19-1980	MOTOROLA INC	Bennett et al.
	2.	4249253	02-03-1981	Gentili et al.	
	3.	4368515	01-11-1983	Nielsen	
	4.	4392212	07-05-1983	Miyasaka et al.	
	5.	4571676	02-18-1986	Mantellina et al.	
	6.	4592011	05-27-1986	Mantellina et al.	
	7.	4633429	12-30-1986	Lewandowski et al.	
	8.	4670748	06-02-1987	Williams	
	9.	4706214	11-10-1987	Mitsubishi Denki Kabushiki Kaisha	Kassai
	10.	4866603	09-02-1989	Chiba	
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EXAMINER /Zubair Ahmed/ (03/13/2016)

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/445,035
				Filing Date	July 28, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2132
				Examiner Name	AHMED, ZUBAIR
Sheet	1	of	2	Attorney Docket Number	NT006US

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Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/445,035	06/18/2019	10324841	129980-5073US	9700

79141 7590 05/29/2019
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 274 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

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APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
14/445,035	07/28/2014	411121

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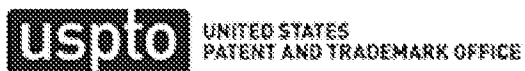
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Provisional Application for Patent Cover Sheet

This is a request for filing a PROVISIONAL APPLICATION FOR PATENT under 37 CFR 1.53(c)

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Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING
DISTRIBUTED DATA BUFFERS

Attorney Docket Number (if applicable)

NT006

Correspondence Address

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☐ Firm or Individual Name

Customer Number

79141

The invention was made by an agency of the United States Government or under a contract with an agency of the United States Government.

☒ No.

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Entity Status

Applicant claims small entity status under 37 CFR 1.27

- ☒ Yes, applicant qualifies for small entity status under 37 CFR 1.27
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MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS

FIELD

[0001] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0002] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0003] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0004] The memory capacity or memory density of a memory module increases with the

number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0005] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0006] For memory devices in such as a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so there is an equal length to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0007] In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0009] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

[0010] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0011] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0012] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0013] FIG. 6 is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0014] FIG. 7 is a diagram illustrating control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 8 is a timing diagram illustrating alignment of module control signals with respect to module clock signals.

[0016] FIG. 9 is a diagram illustrating a metastability detection circuit and signal adjustment circuit in a data buffer according to certain embodiments.

[0017] FIGS. 10A-10C are diagrams illustrating a metastability detection circuit according to certain embodiments.

[0018] FIGS. 10D is a diagram illustrating a signal adjustment circuit according to certain embodiments.

[0019] FIGS. 11A-11B are diagrams illustrating a metastability detection circuit and signal adjustment circuit, respectively, according to certain embodiments.

[0020] FIGS. 12A-12B are a timing diagrams illustrating a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0021] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0022] FIG. 14 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0023] FIG. 15 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0024] FIG. 16 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0025] FIG. 17 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0026] FIG. 18 is a flowchart illustrating a method for data edge alignment according to embodiments.

[0027] FIG. 19 is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

DESCRIPTION OF EMBODIMENTS

[0028] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation devices (I.D.) or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the

memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module command signals and a set of module control signals in response to each memory command from the memory controller. The module command signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the module control signals (referred sometimes herein as module control signals) are transmitted by the module control device to the buffer circuits via module control signal lines.

[0029] The buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the respective groups of memory devices. Thus, during certain high speed operations, each module control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to select a subgroup in the respective group of memory devices to communicate data with the memory controller in response to the module control signals. Thus, the memory module can have more ranks of memory devices than what is supported by the memory controller.

[0030] In one embodiment, each buffer circuit includes metastability detection circuits to detect metastability condition in the module control signals and signal adjustment circuits to adjust the module control signals and/or a module clock signal to mitigate any metastability condition in the module control signals.

[0031] Further, in one embodiment, each buffer circuit includes signal alignment circuits that determine, during a write operation, a time interval between a time when one or more module control signals are received from the module control circuit and a time when a strobe or data signal is received from the memory controller. This time interval is used

during a subsequent read operation to time transmission of read data to the memory controller, such that the read data arrives at the memory controller within a time limit in accordance with a read latency parameter associated with the memory system.

[0032] FIG. 1 shows a system 100 including a memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or isolation devices 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the isolation devices 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119.

[0033] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0034] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any

other components or connections that provide connections for signal transmission.

[0035] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0036] Referring to FIG. 2A, which illustrates one memory module 110 according to an embodiment, the module control device 116 receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120 and generates module command signals and module control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0037] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0038] Examples of module command signals include, but are not limited to module /CS signals, which can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column

address signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module command signals may also include the module clock signal CK.

[0039] Examples of module control signals include, but are not limited to a mode signal (MODE), which specifies a mode of operation (e.g., test mode or operating mode) for the isolation devices 118; one or more enable signals, which are used by an isolation device to select one or more subgroups of memory devices to communicate data with the memory controller; and one or more ODT signals, which are used by the isolation devices to set up on-die termination for the data/strobe signals. In one embodiment, the module control signals are transmitted to the isolation devices 118 via respective module control signal lines 230. Alternatively, the module control signals can be packetized before being transmitted to the isolation devices 118 via the module control signal lines and decoded/processed at the isolation devices.

[0040] Module control device 116 transmits the module command signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module command signals to receive write data or output read data as if the module command signals were from a memory controller. The module control device transmits the module control signals together with the module clock signal CK to the isolation devices 118 via module control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the isolation devices 118 share a same set of module control signal lines 230.

[0041] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N,1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N,2}$,

and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11} , M_{12} , M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0042] As shown, the isolation devices 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, isolation device ID-1 among the isolation devices 118 is associated with the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, isolation devices ID- i among the isolation devices 118 is associated with the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on.

[0043] In one embodiment, each group or sets of memory devices are coupled to the associated isolation device 118 via a set of module data/strobe lines 210. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated isolation device 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated isolation device 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated isolation device via an associated second subset of module data/strobe lines YB, as shown. For example, memory

devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the isolation device ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the isolation device ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0044] In one embodiment, the isolation devices 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each isolation device 118 is configured to select a subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the module control signals, such that the memory module can include more ranks than what is supported by the MCH 101. Further, each isolation devices 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0045] In one embodiment, the isolation devices 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to the respective groups of memory devices. For example, isolation device ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , and isolation device ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} . In one embodiment, the first position is between the first group of memory devices and an edge 201 of the module board 119 where connections (not shown) to the data/strobe signal lines 130 are placed, and i^{th} position is between the i^{th} group of memory devices and the edge 201 of the module board 119. In one embodiment, the isolation devices 118 are distributed along the edge 201 of the memory module 110. In one embodiment, each isolation device 118 is a separate integrated circuit device packaged either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the module control signal lines 230 include signal traces

formed on and/or in the module board 119.

[0046] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0047] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, input and output signals electrical and logical level control information for the control circuit 116, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0048] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the isolation devices 118 before being transmitted to the memory devices 112 by the isolation devices 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the isolation devices before being transmitted to the MCH via the system

data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the isolation devices 118, and the system 100 depends on the isolation devices 118 to properly time the transmission of the read data and strobe signals to the MCH 101.

[0049] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0050] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112, and transmits module control signals and a buffered clock signal to the isolation devices 118, in a fly-by configuration. As the speed of memory operations increase, issues can arise with respect to signal alignment for input, output delay variation due process, voltage and temperature (PVT) variations, synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation due to charge build up and timing interface calibration drift during operation due to environment change can also create issues.

[0051] For example, load reduction mechanism in the isolation devices 118 would provide a single data bus interface to the respective set of memory devices, which is hidden from the system memory controller 101. Thus, a long sequence of interface timing training may be required due to limited controllability of the system memory controller 101 over the interface between the memory devices 112 and the associated isolation devices 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0052] Moreover, clock skew amongst the memory devices 112 and the associated isolation devices 118 due to the distributed architecture of the memory module 110 can cause synchronization issues. As the speed of memory operation increase, data period can become very close to the signal propagation delay time. Thus, such issues cannot simply be addressed by pipelining the data paths, as variation of the signal propagation time through I/Os becomes a very significant portion of a data period.

[0053] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112, and transmits the module control signals and the buffered clock signal to the data buffers 118, in a fly-by arrangement. The memory devices 112 do not receive the buffered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by the respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 is also responsible for providing the correct control signal timing between the control circuit 116 and the respective set of memory devices 112.

[0054] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each respective set of memory devices 112, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals coupled to the memory devices 112, programmable phase adjustment for the data and data-strobe signals coupled to the system memory controller 101, programmable phase adjustment related to at least one control signal that is coupled to the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data

buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0055] In certain embodiments, operations of the isolation devices 118 are controlled by the module control signals from the module control circuit 116, which generates the module control signals according to the C/A signals received from the MCH. Thus, the module control signals need to be properly received by the isolation devices 118 to insure their proper operation. In one embodiment, the module control signals are transmitted together with the module clock signal CK, which is also generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the module control signals. Since the isolation devices 118 are distributed across the memory module, the module control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the module control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received module control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the module control signals and signal adjustment circuits to adjust the module control signals and/or the module clock signal to mitigate any metastability condition in the module control signals, as explained in further detail below.

[0056] Because the isolation devices 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the module control signals to travel along the module control signals lines 230 from the module control device 116 to the farthest positioned isolation devices 118, such as isolation device ID-1 and isolation device ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of module control signals may reach different isolation devices 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a module control signal would travel about 15 cm during one

clock cycle. When the clock frequency increases to 1600 MHz, a module control signal would travel less than 8 cm during one clock cycle. Thus, a module control signal line can have multiple module control signals on the line at the same time, i.e., before one module control signal reaches an end of the signal line, another module control signal appear on the signal line.

[0057] With the isolation devices 118 receiving module control signals at different times across more than one clock cycle, the module control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the isolation devices 118. In one embodiment, each isolation devices includes signal alignment circuits that determine, during a write operation, a time interval between a time when one or more module control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used during a subsequent read operation to time the transmission of read data to the MCH 101, such that the read data follows a read command by a read latency value associated with the system 100, as explained in more detail below.

[0058] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0059] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of data signals $DQ_0, DQ_1, \dots, DQ_{n-1}$, and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated isolation device 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple

subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0060] Each isolation device 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each isolation device 118 further includes an ID control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} . The ID control circuit 310 also receives the module clock signal CK and the module control signals via the module control signal lines 230, and outputs ID control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed module control signals, a delay signal DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the module control signals, as explained in more detail below.

[0061] In certain embodiments, the ID control circuit 310 also provides a delay signal DS, which is used by the DQ routing circuits 320 to align read data output by the isolation device 118 with read data output by the other isolation devices 118, as explained in further detail below. In certain embodiments, the ID control circuit 310 regenerates a clock signal from the module clock signal CK, which can have a programmable delay from the module clock signal. The regenerated clock signal is used as the clock signal CK0 and a clock signal CKM that is provided to the corresponding set of memory devices, as explained in more detail below.

[0062] The memory devices 112 are coupled to the isolation devices 118 via a same set of module data/strobe signal lines or different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the isolation device ID-1 via a same set of module data lines $Y-1_0$, $Y-1_1$, ..., $Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the isolation devices to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0063] Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup in the first group of memory devices, are coupled to the isolation device ID-1 via the module data lines $YA-1_0$, $YA-1_1$, ..., $YA-1_n$ and module strobe line $YA-1_{DQS}$ and memory devices M_{12} and M_{14} , which form another subgroup in the first group of memory devices, are coupled to the isolation device ID-1 via the module data lines $YB-1_0$, $YB-1_1$, ..., $YB-1_n$ and module strobe line $YB-1_{DQS}$. Memory devices coupled to the same isolation devices can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same isolation devices may be placed side by side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated isolation device.

[0064] Multiple memory devices having a data width that is less than a data width of the isolation devices 118 may be used in place of one of the memory devices 112, which has the same data width as that of the isolation devices. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the isolation device ID-1 via module data lines $YA-1_0$, ..., $YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory circuit M_{11-2} is coupled to the isolation device ID-1 via module data lines $YA-1_4$, ..., $YA-1_7$ and module strobe line $YA-1_{DQS-2}$.

[0065] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4}

may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the isolation device ID-1 via module data lines $YA-1_0, \dots, YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory device M_{11-2} is coupled to the isolation device ID-1 via module data lines $YA-1_4, \dots, YA-1_7$ and module strobe line $YA-1_{DQS-2}$, and so on.

[0066] FIG. 6 illustrates the ID control circuit 310 in an isolation device 118. As shown, the ID control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, and to output a buffered clock signal CK0. The ID control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to the corresponding system DQS signal line 324 and on another side to the corresponding module DQS signal lines YA_{DQS} and YB_{DQS} . The ID control circuit 310 further includes a receiver circuit 630 with respect to each of at least some of the module control signals (MCS) to receive a respective one of the module control signals. The ID control circuit 310 further includes a command processing circuit 640 that provides the received, decoded, and/or otherwise processed module control signals 330 to the DQ routing circuits 320 and the strobe routing circuit 620 either directly or after further processing, if needed. The received/decoded/processed module control signals may include, for example, one or more enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the isolation device is associated.

[0067] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and output either a write strobe $WDQS$ or read strobe $RDQS$ to the DQ routing circuits 320. In one embodiment, the ID control circuit 310 further includes a delay control circuit 650 that receives one of the module control signals and either a data signal or a strobe signal and determines a delay amount to be used by the DQ routing circuit 320 and the strobe routing circuit 620. The delay amount is provided to the DQ routing circuit 320 and the strobe routing circuit in a delay signal DS.

[0068] In a receiver circuit 630, the respective MCS is received in accordance with the buffered clock signal CK0. In one embodiment, receiver circuit 630 samples the respective MCS using rising (or falling) edges of the buffered clock CK0. Since the isolation devices 118 are distributed across the memory module 110 at positions corresponding to the respective groups of memory devices, the module control signal lines 230 that carry the MCS to the isolation devices can stretch over a distance of more than 10 centimeters, as shown in FIG. 7. As the MCS and CK0 travel along their respective module control signal lines 710 and 720, they can become misaligned with each other when they reach the input pins 730 of an isolation device 118.

[0069] For example, a module control signal, like the MCS 810 shown in FIG. 8, can be perfectly aligned with the module clock signal CK, with a rising edge 801 of the module clock signal CK being at a center of a data eye 802, when the MCS signal and the clock signal leave the module control circuit 116. When the module control signal and the module clock signal reach an isolation device, however, their alignment can become shifted like the MCS 820 with respect to the CK signal, i.e., the rising edge 801 of the clock signal is near a left edge of a data eye of the MCS 820, barely providing enough set up time for proper sampling of the module control signal. Or, the module control signal, like the MCS 830, can be shifted with respect to the module clock signal such that a rising edge 801 of the clock signal is near a right edge of a data eye of the MCS, barely providing enough hold time for proper sampling of the module control signal. Or, even worse, the module control signal, like the MCS 840, can be so shifted with respect to the module clock signal such that a rising edge 801 of the clock signal falls in the glitches 803 at the edge of a data eye of the MCS, meaning that the sampled results could be metastable.

[0070] In one embodiment, as shown in FIG. 9, a receiver circuit 630 includes a metastability detection circuit (MDC) 910 to determine a metastability condition in a corresponding module control signal MCS0. In one embodiment, the MDC 910 generates at least one delayed version of the buffered clock signal CK and at least one delayed version of the corresponding MCS0. The MDC 910 also generates one or more metastability indicators and outputs the one or more metastability indicators via lines 912

and/or 914.

[0071] The receiver circuit 630 further includes a signal selection circuit 920 that receives the buffered clock CK and the at least one delayed version of the buffered clock via signal lines 916. The signal selection circuit 920 also receives the corresponding MCS and the at least one delayed version of the corresponding MCS via signal lines 918. The signal selection circuit 920 selects a clock signal CK_i from among the buffered clock CK and the at least one delayed version of the buffered clock based on one or more of the metastability indicators. The signal selection circuit 920 may also select an MCS signal MCS_i from among the corresponding MCS and the at least one delayed version of the corresponding MCS based on at least one other metastability indicator.

[0072] The receiver circuit 630 further includes a sampler or register circuit 930 that samples the selected module control signal MCS_i according to the selected clock signal CK_i and outputs the sampled signal as the received module control signal, which is provided to the command processing circuit 640 for further processing (if needed) before being provided to the DQ routing circuits 320 and DQS routing circuit 620.

[0073] FIG. 10A illustrates an MDC 910 according to one embodiment. As shown, the MDC 910 includes a delay circuit 1012 that generates a delayed version MCS_1 of the corresponding MCS_0 by adding a predetermined amount of delay (e.g., 10ps) to MCS_0 . MDC 910 also includes a delay circuit 1016 that generates a delayed version CK_1 of the clock signal CK_0 by adding a predetermined amount of delay to CK_0 . In one embodiment, CK_1 is delayed from CK_0 by about 1/10th of a clock cycle, e.g., 50-70 ps for an operating frequency of about 1600 MHz. The MDC 910 further includes a sampler circuit 1042 that samples MCS_1 according to CK_0 and outputs a sampled result A, a sampler circuit 1044 that samples MCS_0 according to CK_0 and outputs a sampled result B, and a sampler circuit 1046 that samples MCS_0 according to CK_1 and outputs a sampled result C. The MDC 910 further includes a logic circuit (e.g., a majority decision circuit) that generates metastability indicators Z1 and Z2 based on the sampled results A, B, and C.

[0074] In one embodiment, Z1 is the result of a logic operation (e.g., an XNOR operation) on the sampled result, e.g., $Z1 = \overline{A \oplus B}$, and Z2 is the result of another logic operation on the sampled results, e.g., $Z2 = \overline{B \oplus C}$. Thus, as shown in FIG. 10B and Table 1 below, when a metastability condition of insufficient hold time occurs, i.e., a rising clock edge 1061 of CK0 is close to the right side of a data eye where glitches at the edges of the data eyes can make C unpredictable, A and B can be in agreement (i.e., Z1 is true) while B and C are likely not in agreement (i.e., Z2 is false). FIG. 10C illustrates a metastability condition when there is insufficient set-up time. As shown in FIG. 10C and Table 1 below, a rising clock edge 1061 of CK0 is close to the left side of a data eye where glitches at the edges of the data eyes can make A unpredictable. Thus, A and B can be in disagreement so Z1 is false while B and C can be in agreement so Z2 is true. Not shown in the figures is the situation that all A, B, and C are in agreement, meaning that both the rising clock edge 1061 of CK0 and the rising clock edge 1062 of CK1 are near the middle of an MCS0 data eye so there is no metastability issues and both Z1 and Z2 are true, as shown in Table 1.

[0075] FIG. 10D illustrates a signal selection circuit 920 according to an embodiment. As shown, in one embodiment, the signal selection circuit 920 includes a first multiplexor 1071 that selects between CK0 and CK1 based on the metastability indicator Z1, and a second multiplexor 1072 that selects between MCS0 and MCS1 based on the metastability indicator Z2. Thus, as shown in Table 1, where a metastability condition of insufficient hold time occurs, Z1=1 and Z2=0, and MCS1 is output from multiplexor 1071 while CK0 is output from multiplexor 1072. Sampler 930 thus samples MCS1 according to the rising edges of CK0. Thus, more hold time is provided to mitigate the metastability condition since MCS1 is shifted from MCS0 toward the right.

[0076] On the other hand, where a metastability condition of insufficient set-up time occurs, Z1=0 and Z2=1, and CK1 is output from multiplexor 1071 while MCS0 is output from multiplexor 1072. Sampler 930 thus samples MCS0 according to the rising edges of CK1. Since CK1 is shifted from CK0 toward the right, more set-up time is provided to mitigate the metastability condition.

[0077] In the case when no metastability is detected, Z1=1 and Z2=1, and CK0 is output from multiplexor 1071 while MCS0 is output from multiplexor 1072. So, the unshifted module control signal is sampled according to the unshifted module clock signal.

Table 1 Metastability Detection and Signal Selection

Sampler Output			MS Indicators		MS Condition	Signal Selection	
A	B	C	Z1	Z2		CK	MCS
D1	D1	D2	1	0	insufficient hold time	CK0	MCS1
D1	D2	D2	0	1	insufficient set-up time	CK1	MCS0
D1	D1	D1	1	1	no metastability	CK0	MCS0

[0078] FIGS. 10A-10D illustrate a relatively simple implementation of the metastability detection circuit (MDC) 910 where only three different sample points are provided to detect metastability condition in the module control signal. In general, the MDC 910 may generate more delayed versions of the module clock signal CK0 and/or the corresponding module control signal MCS0, and may include more sampler circuits to sample any additional delayed versions of the module control signal according to either the module clock signal or one of the delayed versions of the module clock signal. For example, as shown in FIG. 11A, the MDC 910 can include a plurality of delay circuits 1102 that generate m delayed versions of MCS0, e.g., MCS1, MCS2, ... MCS m , and m delayed versions of CK0, e.g., CK1, CK2, ..., CK m . The MDC 910 can include sampler circuits 1104 that sample MCS0 according to CK0, CK1, ..., CK m , respectively, and sampler circuits 1104 that sample MCS0, MCS1, MCS2, ... MCS m according to CK0, respectively. The outputs of the samplers 1104 are provided to a logic circuit 1120, which determines a metastability condition in MCK0 based on the sampler outputs using, for example, a majority decision logic. The logic circuit 1120 outputs a first metastability indicator on line(s) 912 and a second metastability indicator on line(s) 914.

[0079] FIG. 11B illustrates a signal selection circuit 920 according to an embodiment. As shown, in one embodiment, the signal selection circuit 920 includes a first multiplexor 1171 that selects between CK0, CK1, ..., CK m based on the metastability indicator

provided on line(s) 912, and a second multiplexor 1172 that selects between MCS0, MCS1, ..., MCS m based on the metastability indicator provided on line(s) 914, such that the rising edges of the selected clock signal, e.g., Ck i , are close to the middle of the respective data eyes in the selected module control signal, e.g., MCS i . The selected signals MCS i and Ck i are provided to the sampler 930, which samples MCS i according to the rising edges of CK i .

[0080] As stated above, in certain embodiments, since the isolation devices 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, the isolation devices 118 includes signal alignment mechanism to time the transmission of read data signals based on timing information derived from a prior write operation, as discussed further below.

[0081] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by an isolation device 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another isolation device 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the isolation device 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the isolation device 118 since both these signals are received by the isolation device. Based on such determination, the isolation device 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the isolation device 118 to properly time transmission of read data to the MCH, as explained further below.

[0082] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an isolation device 118 at time t7, which afterwards receives at time t8 read data signals (not shown) and one or more strobe signal DQS from the respective group of memory devices. Note that the same enable signal may be received by another isolation device 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the isolation devices 118.

[0083] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in certain embodiments, the isolation device can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the isolation device to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0084] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the ID control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the module control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the module control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit, i.e., the delay signal DS, would indicate a time interval from when the write preamble is detected and when the one of the module control signal is received.

[0085] FIG. 14 illustrates a DQ or DQS routing circuit 320 or 620 according to an

embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 1401 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 1402 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS} , or YA/YA_{DQS} and YB/YB_{DQS} . The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 1410 that buffers write data/strobe, and a write data/strobe receiver 1420 that samples the write data/strobe. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 1430 that are selectable or can be selectively enabled by one or more of the module control signals, such as the enable signals ENA and ENB.

[0086] The DQS routing circuit further includes a plurality of read paths 1450 that are selectable by the one or more of the module control signals. Output from the selected read path is delayed in a delay circuit 1460 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 1470. The sampled read data/strobe is transmitted by transmitter 1480 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 1401.

[0087] FIG. 15 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 1501 that is coupled to a corresponding DQS signal line 324, a second DQS pin 1502A that is coupled to a corresponding module DQS line YA_{DQS} , a third DQS pin 1502B that is coupled to a corresponding module DQS line YB_{DQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 1501 and the second DQS pin 1502A and a second write strobe path coupled between the first DQS pin 1501 and the third DQS pin 1502B. The first write strobe path includes a write strobe buffer 1510 that buffers a write strobe, a write strobe receiver 1520 that samples the write strobe according to the buffered module signal CK0. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 1530A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line YA_{DQS} . The second write strobe path includes the write strobe buffer 1510, the write strobe receiver 1520, and a second write strobe transmitter 1530B that transmits the write strobe to one or more memory devices

112 coupled to the module strobe line YB_{DQS} . The first and second write strobe transmitters, 1530A and 1530B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0088] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 1501 and a selected one of the second and third DQS pins 1502A and 1502B. In the read strobe path, a select circuit 1550 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 1502A or a read strobe signal received via DQS pin 1502B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 1560 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 1570 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 1580 onto the corresponding strobe signal line 324 via the first DQS pin 1501.

[0089] FIG. 16 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1601 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1602A that is coupled to a corresponding module DQ line YA_{DQ} , a third DQ pin 1602B that is coupled to a corresponding module DQ line YB_{DQ} . The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1601 and the second DQ pin 1602A and a second write data path coupled between the first DQ pin 1601 and the third DQ pin 1602B. The first write data path includes a write data buffer 1610, a write data receiver 1620 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, and a first write data transmitter 1630A that transmits the write data to one or more memory devices 112 coupled to the module data line YA_{DQ} . The second write data path includes the write data buffer 1610, the write data receiver 1620, and a second write data transmitter 1630B that transmits the write data to one or more memory devices 112 coupled to the module data line YB_{DQ} . The first and second write data transmitters, 1530A and 1530B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first

write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0090] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1601 and a selected one of the second and third DQ pins 1602A and 1602B. In the read data path, a select circuit 1650 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1602A or a read data signal received via DQ pin 1602B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1660 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1670 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1680 onto the corresponding data signal line 130 via the first DQ pin 1601.

[0091] FIG. 17 illustrate a delay circuit 1560 or 1660 according to an embodiment. As shown, the delay circuit 1560 or 1660 includes a plurality of delay stages, such as delay stages 1710, 1720, and 1730, each delaying a read data or read strobe signal from the select circuit 1550/1650 by a predetermined amount. The delay circuit 1560 or 1660 further includes a select circuit 1740 (e.g., a multiplexor) that selects from among the read data or read strobe signal and the outputs from the delay stages according to the delay signal DS. The output of the select circuit 1740, is provided to the sampler circuit 1570 or 1670, either directly or after being buffered by a buffer circuit 1750.

[0092] Thus, as shown in FIG. 18, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1800. In the method, during a write operation, one or more module control signals are received by an isolation device 118 from a module control circuit or module controller 116 (1810). The module controller 116 generates the one or more module control signals in response to C/A signals representing a write command from the MCH 101. The one or more module control signals are used to control the isolation device 118. For example, the one or more module control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory

devices coupled to the isolation device 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the isolation device 118 from the MCH 101 (1820). In one embodiment, upon receiving the one or more first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0093] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is a set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the isolation device 118. The time interval CED can be used by the isolation device 118 to properly time the transmission of read data to the MCH 101, as described above and explained further below.

[0094] As shown in FIG. 18, a delay signal DS is generated according to the time interval EWD (1830). Concurrent to receiving the write strobe signal DQS, the isolation device 118 also receives a set of write data signals DQ (1840). The received write data signals are transmitted to the subgroup of memory devices (1850), which are selected from the group of memory devices coupled to the isolation device 118 by the one or more first enable signals.

[0095] During a read operation, another set of module control signals including, for example, one or more second enable signals, are received by the isolation device 118 from the module controller 116 (1860). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the isolation device 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1870). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ

signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

[0096] In certain embodiments, especially the embodiments shown in FIG. 2D, the delay circuits 1560 and 1660 shown in FIGS. 15 and 16 are not needed to provide alignment of the read data. As shown in FIG. 19, the ID control circuit 310 includes a clock regeneration circuit 1920 that regenerates the clock signal CK received from the control circuit 116, according to the delay signal DS. The regenerated clock signals CK0 and CKM each includes a proper amount of delay as compared to the clock signal CK. The clock CK0 is provided to the strobe routing circuit 620 so that the strobe signals are properly timed to result in proper data alignment. The regenerated clock signal CKM is provided to the respective set of memory devices so that the respective data buffer 118 and the respective set of memory devices are locally synchronized.

WE CLAIM:

1. A memory module to operate in a memory system with a memory controller, the memory system operating according to a system clock, comprising:

memory devices organized in groups, each group including at least one memory device; and

a plurality of buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices, the respective buffer circuit including data paths for communicating data between the memory controller and the respective group of memory devices; and

wherein the respective group of memory devices operate in accordance with a clock signal received from the respective buffer circuit.

2. A memory module to operate in a memory system with a memory controller, the memory system operating according to a system clock, comprising:

a module control device to receive command signals from the memory controller and to output module command signals, module control signals and a module clock signal, the module clock signal being derived from the system clock signal;

memory devices organized in groups, each group including at least one memory device, the memory devices receiving the module command signals from the module control device; and

a plurality of buffer circuits to receive the module control signals, a respective buffer circuit corresponding to a respective group of memory devices, the respective buffer circuit including data paths for communicating data between the memory controller and the respective group of memory devices, the data paths being controlled by at least one of the module control signals; and

wherein the plurality of buffer circuits are distributed across a surface of the memory module such that each module control signal arrives at the plurality of buffer circuits at different points in time, and

wherein the respective buffer circuit is configured to regenerate a local clock signal from the module clock signal and provide the local clock signal to the respective group of memory devices.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS

ABSTRACT OF THE DISCLOSURE

A memory module is operatable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals from the memory controller and to output module command signals and module control signals. The module command signals are provided to memory devices organized in groups, each group including at least one memory device, while the module control signals are provided to a plurality of buffer circuits to control data paths in the buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices. The plurality of buffer circuits are distributed across a surface of the memory module such that each module control signal arrives at the plurality of buffer circuits at different points in time. The plurality of buffer circuits include clock regeneration circuits to regenerate a clock signal received from the module control device and to provide regenerated clock signals to respective groups of memory devices.

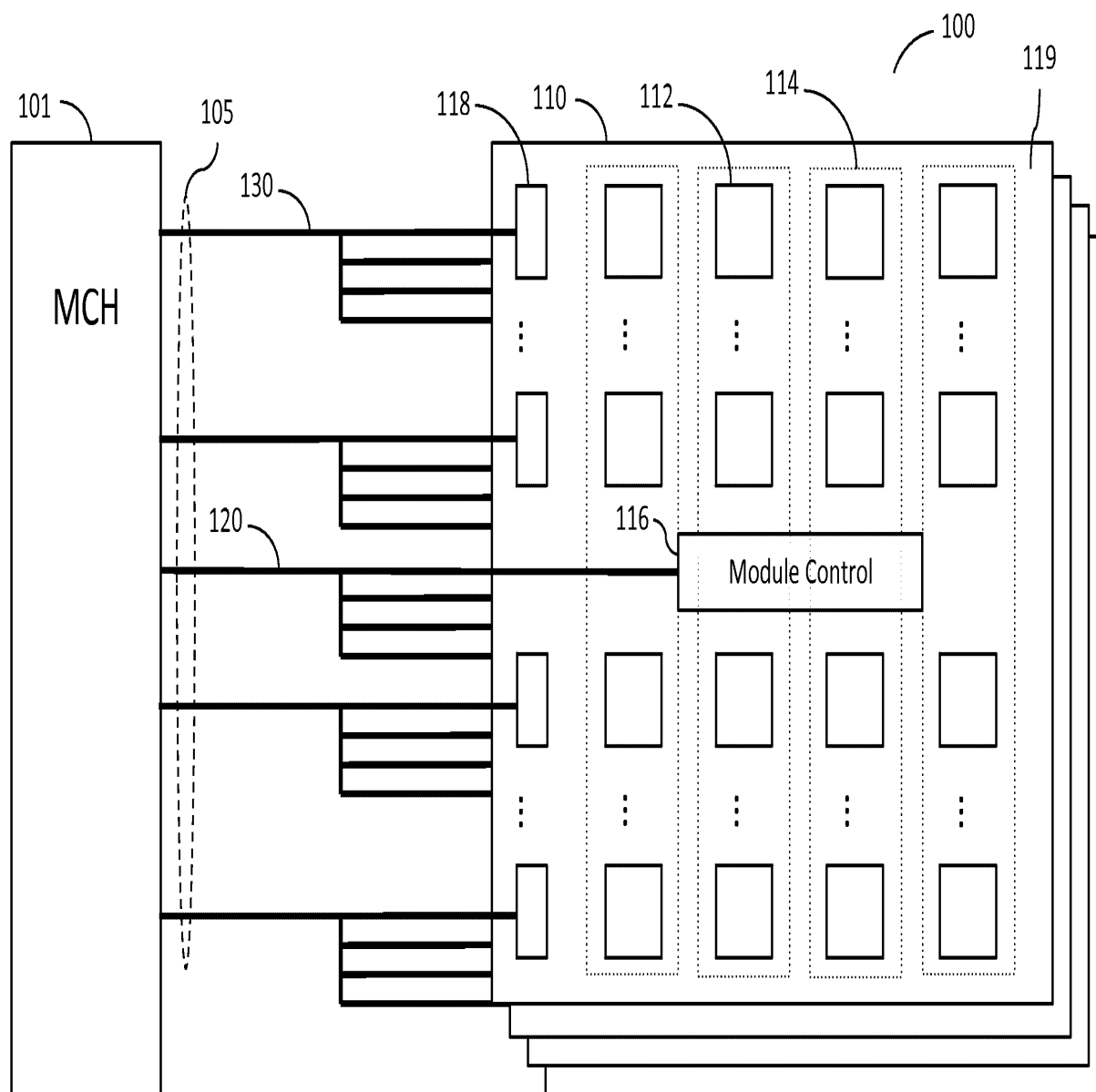


FIG. 1

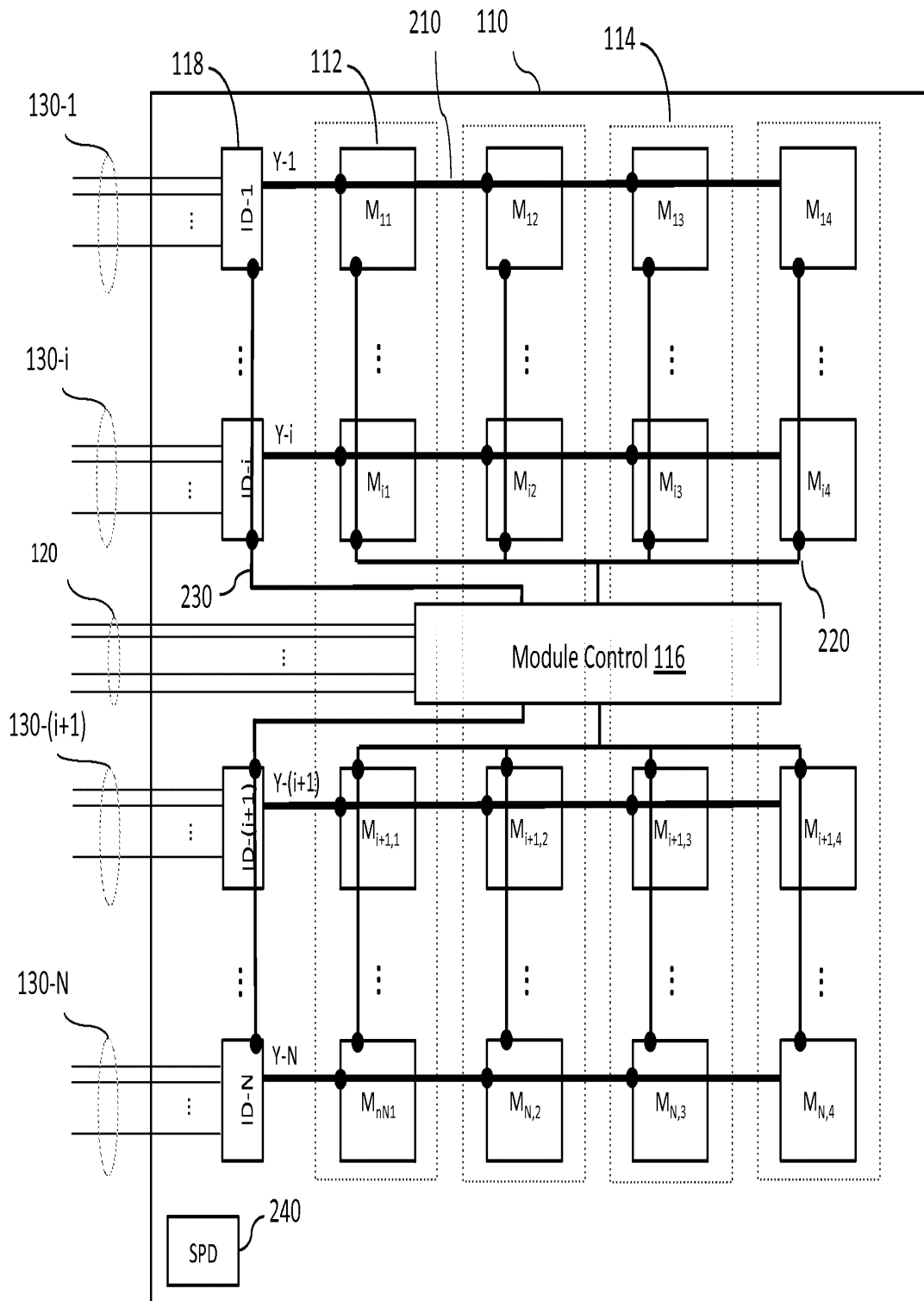


FIG. 2A

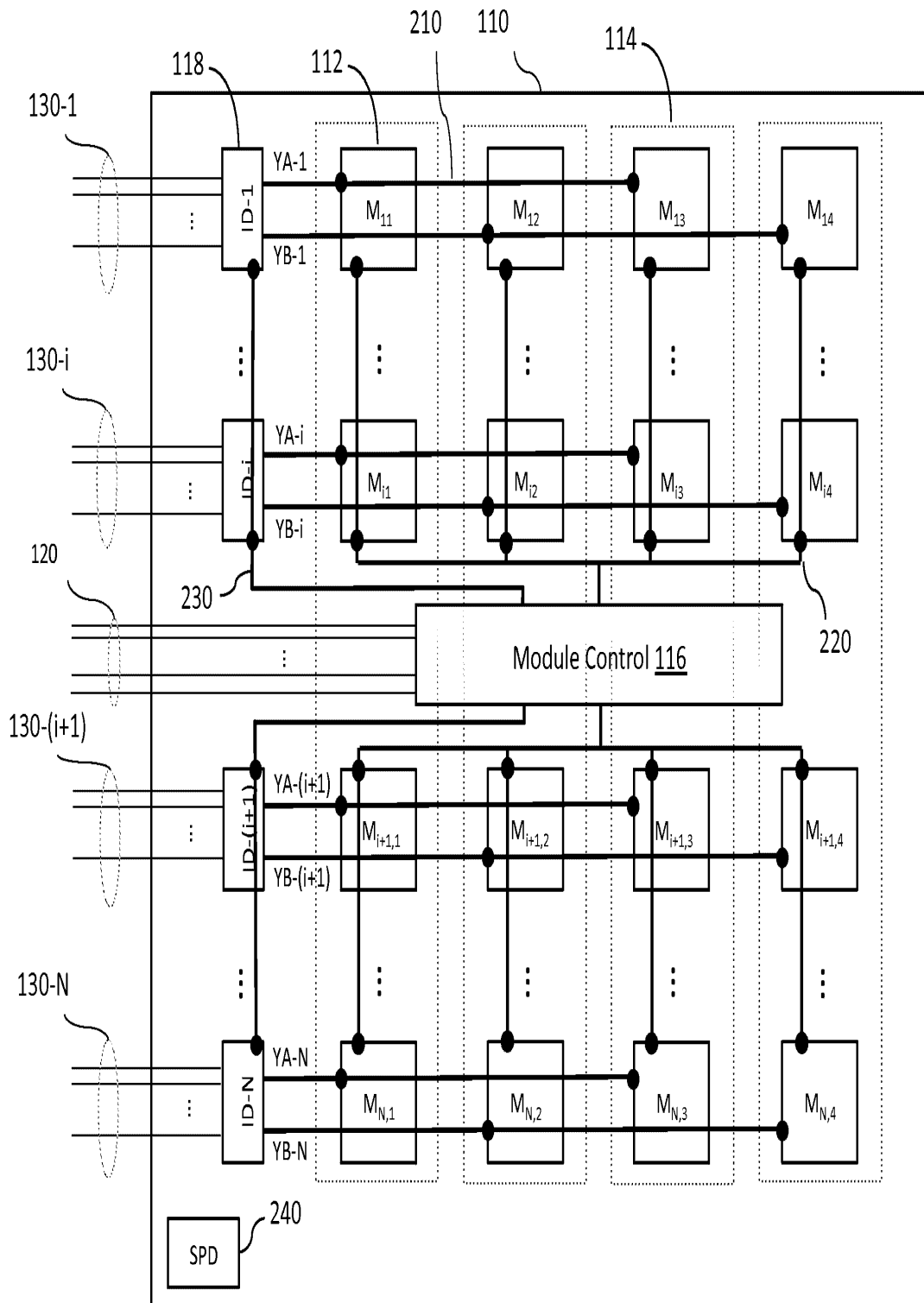


FIG. 2B

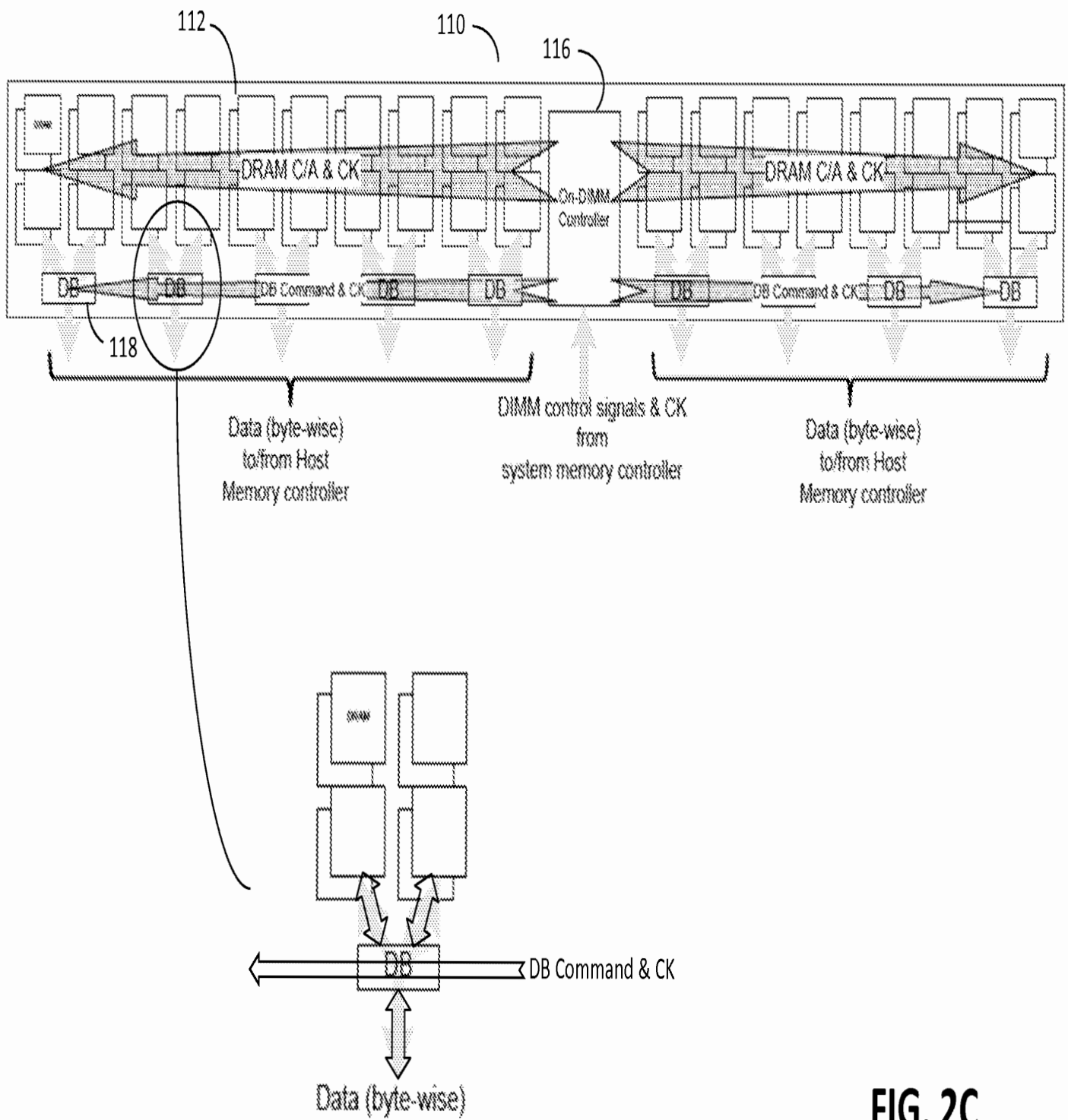


FIG. 2C

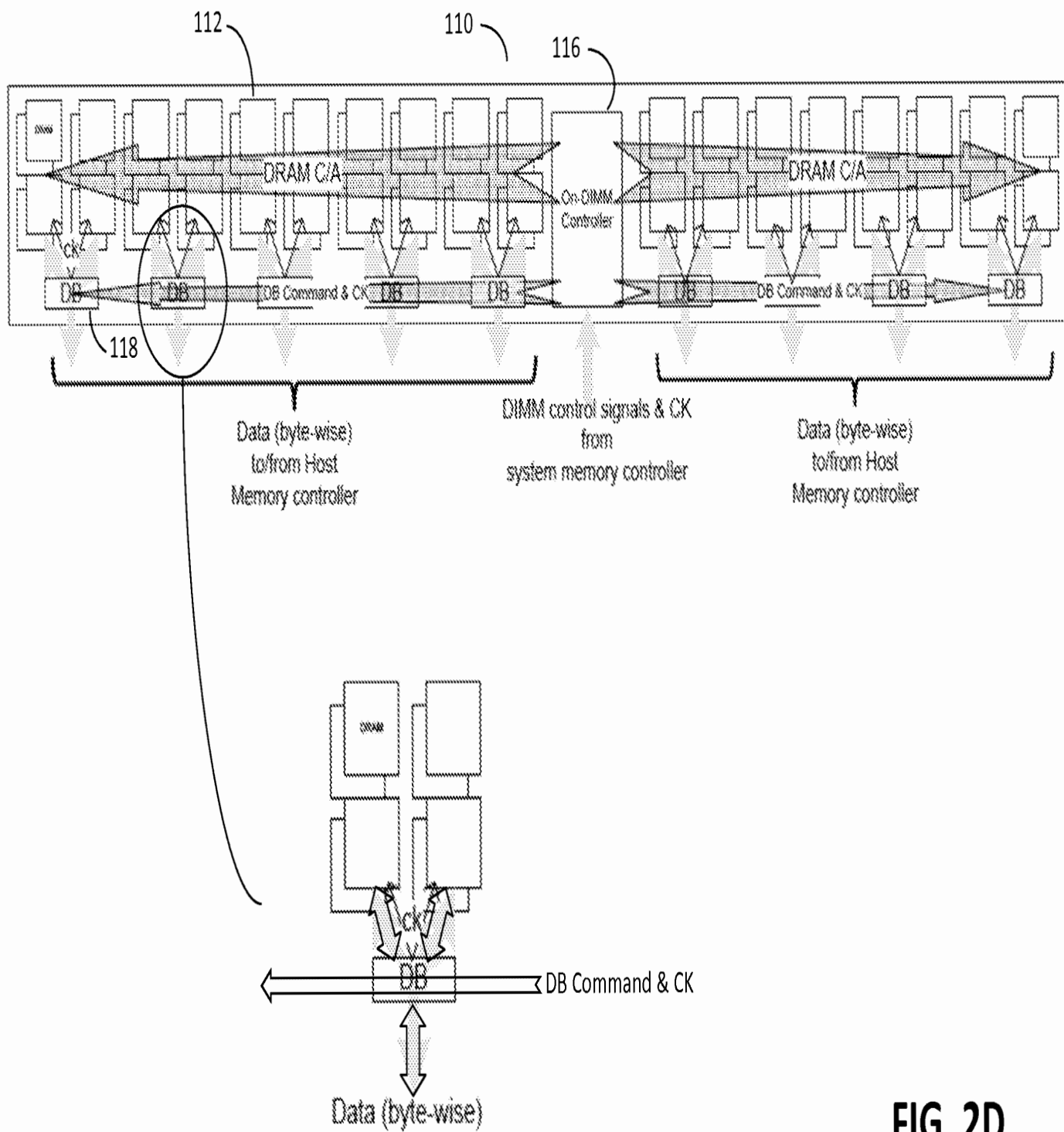


FIG. 2D

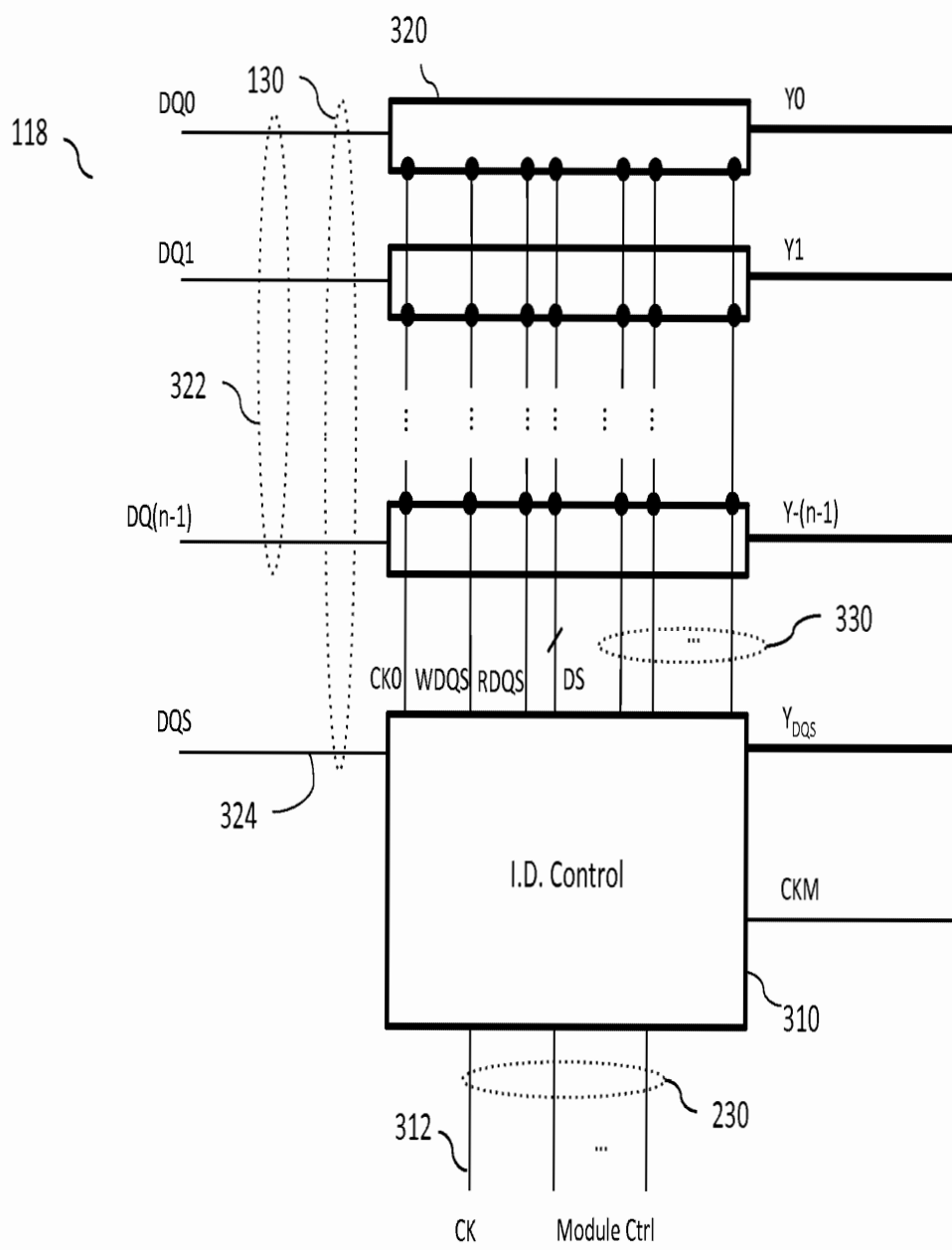


FIG. 3

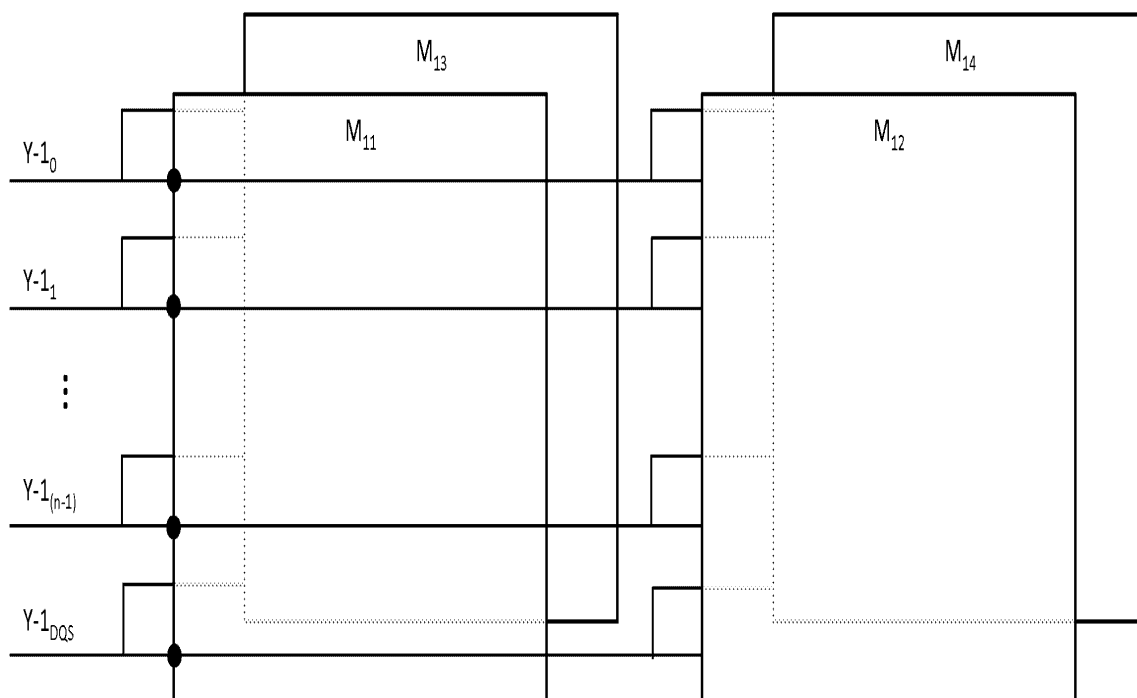


FIG. 4A

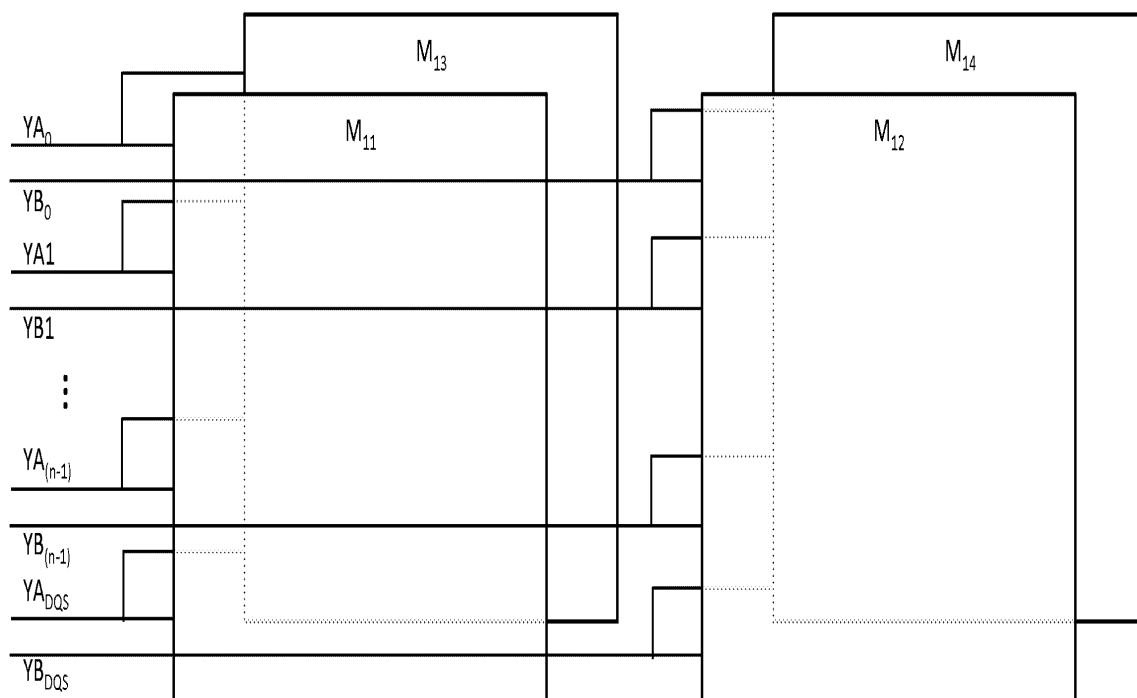


FIG. 4B

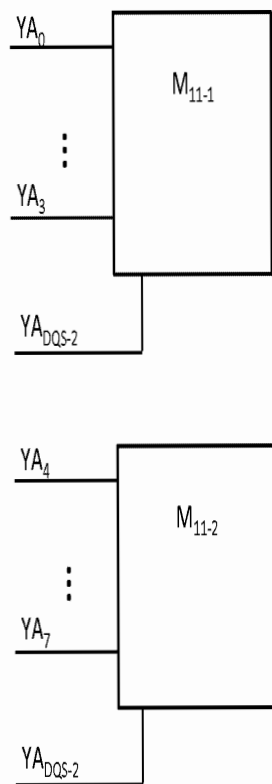


FIG. 5A

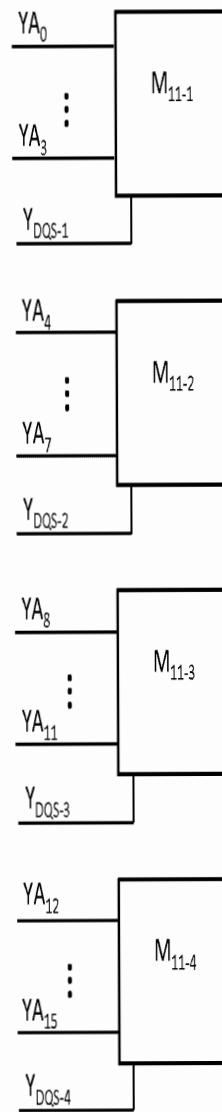


FIG. 5B

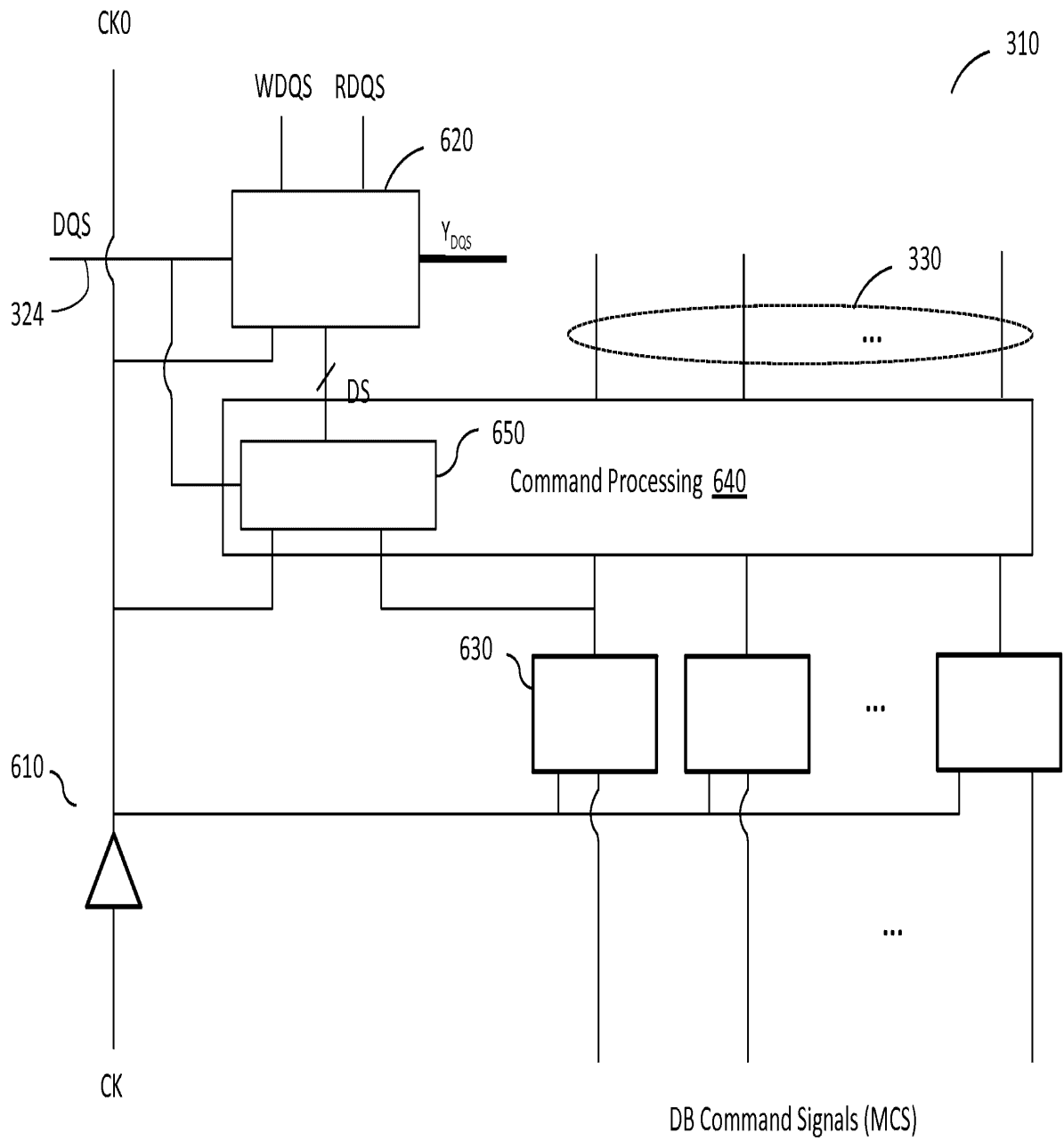


FIG. 6

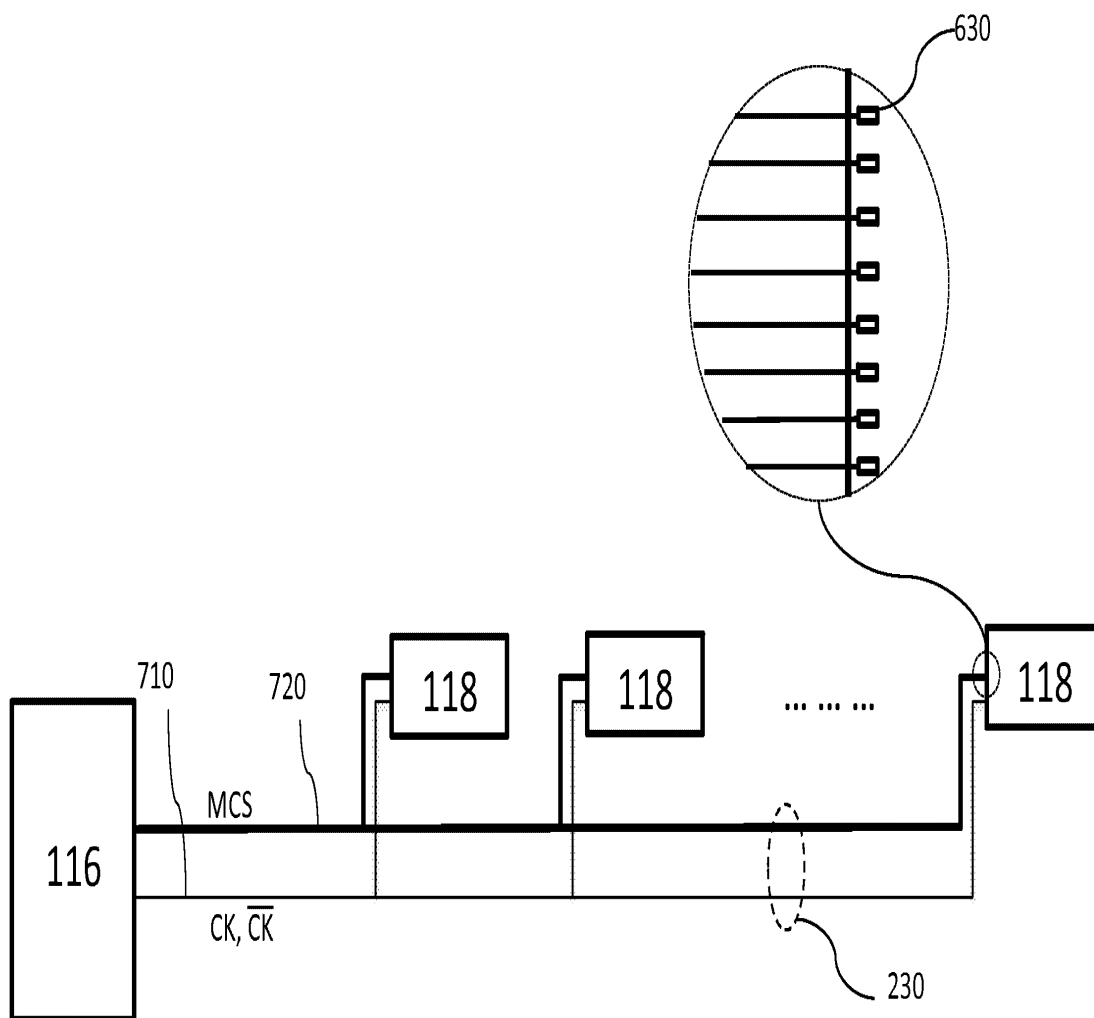


FIG. 7

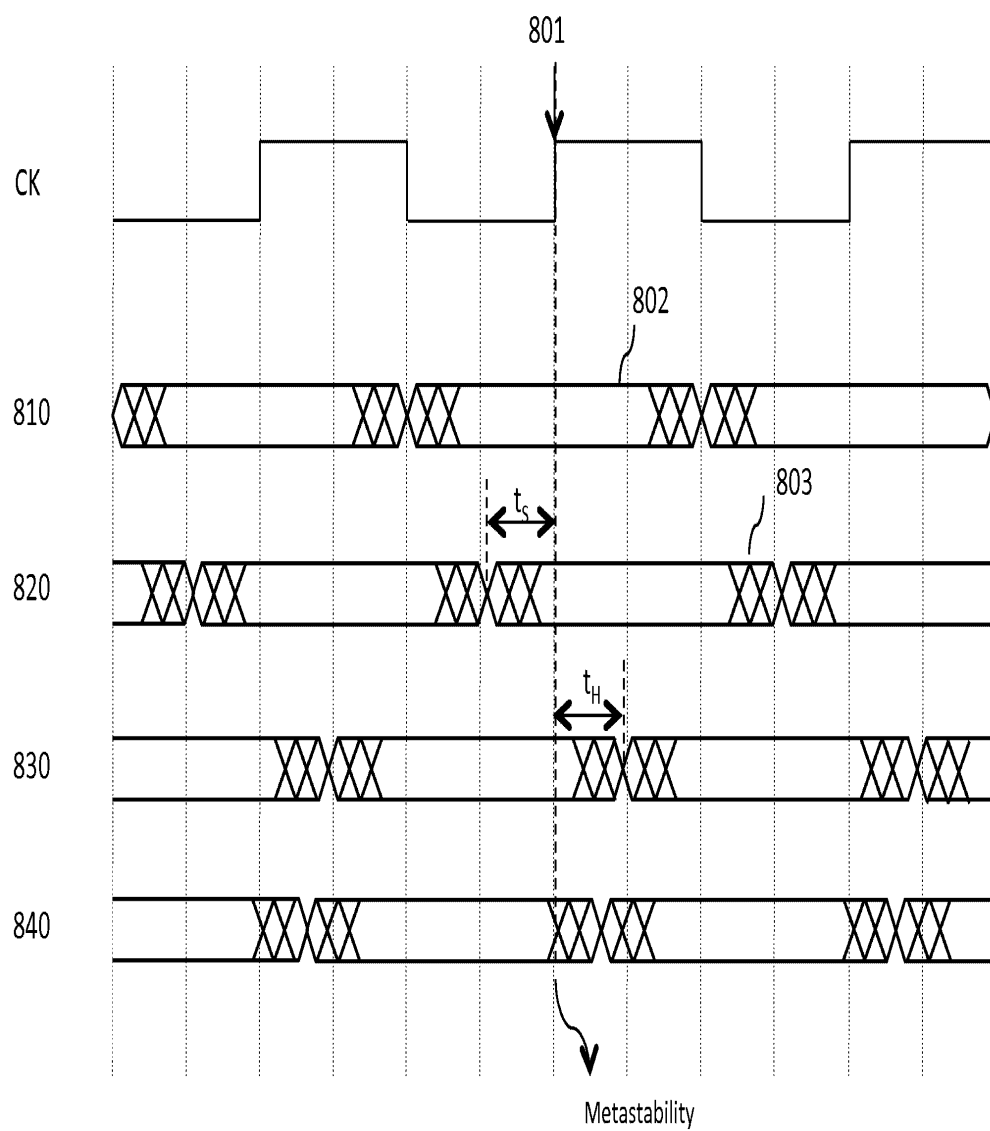


FIG. 8

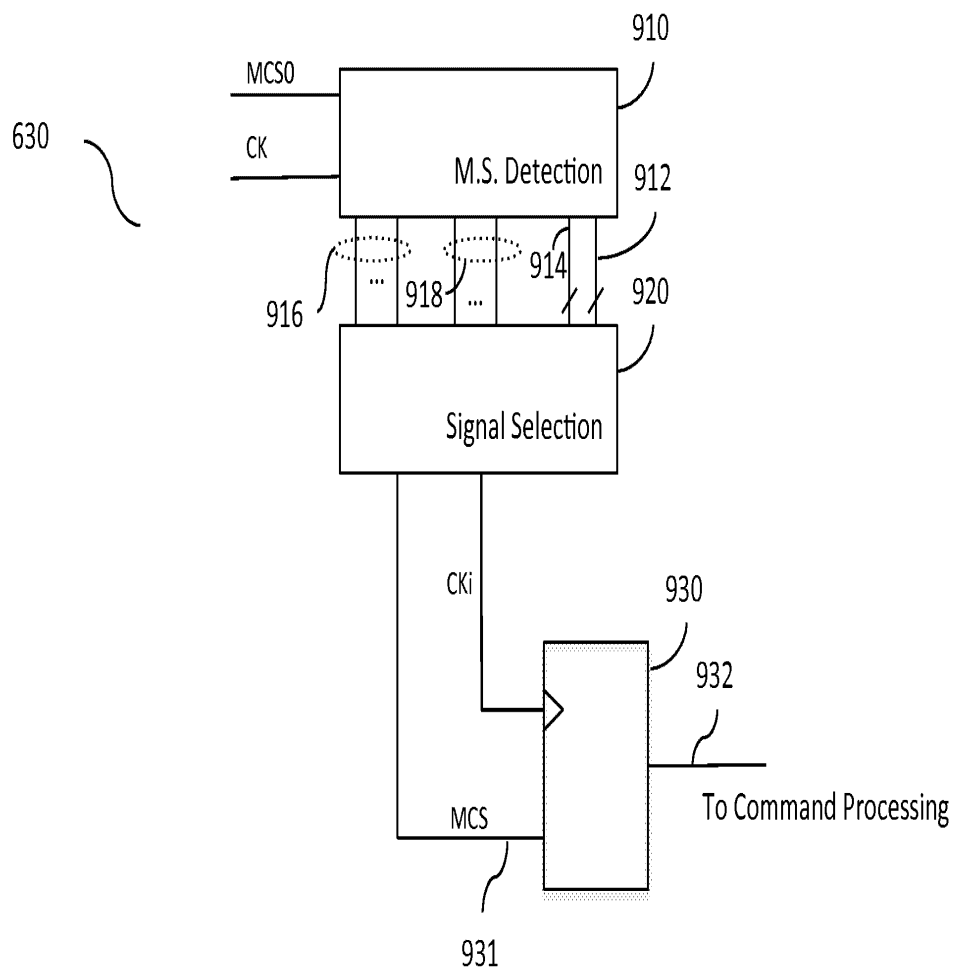


FIG. 9

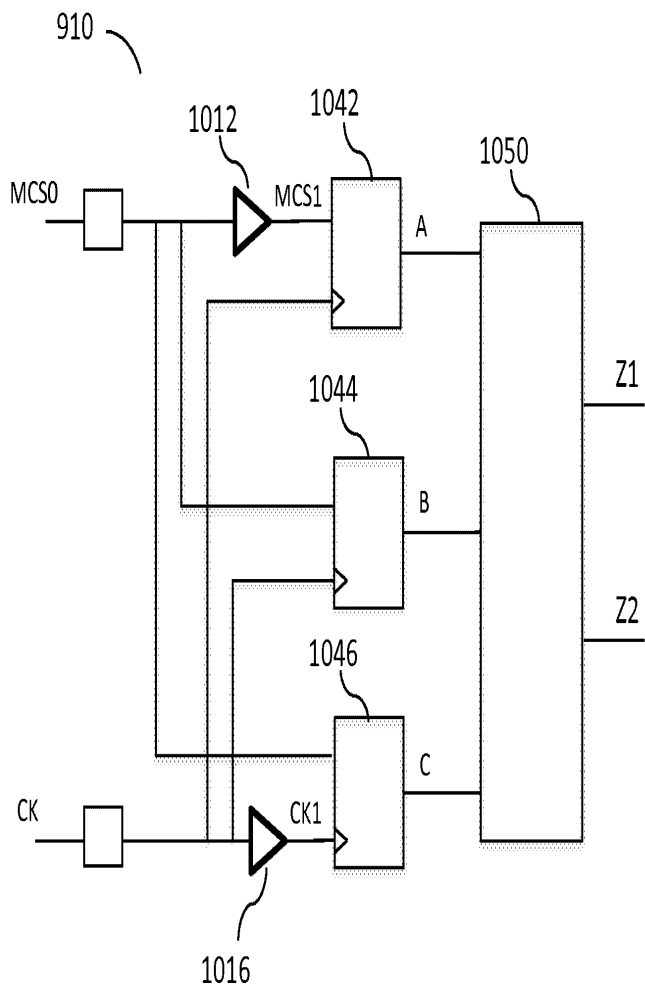


FIG. 10A

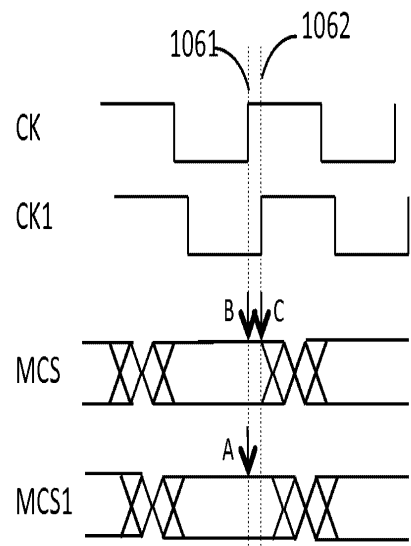


FIG. 10B

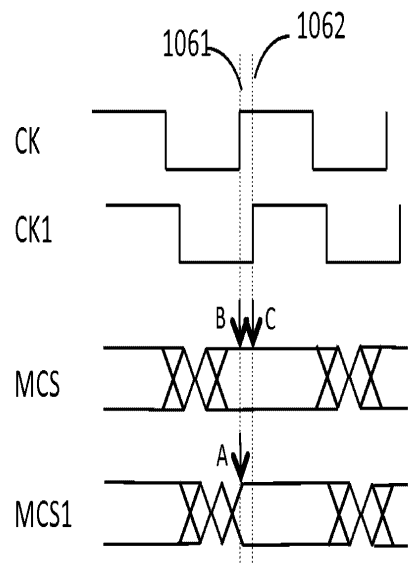


FIG. 10C

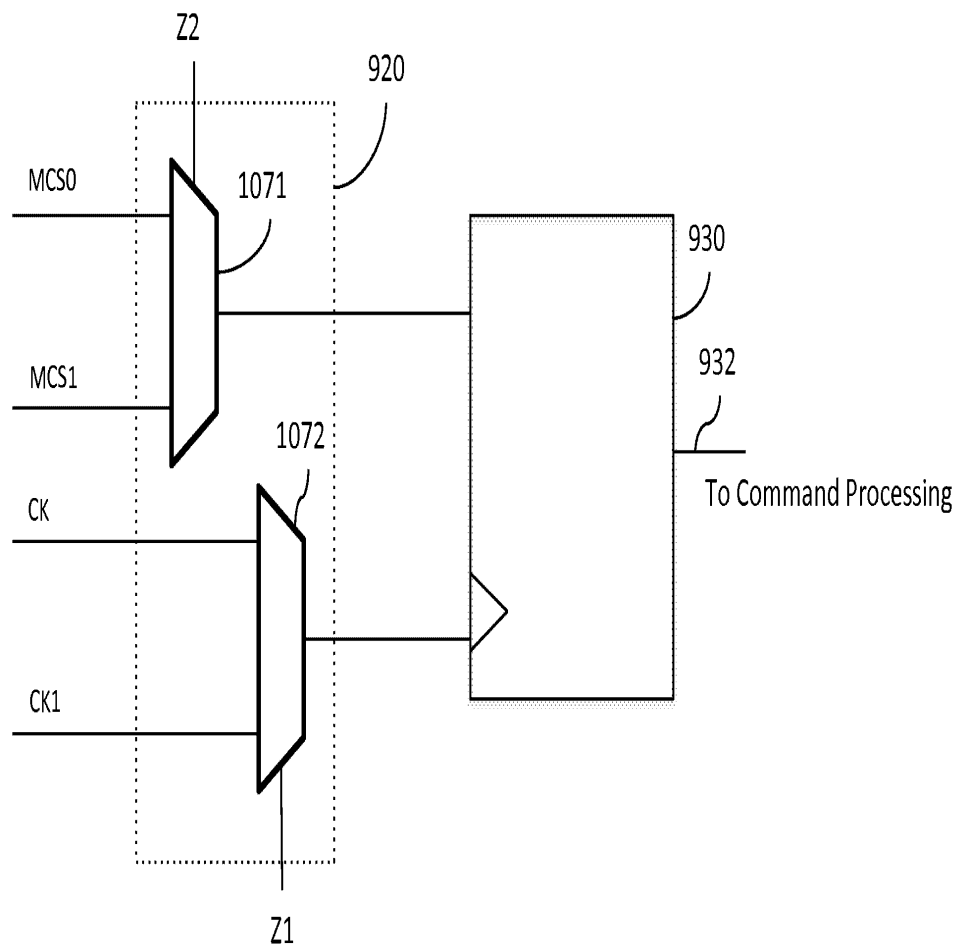


FIG. 10D

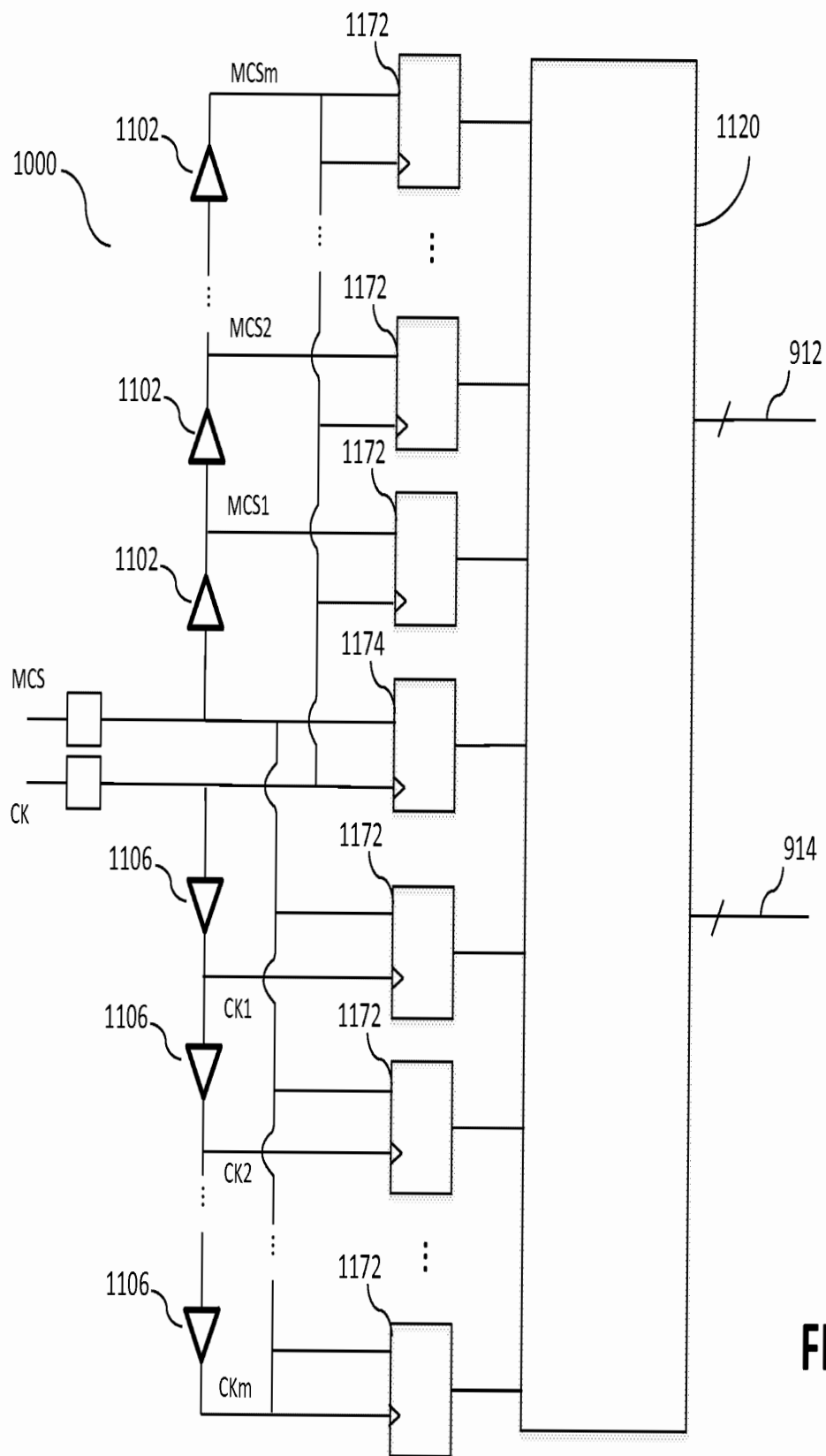


FIG. 11A

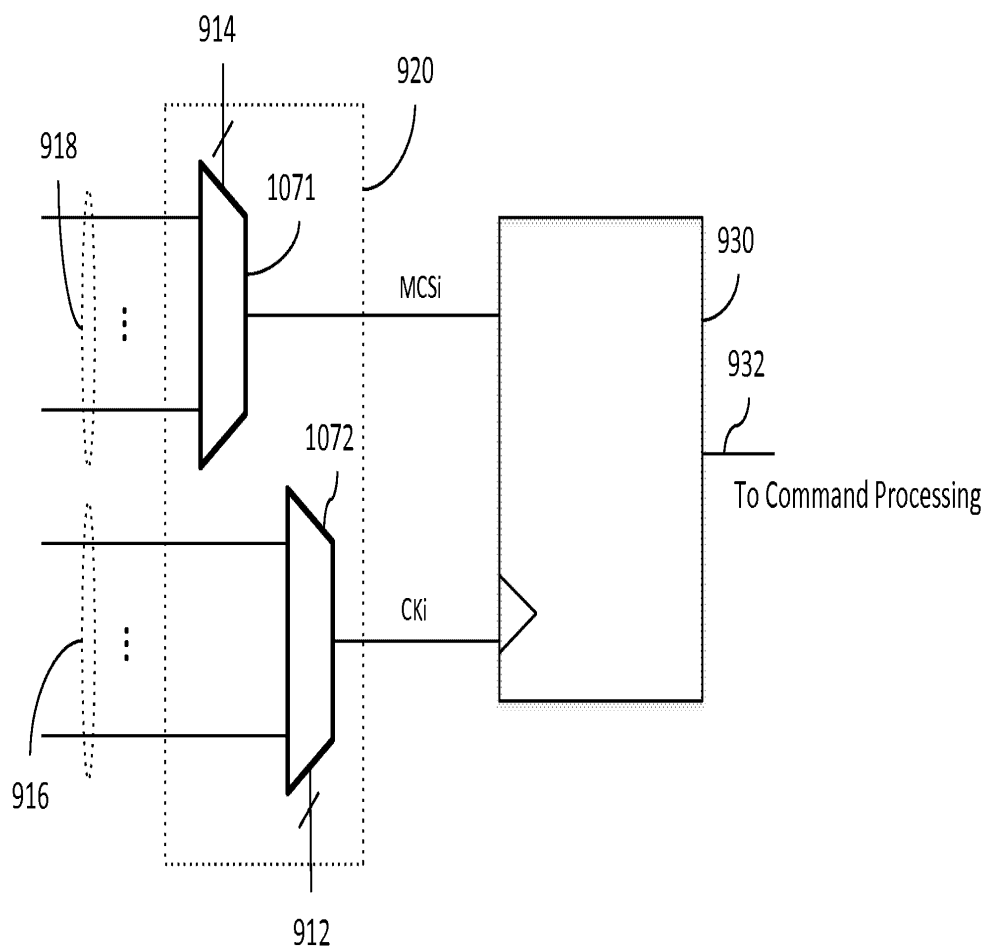


FIG. 11B

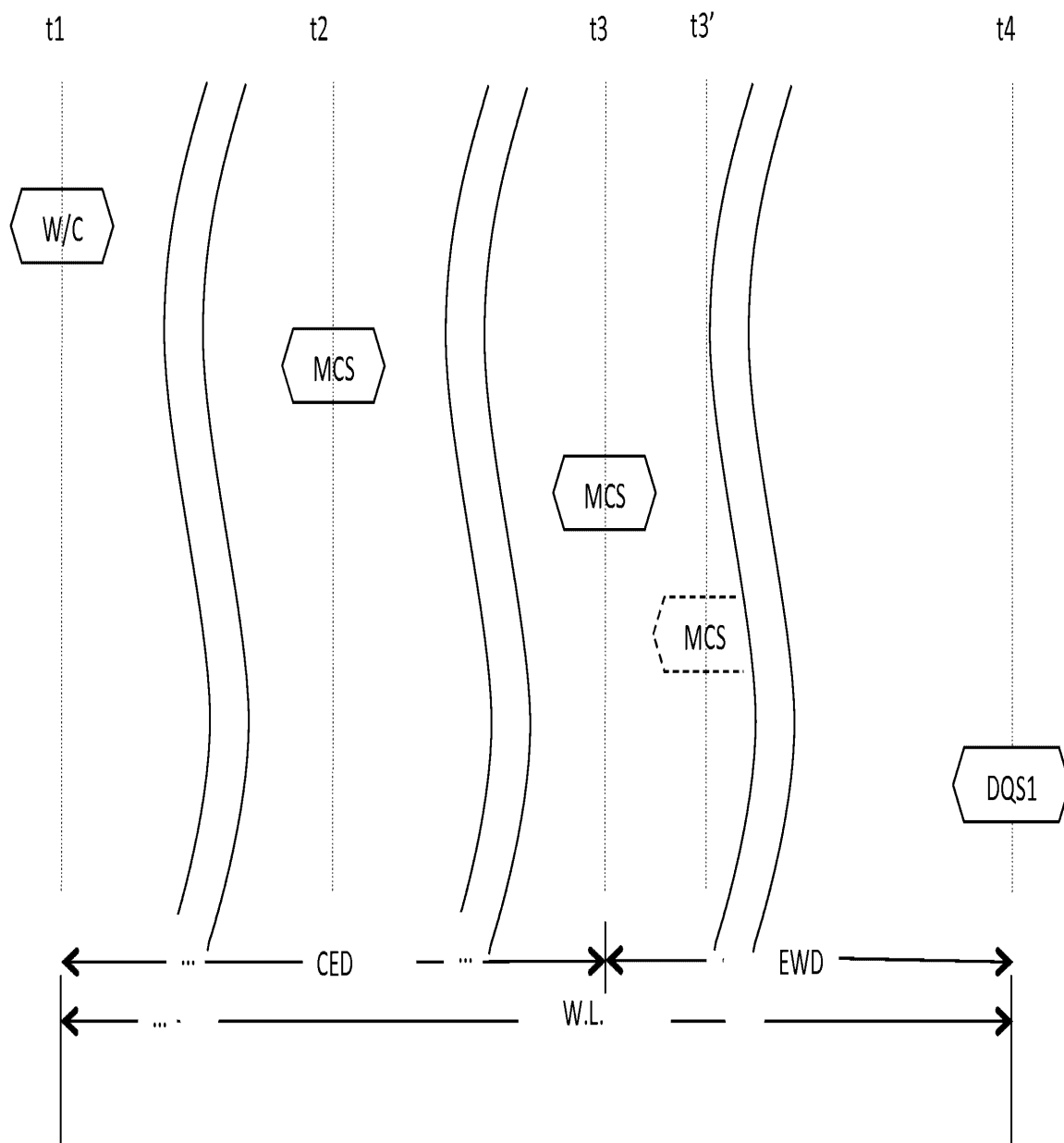


FIG. 12A

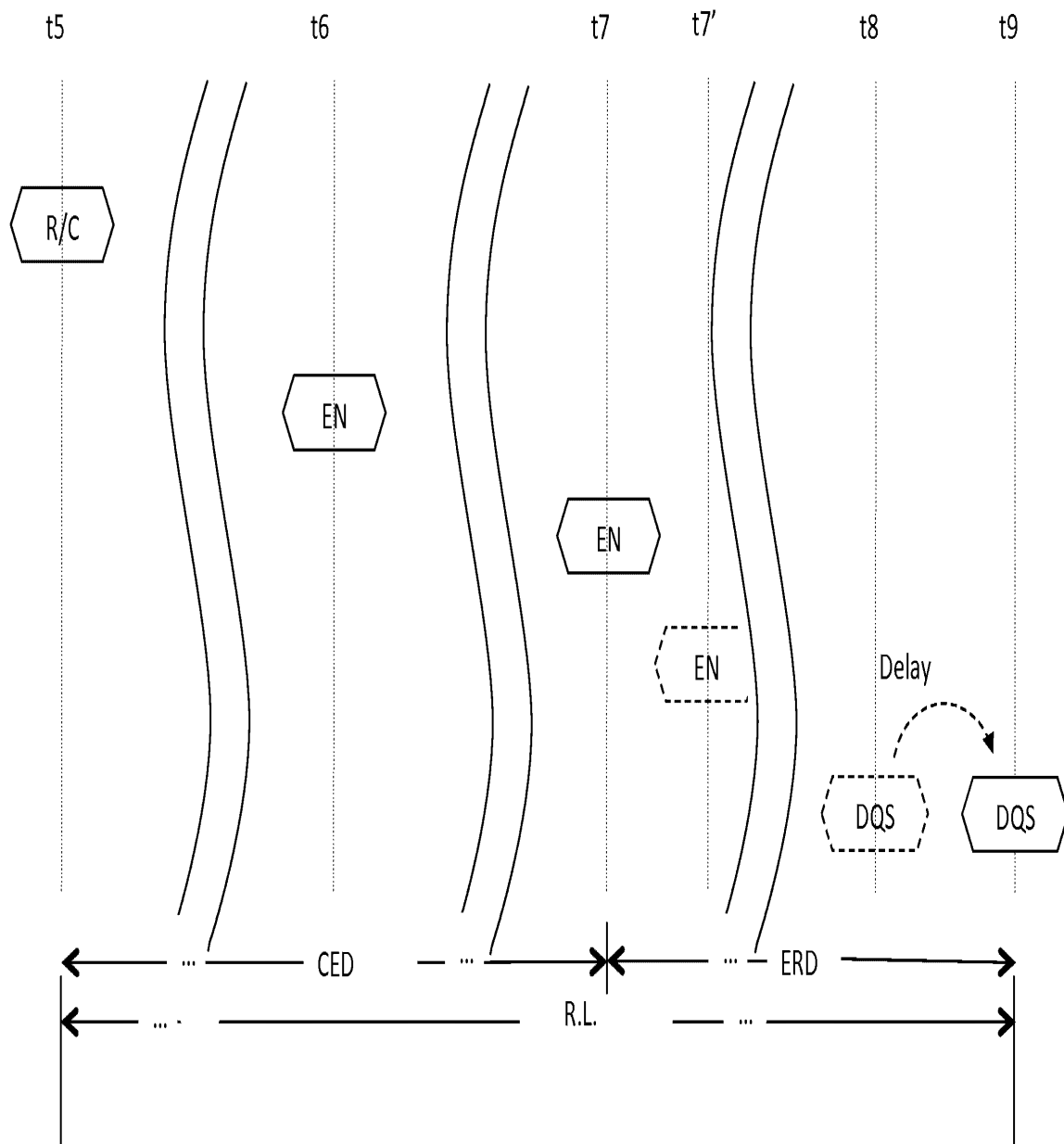


FIG. 12B

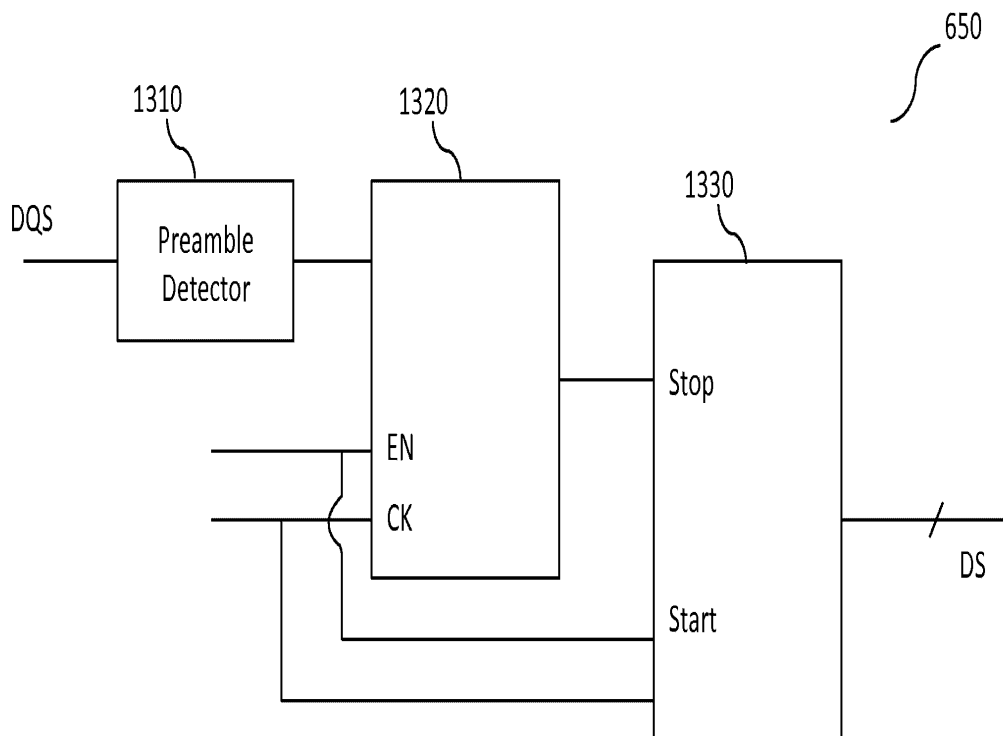


FIG. 13

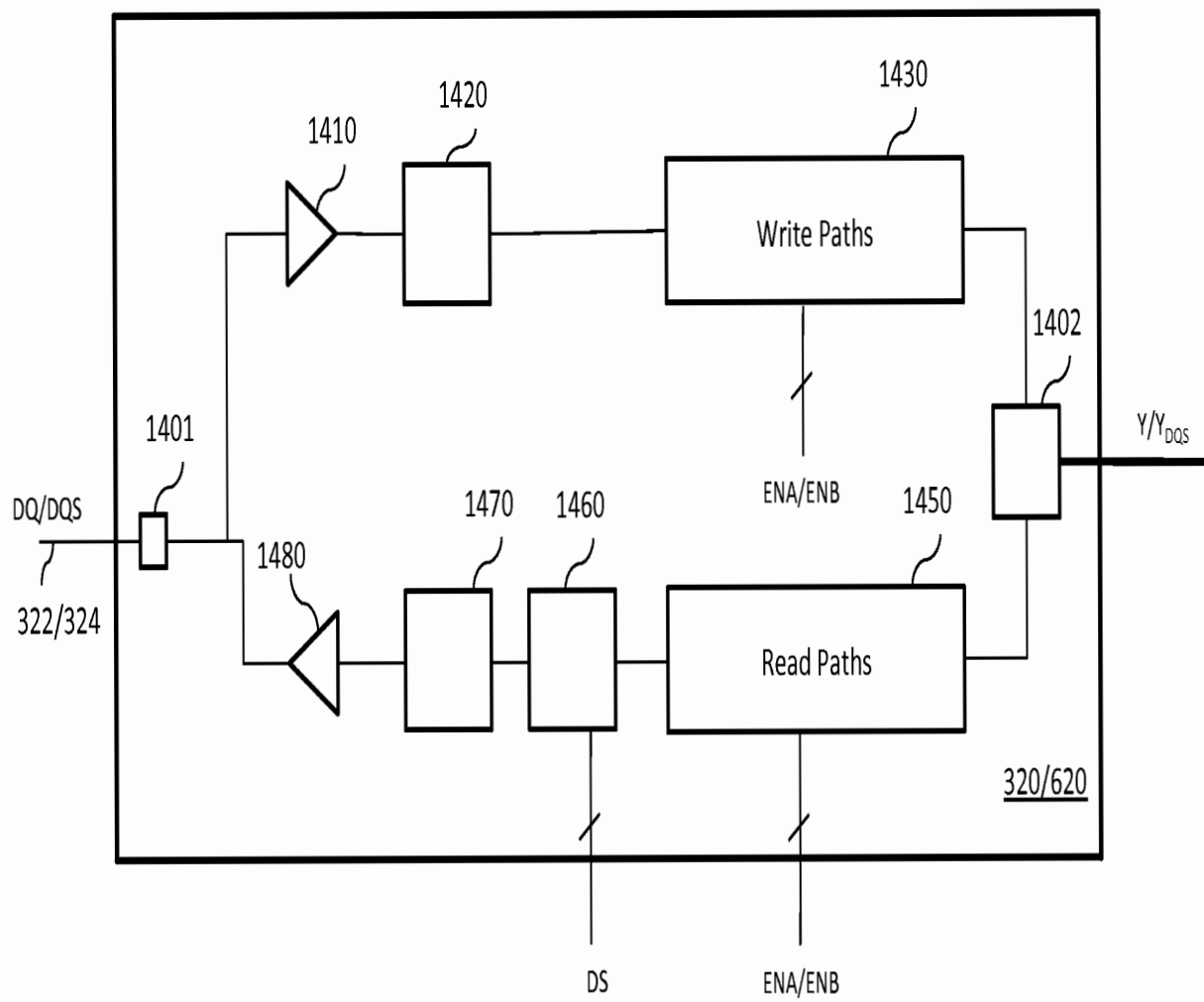


FIG. 14

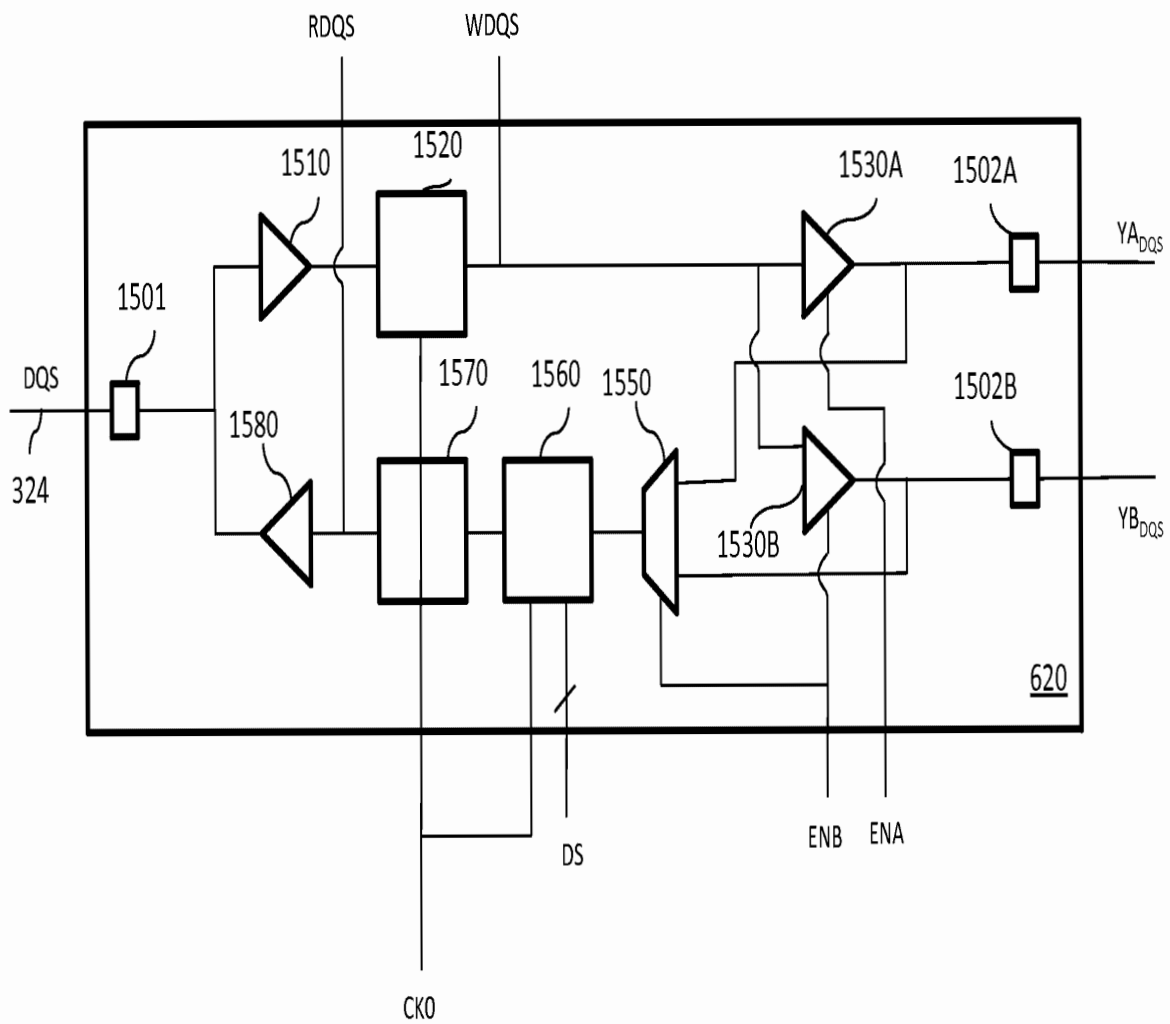


FIG. 15

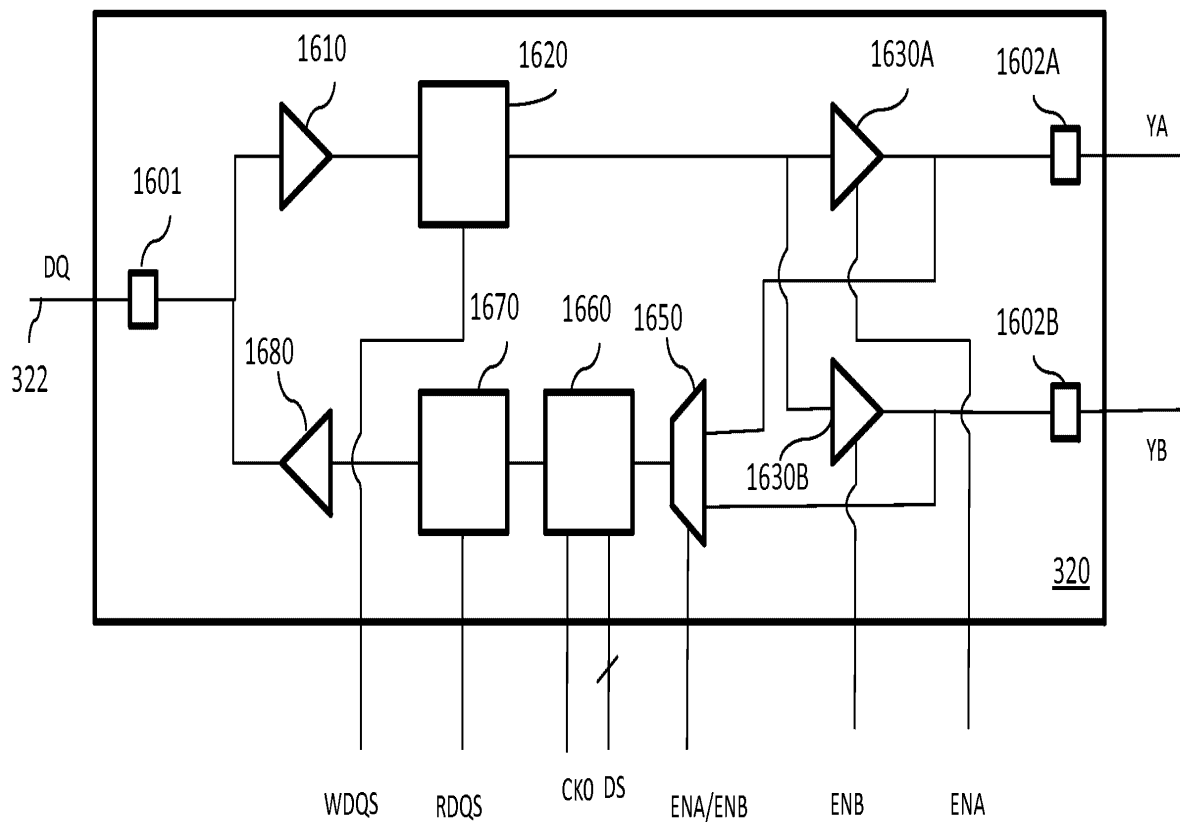


FIG. 16

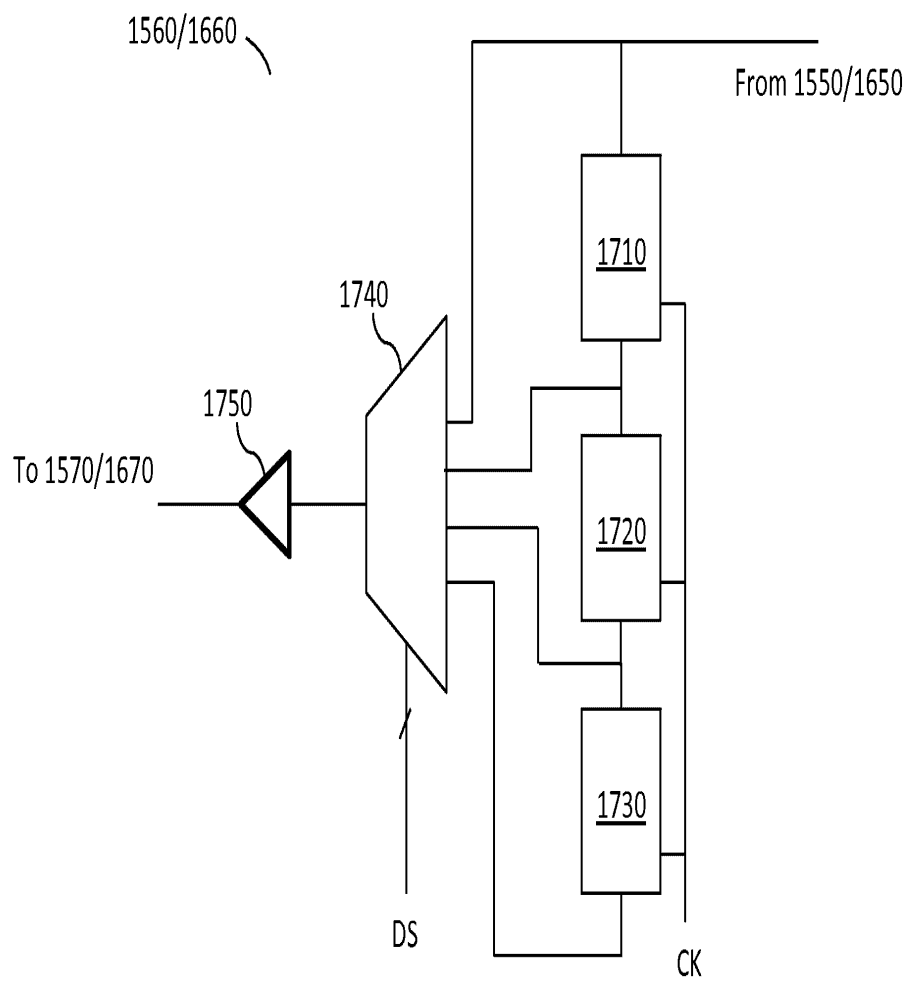


FIG. 17

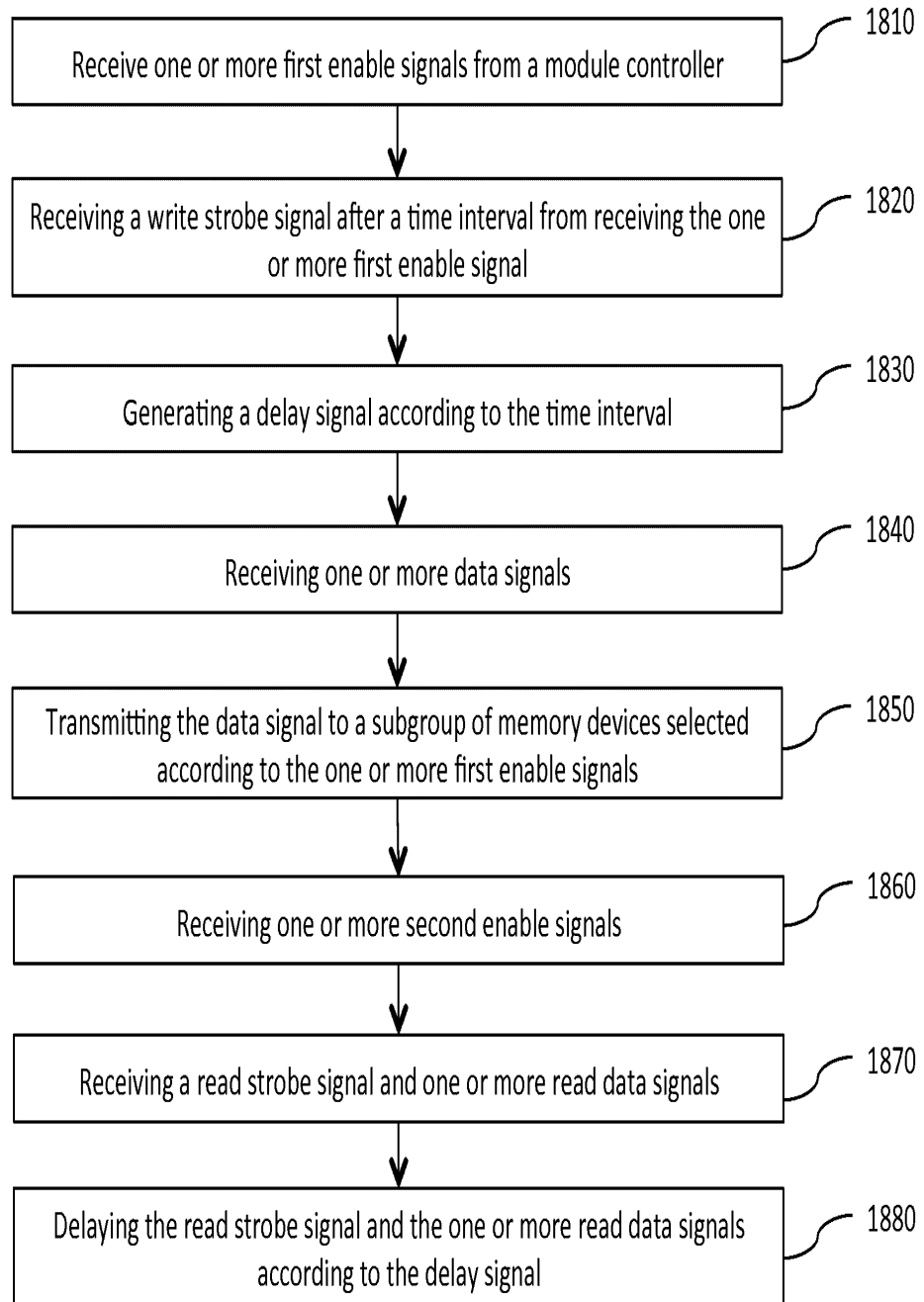


FIG. 18

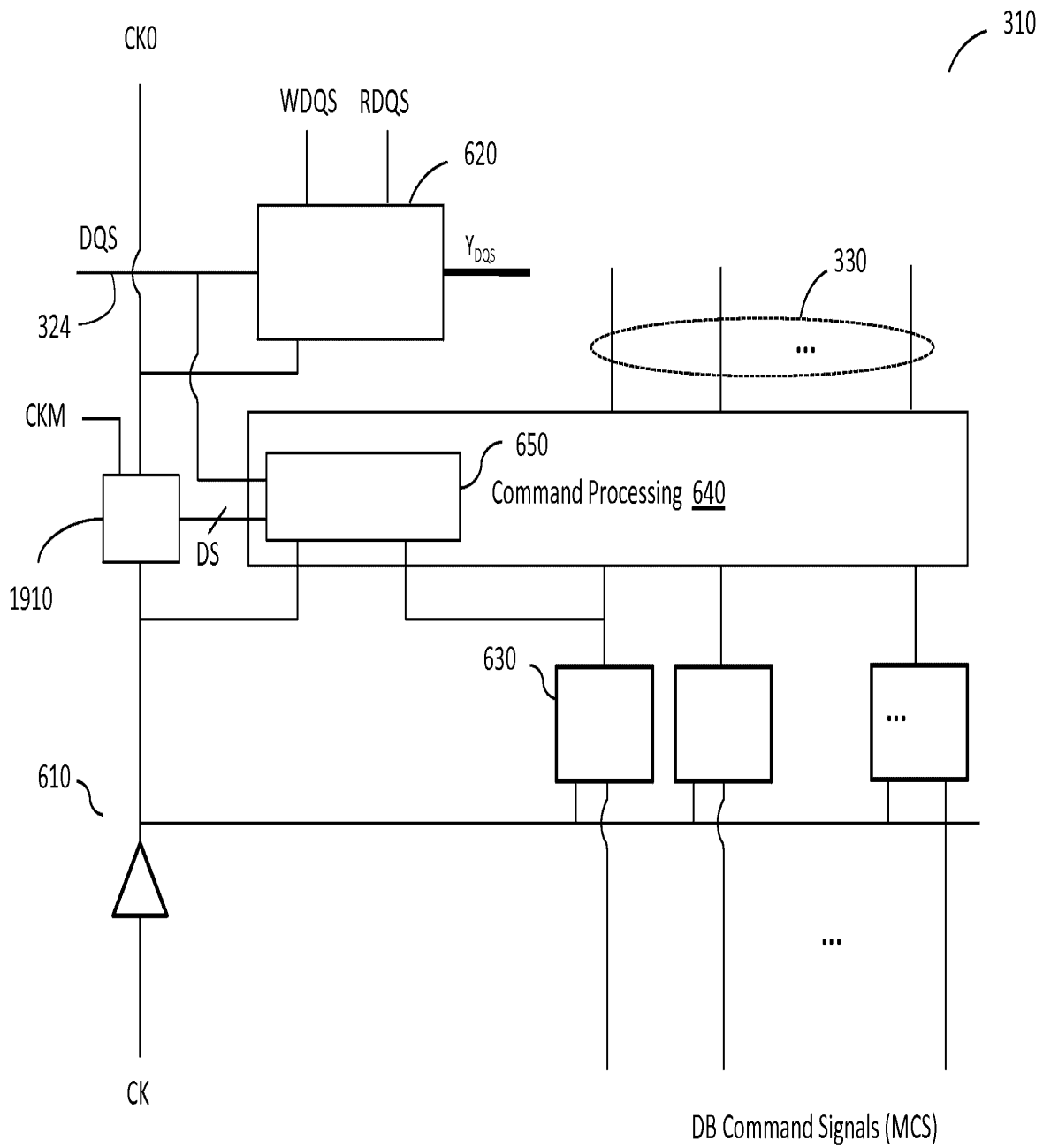


FIG. 19

Electronic Acknowledgement Receipt

EFS ID:	16434404
Application Number:	61859215
International Application Number:	
Confirmation Number:	4475
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First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
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Attorney Docket Number:	NT006
Receipt Date:	27-JUL-2013
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Time Stamp:	17:14:12
Application Type:	Provisional

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Provisional Cover Sheet (SB16)	2013-07-27_ProvisionalCoverSheet_NT006.pdf	2071341 b6679a3b1ddaedbc8d629349c659d6a53c8c94df	no	3

Warnings:

Information:

2		2013-07-27_Specification_NT006.pdf	341047 62211ebb833ab5bff9a9699f63982e7251912622	yes	32
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	29	
	Claims		30	31	
	Abstract		32	32	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	2013-07-27_Drawings_NT006.pdf	339506 5294bbbf27f6cf07f7e5350504249fbd7750e5c2	no	26
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61/859,215	07/27/2013	Hyun Lee	NT006

CONFIRMATION NO. 4475

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P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 08/05/2013

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.

Applicant must submit \$260 to complete the basic filing fee for an undiscounted entity. If appropriate, applicant may make a written assertion of entitlement to small entity status and pay the small entity filing fee (37 CFR 1.27) or make a certification of entitlement to micro entity status and pay the micro entity filing fee (37 CFR 1.29).

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$ **60** for an undiscounted entity, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$ **320** for an undiscounted entity

- \$ **260** Statutory basic filing fee.
- \$ **60** Surcharge.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.

<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/nhnguyen/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/859,215	07/27/2013		0.00	NT006		

CONFIRMATION NO. 4475

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 08/05/2013

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Power of Attorney:

Jamie Zheng--51167

If Required, Foreign Filing License Granted: 08/01/2013

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/859,215**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent

in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

ASSERTION OF ENTITLEMENT TO SMALL ENTITY STATUS UNDER 37 CFR 1.27

Applicant	: Lee et al.
App. No.	: 61/859,215
Filed	: July 27, 2013
For	: MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS
Examiner	: Unknown
Art Unit	: Unknown
Conf. No.	: 4475
Docket No.	: NT006

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The Assignee of all rights and title in the above-identified patent, Netlist, Inc., is a small entity concern as defined in 37 C.F.R. 1.27(a). Assertion of entitlement to small entity status under 37 C.F.R. 1.27(c) is hereby made in this application.

Applicants acknowledge the duty to file notification of a loss of entitlement to small entity status in the patent prior to paying, or at the time of paying, any maintenance fee due after the date on which status as a small entity is no longer appropriate.

Respectfully submitted,

Date: September 23, 2013

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	61859215			
Filing Date:	27-Jul-2013			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NT006			
Filed as Small Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Provisional Application Filing Fee	2005	1	130	130
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Provisional Filing Fee/Cover Sheet	2052	1	30	30
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	16931395
Application Number:	61859215
International Application Number:	
Confirmation Number:	4475
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NT006
Receipt Date:	23-SEP-2013
Filing Date:	27-JUL-2013
Time Stamp:	20:02:58
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$160
RAM confirmation Number	6910
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Assertion of entitlement to small entity status	2013-09-23_Small_Entity_Status_NT006.pdf	73365 f755287671f8636289263bb5ca72a3dabca92fce	no	1
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	31862 925c321a0396baeab739e6f80ef15e28aeebc0b1	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			105227		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/859,215	07/27/2013		160	NT006		

CONFIRMATION NO. 4475

UPDATED FILING RECEIPT

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



Date Mailed: 10/01/2013

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Power of Attorney:

Jamie Zheng--51167

If Required, Foreign Filing License Granted: 08/01/2013

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/859,215**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

**** SMALL ENTITY ****

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION USING DISTRIBUTED DATA BUFFERS

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.